ANNEX 1 (part A)

Bio-based Industries Research and Innovation action

NUMBER — 745828 — PERCAL

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1.1. The project summary

Project Number [1]	745828	Project Acronym [2]	PERCAL

One form per project

General information	
Project title [3]	Chemical building blocks from versatile MSW biorefinery
Starting date [4]	01/07/2017
Duration in months [5]	36
Call (part) identifier [6]	H2020-BBI-JTI-2016
Topic	BBI-2016-R08 Emerging technologies for conversion of the organic content of Municipal Solid Waste and improving waste-to-chemicals value chains
Fixed EC Keywords	
Free keywords	Bioethanol, Municipal Solid Waste, Succinic acid, Lactic acid, surfactants, adhesives, solvents, polyester polyols, fermentation, membrane electrolysis, enzymatic hydrolysis

Abstract [7]

PERCAL will exploit Municipal Solid Waste (MSW) as feedstock to develop intermediate chemical products at high yield and low impurity level with huge industrial interest. These will be complementary to the bioethanol (current PERSEO technology), to achieve a cascade valorisation of the MSW components, i.e.:
• Lactic acid (LA) to produce: 1) Eco-friendly ethyl lactate solvents by reactive distillation from lactic acid & bio-ethanol to be used in cleaning products and inks and 2) hot-melt adhesives for cardboard and other non-food applications in combination with maleic anhydride by reactive extrusion.
• Succinic acid (SA) as an intermediate building blocks to production of polyols for the polyurethane industry.
• Biosurfactants by chemical and/or microbiological modification of protein and lipid fraction from remaining fraction of MSW fermentation.
IIn order to minimize the MSW heterogeneous composition (due to seasonal and origin variability driven by local economic, social and climate conditions) limitations as a fermentation feedstock, four main innovations will be proposed: i) New enzymatic cocktails to maximize hydrolysis of fermentable organic matter with low inhibitors production, ii) the use of high yield, specific and robust strains for each selected acid, iii) the extraction of fermentation by-products acting as inhibitors to succinic acid production via novel membrane electrolysis employing an integrated continuous fermentation coupled with simultaneous organic acid removal in comparison with SA sequential fermentation followed by its separation using an electrodialysis-based and iv) optimize simultaneous saccharification and fermentation for lactic acid production followed by a downstream separation process based on membrane electrodialysis. To maximize the yield and purity of target organic acids, continuous and single step fermentation process will be prioritized in order to allow their integration in the PERSEO plant.

1.2. List of Beneficiaries

Project Number [1]	745828	Project Acronym [2]	PERCAL

List of Beneficiaries

No	Name	Short name	Country	Project entry month[8]	Project exit month
1	INDUSTRIAS MECANICAS ALCUDIA SA	IMECAL	Spain	1	36
2	AIMPLAS - ASOCIACION DE INVESTIGACION DE MATERIALES PLASTICOS Y CONEXAS	AIMPLAS	Spain	1	36
3	LEIBNIZ-INSTITUT FUER AGRARTECHNIK UND BIOOEKONOMIE EV	ATB	Germany	1	36
4	AGRICULTURAL UNIVERSITY OF ATHENS	AUA	Greece	1	36
5	UNIVERSITEIT GENT	UGent	Belgium	1	36
6	FUNDACION CENER-CIEMAT	CENER	Spain	1	36
7	TBW RESEARCH GESMBH	TBWR	Austria	1	36
8	IRIS ADVANCED ENGINEERING LIMITED	VISUM	Ireland	1	36
9	SO.F.TER. SPA	SOFTER	Italy	1	36
10	HAYAT KIMYA SANAYI ANONIM SIRKETI	HAYAT	Turkey	1	36
11	COVESTRO DEUTSCHLAND AG	COVESTRO	Germany	1	36
12	EXERGY LTD	EXERGY	United Kingdom	1	36

1.3. Workplan Tables - Detailed implementation

1.3.1. WT1 List of work packages

WP Number[9]	WP Title	Lead beneficiary[10]	Person-months[11]	Start month[12]	End month[13]
WP1	Characterization, handling and pre-processing of MSW from different localizations and seasons. Final products requirements definition	1 - IMECAL		1	18
WP2	Microbial conversions of MSW to lactic acid.	3 - ATB		2	32
WP3	Microbial conversions of MSW to succinic acid.	5 - UGent		3	34
WP4	Valorisation of fermentation by-products	6 - CENER		8	34
WP5	Green chemical routes to produce solvents, adhesives & polymers from developed acids.	11 - COVESTRO		10	34
WP6	Study of adaptation and monitoring of PERSEO plant for the production of ethanol, LA & SA with suitable quality to produce the selected final products.	8 - VISUM		12	36
WP7	Environmental, regulatory & economic analyses.	4 - AUA		1	36
WP8	Communication, Dissemination, Exploitation, Business plan and Technology Transfer	12 - EXERGY		1	36
WP9	Project Management	1 - IMECAL		1	36
WP10	Ethics requirements	1 - IMECAL	N/A	1	36
		Total	447.50		

1.3.2. WT2 list of deliverables

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.1	Selected final products and intermediates requirements.	WP1	10 - HAYAT	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D1.2	Variability of biogenic fraction of MSW.	WP1	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	8
D1.3	Report on hydrolysis improvements.	WP1	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	12
D1.4	Report on selected feedstocks.	WP1	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	14
D1.5	Report on validation of fermentation processes	WP1	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D2.1	Lactic acid fermentation process	WP2	3 - ATB	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D2.2	Down-stream processing	WP2	7 - TBWR	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D2.3	Scaling of the full process.	WP2	3 - ATB	Report	Confidential, only for members of the consortium (including the Commission Services)	32

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D3.1	Selection of optimal bacterial strain using different MSW hydrolysates.	WP3	4 - AUA	Report	Confidential, only for members of the consortium (including the Commission Services)	8
D3.2	Report on the integrated SA production with electrolysis membrane extraction.	WP3	5 - UGent	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D3.3	Report on production and purification of SA using electrolysis or electrodialysis based processes	WP3	4 - AUA	Report	Confidential, only for members of the consortium (including the Commission Services)	28
D3.4	Report on process scale-up	WP3	5 - UGent	Report	Public	32
D4.1	Characterization of the fermentation residue.	WP4	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D4.2	Pre-treatments for revalorization of the protein fraction.	WP4	6 - CENER	Report	Confidential, only for members of the consortium (including the Commission Services)	26
D4.3	Pre-treatments for revalorization of the lipid fraction.	WP4	6 - CENER	Report	Confidential, only for members of the consortium (including the Commission Services)	26
D4.4	Biosurfactant production from lipids and proteins.	WP4	10 - HAYAT	Report	Confidential, only for members of the consortium (including the Commission Services)	34
D5.1	Production of Ethyl lactate at laboratory level.	WP5	7 - TBWR	Report	Confidential, only for members of the consortium (including the	24

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
					Commission Services)	
D5.2	Ethyl-Lactate scale up and industrial validation.	WP5	10 - HAYAT	Report	Public	34
D5.3	Production of HMA at laboratory level.	WP5	2 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D5.4	HMA scale up and industrial validation	WP5	9 - SOFTER	Report	Public	34
D5.5	Production of polyols at laboratory level.	WP5	2 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D5.6	Polyols scale up and industrial validation	WP5	11 - COVESTRO	Report	Public	34
D6.1	Adaptation of PERSEO plant for the versatile production of chemical intermediates & identification of key parameters	WP6	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D6.2	Monitoring technologies and testing results.	WP6	8 - VISUM	Report	Confidential, only for members of the consortium (including the Commission Services)	30
D6.3	Integration of the production in a biotechnological plant.	WP6	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	34
D7.1	Draft report on economic analysis	WP7	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D7.2	Simulation of the system and M&E balances	WP7	12 - EXERGY	Report	Confidential, only for members of the consortium (including the	32

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
					Commission Services)	
D7.3	Report on Life Cycle Assessment (LCA).	WP7	4 - AUA	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D7.4	Report on Social Life Cycle Assessment (S-LCA).	WP7	12 - EXERGY	Report	Public	34
D7.5	Report on regulatory analysis and safety aspects	WP7	2 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D7.6	Final report on economic analysis	WP7	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D8.1	Project web site	WP8	2 - AIMPLAS	Report	Public	3
D8.2	Initial Communication and dissemination Plan	WP8	2 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D8.3	Draft communication activities & dissemination Plan updates	WP8	2 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D8.4	IPR policy implementation in line with Exploitation Plan	WP8	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D8.5	Report of the IPR management activities	WP8	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	36

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D8.6	Final Exploitation Plan including Business Plans of key results	WP8	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D8.7	Technology Transfer Activities, including training to SMEs, other activities and workshops from M1 to M18	WP8	12 - EXERGY	Report	Public	18
D8.8	Guide to explore the conversion routes of MSW in high added value products.	WP8	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	34
D8.9	Final version of the PEDR	WP8	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D8.10	Technology Transfer Activities, including training to SMEs, other activities and workshops from M18 to M36	WP8	12 - EXERGY	Report	Public	36
D8.11	Final communication activities & dissemination Plan updates	WP8	2 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D9.1	Risk indicators and Risk assessment	WP9	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	9
D10.1	NEC - Requirement No. 1	WP10	1 - IMECAL	Ethics	Confidential, only for members of the consortium (including the Commission Services)	1

1.3.3. WT3 Work package descriptions

Work package number [9]	WP1	**Lead beneficiary** [10]	1 - IMECAL
Work package title	Characterization, handling and pre-processing of MSW from different localizations and seasons. Final products requirements definition		
Start month	1	**End month**	18

Objectives

- Definition of intermediates requirements and related standard to effectively produce the targeted final products.
- Characterization of representative organic MSW samples to be used as substrates in the project.
- Improvement of enzymes cocktails to maximize organic hydrolysis for MSW compositions. Target to produce up to 85% of potential fermentable sugars contained in the organic MSW.

Description of work and role of partners

WP1 - Characterization, handling and pre-processing of MSW from different localizations and seasons. Final products requirements definition [Months: 1-18]
IMECAL, AIMPLAS, ATB, AUA, UGent, CENER, TBWR, VISUM, HAYAT, COVESTRO, EXERGY
Task 1.1. Final products and intermediates requirements Task Leader: HAYAT; Other partners: SOFTER, COVESTRO, IMECAL, EXERGY and AIMPLAS.
Based on a review of the current petrol based or renewable selected products and intermediates properties, environmental requirements and especially economic limitations, a list of specific requirements will be defined for each one. In a further evaluation step these requirements will be summarised to specific requirement-lists that serve as a basis for future development. The aim here is to group requirements so that more general target areas can be defined that allow the use of material developments in more than one application. The report will also include the specific standards required for each case-study selected.

Task 1.2. Study of variability of organic MSW. Task Leader: IMECAL Other partners: all.
The main challenge to use the organic fraction of MSW as feedstock for bio-based industries lies in its high complexity and variability in composition. This heterogeneous composition depends on three main parameters:
• The waste collection system. The organic fraction of MSW contains less amount of inserts in function of the MSW collection method employed (door-to-door, municipal bring points…)
• Geographical location. Differences in climate conditions and nutrition culture of EU-countries affects the specific contents of food and therefore its chemical composition. Bio-waste coming from rural or urban areas also exhibits different compositional characteristics.
• Seasonality. Climate variations during the year have also shown an important impact in the food content and bio-waste composition.
In this task inputs regarding MSW composition variation from all the partners will be shared and bibliographic data will be analysed in order to better know the bio-waste composition and variability across Europe, PERCAL has partners and support companies of 12 different countries, including Mediterranean, Atlantic, continental and warm and cold countries. This analysis will serve as a basis to select ten MSW samples that will be characterized in task 1.3.

Task 1.3. Characterization of MSW samples. Task Leader: IMECAL Other partners: ATB & AUA.
In this task the selected substrates of task 1.2 will be fully analysed following the NREL methodology for lignocellulosic materials in order to know their potential for the intermediates production.
• The glucan and hemicellulose content, which are the polysaccharides that contain the fermentable sugars, will be determined by acid hydrolysis and subsequent HPLC analysis.
• Humidity degree will be measured by dying in oven at 105ºC and the ash content by kiln calcination at 575ºC.
• Fats and waxes will be analysed by means of Soxhlet extraction with chloroform.
• Soluble extractives will be analysed by water incubation.
• Pectin and Lignin content will be determined by extraction with EDTA or with acetic acid and NaClO2.
• Protein content will be determined by specific staining and spectrophotometry against bovine serum albumin (bicinchoninic acid assay) as well as by Total Nitrogen analysis (Kjeldahl method).

This analysis will serve as a basis to select three representative MSW samples that will be used as feedstocks in the project.

Task 1.4. Pre-treatment and enzymatic hydrolysis improvement from PERSEO bioethanol® process. Task Leader: IMECAL Other partners: NOVO.
The selected samples will be pre-treated by means of thermo-chemical processes in order to liberate glucan and hemicellulose content for subsequent enzymatic hydrolysis. Both treatments will be based on state-of-the art processes performed currently at PERSEO semi-industrial plant. Current saccharification processes are able to liberate around 65% of all potential fermentable sugars present in the organic fraction of MSW. New enzymes cocktails developed by NOVO will be tested with the aim of producing more than 85% of all the potential fermentable sugars present, while minimising the generation of inhibitors.
The dosage of the enzymatic cocktail will be calculated based on the present protein content. Five dosages will be evaluated for each cocktail, in duplicate. Fermentable sugars will be analysed at the beginning of the enzymatic hydrolysis (t=0), at 24 and 48 h.
In order to diminish the inhibitory effects of some compounds (weak acids, furfural, hydroxymethyl furfural) towards the fermentative microorganisms in the pre-treatment strategy might be modulated depending on the intrinsic resistance of microbial strains to these hydrolysates. The amount inhibitors produced will be determined by liquid chromatography.

Task 1.5. Validation of fermentation processes. Task Leader: IMECAL; Other partners: AUA & ATB.
The hydrolysates generated in Task 1.3 with the three representative samples will be validated in terms of toxicity and potential carbon sources for the ethanol, LA and SA production. Tolerance tests will be performed with the yeast responsible of ethanol fermentation, and the different bacteria employed for the production of LA and SA. The tolerance of the selected strains will be evaluated in 96-well growth experiment assays in ten different substrate concentrations. Microbial growth will be measured spectrophotometrically. Flask scale fermentation reactions will be performed taking into account the tolerance of the different strains. Production of the target molecules will be tested following the indications of the different partners involved in fermentation. Production of the target compounds, together with other metabolites, will be followed by HPLC (RID detector). Interaction with task 1.3 will be required if inhibition issues arise. Detoxification or feed strategies will be proposed if required to cope with tolerance or inhibition problems.

Participation per Partner

Partner number and short name	WP1 effort
1 - IMECAL	
2 - AIMPLAS	
3 - ATB	
4 - AUA	
5 - UGent	
6 - CENER	
7 - TBWR	
8 - VISUM	
10 - HAYAT	
11 - COVESTRO	
12 - EXERGY	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.1	Selected final products and intermediates requirements.	10 - HAYAT	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D1.2	Variability of biogenic fraction of MSW.	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	8
D1.3	Report on hydrolysis improvements.	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	12
D1.4	Report on selected feedstocks.	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	14
D1.5	Report on validation of fermentation processes	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	18

Description of deliverables

D1.1. Report of selected final products and intermediates requirements.
D1.2. Report on variability of biogenic fraction of MSW.
D1.3. Report on hydrolysis improvements.
D1.4. Report on Selected feedstocks.
D1.5. Report on validation of fermentation processes

D1.1 : Selected final products and intermediates requirements. [6]

Selected final products and intermediates requirements.

D1.2 : Variability of biogenic fraction of MSW. [8]

Variability of biogenic fraction of MSW.

D1.3 : Report on hydrolysis improvements. [12]

Report on hydrolysis improvements.

D1.4 : Report on selected feedstocks. [14]

Report on selected feedstocks.

D1.5 : Report on validation of fermentation processes [18]

Report on validation of fermentation processes

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS1	Composition of MSW available	1 - IMECAL	8	Comparison with reported values in literature for similar feedstocks.
MS2	Improved enzymatic hydrolysis.	1 - IMECAL	12	Comparison of hydrolysis yields
MS3	Samples of hydrolysed MSW available for the different subsequent tasks	1 - IMECAL	20	Sufficient amounts of different fractions for lab tests within WP2, WP3 and WP4

Work package number [9]	WP2	**Lead beneficiary** [10]	3 - ATB
Work package title	Microbial conversions of MSW to lactic acid.		
Start month	2	**End month**	32

Objectives

This WP2 aims at bioconversion of the hydrolysed carbohydrates from organic municipal solid waste to lactic acid (LA). The utilisation and upgrading of organic municipal solid waste streams and the purification of lactic acid as a monomer for green solvents will be addressed. Biotechnological tools for the development of non-food industrial products and services and/or biogenic sources as feedstock are applied.

Description of work and role of partners

WP2 - Microbial conversions of MSW to lactic acid. [Months: 2-32]
ATB, IMECAL, AUA, TBWR, HAYAT
Lactic acid (LA) from organic municipal solid waste streams (MSW) feedstocks will be produced by ATB at laboratory and pilot-scale using sequential saccharification and fermentation approaches. In another valorization path, suitable raw material will be applied for lactic acid production via simultaneous saccharification and fermentation approaches using newly isolated strains. These complementary processes will be evaluated in parallel and assessed regarding optimum LA production that is suitable for subsequent processing.

Task 2.1. Screening and selection of new/alternative bacterial strains. Task Leader: ATB.
Screening and selection of new/alternative bacterial strains with respect to high product yields, sufficient substrate uptake, tolerance against eventually remaining impurities/inhibitors and product quality. One of the first targets will be the selection of the most efficient and robust LA producing microorganism among the ones mentioned in the state of the art or beyond. ATB will employ several screening algorithms to identify microorganisms that satisfy the performance criteria (e.g. high specific growth rate, high productivity, maximum sugar consumption, optical purity, high process stability etc.) with regard to industrial application. We will screen the most promising microorganisms in first simple fermentations in tubes, bottles, shake flasks and lab-scale bioreactors.

Task 2.2. Fermentation processes for converting biogenic fraction MSW into lactic acid. Task Leader: ATB Other partners: AUA.
Develop efficient fermentation processes for converting bio waste into lactic acid (L(+) isomer). Mixtures and/or single fractions of the (C6/C5) sugar solution from pre-treated fractions will be converted to lactic acid in order to provide compounds for further processing steps.
Depending on the steps before (WP1/sub-tasks for pre-treatment and/or hydrolysis) and the available analytical information the different materials, hydrolysates, sugars, mixtures etc. have to be characterized in view of their main components and possible impurities. If necessary additional enzymatic treatment could be adapted for a better release of sugars and/or nutrients. Finally a fermentation broth has to be designed which represents an optimal mixture of all necessary carbohydrates, macro- and micro-nutrients for the bacteria.
ATB will develop an efficient LA fermentation process through the optimization and elucidation of various parameters. Optimization of fermentation conditions will focus on initial sugar concentration, addition of commercial (e.g. yeast extract) or crude nutrient supplements (e.g. oilseed meal hydrolysates, grass juice), and addition of pH correcting agents (e.g. ammonia, caustic etc.). Further studies will focus on the evaluation of kinetic parameters including the inhibitory effect of by-products and/or impurities to microbial growth and product formation. Fermentation runs will be carried out in batch, fed-batch and repeated batch fermentation mode in order to compare productivity, yield, final LA concentration, cost aspects and environmental criteria. The adapted strain(s) may be able to sustain inhibitory conditions during continuous fermentation (optional).

Task 2.3. Downstream processing (DSP) for recovery of lactic acid Task Leader: ATB; Other partners: TBWR
ATB will adapt and/or develop LA purification considering process scalability, purity/quality, robustness, separation yield and cost-competitiveness. One criterion should be the comparison with already existing figures related to other fermentation feedstocks, whereas the down-stream procedure for eventually unknown MSW-related components and impurities needs to be adapted due to missing experiences. In the case that there are still residual sugars in the broth, a nanofiltration step could be integrated in order to recover unconsumed substrate to enhance the overall process productivity. All the necessary components (filtration, electrodialysis, ion exchange, evaporation etc.) are available on bench (20-50 litres) as well as pilot scale (800-1000 litres).

It is foreseen that the fermentation broth will also contain a variety of other compounds such as a significant amount of minerals and remaining nitrogen (most likely no protein but polypeptides or single amino acids. In addition other organic acids which have its origin from the waste (such as acetic acid, propionic acid or butanolic acid) are also expected as impurities. Within this task different DSP- configurations will be tested for the recovery of the key product of lactic acid. Partner TBWR has piloted a separation technology to recover LA and amino acid (AA) in a combined processing which is currently demonstrated at production scale. In case the feedstock origin can also deliver AA at significant level, a transfer of technology from grass silage juice to PERCAL intermediates will be tested and assessed. This task is closely connected to WP5 the esterification of lactic acid with ethanol to ethyl lactate and the production of hot-melt adhesives by reactive extrusion. There is potential to compromise on quality of LA (e.g. use a racemic mixture, some remaining ash etc…) if it is used as intermediate for ethyl lactate.

Task 2.4. Up-scaling of the entire process Task Leader: ATB; Other partners: IMECAL
Up-scaling of the entire process chain including downstream processing for the provision of product samples in high quality. The optimised process in lab-scale (1-3 L bioreactors) will be scaled up in pilot-scale at 50 L bioreactor system. Process operation will be mainly dependent on the fermentation mode that will be selected in previous tasks 2.2 and 2.3. The pilot facility for the production of lactic acid at ATB and the corresponding pre-treatment and down-stream unit operations will be adapted to the waste materials under investigation to compare the process performance with previous feedstocks. General parameters such as utilisation of carbon sources, possible limitations by lacking nutrients, response/stability of microorganism and bioreactor and downstream separation performance at higher scales will be elucidated. This task will also provide several product samples in certain qualities for further processing and useful data for economic and environmental assessment, respectively. From the evaluation view not only the quantitative estimation but also the qualitative (impurities, optical purity) validation of the “waste-based lactic acid” will be an important part of the project.
The overall bioprocess design for lactic acid production at the pilot scale will be proposed, including up- and down-stream processes. Overall process testing will include all accompanying steps, such as hydrolysis, filtration, fermentation, full downstream processing based on membrane processes (i.e. electrodialysis) achieving a highly pure concentrated lactic acid, and will rely on previous results generated within WP2. Moreover, the wastewater from the pilot scale down streaming is collected to investigate its reutilization in the preceding processes.

Participation per Partner

Partner number and short name	WP2 effort
1 - IMECAL	
3 - ATB	
4 - AUA	
7 - TBWR	
10 - HAYAT	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D2.1	Lactic acid fermentation process	3 - ATB	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D2.2	Down-stream processing	7 - TBWR	Report	Confidential, only for members of the	24

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
				consortium (including the Commission Services)	
D2.3	Scaling of the full process.	3 - ATB	Report	Confidential, only for members of the consortium (including the Commission Services)	32

Description of deliverables

Responsible Deliverables Due
ATB D2.1. Report of lactic acid fermentation process. M18
TBWR D2.2 Report of down-stream processing. M24
ATB D2.3 Report of up scaling of the full process. M32

D2.1 : Lactic acid fermentation process [18]

Lactic acid fermentation process

D2.2 : Down-stream processing [24]

Down-stream processing

D2.3 : Scaling of the full process. [32]

Scaling of the full process.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS4	LA with purity higher than 97% for L-LA .	3 - ATB	22	Application of chiral HPLC to estimate the optical purity of all LA samples
MS5	Yield at pilot plant facilities similar to other feedstocks already tested.	3 - ATB	30	Max. 10% losses compared to current process using pure lignocellulosic biomass feedstock.

Work package number [9]	WP3	**Lead beneficiary** [10]	5 - UGent
Work package title	Microbial conversions of MSW to succinic acid.		
Start month	3	**End month**	34

Objectives

• Optimisation of fermentation media composition for industrially competitive succinic acid production using MSW hydrolysates as the major source of nutrients.
• Development and optimisation of an integrated bioprocess combining SA production with separation from the fermentation broth via electrolysis membrane extraction targeting maximisation of productivity and yield.
• Comparison of succinic acid separation and purification based on a electrolysis membrane extraction system or an electrodialysis-based downstream separation system.
• Optimisation of continuous cultures for succinic acid production and separation using immobilised cultures and a low cost MSW hydrolysate as fermentation feedstock.
• Scale-up of the most efficient succinic acid production process.

Description of work and role of partners

WP3 - Microbial conversions of MSW to succinic acid. [Months: 3-34]
UGent, IMECAL, ATB, AUA, CENER
Task 3.1. Evaluation of MSW as fermentation feedstock. Task Leader: AUA Other partners: CENER.

Subtask 3.1.1 – Selection of bacterial strain and optimisation of fermentation media: The primary target will be the selection of the most efficient and robust succinic acid producing bacterial strain when the different MSW hydrolysates produced in WP1 are used. AUA will screen Basfia succiniciproducens, Actinobacillus succinogenes and Escherichia coli AFP184 that have been previously identified by AUA as the most efficient bacterial strains depending on the feedstock employed. AUA will screen the bacterial strains in single or two-stage (only in the case of E. coli AFP184) fermentations in Duran bottles and bench-scale bioreactors with constant supply of CO2 in the succinic acid production stage.
The optimal bacterial strain will be employed in the optimisation of the fermentation medium. The hydrolysis of polysaccharides, proteins (the supply of crude N sources will reduce the supply of yeast extract) and phytic acid (source of phosphorus) contained in MSW will lead to the production of a nutrient-rich MSW hydrolysate. AUA will employ ICP-MS analysis in order to identify the content of minerals and thus minimise the supply of commercial salts. Preliminary fermentations in Duran bottles will be carried out in order to assess the inhibitory effect of the MSW hydrolysates (produced in WP1) on bacterial growth and succinic acid production as compared to control fermentations carried out with commercial sugars (using the same mixture of sugars present in MSW hydrolysates), yeast extract and corn steep liquor. In the case that lower efficiency is observed when the MSW hydrolysates are used, AUA will study the effect of potential inhibitors present in MSW hydrolysates (e.g. furfural, 5-hydroxy-methyl-furfural, acetic acid, phenolics etc) and the effect of commercial nutrient supplementation (e.g. yeast extract, corn steep liquor) using a 96-well microplate reader followed by design of experiments (DoE) carried out in Duran bottles. The DoE will focus on the optimisation of nutrient composition. AUA will optimise the fermentation medium regarding nutrient content (addition of yeast extract, salts, MgCO3 and corn steep liquor) and will propose pre-treatment strategies for the removal of potential inhibitors.

Subtask 3.1.2 – Optimisation of lab-scale fed-batch and continuous cultures: AUA will optimise fed-batch and continuous cultures using the selected bacterial strain and fermentation media identified in Task 3.1.1. Optimisation of fed-batch cultures will focus on initial sugar concentration, addition of MgCO3 or Na2CO3 and feeding strategies. Continuous cultures will be optimised regarding initial sugar concentration, dilution rate and bacterial cell immobilisation. Fermentation modes of operation will be compared regarding productivity, yield, final sucinic acid concentration, capital/operating cost requirements and environmental criteria (e.g. non-renewable energy utilisation, waste generation).

Task 3.2. Optimisation of integrated bioprocess for succinic acid production and in situ separation. Task Leader: CMET Other partners: AUA.

Subtask 3.2.1 – Study of electrolysis membrane extraction using a synthetic system: A number of commercially available anion exchange membranes will be assessed according to electrochemical parameters such as applied current, resistance and potential across the electrochemical cell, and subsequent flux of succinic acid and critical co-products (e.g. acetic

acid, lactic acid). After an appropriate membrane is chosen, the ideal operational parameters can then be assessed according to other process parameters, including extractant concentration, pH and conductivity. This will be critical towards minimising the input power per extracted kg of succinic acid, and equivalent masses of OH- generated to manage the fermentation.

Subtask 3.2.2 – Optimisation of the integrated system at lab-scale: AUA will use the data from subtask 3.2.1 for the development of the integrated system, which will consist of 7 L bioreactor coupled to a high-rate electrochemical cell performing the extraction of organic acids produced during fermentation via recycling of fermentation broth through the extraction cell. This novel system will be developed at AUA in lab-scale in close collaboration with visiting researchers from CMET. AUA and CMET will study the effect of organic acid extraction on productivity and yield during fermentation. The effect of the applied current will be elucidated along with the effect of the produced H2 on bacterial metabolism. The integrated system will be evaluated using either fed-batch or continuous cultures using in the second case either free cells or immobilised cultures. The nutrients that pass through the membrane together with organic acids will be identified in order to assess the potential effect on fermentation efficiency.

Task 3.3. Succinic acid purification. Task Leader: AUA Other partners: CMET
Conventional purification of succinate salts from fermentation broths requires cell removal via filtration, colour and protein removal via activated carbon treatment, acidification of organic acids using resins, concentration of succinic acid and removal of acetic and formic acids via evaporation, crystalisation of succinic acid followed by drying. AUA has optimised the above process to yield succinic acid crystals with purity higher than 99.5% and recovery yield higher around 80% when fermentations were carried out in spent sulphite liquor based media. This downstream process will be used as base-case scenario. AUA will optimise the purification of succinic acid produced by the electrolysis extraction system. The optimal concentration of succinic acid in the extracted aqueous solution will be identified in order to minimise energy consumption during evaporation which is required in order to remove the acetic and formic acids that also pass through the extraction electrolysis cell. The composition of the aqueous solution after the extraction will be identified in order to develop the appropriate separation scheme. Any sufficiently small anion will cross the membrane (e.g. chloride, sulfate, phosphate, formate, and more), and effective recovery and recycle of these compounds will be investigated.

Task 3.4. Process scale-up Task Leader: CMET; Other partners: AUA
CMET will use the data from tasks 3.1-3.3 regarding the lab-scale operation of the fermentation and in situ membrane electrolysis extraction in order to move the operation over to a pilot scale unit of > 50 L. The aim will be to control the succinic acid production via fermentation exclusively with the applied current (i.e. the electrolysis of water to hydroxide), and develop a system to continuously extract the succinic acid crystals from the acidic extract, such as a cooling and settling unit. The process can be investigated under conditions for maximum production rate, versus maximum yield, versus minimal power input to develop a set of parameters to compare against existing processes in succinic acid production. Other operational parameters more unique to membrane electrolysis, (e.g. membrane resistance, electrode potential, electrode fouling, voltage losses) will be investigated in detail, to compare the membrane electrolysis lifetime and operation against the other existing technologies. The process scale up will mimic the fermentation, extraction and purification of succinic acid, with the goal of minimising product output according to added chemicals as a first goal (i.e. completely replacing chemical caustic and acid control with electrochemistry), and followed by minimising the power investment per kg of succinic acid product. At an anticipated production rate of 20 g/L/d, here we aim to produce 1 kg/d in continuous, steady state operation, which should allow sufficient product for assessment and benchmarking against existing processes.

Task 3.5. Optimisation of bioprocess based on electrodialysis downstream separation system. Task Leader: ATB Other partners: AUA
ATB and AUA will optimise the production and purification of succinic acid in 20 – 50 L scale experiments using the unit operations (bioreactor, ultra/nanofiltration, softening, bipolar membrane electrodialysis (BMED), cation/anion exchange resins, activated carbon, vacuum evaporation) available at ATB. Succinic acid separation will commence with cell removal using hollow fiber membranes. Softening is then carried out by a series of anion and cation exchange resins to remove various impurities and metals that could cause membrane fouling, pipe jam or pump erosion. BMED will be employed to convert the succinate (and other organic acid by-products) into the respective acid form. In the BMED configuration, which is composed of three compartments, both the anions of the organic acids and the sodium cations are transferred across the anion and cation exchange membranes, and then they are combined with H+ and OH- respectively, which are produced in the bipolar membrane intermediate layer, to form the organic acids and NaOH in the acid compartment and base compartment, respectively. The process parameters to be evaluated and optimized are the initial organic acid feed, the voltage of BMED process as well as the conductivity and the pH. Another important factor is to reduce the water transport per molarity succinate treated. The most important hurdle in this process is to prevent membrane fouling. In order to do so, the adequate pretreatment methods should be carried out in conjunction to

the following measures: a) reversing the polarity of electrodes, b) decrease operating current, c) improve the hydraulic conditions in stack compartments, d) add chemicals to prevent the formation of precipitates or microbial growth, e) extract products before saturation. After electrodialysis, the product stream is processed via cationic/anionic resin or activated carbon in order to remove colour impurities, residual carbon or/and nitrogen sources. The final purification steps involve vacuum evaporation, crystalisation and drying. This process will be carried at ATB in collaboration with researchers from AUA. The results will be compared with the results achieved in previous tasks by electrolysis membrane extraction.

Participation per Partner

Partner number and short name	WP3 effort
1 - IMECAL	
3 - ATB	
4 - AUA	
5 - UGent	
6 - CENER	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D3.1	Selection of optimal bacterial strain using different MSW hydrolysates.	4 - AUA	Report	Confidential, only for members of the consortium (including the Commission Services)	8
D3.2	Report on the integrated SA production with electrolysis membrane extraction.	5 - UGent	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D3.3	Report on production and purification of SA using electrolysis or electrodialysis based processes	4 - AUA	Report	Confidential, only for members of the consortium (including the Commission Services)	28
D3.4	Report on process scale-up	5 - UGent	Report	Public	32

Description of deliverables

D3.1. Selection of optimal bacterial strain using different MSW hydrolysates. M18
D3.2. Report on the integrated SA production with electrolysis membrane extraction. M24
D3.3 Report on production and purification of SA using electrolysis or electrodialysis based processes M28
D3.4. Report on process scale-up M32

D3.1 : Selection of optimal bacterial strain using different MSW hydrolysates. [8]

Selection of optimal bacterial strain using different MSW hydrolysates.

D3.2 : Report on the integrated SA production with electrolysis membrane extraction. [24]
Report on the integrated SA production with electrolysis membrane extraction.

D3.3 : Report on production and purification of SA using electrolysis or electrodialysis based processes [28]
Report on production and purification of SA using electrolysis or electrodialysis based processes

D3.4 : Report on process scale-up [32]
Report on process scale-up

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS6	Production of succinic acid in fed-batch and continuous processes	4 - AUA	15	Report on lab-scale experiments and comparison with preliminary experiments with the integrated system.
MS7	Extraction of succinic acid directly from the extraction cell at 99% purity.	4 - AUA	24	Report on the purity of succinic acid crystals extracted directly from the concentrated solution produced via membrane electrolysis.

Work package number [9]	WP4	**Lead beneficiary** [10]	6 - CENER
Work package title	Valorisation of fermentation by-products		
Start month	8	**End month**	34

Objectives

The objective of WP4 is to develop novel biosurfactants by employing the lipid and protein fraction remaining after the fermentation processes to ethanol, LA or SA, of pre-treated and hydrolysed MSW.

Description of work and role of partners

WP4 - Valorisation of fermentation by-products [Months: 8-34]
CENER, IMECAL, HAYAT
Task 4.1 Characterization of the fermentation residue of the ethanol/LA and SA processes Task Leader: CENER Other partners: IMECAL
Fermentation residues from the fermentation of ethanol, SA and LA processes will be analyzed employing the characterization techniques described in Task 1.2. Those residues with the highest content in protein and lipids will be selected for the revalorization of these fractions within the following tasks. Depending on the composition the strategy for the lipid and protein fraction will be evaluated. Processes for the fractionation of lipids and proteins rich fractions are well established in the literature, but adaptations and optimizations to the raw material are needed.

Task 4.2 Hydrolysis of protein fraction Task Leader: CENER Other partners: IMECAL
Hydrolysis of proteins into amino acids is the first step of converting proteins into surfactants. It is necessary to work with concentrated biomass, therefore if its water content is very high concentration by centrifugation or filtration will be applied. In order to increase the yield of the enzymatic hydrolysis some pretreatment processes will be studied: extraction with ethanol to remove color, previous lipid extraction, milling, heating, additional enzymes. Test conditions will be selected from literature in term of solid loads and other variables. The enzymatic processes will be evaluated on these pre-treated samples. Enzymes working at neutral or basic pH will be preferred such as papain (pH 7.0), pancreatin (pH7.5), Alcalase® 2.5L (pH 8) and Flavourzyme®1000L (pH 7). Working temperature will be around 50-60ºC, and 3-4 hours of incubation. Optimal dosage described in literature will be tested. Finally, free aminoacids will be quantified by a colorimetric test based on nihidrin reaction and its aminogram will be analyzed by HPLC from samples treated with phenylthio isocyanate
Chemical hydrolysis methods (based on acid or alkali) will be revised from. Acid hydrolysis completely destroys tryptophan and partially damages tyrosine, serine and threonine. Alkaly hydrolysis can modify arginine, lysine, cystein, serine and threonine. Therefore, pure chemical hydrolysis will be avoided. The enzymatic approach will be the preferred one, although, combination with mild acid hydrolysis will be evaluated in order to maximize yield and minimize processing time. The protein hydrolyzate will be employed as a Nitrogen source for fermentation procedures and as synthons for biosurfactant production.

Task 4.3 Extraction and hydrolysis of the lipid fraction Task Leader: CENER Other partners: IMECAL
There are two main approaches for lipid valorisation: the first one is a first step of extraction, followed by the subsequent esterification or saponification reactions; the second approach is the direct treatment without separation. In the second case extreme chemical agents are required, and it can be harmful for the protein fraction, therefore, within the framework of this project the lipids will be extracted as the first step in revalorization process. Lipid extraction will be performed on the dried fermentation residue because water can affect the subsequent saponification reactions. It can be performed with a single solvent or with solvent mixtures. Extraction tests will be performed in a soxhlet system with organic solvents (dichloromethane, ethyl acetate: ethanol, or mainly ethyl lactate: ethanol) for 18 hours. After the extraction, solvent will be removed and the degree of extraction will be quantified for each case. The lipid fraction will be analyzed in order to know the content in acylglycerols (by GC-FID). Analysis of fatty acid composition will be performed by GC-FID or GC-MS of methanolized samples.
Task 4.4. Synthesis of the amino-lipid biosurfactant Task Leader: CENER Other partners: IMECAL
By the combination of chemical, enzymatic or a combination of both strategies, surfactants will be obtained by the condensation of amino acid and lipids. The N-acylation of the amino group by condensation with fatty acids from the saponified lipid fraction or directly from the acylglycerides, will be prepared by either chemical methods or by lipases (from Candida antarctica & Rhizomucor meihei). Aqueous or solvent reaction systems will be evaluated. For the chemical approach the activation of the acyl carbon of the fatty acids will be performed by the formation of acyl chlorides in strongly alkaline aqueous medium containing the aminoacids. Protection of Nitrogen-amino group will be taken into

account for some amino acids. A mixture of different molecules is expected, so therefore state of the art isolation and characterization techniques will be applied. Surface active properties of the surfactants will be tested such as surface tension at the critical micelle concentration, critical micelle concentration, maximum surface excess concentration at the air/aqueous solution interface and minimum area per surfactant molecule at the air/aqueous solution interphase.
Task 4.5. Validation of the biosurfactants in relevant industrial applications Task Leader: HAYAT.
HAYAT uses Linear Alkyl Benzene Sulphonic Acid (LABSA), it is an anionic surfactant widely used in the production of varieties of detergents and emulsifiers. HAYAT will evaluate the performance of the new surfactant from MSW in a standard formulation in one of their detergents following their own internal methodology and suitable EN/ISO/ASTM/ DIN standards such as ISO 4319, ASTM D2960-98, ASTM D2960-05, and DIN 44983. If required, additional additives or blends with LABSA or other commercial surfactants will be employed. Full validation and new products data-sheets will be provided at the end of the project.

Participation per Partner

Partner number and short name	WP4 effort
1 - IMECAL	[illegible]
6 - CENER	[illegible]
10 - HAYAT	[illegible]
Total	[illegible]

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D4.1	Characterization of the fermentation residue.	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D4.2	Pre-treatments for revalorization of the protein fraction.	6 - CENER	Report	Confidential, only for members of the consortium (including the Commission Services)	26
D4.3	Pre-treatments for revalorization of the lipid fraction.	6 - CENER	Report	Confidential, only for members of the consortium (including the Commission Services)	26
D4.4	Biosurfactant production from lipids and proteins.	10 - HAYAT	Report	Confidential, only for members of the consortium (including the Commission Services)	34

Description of deliverables

D4.1. Report on characterization of the fermentation residue. M18
D4.2. Report on pre-treatments for revalorization of the protein fraction. M26
D4.3 Report on pre-treatments for revalorization of the lipid fraction. M26

D4.4. Report on biosurfactant production from lipids and proteins. M34

D4.1 : Characterization of the fermentation residue. [18]
Characterization of the fermentation residue.

D4.2 : Pre-treatments for revalorization of the protein fraction. [26]
Pre-treatments for revalorization of the protein fraction.

D4.3 : Pre-treatments for revalorization of the lipid fraction. [26]
Pre-treatments for revalorization of the lipid fraction.

D4.4 : Biosurfactant production from lipids and proteins. [34]
Biosurfactant production from lipids and proteins.

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS8	A method for the production of amino acid and lipid surfactants from MSW by-products.	6 - CENER	32	Comparison of surface active properties with those described in literature.
MS9	Biosurfactants has fulfil the HAYAT internal standards.	10 - HAYAT	34	Comparison of bio-surfactant properties with commercial ones using DIN/EN/ISO standard.

Work package number [9]	WP5	**Lead beneficiary** [10]	11 - COVESTRO
Work package title	Green chemical routes to produce solvents, adhesives & polymers from developed acids.		
Start month	10	**End month**	34

Objectives

The main objectives of WP5 are the production and industrial validation (TRL5) of the follow products using the monomers developed in WP2 &3: 1) Ethyl lactate by reactive distillation of LA and ethanol, 2) Hot Melt Adhesives by reactive extrusion of LA, and 3) Polyester-Polyols by conversion of SA with diamines. Polyester Polyols will be validated by PU formation using Di-isocyanates.

Description of work and role of partners

WP5 - Green chemical routes to produce solvents, adhesives & polymers from developed acids. [Months: 10-34]
COVESTRO, AIMPLAS, TBWR, SOFTER , HAYAT
Task 5.1. Ethyl lactate as a solvent. Task Leader: TBWR, Other partners: HAYAT.
Lactic acid and ethanol will used for esterification to produce ethyl lactate. The main technological focus is to test and further develop a reactive distillation process for simultaneous esterification and separation of product in one column. TBWR is able to bring in a reactive distillation prototype into the project which was designed specifically for ethyl lactate within an ongoing national R&D project. (Column DN 50mm; height 4.5 meters).
Subtask 5.1.1 Esterification test in lab scale for validation of PERCAL intermediates. This task involves batch esterification test in lab and at bench scale to test different solid catalyst designs. Also potential degeneration of catalyst resigns need to be tested and evaluated which is related to PERCAL - LA quality and potential impurities.
Subtask 5.1.2 Design, validate and modify a chemical model for the esterification process. The esterification process and simultaneous product separation with a reactive distillation will be described by a multi parameter chemical model. The specific model will be used to simulate processing to get first results on the influence of the feedstock characteristic (e.g. concentration of LA and ethanol) and its interaction on esterification performance. The model is used as a tool to plan systematic experiments in 5.1.3 and can also be adjusted by results of the experiments.
Subtask 5.1.3 Upscale reactive distillation to produce ethyl lactate. Within this task an existing reactive distillation pilot prototype column will be used to upscale the esterification and separation process. This task involves a series of systematic trials to run the reactive distillation at various process parameters. The esterification test/process is assessed using analytical data of various streams of the reactive distillation. Results are used to adapt and fine tune the chemical model and refine the set of experiments. The goal is to identify optimized operation parameters for the production of ethyl lactate by means of reactive distillation. Also this task will deliver first sets of products as samples to be provided to industrial partners.
Subtask 5.1.4 Assessment of the esterification process and up-scaling. This task uses and analyses the results generated in Task 5.1 to refine key information and data on esterification process performance (efficiency, energy demand, product recovery rate, potential technical bottlenecks etc.) and also characterise the quality of ethyl lactate generated. The objective is to compile concrete data for the technical and economic analysis on the process and product to establish the basis for a sound evaluation of further upscaling of reactive distillation.
Subtask 5.1.5 Ethyl Lactate Validation. HAYAT will validate the Ethyl Lactate in two products of their current products using their internal methodology and the suitable EN/ISO/ASTM/DIN tests such as ISO 4319, ASTM D2960-98, ASTM D2960-05, and DIN 44983. In concrete, they studied the substitution of propylene glycol monomethyl ether in inks for printing and isopropanol in windows cleaners. Full validation and new products data-sheets will be provided at the end of the project.
Task 5.2. LA based hot melt adhesives by reactive extrusion. Task Leader: AIMPLAS Other partners: SOFTER.
Subtask 5.2.1 Production and purificaction of lactide from lacti acid. The procedure of lactide obtention from lactic acid consists basically of heating up lactic acid in a glass vessel under vacuum whilst steering and distilling off water in order to promote a polycondesation reaction to produce a lactic acid polycondesate. Once the polyester is formed an addition of a catalyst will lead us to transform the polycondensate into lactide. The dimer (lactide) will be purified from the crude by distillation and by solution crystallization. For the production of up several kilos, a 15L reactor of AIMPLAS's kilolab facilities will be employed.
Subtask 5.2.2 Production of Low molecular PLA and PLA-copolymers by reactive extrusion. Lactides will be used for their Ring Opening polymerization using a large L/D twin screw extruder as a continuous reactor. In order to optimize the reaction conditions in terms of reaction time, initiatior, ratio of copolymer, catalyst and temperature a pre-optimization employing a Haake minilab extruder will be carried out.

Once the different polymers and copolymers have been obtained by reactive extrusion and by solution procedures, those compounds will be characterized employing several techniques as Differential scanning calorimetry (DSC), thermogravimetry (TGA), Gel permeation chromatography (GPC) and rheological tests which are currently available in AIMPLAS. In addition the molecular weight of the polyester will be analyzed employing a new NMR method based in the correlation of diffusion coefficients acquired by proton NMR spectroscopy and the MW. The method lead us to explore and analyze with just some milligrams of sample properties as MW & polidispersity index

Once the best conditions has been optimized several tests will be done to adjust the reactive extrusion conditions to obtain a PLA grade with a low molecular weight suitable to be used in hot melt adhesives. Special attention will be done in the selection of type and amount of the catalyzer. The reaction will be designed to be done in the residence time of conventional extrusion process. The PLA will be characterized.

Subtask 5.2.3 HMA by PLA grafting. Initial tests to PLA graft will be done at lab scale in a discontinuous 1 L reactor using some reactive substances (glycolic acid, maleic acid, methacrylate or glycidyl acrylate) and suitable peroxides to create a chemical graft bonds between the matrix and selected monomers. The objective is to achieve a percentage of graft superior to 1% w/w. The percentage of co-monomer will be determinate by Fourier-transform infrared spectroscopy (FT-IR) and their thermal stability by differential scanning calorimetry (DSC) and thermogravimetric analysis (TGA).The best results obtained in solution will be transferred to a reactive extrusion process employing a Haake minilab extruder.Once the reaction conditions have been optimized the process will be scale up using a twin esxtrusion processing parameters (screw design, residence time, feeding of each component, etc..) will be optimized.

The best PLA grafted grade obtained will be stabilized and processing improved by adding by melt compounding the suitable additives to be used as a HMA (e.g., antioxidants, UV stabilizers, plasticizers, tackifiers, waxes, etc…).

Finally, in order to have a cost competitive product, we the possibility to produce the HMA in one compounding steps, using lactides and reactive co-monomers will be studied, i.e., lactide polymerization, reactive commoner grafting and adding and mix the rest additives for final HMA formulation.

Subtask 5.2.4 HMA scale-up and Validation. The production of HMA will be scaled-up and validated in the SOFTER industrial facilities. The company is producing several tons of petrol based HMA each year and they will be used the same methodology to validate the new renewable source. Other partners such as HAYAT, COVESTRO and ENERGO will use the lactic acid to sealing delivery box cardboard or labels prototypes and validated according their internal procedures.

Task 5.3. Butanediol from succinic acid using green chemistry routes. Task Leader: COVESTRO Other partners: AIMPLAS.

COVESTRO has a pilot plant equipped with several reactors of different sizes to produce different type of building blocks from several monomers. Using these facilities, the SA obtained in WP3 will be combined with several diamines to produce high molecular weight Polyester-Polyols. Processing parameters such as temperature, pressure, residence time and other chemical to control the final polyester chain length. Polyester Polyols will be validated by PU formation using Di-isocyanates using an internal methodology that use own defined tests and suitable EN/ISO standards.

Participation per Partner

Partner number and short name	WP5 effort
2 - AIMPLAS	
7 - TBWR	
9 - SOFTER	
10 - HAYAT	
11 - COVESTRO	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D5.1	Production of Ethyl lactate at laboratory level.	7 - TBWR	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D5.2	Ethyl-Lactate scale up and industrial validation.	10 - HAYAT	Report	Public	34
D5.3	Production of HMA at laboratory level.	2 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D5.4	HMA scale up and industrial validation	9 - SOFTER	Report	Public	34
D5.5	Production of polyols at laboratory level.	2 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D5.6	Polyols scale up and industrial validation	11 - COVESTRO	Report	Public	34

Description of deliverables

D5.1. Report on production of Ethyl lactate at laboratory level. M24
D5.2. Report on Ethyl-Lactate scale up and industrial validation. M34
D5.3. Report on production of HMA at laboratory level. M24
D5.4. Report on HMA scale up and industrial validation. M34
D5.5. Report on production of polyols at laboratory level. M24
D5.6. Report on polyols scale up and industrial validation. M34

D5.1 : Production of Ethyl lactate at laboratory level. [24]

Production of Ethyl lactate at laboratory level.

D5.2 : Ethyl-Lactate scale up and industrial validation. [34]

Ethyl-Lactate scale up and industrial validation.

D5.3 : Production of HMA at laboratory level. [24]

Production of HMA at laboratory level.

D5.4 : HMA scale up and industrial validation [34]

HMA scale up and industrial validation

D5.5 : Production of polyols at laboratory level. [24]

Production of polyols at laboratory level.

D5.6 : Polyols scale up and industrial validation [34]

Polyols scale up and industrial validation

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS10	Simultaneous esterification & products separation (reactive distillation) for ethyl lactate established.	7 - TBWR	34	Continuous mode reactive distillation established and validated at scale 3-5 L/h.
MS11	Renewable HMA has fulfil the end-user requirements or SOFTER internal standards.	2 - AIMPLAS	34	Comparison of HMA main properties with commercial EVA and renewable source HMA using the suitable EN/ ISO.
MS12	Polyester-Polyols by conversion of SA with diamines have similar reactivity than using commercial bio-based SA (less than 20% time increase) and the PU formation obtained have at least 80% of performance than commercial ones.	11 - COVESTRO	34	Comparative analysis using internal and well-defined methodology of COVESTRO.

Work package number [9]	WP6	**Lead beneficiary** [10]	8 - VISUM
Work package title	Study of adaptation and monitoring of PERSEO plant for the production of ethanol, LA & SA with suitable quality to produce the selected final products.		
Start month	12	**End month**	36

Objectives

The main objectives of WP6 are to study the adaptability of a versatile biotechnological plant to obtain different chemical intermediates and to determine the most relevant parameters to control and monitor the PERSEO's process at industrial level.

Description of work and role of partners

WP6 - Study of adaptation and monitoring of PERSEO plant for the production of ethanol, LA & SA with suitable quality to produce the selected final products. [Months: 12-36]
VISUM, IMECAL, ATB, AUA
Description of work.
Process parameters of PERSEO Bioethanol® semi-industrial plant are currently controlled with a SCADA technology. New monitoring technologies developed by VISUM will improve the performance of the plant control by obtaining relevant parameter information through advanced sensors and adding an intelligent plant control platform with decision support models. The objectives of this WP consist of studying the adaptability of PERSEO Bioethanol® plant to produce lactic acid and succinic acid as well as the design and testing of new sensors and control platform for the monitoring of the process parameters of interest in a laboratory scale and the study of the scalability of the processes and control system at full plant level.

Task 6.1. Study of parameters of the pre-treatment and fermentation process for the production of bioethanol, lactic & succinic acid and the by-products stream. Task Leader: IMECAL Other partners: VISUM.
In order to determine the main parameters in the fermentative processes, current control parameters measured on line in PERSEO pilot plant during the simultaneous saccharification and fermentation processes for the production of bioethanol are pH, temperature, pressure and material flow. Additional parameters that would be useful to monitor in real time for the better control of the process progress are the viscosity and homogeneity of the feedstock, quantification of liberated sugars and quantification of generated ethanol, which are currently measured off line.
Moreover, taking into account the optimization for the microbial conversion of MSW to lactic and succinic acid (WP 2 and WP3) the adaptation of PERSEO plant will be studied for the production of both intermediates. In particular, the required conditions of the fermentation unit (temperature, retention time, pH control, addition of nutrients…) for the optimum production of the organic acids at semi-industrial scale will be evaluated, studying the main parameters to be controlled for the respective fermentation processes. Control of the substrate viscosity progress and quantification of liberated sugars will be common to the three different type of fermentations. Quantification in real time of lactic and succinic acid produced will be also evaluated.

Task 6.2. Study of parameters of the separation/purification processes for the production of bioethanol, lactic acid and succinic acid. Task Leader: IMECAL Other partners: VISUM
To determine the main parameters in the separation/fermentation processes. Each of the target intermediates require different separation and purification techniques. Bioethanol is purified by means of distillation. The range of temperature, pressure, flows and amount of ethanol at the top of distillation columns are all important parameters to be studied. In case of lactic acid, the integration of all the necessary chemical purification systems (filtration, electro-dialysis, ion exchange, evaporation units etc.) for the achievement of the required quality will be evaluated and the relevant parameters to control will be identified. Similarly, the adaptation of the electrochemical separation described in WP3 at laboratory scale for the purification of SA will be also studied to be implemented in PERSEO plant.

Task 6.3. Monitoring of the processes parameters. Task Leader: VISUM Other partners: IMECAL.
Based on the parameters studied by IMECAL in Tasks 6.1 and 6.2, VISUM will build at least 1 monitoring system with adjusted sensors, possibly a sampling system, and a data acquisition and treatment platform to be able to track the Critical Control Parameters (CCPs) and the Critical Quality Attributes (CQAs). Potential sensors to be used include but are not limited to NIR or other optical technologies. This system will be applied at laboratory scale where the different processes for the production of bioethanol, lactic acid and succinic acid will be studied. These tests will allow to study the feasibility of monitoring the parameters listed as critical to quality of the processing of OMSW and will provide

deep understanding of the processes in real time. VISUM will also develop the associated calibration models to analyse the measurements of the sensors in order to obtain a tailored and fit for purpose monitoring system. These models will be developed if possible, from existing similar models or in case there are no suitable previous studies, new chemical analysis will be carried out to establish or validate the spectroscopic calibration models.

Task 6.4. Integration and adaptation of processes to a biotechnological plant configuration Task Leader: IMECAL; Other partners: VISUM
In this task, we plan to study the adaptation and monitoring of the intermediates production processes to a unified biotechnological plant. Based on the different processes studied in previous tasks T6.1 and T6.2, IMECAL will study their integration and adaptation to a MSW biotechnology pilot plant for the versatile production of bioethanol, lactic and succinic acid. VISUM will study the scaling of the monitoring system to a unified plant configuration in which all process steps of the production of bioethanol, lactic acid and succinic acid are performed continuously in a single biotechnological plant. To this end, a plant control system will be studied and evaluated and a configuration for controlling the processes of the unified plant will be reported in Deliverable 6.3. To this end a process modelling task will be required which will be done in conjunction with IMECAL.

Participation per Partner

Partner number and short name	WP6 effort
1 - IMECAL	
3 - ATB	
4 - AUA	
8 - VISUM	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D6.1	Adaptation of PERSEO plant for the versatile production of chemical intermediates & identification of key parameters	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D6.2	Monitoring technologies and testing results.	8 - VISUM	Report	Confidential, only for members of the consortium (including the Commission Services)	30
D6.3	Integration of the production in a biotechnological plant.	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	34

Description of deliverables

D6.1 Report on the adaptation of PERSEO plant for the versatile production of chemical intermediates and identification of key process parameters. M18

D6.2 Report on the monitoring technologies and testing results. M32
D6.3 Report on the integration of the production in a biotechnological plant configuration. M36

D6.1 : Adaptation of PERSEO plant for the versatile production of chemical intermediates & identification of key parameters [18]

Adaptation of PERSEO plant for the versatile production of chemical intermediates & identification of key parameters

D6.2 : Monitoring technologies and testing results. [30]

Monitoring technologies and testing results.

D6.3 : Integration of the production in a biotechnological plant. [34]

Integration of the production in a biotechnological plant.

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS13	Main process parameters identified.	1 - IMECAL	18	Study of a versatile plant for bioethanol, LA and SA production.
MS14	Monitoring system built and tested.	8 - VISUM	30	The sensors and control system will be available for lab validation and suitable results will be reported.

Work package number [9]	WP7	**Lead beneficiary** [10]	4 - AUA
Work package title	Environmental, regulatory & economic analyses.		
Start month	1	**End month**	36

Objectives

. The objectives of WP7 are summarized in the follow points:
• Study the environmental suitability of the processes by a Life Cycle Assessment (LCA).
• To demonstrate the technical integration of the modelled processed through virtual simulation
• Assess the economic viability each the product stream through the development of a technical and economic evaluation to guide the decision points throughout the project and assess economic viability.
• Ensure that the new material and their processing technology complies with the relevant regulatory landscape.

Description of work and role of partners

WP7 - Environmental, regulatory & economic analyses. [Months: 1-36]
AUA, IMECAL, AIMPLAS, ATB, UGent, CENER, TBWR, VISUM, HAYAT, COVESTRO, EXERGY
Description of work.
Task 7.1 Evaluation of synergies with other projects involved in MSW valorisation. Task Leader: BIOPOLIS, Other partners: IMECAL, VISUM.
There are several projects aiming to make biofuels and/or bio-based products from MSW (FP6, FP7 and H2020). Some of them feed the waste to microorganism; some others first go via a hydrothermal liquefaction process or a gasification. In the second case, it is important to check that no chemical by-products will affect the downstream microorganism process. Synergies with current or past fuel related projects will be evaluated using the CORDIS database, personal contacts and information from the different partners will be employed. A stakeholder board, or an advisory board will be proposed in the first months of the project that include mainly companies and research centres from the biofuel sector (such as Lanzatech, Enerkem, French institute of Petroleum and New Energies, BP, Shell or Total). In terms of Board members, we plan to involve several related European associations.
Task 7.2. Modelling and simulation of the overall process. Mass and energy balances. Leader EXERGY, with the support of all partners
In order to evaluate the environmental impact of the project in terms of energy and mass streams, including the actual CO2 balance which aims to be positive, virtual simulations of the entire PERCAL process at the process scale and/or larger scale (as a case study) will be run in order to evaluate the different balances. The results will be complemented with the LCA outcomes in order to evaluate the real environmental impact of the project.
Additionally, simulations will be used for the optimisation of the process and the identification of potential hotspots. Results will be reported along with potential enhancements of the overall PERCAL process and will be used for the evaluation of the exergy analysis. The capacity for the up-scaled simulation will be decided during the project's execution.
Task7.3. Product Life-Cycle-Assessment. Leader AUA, with the support of all partners.
Life Cycle Assessment considers the entire life cycle of a product to assess the environmental impacts associated with all the stages of the product's life from-cradle-to-grave (i.e. from raw material extraction through material processing, manufacture, distribution, use, repair and maintenance, and disposal or recycling).
To ensure the consistency of the LCA studies performed in PERCAL, Task 7.3 will build upon existing and ongoing LCA activities on bio-based products and processes, and will take into account the guidelines of the ILCD handbook project , which has developed guidelines to make LCA studies more coherent and comparable. Furthermore, the LCA approach will be used to compare bio-based and non-biobased products within the same LCA framework, this comparative approach generates results which are more robust than a reliance on absolute values and will help in the marketing claims of the advantage of the new solutions vs. conventional synthetic counterparts.
The LCAs will be performed using dedicated software (e.g. Umberto NXT software) and the Ecoinvent database for a selected set of raw material/production process/product, taking into account use and disposal or re-use, i.e. 'from cradle-to-grave'. The LCA will follow the ILCD guidelines and ISO 14040 methodologies. This will include the definition of the following aspects: goal, scope and system boundaries; life cycle inventory; life cycle impact assessment and interpretation of the results.

A dynamic approach with feedback to the process development activities will be implemented, using the outputs of the LCA analysis. The pilot tests and demonstrators will likewise provide feedback to the Life Cycle Inventory, with up-to-date field data.
The LCA will serve both as a characterisation tool for the performance of the developed processes, and as a means of identifying areas where process optimisation can make the largest environmental savings. This information will be fed back to the RTD development and scale-up activities so that effective process optimisations may be realised.
Task 7.4 Economic analysis. Lead partner: EXERGY, Other partners: All partners
An economic study is required to validate the economic feasibility of the improved materials and technologies developed in the project and take further decisions. Parallel to the environmental analysis, an economic analysis of the materials will be conducted using life cycle costing (LCC) and will be complemented with techno-economic assessment (TEA). LCC is a methodology for the systematic evaluation of the economic costs associated with the entire life cycle of a product or process, including not only direct costs but also external relevant costs and benefits. LCC also follows a life cycle approach, but it assesses the economic costs of the product or process instead of the environmental impacts as in LCA. The procedure used for the economic analysis will be similar to that used for the environmental analysis. The economic costs of current scenarios will be assessed and compared with the costs of alternative scenarios and current solutions. EXERGY will be responsible for conducting the LCC studies, which will take into account not only investment and operating costs but also other external relevant costs, such as environmental costs.
Task 7.5 Social analysis. Lead partner: EXERGY, Other partners: All partners
Social impacts must also be taken into account when developing new technologies. A social analysis is necessary to determine whether the products coming from MSW can be accepted by customers.
A social analysis of the materials and technologies will be conducted using social life cycle assessment (S-LCA), which will be performed in parallel to the environmental (LCA) and economic analyses (LCC). S-LCA is a methodology to assess the social and socio-economic aspects of materials or processes and their potential positive and negative impacts along their life cycle. These aspects assessed in S-LCA are those that may directly or indirectly affect stakeholders, including workers, local community, consumers, value chain actors and society. S-LCA also follows a life cycle approach, but the impacts in this case are related to socio-economic processes and comprise human rights, working conditions, health and safety, socio-economic repercussion, and so forth.
As in previous analyses, the procedure used for the social analysis will involve the assessment and comparison of social impacts between current scenarios and alternative scenarios, these latter being based on the new developments from this project.
EXERGY will be responsible for conducting the S-LCA studies and the data required for the development of the S-LCA studies will be provided by other project partners and by organise some discussions along the project with engaging researchers, industry, policymakers, NGOs and Trade Unions, as well as desk research on identifying potential issues.
Task 7.6: Process inefficiencies Analysis Leader: EXERGY.
This task will study process life cycles from a thermodynamic perspective in order to analyse inefficiencies at different stages in process and propose potential improvements. An exergy assessment methodology to determine specific exergy contents to all mass and energy in and outflows will be developed. Where necessary, computerized simulation tools (i.e. COMSOL or TRNSYS) will be utilized.
The potential for improvements will be assessed by the overall exergy efficiency, which is the real exergy over the exergy used in the process. This study will compare: (1) the efficiency of the processes before and after the implantation of PERCAL innovations and (2) the efficiency with regards to other similar biological processes.
Aside from the determination of the efficiency for the different heat transfers along the process, exergy analysis will establish the potential causes for the detected inefficiency and will decrease cost associated to operation and maintenance of the system by means of a suitable assignment of energy optimisation measures at different levels of the process.
Task 7.7. Regulatory and safety analysis. Task Leader: AIMPLAS, Other participants: All partners.
The verification that the products comply with the existing regulatory landscape for the use of the rest raw materials and for the use of the final products on the envisaged applications and fulfil EC Directives and specific national laws in the EU28, regulatory compliance will be regularly checked throughout the project . The assessment should be performed before the end of the project in order to enable any processing/product specification changes. Input from final users is required. A regulatory compliance report will be written to make sure that the products and technologies also fulfil all standards and legislation in force with regards to safety criteria.

Participation per Partner

Partner number and short name	WP7 effort
1 - IMECAL	

Partner number and short name	WP7 effort
2 - AIMPLAS	[illegible]
3 - ATB	[illegible]
4 - AUA	[illegible]
5 - UGent	[illegible]
6 - CENER	[illegible]
7 - TBWR	[illegible]
8 - VISUM	[illegible]
10 - HAYAT	[illegible]
11 - COVESTRO	[illegible]
12 - EXERGY	[illegible]
Total	[illegible]

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D7.1	Draft report on economic analysis	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D7.2	Simulation of the system and M&E balances	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	32
D7.3	Report on Life Cycle Assessment (LCA).	4 - AUA	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D7.4	Report on Social Life Cycle Assessment (S-LCA).	12 - EXERGY	Report	Public	34
D7.5	Report on regulatory analysis and safety aspects	2 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D7.6	Final report on economic analysis	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	36

Description of deliverables

D7.1 Report on economic analysis (Preliminary month 18 and Final Month 36). M18/36
D7.2. Final report on simulation of the system and M&E balances M32
D7.3 Report on Life Cycle Assessment (LCA). M36
D7.4 Report on Social Life Cycle Assessment (S-LCA). M34
D7.5 Report on regulatory analysis and safety aspects. M36

D7.1 : Draft report on economic analysis [18]

Draft report on economic analysis

D7.2 : Simulation of the system and M&E balances [32]

Simulation of the system and M&E balances

D7.3 : Report on Life Cycle Assessment (LCA). [36]

Report on Life Cycle Assessment (LCA).

D7.4 : Report on Social Life Cycle Assessment (S-LCA). [34]

Report on Social Life Cycle Assessment (S-LCA).

D7.5 : Report on regulatory analysis and safety aspects [36]

Report on regulatory analysis and safety aspects

D7.6 : Final report on economic analysis [36]

Final report on economic analysis

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS15	Environmental performance of the new materials and processes according to standards and EU legislation in force.	2 - AIMPLAS	36	Check the environmental regulatory assessments that affect the product.
MS16	Final cost of intermediates and products at least 15% cheaper than current biobased market analogues.	6 - CENER	34	Final costs of the products taking into account all process steps.
MS17	The new bio-based products and intermediates fulfil the related regulatory requirements.	1 - IMECAL	36	Check the regulatory assessments that affect the product.

Work package number [9]	WP8	**Lead beneficiary** [10]	12 - EXERGY
Work package title	Communication, Dissemination, Exploitation, Business plan and Technology Transfer		
Start month	1	**End month**	36

Objectives

This WP encompasses the development and delivery of the project Plan for Exploitation of the Results and Dissemination (PERD).WP8 comprises of six interlinked actions designed to ensure successful exploitation of the projects technical innovations: 1) Communication, 2) Dissemination, 3) Exploitation, 4) Business Models and Plans, 5) Intellectual Property Management and 6) Technology Transfer.

Description of work and role of partners

WP8 - Communication, Dissemination, Exploitation, Business plan and Technology Transfer [Months: 1-36]
EXERGY, IMECAL, AIMPLAS, ATB, AUA, UGent, CENER, TBWR, VISUM, HAYAT
Description of work.
Task 8.1 Communication. Task Leader: AIMPLAS with the support of all partners.
The purpose of the communication plan is to maximise awareness of project aims and outputs amongst a broad stakeholder base. Communication activities complement dissemination actions as they add public value to the achievements of the project by transforming the sometimes-complex scientific and technological results into media resources. The communication plan will identify channels for tailored, target-specific communication and will include guidelines for using and disseminating the knowledge gained during the project lifetime. This plan will identify the relevant trade media (e.g. bio-based chemicals and materials, sustainability, industrial biotechnology as well as household products, adhesives, paints, etc. and the relevant specialty chemicals publications).
Stakeholder mapping - At the start of the project, the Consortium will review and expand the list (email pool) of groups, projects, experts and other stakeholders, identified during the proposal phase, and develop a stakeholder mapping to use in its communication activities.
Messaging – WP8 will work on defining the key messages that need to be used in the communications materials. The messages, based on scientific information, will be concise, clear and easy to understand, and tailored to the target stakeholder group. The project communications tools will include: a) Project logo and marketing communications tools - A PERCAL logo along with the project's visual identity will be designed and produced. The logo and the project's visual identity will be rolled out to the different communications tools that will be produced to engage consumers; b) Project website and communication channels - At the beginning of the project, WP8 will set up the PERCAL website. The website (in English language) will be developed to provide an overview of the project objectives and programme of work (i.e project's workshop, events and other activities), to track its progress and present relevant results of individual work packages in near real time (public deliverables). Another channel for dissemination of the PERCAL objectives, organisation and results is through social media. WP8 will set-up and maintain a presence in social media networks (Facebook, Twitter, YouTube). The PERCAL social media presence will broaden the reach to the public and experts interested in following the project and offering feedback.
Task 8.2 Dissemination. Task Leader: AIMPLAS with the support of all partners.
The purpose is to support project exploitation through technical dissemination and industrial outreach. The PERCAL Consortium will use a variety of platforms tailored, when possible, to reach particular industry and research groups and maintain communication with them. Press releases will be drafted and disseminated and opportunities to publish through the trade media will be compiled. The consortium will also carefully select events organised by others to participate in and present the project results towards the end of it.
A dissemination plan (a Working Document among the partners) with tailored means to reach main target audiences will be developed and will designate specific points of contact for value chain participants, research communities, press, the general public and the European bioeconomy process industry that will be maintained during the course of the project, so that the PERCAL consortium can always be contacted by relevant actors in the field and interested persons for any query on content and/or findings but also as opportunities for possible collaborations.
Task 8.3 Exploitation. Task Leader: EXERGY with the support of all partners.
The Exploitation Plan will be designed at the beginning of the project as a working document and updated constantly throughout the project's life cycle. The plan will be reviewed yearly with progress and identified issues recorded and exploitations actions updated. The Exploitation Plan will address the following aspects: 1) List of project results and identification of key exploitable results (KER i.e. most promising results), 2) the final exploitation plan will incorporate

the information and knowledge developed in Task 8.4 and also an exploitation risk analysis, 3) an initial version of the exploitation plan will be drawn with the initial results identified in the proposal stage.
At the end of the project, a detailed exploitation plan will be finalised. Exploitable outputs will be identified and discussed at regular exploitation workshops;
Exploitation workshops - The whole consortium will be involved in exploitation workshops that will be led by EXERGY. The objective of the workshops will be the characterisation of the exploitable results and identification of new results as the project progresses. The workshops will serve as screening mechanism to identify the key exploitable results to be taken through to business model and plan development (Task 8.4). The exploitation workshops will be organised on a yearly basis, and whenever possible just before/after the General Assemblies. The outcomes of these workshops will feed into the Exploitation Plan.
Exploitation risk analysis will lead to a risk management plan, taking into account the internal and external factors which may prevent the successful exploitation of the project results by follows three steps: 1) the identification of exploitation risks. They may include technical and non-technical; the risk register will draw from the information and knowledge gathered in all work packages & tasks, 2) the assessment of these exploitation risks along two dimensions: probability of occurrence & potential impact. A matrix combining these two dimensions then allows ranking the different risks and 3) the design of a risk management plan detailing the actions to put any type of exploitation risk under proper control. The risk management will be implemented during the project with recommendations for the pre-exploitation phase in the final exploitation plan.
Task 8.4 Business Models and Plans for KER. Task leader: EXERGY with the support of all partners.
Development of business models: EXERGY will develop business models to support the continued exploitation and eventual commercialisation of each key project result (please see Impact section for more details). This task will be led in coordination with the dissemination/communication (Tasks 8.1/8.2) activities to ensure the coherence of the project's messages towards the outside world.
The task will involve outlining, refining and defining the business models through two actions:
Action 1. Business Model Development. A two-step process will be used to build an initial plan. Step 1) Understand, refine and define the innovations value proposition through the analysis of, and discussion with consortium partners and through literature review considering 1) Policy/Environmental/Social/Technological (PEST) issues, 2) competitive intensity analysis, 3) the need for alliances and collaboration and finally a Strengths/Weaknesses/Opportunities/Threats (SWOT) analysis. Step 2) Build economic models describing the cost and revenue projections for the innovations, this allows profit margins to be calculated and business sensitivities to be identified. This action will draw on information produced in Task8.2. Estimating revenue potential draws on the knowledge gathered in the development of the value proposition. The economic model will allow understanding of future cash flows and highlight the need for additional funding or external investment to reach the first sales.
Action 2. Business Model Validation. A two-step process will be used to validate the business model. Step 1) To ensure valuable feedback, careful stakeholder identification and recruitment is required. In collaboration with project partners, a targeted list based on stakeholder profiles will be drawn-up. The target list will ensure appropriate value chain coverage with an emphasis on the potential to become early innovation adopters. Stakeholders will be recruited through an initial contact, introduction of the value proposition and request for initial feedback, recruitment will draw on the stakeholder database developed for dissemination activities. Step 2) Feedback on the business model will be captured through interviews, questionnaires and, if relevant, visits to the experimental demonstration sites, depending on the targets' profiles. The feedback analysis will lead to a review and update of the value proposition and revenue/cost models within the business models.
Business Plan: The developed business models will be detailed in a business plan that will be the central part of the exploitation plan. Business Plan will be developed as a sustainability strategy outlining what should happen to the project outputs at the end of the project, and to explore how they can be sustained. The business plan will include client definition, a strategy for marketing, sales and operations financial models and go-to market strategies.
New business opportunities will be assesed during the drafting of the different business models and the business plan. The new value chains (three at least) will be analysed from a business approach along with the real opportunities of penetration in the market from MWS as raw material. The impact on the actual industrial sectors where PERCAL is envisaged to be introduced (See Section 2.1) will be determined and compared to current industries practises. This activity, in alignment with the exploitation activities, will set the most approapiate route for the results: licensing, marketing/direct sales or vending. The analysis of the new business opportunities will be performed taking into consideration these options for the consortium members as well as other biobased stakeholders which might be involved in the new value chains.
Task 8.5 Intellectual Property Management. Task leader: EXERGY with the support of all partners.
This task will undertake a regular review and assessment of the consortiums 'Freedom to Operate' in the areas of exploitable outputs. Reviewing 'Freedom to Operate' throughout the project will ensure that the consortium is safe to produce a product or use a production technology without infringing or violating third party intellectual property

rights or highlight the need for mitigating actions by the consortium (eg. acquisition of license, which would impact the exploitation).
Furthermore, IPR management will include the mapping of existing patents and potentially overlapping IPR; and when the developing partner identifies possible novelty in a KER being developed, an assessment of the new knowledge generated during the project (outputs and other knowledge such as data); a proposition of optimal IPR protection options (e.g. patent, copyright, trademark, confidentiality), in line with the selected business model options (Task 8.4) and taking into account possible co-ownership; the proper implementation of the IPR protection measures during the project. The findings and recommendations for IPR protections will be included in the final exploitation plan (D8.3). VISUM will lead the IPR management by providing guidance to the partners for their FTO and patentability search but each partner will be responsible of the right management of his result. When necessary, after the initial identification of a promising (patentable) idea, external professional support from a patent agent will be sought.
Task 8.6 Technology Transfer. Task Leader: EXERGY with the support of all partners.
Technology transfer will target the wide range of production companies in the interest areas of the project e.g. protein/ lipid separation and chemical building blocks for a wide range industrial applications. The activities planned to achieve a successful technology transfer are: a) Workshops/seminars oriented to different sectors (microbiology, green chemistry, engineering, bioprocesses, biorefinery, etc.). The two workshops will be organised in M24 and M36, in two different European countries (TBD), in cooperation with Task 8.2. The subject will be selected at M18 and M30, depending on the intermediate results; b) Training at companies' facilities. Technology transfer will be facilitated through the arrangement of 'short stay' training courses designed for scientists/Technicians to take place at partner companies' facilities; c) Technical Guidance Document: A 'Guide to the production and use of green chemical intermediates using MSW as a main raw material' will be prepared and published after completion of the project, in order to contribute to the technology transfer, dissemination and training. According to best industrial and environmental practices, all partners will co-operate in the elaboration of this Guide, which will contribute to an optimal profit of the new chemical products advantages. This guide will be developed by all Partners (D8.8) and will include the safety, regulatory, economic and environmental issues identified in WP8, and will be made public in the second Project Workshop (M36); d) Publishing scientific articles in journals and conferences (subject to the consortium's IP protection publication policy) and detailed in table 2.2.c. (section 2.2); e) Development of policy briefs and case studies targeting policy makers responsible for drafting EU or national legislation, regulations and guidelines affecting MSW based products valorisation, including sustainability guidelines and standards. This will also be fuelled by the results of task 8.5 in terms of influence of current policies on the market deployment of the project results (eg. barrier or gaps).
Task 8.7 Set up of PERCAL Stakeholders' Community of Interest. Task Leader: IMECAL with the support of all partners.
This task aims at tying up a critical mass of stakeholders to i) amplify the impacts of PERCAL at EU and international level by involving them in project activities performed in previous tasks ii) understand their specific market demands and properly address to their needs. This implies the execution of a stakeholder analysis (identify and prioritize who to mobilise and engage with) approaching key actors across all stages of the value chain. The analysis will be based on a 3-step process: i) Identify additinonal Stakeholders ii) Create a Stakeholder Map iii) Create a Stakeholder Management Strategy considering their knowledge, interests, positions, alliances, relationships, key roles and responsibilities. Target stakeholders will include chemical companies, MSW managers companies, policy makers, industrial associations, companies/reseach centres/universities that are leading projects with the same target (identified in task 7.1) and potential users of target chemical intermediates.
Measures for establishing and maintaining stakeholder liaisons include: 1) Bilateral and multilateral exchange of information, participation in roundtables, open days, action groups. 2) Open stakeholder consultation by means of the internet; submitting surveys and interviews in order to gather relevant information to feed into subsequent activities of this WP. The stakeholders' community will be approached by providing them all the available knowledge on PERCAL and other potential paralel routes of research identified during the project.

Participation per Partner

Partner number and short name	WP8 effort
1 - IMECAL	5.00
2 - AIMPLAS	2.00
3 - ATB	2.00
4 - AUA	2.00

Partner number and short name	WP8 effort
5 - UGent	2.00
6 - CENER	2.00
7 - TBWR	2.00
8 - VISUM	2.00
10 - HAYAT	0.30
12 - EXERGY	10.00
Total	29.30

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D8.1	Project web site	2 - AIMPLAS	Report	Public	3
D8.2	Initial Communication and dissemination Plan	2 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D8.3	Draft communication activities & dissemination Plan updates	2 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D8.4	IPR policy implementation in line with Exploitation Plan	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D8.5	Report of the IPR management activities	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D8.6	Final Exploitation Plan including Business Plans of key results	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D8.7	Technology Transfer Activities, including training to SMEs, other activities and workshops from M1 to M18	12 - EXERGY	Report	Public	18
D8.8	Guide to explore the conversion routes of	1 - IMECAL	Report	Confidential, only for members of the	34

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
	MSW in high added value products.			consortium (including the Commission Services)	
D8.9	Final version of the PEDR	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D8.10	Technology Transfer Activities, including training to SMEs, other activities and workshops from M18 to M36	12 - EXERGY	Report	Public	36
D8.11	Final communication activities & dissemination Plan updates	2 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	36

Description of deliverables

D8.1 Project web site. M3
D8.2 Initial Communication and dissemination Plan. M6
D8.3 Report on performed communication activities and dissemination Plan updates. M18
D8.4 IPR policy implementation in line with the Exploitation Plan. M6
D8.5 Report of the IPR management activities. M36
D8.6 Final Exploitation Plan including Business Plans of key results. M36
D8.7 Report of Technology Transfer Activities, including training to SMEs, other activities and workshops. M18
D8.8 Guide to explore the conversion routes of MSW in high added value products. M34
D8.9. Final version of the PEDR. M36
D8.10 Technology Transfer Activities, including training to SMEs, other activities and workshops from M18 to M36 M36
D8.11 Final communication activities & dissemination Plan updates M36

D8.1 : Project web site [3]

Project web site

D8.2 : Initial Communication and dissemination Plan [6]

Initial Communication and dissemination Plan

D8.3 : Draft communication activities & dissemination Plan updates [18]

Draft communication activities & dissemination Plan updates

D8.4 : IPR policy implementation in line with Exploitation Plan [6]

IPR policy implementation in line with Exploitation Plan

D8.5 : Report of the IPR management activities [36]

Report of the IPR management activities

D8.6 : Final Exploitation Plan including Business Plans of key results [36]

Final Exploitation Plan including Business Plans of key results

D8.7 : Technology Transfer Activities, including training to SMEs, other activities and workshops from M1 to M18 [18]

Technology Transfer Activities, including training to SMEs, other activities and workshops from M1 to M18

D8.8 : Guide to explore the conversion routes of MSW in high added value products. [34]

Guide to explore the conversion routes of MSW in high added value products.

D8.9 : Final version of the PEDR [36]

Final version of the PEDR

D8.10 : Technology Transfer Activities, including training to SMEs, other activities and workshops from M18 to M36 [36]

Technology Transfer Activities, including training to SMEs, other activities and workshops from M18 to M36

D8.11 : Final communication activities & dissemination Plan updates [36]

Final communication activities & dissemination Plan updates

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS18	Project website running.	2 - AIMPLAS	36	Website uploaded content & running.
MS19	Advanced exploitation strategy defined.	12 - EXERGY	18	Previous to issue the draft PEDR, the exploitable results are completely agreed, as well as co-ownership and preliminary business plans for each result outlined.
MS20	Final Business/ Exploitation & Communication Plans agreed.	12 - EXERGY	36	Agreement on final Business and Communication Plans.
MS21	Up to eight peer-reviewed papers and two patents ready to be submitted.	12 - EXERGY	36	Paper submitted to different journals and patents prepared to be submitted.
MS22	Two workshops successfully organized.	12 - EXERGY	36	Assistance of 50 firms in each one.

Work package number [9]	WP9	**Lead beneficiary** [10]	1 - IMECAL
Work package title	Project Management		
Start month	1	**End month**	36

Objectives

Overall project co-ordination and management. To optimize the resources and to ensure that all BBI JU requirements for communication and reporting are met. Specific objectives:
• Setting up of Project Management and Administration: Definition of Management Procedures; development of procedures, rules and templates for the consortium reporting activity, management platform through web-based project intranet.
• Management Co-ordination of the Project: Administrative and financial project management; legal representation of the project.
• Project Administration, control and support activities: Co-ordination and development activity reporting; Requesting and processing cost statements and cost certificates; Follow-up the progress of the project; Provide generic support at project level.
• Project Communication: Define, provide, set-up and maintain the necessary communication procedures and tools between all the partners of the project to ensure an efficient communication flow. Communication with the BBI JU representatives.
Risk management: to carry out a continuous (ongoing) assessment of the progress of the project. To review the progress of the work in relation to the defined success measures and to report on this progress. Where appropriate, contingency plans will be identified and implemented.

Description of work and role of partners

WP9 - Project Management [Months: 1-36]
IMECAL, AIMPLAS, ATB, AUA, UGent, CENER, TBWR, VISUM, COVESTRO, EXERGY
Description of work.
Task 9.1 Management of the Resources. Task Leader: IMECAL Involved partners: All WP leaders.
The project will be managed on a daily basis by the lead partner IMECAL with the support of AIMPLAS. Full consortium project management meetings will be held every quarter to discuss technical progress and contractual issues (conference calls). Project management duties will include a dedicated project intranet which will act as a project management and communication tool and will include a password protected area for the sharing of meeting arrangements, minutes, reports and financial statements.
The Project Coordinator has the responsibility of managing the project cost effectively, including day to day management aspects, distribution of work within project partners, assessment of results, risk management, etc. Due to the number of partners involved (12) and the number of Member States (9), the management effort required should not be underestimated, requiring of an efficient and well-designed management structure. The project management structure, i.e. Project Coordinator, General Assembly (GA) and Executive Board (EB) is fully described in section 3.2. The General Assembly, representing the voice of the whole consortium, will form the highest decision making body. The EB will form the decision-making and arbitration body (on behalf of the GA) and will be responsible for the overall strategic direction of the project, in order to ensure that the project progresses properly and the risks are properly identified and managed. Intellectual/Industrial Property Rights (IPR) are managed by the IPR & Exploitation Board (IPREB), whose responsibilities are also described in section 3.2. In all these boards, the membership ensures that there is a strong industrial leadership in all Project activities. The TMC, formed by the WP leaders, who are in turn responsible for the coordination of all the activities performed by each participant within the work package and for on time delivery of their specific tasks and the subsequent report of problems or special issues, will assess the technical progress of the Project reporting to the EB and the GA.
Management Co-ordination. The Project Co-Coordinator is IMECAL, supported by AIMPLAS, who will assume the administrative management of the project, assuring the project's progress. The Coordinator supports the Project Consortium in legal, administrative and financial issues. This co-ordination will lead the strategic, political, economic and operational of the project, articulated through the different Boards. AIMPLAS will thus be in charge of the preparation, organization, administration, minuting and follow up of the main project meetings (GA, EB, IPREB, TMC) . The tasks are fully detailed in section 3.2.2.

Project administration. Progress of each task will be assessed every three months to closely control the development activities of the partners in order to ensure that there is no deviation from the original plan or that corrective actions are taken in case of deviations. Financial control will be performed for the official reporting periods (requesting and processing the cost statements and the necessary cost certificates) as well as periodic Intermediate internal financial controls. It is foreseen to send to the BBI JU, two official progress reports (M15, M36) and a final report in M48. Details about the methods for monitoring and reporting progress, and documenting results, are provided in Section 3.2. An internal project web site will be created at the beginning of the project, in order to facilitate the monitoring and reporting of progress and documenting of results (see WP9). The website will also allow interactive internal communication between the partners involved in the different tasks. An overall description of all the Committees, their membership and procedures is provided in section 3.2.
Task 9.2 Risk management. Task Leader: IMECAL. Involved partners: TMC Members
Based on the initial risk identification reported in table 3.1.c., at the beginning of the project and for each work package, the risk indicators will be defined taking into account the overall project progress, and will be reported in M9.1 and D9.1 Each WP leader will collect the risk indicators and will monitor technical progress at WP level along the project implementation. Progress of each task will be assessed every 3 months to control the developed activities of the partners in order to ensure that there is no deviation from the original plan or that corrective actions are taken when appropriate. The TMC and EB will review and assess the progress of each WP in relation to the identified risk indicators and assist the WP leaders. When appropriate, contingency plans and countermeasures will be developed and when approved by the EB, will be discussed with the BBI JU Scientific Officer. If approved, they will be implemented. The progress will be reported periodically, in the contractual reports (month 15, 36) and in the internal intermediate progress reports, where deviations from the work programme objectives (Deliverables and Milestones) are identified, they will be addressed and corrective actions/plans and countermeasures implemented to ensure that the Project reaches its planned outcomes.
Task 9.3. Communication with BBI JU. Task Leader: IMECAL.
The coordinator is the sole point of contact between the BBI JU and the project consortium. IMECAL will ensure that all milestones are met and that all reports are prepared in accordance with the contractual requirements. A Milestone report will be issued and sent to the BBI JU describing the Milestone completion and its implication in the project progress. The structure for monitoring and reporting progress will consist of a series of reports and meetings. Technical reports will be sent to the BBI JU to meet the reporting pattern expected. Documentation will be in English and in Microsoft Word format. All reports will be assembled by the Project Coordinator in a master file, which will be uploaded to the project intranet. Review procedures will be agreed and included in the Consortium Agreement to ensure that all the project deliverables reach a satisfactory quality standard (e.g. technical deliverables will be assessed by the TMC or certain appointed members and the Project Coordinator prior to submitting them to the EC).

Participation per Partner

Partner number and short name	WP9 effort
1 - IMECAL	14.00
2 - AIMPLAS	7.00
3 - ATB	1.00
4 - AUA	1.00
5 - UGent	1.00
6 - CENER	1.00
7 - TBWR	0.50
8 - VISUM	1.00
11 - COVESTRO	0.50
12 - EXERGY	1.00
Total	28.00

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D9.1	Risk indicators and Risk assessment	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	9

Description of deliverables

D9.1 : Risk indicators and Risk assessment [9]
Risk indicators and Risk assessment

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS23	WPs risk indicators identified.	1 - IMECAL	3	Risk indicators for all WPs identified in order to start Risk Management procedures.

Work package number [9]	WP10	**Lead beneficiary** [10]	1 - IMECAL
Work package title	Ethics requirements		
Start month	1	**End month**	36

Objectives

The objective is to ensure compliance with the 'ethics requirements' set out in this work package.

Description of work and role of partners

WP10 - Ethics requirements [Months: 1-36]
IMECAL
This work package sets out the 'ethics requirements' that the project must comply with.

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D10.1	NEC - Requirement No. 1	1 - IMECAL	Ethics	Confidential, only for members of the consortium (including the Commission Services)	1

Description of deliverables

The 'ethics requirements' that the project must comply with are included as deliverables in this work package.

D10.1 : NEC - Requirement No. 1 [1]

Partner 10: Hayat Kimya San. A.S. is based in Turkey. As such, the applicants should confirm to the BBI that the ethical standards and guidelines of Horizon2020 will be rigorously applied, regardless of the country in which the research is carried out. It is recommended that the applicants should ensure that appropriate health and safety procedures conforming to relevant local/national guidelines/legislation are followed for staff involved in this project.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

1.3.4. WT4 List of milestones

Milestone number[18]	Milestone title	WP number[9]	Lead beneficiary	Due Date (in months)[17]	Means of verification
MS1	Composition of MSW available	WP1	1 - IMECAL	8	Comparison with reported values in literature for similar feedstocks.
MS2	Improved enzymatic hydrolysis.	WP1	1 - IMECAL	12	Comparison of hydrolysis yields
MS3	Samples of hydrolysed MSW available for the different subsequent tasks	WP1	1 - IMECAL	20	Sufficient amounts of different fractions for lab tests within WP2, WP3 and WP4
MS4	LA with purity higher than 97% for L-LA .	WP2	3 - ATB	22	Application of chiral HPLC to estimate the optical purity of all LA samples
MS5	Yield at pilot plant facilities similar to other feedstocks already tested.	WP2	3 - ATB	30	Max. 10% losses compared to current process using pure lignocellulosic biomass feedstock.
MS6	Production of succinic acid in fed-batch and continuous processes	WP3	4 - AUA	15	Report on lab-scale experiments and comparison with preliminary experiments with the integrated system.
MS7	Extraction of succinic acid directly from the extraction cell at 99% purity.	WP3	4 - AUA	24	Report on the purity of succinic acid crystals extracted directly from the concentrated solution produced via membrane electrolysis.
MS8	A method for the production of amino acid and lipid surfactants from MSW by-products.	WP4	6 - CENER	32	Comparison of surface active properties with those described in literature.
MS9	Biosurfactants has fulfil the HAYAT internal standards.	WP4	10 - HAYAT	34	Comparison of bio-surfactant properties with commercial ones using DIN/EN/ISO standard.
MS10	Simultaneous esterification & products separation (reactive distillation) for ethyl lactate established.	WP5	7 - TBWR	34	Continuous mode reactive distillation established and validated at scale 3-5 L/h.
MS11	Renewable HMA has fulfil the end-user requirements or SOFTER internal standards.	WP5	2 - AIMPLAS	34	Comparison of HMA main properties with commercial EVA and renewable source HMA using the suitable EN/ ISO.

Milestone number[18]	Milestone title	WP number[9]	Lead beneficiary	Due Date (in months)[17]	Means of verification
MS12	Polyester-Polyols by conversion of SA with diamines have similar reactivity than using commercial bio-based SA (less than 20% time increase) and the PU formation obtained have at least 80% of performance than commercial ones.	WP5	11 - COVESTRO	34	Comparative analysis using internal and well-defined methodology of COVESTRO.
MS13	Main process parameters identified.	WP6	1 - IMECAL	18	Study of a versatile plant for bioethanol, LA and SA production.
MS14	Monitoring system built and tested.	WP6	8 - VISUM	30	The sensors and control system will be available for lab validation and suitable results will be reported.
MS15	Environmental performance of the new materials and processes according to standards and EU legislation in force.	WP7	2 - AIMPLAS	36	Check the environmental regulatory assessments that affect the product.
MS16	Final cost of intermediates and products at least 15% cheaper than current biobased market analogues.	WP7	6 - CENER	34	Final costs of the products taking into account all process steps.
MS17	The new bio-based products and intermediates fulfil the related regulatory requirements.	WP7	1 - IMECAL	36	Check the regulatory assessments that affect the product.
MS18	Project website running.	WP8	2 - AIMPLAS	36	Website uploaded content & running.
MS19	Advanced exploitation strategy defined.	WP8	12 - EXERGY	18	Previous to issue the draft PEDR, the exploitable results are completely agreed, as well as co-ownership and preliminary business plans for each result outlined.
MS20	Final Business/ Exploitation & Communication Plans agreed.	WP8	12 - EXERGY	36	Agreement on final Business and Communication Plans.

Milestone number[18]	Milestone title	WP number[9]	Lead beneficiary	Due Date (in months)[17]	Means of verification
MS21	Up to eight peer-reviewed papers and two patents ready to be submitted.	WP8	12 - EXERGY	36	Paper submitted to different journals and patents prepared to be submitted.
MS22	Two workshops successfully organized.	WP8	12 - EXERGY	36	Assistance of 50 firms in each one.
MS23	WPs risk indicators identified.	WP9	1 - IMECAL	3	Risk indicators for all WPs identified in order to start Risk Management procedures.

1.3.5. WT5 Critical Implementation risks and mitigation actions

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
1	Failure to accomplish deadlines and BBI JU procedures	WP1, WP10, WP2, WP3, WP4, WP5, WP6, WP7, WP8, WP9	Risk: Delays Mitigation: Experienced Coordinator that will keep continuously updated about new procedures.
2	IPR/innovation conflicts	WP8	Risk: lack of cooperation, delays Mitigation: a solid Consortium Agreement will be signed before accessing to the G.A, which will include clear rules, Background included/excluded, clear IPR ownership & consensus and conflict resolution mechanisms through the IPREB, external arbitration if necessary.
3	Partners not committed, under-performing, lack capabilities or leaving	WP9	Risk: Work not done, delays Mitigation: The Partners have a strong network of contacts that can replace such a partner and the RO have broad capabilities to fill some temporary gaps.
4	Insufficient cooperation among Partners	WP9	Risk: Work not done, delays Mitigation: frequent meetings to keep partners connected. Strong and proven Organizational structure to ensure workflow.

1.3.6. WT6 Summary of project effort in person-months

	WP1	WP2	WP3	WP4	WP5	WP6	WP7	WP8	WP9	WP10	Total Person/Months per Participant
1 - IMECAL	19	3.	3	18	6	23	3	5	14		88
2 - AIMPLAS	5	0·	0	0	46	0	3	2	7		63
3 - ATB	2	30	4	6	0	1	1.	2	1		41
4 - AUA	0.20	1	50	0	0	1·	19	2	1		74.20
5 - UGent	0.20	0	21	0	0	0	1	2	1		25.20
6 - CENER	1	0	7	25	0	0	3	2	1		39
7 - TBWR	0.20	2.50	0	0	35	0	1	2	0.50		41
8 - VISUM	2	0	0	0	0	25	2	2	1		32
9 - SOFTER	0	·0	0	0	0.70	0	0	0	0		·0.70
10 - HAYAT	1	2	0	2	3.50	0	0.30	0.30	0		·9.10
11 - COVESTRO	0.30	0	0	0	4	0	0.30	0	0.50		5.10
12 - EXERGY	0.20	0	0	0	6	0	18	10	4		29.20
Total Person/Months	31.10	38.30	85	45	89.20	50	51.60	29.30	28		447.50

1.3.7. WT7 Tentative schedule of project reviews

Review number [19]	**Tentative timing**	**Planned venue of review**	**Comments, if any**
RV1	18	venue tbc, probably BBI JU offices in Brussels	This review will be based on the documents submitted within the period in question: periodic technical report, financial statements, and all deliverables due in the reporting period plus possible other documents (e.g. needed for the interim impact evaluation). The reference document for the progress assessment is the GA and especially Annex I (Description of Action, DoA)

1. Project number

The project number has been assigned by the Commission as the unique identifier for your project. It cannot be changed. The project number **should appear on each page of the grant agreement preparation documents (part A and part B)** to prevent errors during its handling.

2. Project acronym

Use the project acronym as given in the submitted proposal. It can generally not be changed. The same acronym **should appear on each page of the grant agreement preparation documents (part A and part B)** to prevent errors during its handling.

3. Project title

Use the title (preferably no longer than 200 characters) as indicated in the submitted proposal. Minor corrections are possible if agreed during the preparation of the grant agreement.

4. Starting date

Unless a specific (fixed) starting date is duly justified and agreed upon during the preparation of the Grant Agreement, the project will start on the first day of the month following the entry into force of the Grant Agreement (NB : entry into force = signature by the JU). Please note that if a fixed starting date is used, you will be required to provide a written justification.

5. Duration

Insert the duration of the project in full months.

6. Call (part) identifier

The Call (part) identifier is the reference number given in the call or part of the call you were addressing, as indicated in the publication of the call in the Official Journal of the European Union. You have to use the identifier given by the Commission in the letter inviting to prepare the grant agreement.

7. Abstract

8. Project Entry Month

The month at which the participant joined the consortium, month 1 marking the start date of the project, and all other start dates being relative to this start date.

9. Work Package number

Work package number: WP1, WP2, WP3, ..., WPn

10. Lead beneficiary

This must be one of the beneficiaries in the grant (not a third party) - Number of the beneficiary leading the work in this work package

11. Person-months per work package

The total number of person-months allocated to each work package.

12. Start month

Relative start date for the work in the specific work packages, month 1 marking the start date of the project, and all other start dates being relative to this start date.

13. End month

Relative end date, month 1 marking the start date of the project, and all end dates being relative to this start date.

14. Deliverable number

Deliverable numbers: D1 - Dn

15. Type

Please indicate the type of the deliverable using one of the following codes:

R	Document, report
DEM	Demonstrator, pilot, prototype
DEC	Websites, patent fillings, videos, etc.
OTHER	
ETHICS	Ethics requirement
ORDP	Open Research Data Pilot

16. Dissemination level

Please indicate the dissemination level using one of the following codes:

PU	Public
CO	Confidential, only for members of the consortium (including the Commission Services)
EU-RES	Classified Information: RESTREINT UE (Commission Decision 2005/444/EC)
EU-CON	Classified Information: CONFIDENTIEL UE (Commission Decision 2005/444/EC)
EU-SEC	Classified Information: SECRET UE (Commission Decision 2005/444/EC)

17. Delivery date for Deliverable

Month in which the deliverables will be available, month 1 marking the start date of the project, and all delivery dates being relative to this start date.

18. Milestone number

Milestone number:MS1, MS2, ..., MSn

19. Review number

Review number: RV1, RV2, ..., RVn

20. Installation Number

Number progressively the installations of a same infrastructure. An installation is a part of an infrastructure that could be used independently from the rest.

21. Installation country

Code of the country where the installation is located or IO if the access provider (the beneficiary or linked third party) is an international organization, an ERIC or a similar legal entity.

22. Type of access

VA	if virtual access,
TA-uc	if trans-national access with access costs declared on the basis of unit cost,
TA-ac	if trans-national access with access costs declared as actual costs, and
TA-cb	if trans-national access with access costs declared as a combination of actual costs and costs on the basis of unit cost.

23. Access costs

Cost of the access provided under the project. For virtual access fill only the second column. For trans-national access fill one of the two columns or both according to the way access costs are declared. Trans-national access costs on the basis of unit cost will result from the unit cost by the quantity of access to be provided.

History of changes

PART B.

#	Description
1	IRIS will be replaced by its Third party VISUM – therefore the name will be VISUM
2	Information on biocatalyst selection provided (page 12)
3	Potential technology and bioproducts customers clearly targeted in communication activities (page 34)
4	Mitigation measure corrected (page 43)
5	Change partner 8 description and remove section 4.2 data, there are no third parties involved.
2nd ROUND	
6	EC has been replaced by BBI JU along the document
7	Table 3.3 has been allocated in section 3.4 and rename as Table 3.4c)
8	Minor changes in the table 3.4b) in order to be consistence with the budget request. The Funding request remains unchanged, it is the same as in the submitted proposal.
9	Ethics requirements section 5 – updated as gathered in the ethics reports
3rd ROUND	
10	Description of CENER section 4
11	Section 3.4: Table 3.4 and section 3.4 details about the breakdown of the project budget and include in IMECAL and CENER subcontracting category
12	Section 3.2: AIMPLAS chair the Executive Board and IMECAL chair the General Assembly
13	Remove BIOPOLIS along the whole part B

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1. EXCELLENCE

1.1. OBJECTIVES.

Cost competitive value chains producing added-value products from heterogeneous and complex waste material create solutions for the environmental problem of increasing waste flows, reducing pressure on virgin resources, and increasing the competiveness of European industry. In this line, **IMECAL partner has developed a biotechnological patented technology** (European Patent EPO 2112226, USA Patent 8399228 B2 and international PCT WIPO 099038for obtaining second generation bioethanol from Organic Municipal Waste. In this project IMECAL has collaborated with CIEMAT and several Universities and private companies such as Ford Spain & NOVO. IMECAL has built a second generation bioethanol plant with capacity to process 25 t/day of organic fraction of MSW (PERSEO Bioethanol, www.imecal.com/perseo/). During more than 9 years of the **PERSEO** project, the hydrolysis enzymatic process has been optimised for different MSW feedstocks. **PERCAL will use this knowledge/background as a starting point to produce new high added value products from MSW**.

PERCAL will exploit Municipal Solid Waste (MSW) as feedstock to develop **intermediate chemical products at high yield and low impurity level with huge industrial interest**. These will be complementary to the bioethanol (current PERSEO technology), to achieve a **cascade valorisation of the MSW components**, i.e.:

- **Lactic *acid (LA)*** to produce: 1) Eco-friendly ethyl lactate solvents by reactive distillation from ***lactic acid*** & ***bio-ethanol*** to be used in cleaning products and inks and 2) hot-melt adhesives for cardboard and other non-food applications in combination with maleic anhydride by reactive extrusion.
- ***Succinic acid (SA)*** as an intermediate building blocks to production of polyols for the polyurethane industry.
- ***Biosurfactants*** by chemical and/or microbiological modification of protein and lipid fraction from remaining fraction of MSW fermentation.

In order to minimize the MSW heterogeneous composition (due to seasonal and origin variability driven by local economic, social and climate conditions) limitations as a fermentation feedstock, **four main innovations will be proposed**: i) ***New enzymatic cocktails*** *to maximize hydrolysis of fermentable organic matter with low inhibitors production*, ii) *the use of high yield, specific and robust* ***strains for each selected acid***, iii) *the* ***extraction of fermentation by-products acting as inhibitors*** *to succinic acid production via novel membrane electrolysis employing an integrated continuous fermentation coupled with simultaneous organic acid removal in comparison with SA sequential fermentation followed by its separation using an electrodialysis-based* and iv) optimize simultaneous saccharification and fermentation for lactic acid production followed by a downstream separation process based on membrane electrodialysis. To maximize the yield and purity of target organic acids, continuous and single step fermentation process will be prioritized in order to allow **their integration in the PERSEO plant**.

Proposed **non-food applications do not require** high purity intermediates, thus their final cost is expected to be competitive, due to the current cost of PERCAL's target final products are 3-4 times higher than bioethanol price.

Figure 1.1.a shows the scheme of the project including the relation between the partners and main final products.

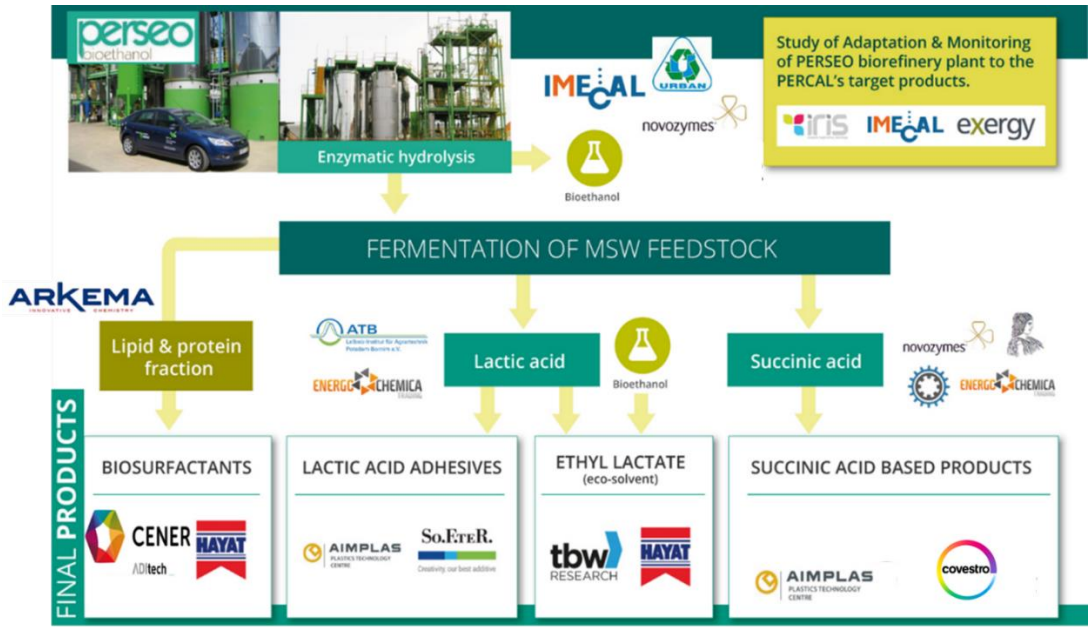


Figure 1.1.a Scheme of PERCAL project including the main chemical as final products and partners & support companies role.

The ***Biogenic fraction of Municipal Solid Waste (MSW)***[1] ranges between 30-50% of total MSW, which represents more than 500 kg/capita (EU-27 average), 300 million tonnes overall every year in the EU-32[2], increasing at an annual rate of 10%. Currently, approximately 50% of this volume is landfilled, 30% is recycled (material separation and bio-waste recycling, including compost and digestion) and the remaining 20% is incinerated. In spite of the strong shift from MSW landfilling to recycling of materials such as plastics, wood or metals and energy production from the organic fraction (mainly by incineration), there is a challenge to turn the biogenic fraction into higher value products than bioenergy.

Partners involved in the project have great experience and international reputations in all of the target materials and processing techniques. They have participated in many previous projects and industrial applications related to the use and production of chemicals from municipal and agroindustrial waste as section 1.3.3 shows

1.1.1. Overall Objectives.

Objective 1: To improve enzymes cocktails to maximize organic hydrolysis for several MSW compositions. Target to produce up to 85% of potential fermentable sugars contained in the organic fraction of MSW *(Linked to WP1)*.
Current state-of-the-art enzymatic hydrolysis processes are able to liberate around 65% of all potential fermentable sugars present in the organic fraction of MSW. Latest developments in the enzymes engineering field have allowed to improve by more than 20 % the amount of fermentable sugars liberated in the hydrolysis step. New ad hoc enzymes cocktails prepared for the different MSW compositions will be tested in order to hydrolyse more than 85% of all the potential fermentable sugars present in the organic fraction of MSW. The hydrolysis of proteins contained in MSW will lead to the provision of complex nitrogen sources (i.e. peptides and amino acids) as substitute for the commercial yeast extract needed for succinic acid and lactic acid production. In this way, further reduction of the cost for fermentation media formulation is expected. ***Performance indicators:*** by M18, we plan to have the new enzymatic cocktails to increase the amount of fermentable sugars and protein hydrolysates over the most representative MSW feedstocks. *Main Partners Involved*: IMECAL. *Support companies involved*: NOVO.
Objective 2: To determine the suitability of a given organic MSW composition for the different chemical intermediate production routes and its specific potential, based in the fermentation results (Linked to WP1).
MSW is a very complex mixture, made of many different materials when it comes to nature, origin, composition and intrinsic and physical properties. Furthermore, modification in composition of MSW may be expected with factors such as wealth, season or consumer's habits. Therefore the components of MSW (cellulose, hemicellulose, lignin, pectin, protein or fat) are found in different amounts and ratios within the different locations, time of the year and sorting and collection method. In PERCAL, the different samples will be analyzed and depending on the major component their transformation potential will be analyzed for the most suitable application. The same will be applied for the fermentation resulting biomass in order to revalorize the remaining fractions *i.e.* proteins and lipids. ***Performance indicators:*** by M6, a report of the main requirements of selected products and intermediates, by M12, a report on **selected feedstocks** for each fermentation process, and by M18, a report con **Validation of fermentation processes** will be prepared. *Main Partners Involved*: IMECAL, ATB & AUA. *Support companies involved*: NOVO & ENERGO.
Objective 3: *To study the production and purification of Lactic acid (LA) from MSW using sequential and simultaneous saccharification and fermentation.* ***Target*** *turnover >90% of fermentable sugars and LA purity >90% (Linked to WP2).*
MSW will be exploited to produce lactic acid (LA). Here, the focus will lie on the development of cost effective and sustainable bioprocess technologies to lactic acid using newly isolated strains. The production of lactic acid will be achieved by sequential and simultaneous saccharification and fermentation followed by lactic acid downstream separation based on membrane electrodialysis in order to achieve the needs of properties and purity for subsequent processing. In this way, the NaOH used during fermentation will be minimised due to re-cycling achieved by the electrodialysis membrane system and the steps required for hydrolysis and fermentation will be reduced due to the simultaneous operation leading to improved process economics. The supply of MSW-derived nitrogen sources will also reduce the lactic acid production cost. ***Performance indicators:*** by M18, a report of preliminary lab studies of the fermentation of lactic acid using MSW treated (WP1) feedstocks, by M24, we plan to define the optimal strategy for LA purification (down-stream) and by M32 a report will be prepared with the full process at pilot plant level. *Main Partners Involved*: ATB, AUA & TBWR. *Support companies involved*: ENERGO.

[1] Biodegradable garden and park waste, food and kitchen waste from households, restaurants, caterers and retail premises.
[2] European Environmental Agency report, 2013.

Objective 4: *To study the production and in-line purification of Succinic Acid (SA) from MSW using membrane electrolysis technology.* ***Target*** *turnover >90% of fermentable sugars and SA purity >99% (Linked to WP3).*

An efficient bioprocess utilising hydrolysed MSW (developed in WP1) with simultaneous succinic acid extraction using electrolytic extraction membrane technology will be developed. A nutrient-complete fermentation feedstock will be produced from MSW exploiting not only the carbohydrate content, but also the protein and mineral content. Proteins will be enzymatically hydrolysed into peptides and amino acids as nitrogen sources. In this way, the cost of media formulation for succinic acid production will be reduced by at least 50%. Pre-treatment of MSW hydrolysates may be carried out in the case that inhibitors are present. Fed-batch and continuous cultures will be developed with emphasis in continuous operation with immobilised cells to reach an overall succinic acid titer up to 60 g/L. We will improve the yield and productivity by extracting the inhibitory carboxylic acids from the fermentation broth using the integrated electrolytic extraction system, while feeding with MSW. Compared to the existing downstream technology several steps of succinic acid separation (cell removal, decolorisation and acidification) will be eliminated using membrane electrolysis (ME) extraction providing significant reduction to the downstream cost. The advantage of this process is the extraction of the ionic form into an acidic extract solution without the addition of chemicals, which will overcome the acidification of neutral pH bacterial fermentations. Indeed, at the cathode side of the ME process the microbial broth is neutralized via water splitting, for which preliminary data shows that there is no negative impact on the bacteria. More than 90 % of the produced succinic acid will be recovered, with high purity (>99%) of succinic acid via evaporation, crystallization and drying as final steps. Electrolytic membrane extraction integrated with fermentation will be compared with the elecrodialysis-based downstream separation process available at ATB. ***Performance indicators:*** by M8, a report on the most significant inhibitory compounds present in MSW and the substrate performance of the succinic acid producers will be addressed. By M18, optimization of succinic acid extraction through membrane electrolysis and by M24 purification of succinic acid will be reported. By M34, process scale-up will be addressed.
Main Partners Involved: AUA & CMET. *Support companies involved*: ENERGO.

Objective 5: *To valorise the fermentation by-products in order to increase the recovery from the organic content of MSW by 25 % (Linked to WP4).*

From the fermentation processes only carbohydrates are converted to bioethanol, LA or SA. Proteins and fats, which account for more than 30% of MSW organic fraction, are still present as by-products. These components will be valorised with the aim of producing new Biosurfactants increasing the yield of recovery of MSW. Surfactants made from aminoacids and lipids mimic the natural amphiphilic structures with surfactant capacity. In this project we aim to obtain the aminoacids from the protein fraction and the lipids from the fat fraction, and to explore the chemical, enzymatic or combination of both methodologies. Hydrolyzed proteins will be used also as a nitrogen source in the LA & SA fermentation to reduce the use of yeast extract. ***Performance indicators:*** by M12, a report will be prepared M26, reports on **pretreatments for revalorization of the protein and lipid fractions** will be prepared; by M34 a report on **biosurfactant** production from lipids and proteins will be prepared.
Main Partners Involved: IMECAL & HAYAT & CENER. *Support companies involved*: ARKEMA.

Objective 6: To produce ethyl lactate solvents for application in cleaning products and inks by simultaneous esterification and product separation process (reactive distillation). *Target: To scale up esterification to a continuous process with a yield of 3-5L/h; product purity >90% (Linked to WP5).*

The objective is to develop a continuous esterification and product separation process applying a reactive distillation process to produce ethyl lactate from two PERCAL bulk chemicals (LA and ethanol). The processing will be adapted and optimized to the selected PERCAL products. This approach will consider any potential impurities and product concentration levels. ***Performance indicators:*** by M18, a report will be prepared with the first esterification test at lab level to evaluate catalysts and raw material quality. A chemical model will be defined and validated with experimental data. From M12 onwards, the esterification will be up scaled by using a recently implemented and tested reactive distillation column (prototype) to produce products in the range of 3-5L/h. By M30, product quality will be characterized, key process data compiled for further analyses (technical an economic analysis) and ethyl lactate samples provided to industrial partners to perform their validation at TRL5.
Main Partners Involved: TBWR & HAYAT.

Objective 7: *To produce lactic acid based hot melts for application in cardboard and labels by reactive extrusion with similar strength than EVA HMA analogous.* ***Target*** *Shear adhesion strength >10 MPa (Linked to WP5).*

PERCAL will develop a medium viscosity hot melt formulated for label and carton sealing. It will have excellent hot tack (Shear adhesion strength >10 MPa ASTM F1921) and fast setting speed (up to 25 m/min). It will maintain high adhesion on un-varnished substrates for applications where the finished goods could be stored from - 15 °C up to + 35°C. ***Performance indicators:*** by M20, a report will be prepared to detail the production of lactide and further polylactic acid from purified lactic acid (LA) obtained by MSW fermentation, in M26 the polymers

obtained will be grafted by reactive extrusion with suitable reactive monomers to improve adhesion capacity. Final LA-HMA formulation will be ready in M28 for final validation in SOFTER and HAYAT facilities.
Main Partners Involved: AIMPLAS & SOFTER.

Objective 8: *To produce polyester and polyether polyols for application in polyurethanes (PUR).* ***Target*** *> 1,000 MW, compatible with current petrol based polyester/polyether polyols (Linked to WP5).*

PERCAL will develop a medium viscosity polyester/polyether polyol for polyurethane formulations with enhanced reactivity. It will have an average molecular weight >1,000 and Hydroxyl Number > 0.02 gKOH /g. ***Performance indicators:*** by M20, a report will be prepared to detail the production of 1,4 butanediol from purified succinic acid (SA) obtained by MSW fermentation, in M26 the butane diol will be polymerized to form a medium molecular weight polyester polyol. Polyester polyol will be integrated and validated in well-known PUR formulations. Main Partners Involved: AIMPLAS & COVESTRO.

Objective 9: To study the integration of in-line control technologies to the PERSEO pilot plant to increase its process control parameters, its versatility & maximize homogeneity of the fermentable feedstock *(Linked to WP6).*

In order to improve the performance of the PERSEO biotechnological plant new monitoring technologies will be studied by obtaining relevant parameter information through advanced sensors and planning an intelligent plant control platform with decision support models. The adaptation of PERSEO Bioethanol® plant to produce LA and SA will be studied. Similarly, design and testing of new sensors and control platform for the monitoring of the process parameters of interest at laboratory scale and the study of the scalability of the processes and control system at full plant level will be performed. ***Performance indicators:*** by M18, a report on the adaptation of PERSEO plant for the versatile production of chemical intermediates and identification of key process parameters will be prepared. A report on the monitoring technologies and testing results will be produced by M30 as well as a report on the integration of the production in a biotechnological plant configuration (M34).
Main Partners Involved: IMECAL & VISUM. Support companies involved: ENERGO.

Objective 10: To demonstrate the safety & regulatory compliance, as well as environmental & economic sustainability of the developed processes and products *(linked to WP7).*

LCA will be performed on the processes developed for the selected intermediate building blocks to reduced environmental impact vs. current traditional materials. Standard and legislative aspects that apply to the project results will be continuously monitored. A safety & Regulatory assessment will be performed during the project. Aspects related to waste management and potential health risks will be addressed. Finally, the techno-economic viability will be evaluated with inputs from all previous WPs. ***Performance indicators:*** by M12, a report analyzing the main regulatory frameworks in place in the EU regions considered in the project will be produced. By M30, a report for the safe operation of the pilot plant will be produced, as well as a recommendations for a DEMO plant. By M36, the LCA and techno-economic assessments will fuel into recommendations for improvements and will be taken into account to assess future commercial viability.

Objective 11: To define sustainable value chains and cross-sectorial connections *(linked to WP8).*

Building upon the variability MSW composition from WP1, a supply chain sustainability analysis will be performed. Based on existing best practices in the field of circular economy, suitable business models for the cross-sectorial valorization of MSW will be recommended. Logistics, resources, markets for the products, as well as the readiness of their potential converters to shift to such feedstocks, but also the consumer perception of such products, the policy and regulatory barriers will also be studied and taken into account into the proposed business models. ***Performance indicator***: at M24, an interim report providing supply and sourcing strategies as well as robust economically, environmentally and socially beneficial business models will be provided, and its final version will be ready in M36. It will also include a thorough market analysis across involved sectors as well as policy and regulatory recommendations

Objective 12: *To maximize the innovation impacts of the project for contributing to the uptake of the project results for companies' business and employment growth (linked to WP8).*

The project knowledge and results will be managed and protected. Business and exploitation plans will be delivered for all Key Exploitable Results (KERs) to map the post-project phase. A comprehensive dissemination and communication campaign will be designed and implemented. Regular knowledge transfer activities will take place among partners. Workshops with researchers and participation in educative actions for stakeholders will be organized. ***Performance Indicators:*** by M12, a robust dissemination plan to support the recognition of PERCAL as a reference in the maximisation of value generated to create new opportunities in the green chemical sector will be in place. By M34, a suite of training activities (>3) will be organized. By M36, final plans will be in place for the transition to the post-project phase to advance the TRLs and sources of further funding to build full DEMO scale processes will be analysed focusing on applications where a return on investment is feasible.

1.2. RELATION TO THE BBI JU ANNUAL WORK PLAN.

The PERCAL project addresses the topic: **BBI 2016.R8 - Emerging technologies for conversion of the organic content of Municipal Solid Waste and improving waste-to-chemicals value chains**. First on a wider point of view, the project is totally aligned with to the BBI road map "The BBI aims at accelerating the building of bio-based value chains, starting from sustainable feedstock production and mobilisation towards the implementation and use of bio-based materials and products (as defined in Value Chain 4- Emergence of new value chains from (organic) waste). We furthermore validate the derived biobased products in most of the strategic area of the Strategic Innovation and Research Agenda (SIRA). The detailed relation of the project to the specific challenge and scope addressed by the topic are justified in table 1.2.a below.

Table 1.2.a How PERCAL relates to the scope and challenges of the BBI 2016 R8 topic

Challenges of BBI 2016 R08	How PERCAL addresses these challenges
Specific challenge: Conversion processes performance is hindered by the high variability of the input feedstock, resulting in the need of tailored sorting and pre-treatment of the feedstock. The latter steps reduce the presence of inhibitors for the downstream steps. However, the costs of the sorting and pre-treatment steps, combined with the need of high performance separation and purification, often hinder the overall economic feasibility of a waste-to-chemicals value chain.	Based on the knowledge of PERSEO bioethanol® plant and proposed innovations in PERCAL project, we plan to obtain three intermediates (LA, SA and ethanol) with suitable purity in order to have a more competitive plant than the current state-of-the-art. The consortium covers all the waste-to-chemicals value chain from MSW management until end-users of biobased products), and their main priority is to obtain high quality products (comparable to petrol-based counterparts) with a cost comparable to current ones obtained from plant sugars or oils. The selected lactic and succinic acid producing strains are expected to tolerate moderate concentrations of inhibitors produced during pre-treatment of MSW. PERCAL will focus on the optimisation of the downstream processes leading to reduced costs (normally the most expensive step). Other challenges proposed in PERCAL are full detailed in the 12 target objectives defined in point 1.1.1.
Scope: Develop and validate at lab scale innovative processes for conversion of the organic fraction of MSW into chemical intermediates aiming at: • Maximising the yield of recovery of intermediates from the substrate; • Minimising the presence of inhibitors in order to enable achieving profitable yields from the conversion of intermediates into products.	Partners such as AUA, CMET and ATB have long-term experience with bioprocess development for the production of bulk chemicals. PERCAL will contribute to the development and optimization of bioprocesses based on MSW as feedstock by: • Enhancing succinic acid production via removal of organic acids produced during fermentation by employing an integrated ME extraction system leading to elimination of product inhibition during fermentation. • Maximising the degradation of remaining oligosaccharides in the MSW-based broth due to the effect of electrolysis leading to increased succinic acid production yield • Reducing the addition of neutrilisers during fermentation due to membrane electrolysis or recycling of NaOH due to membrane electrodialysis. • Reduce downstream separation costs for acid production due to the application of membrane electrolysis that eliminates the need for expensive unit operations required for cell removal, colour/protein removal and acidification of succinate. • Compare integrated electrolytic extraction system with electrodialysis-based downstream separation (ATB) to select the most sustainable process for succinic acid production. The electrolysis extraction system for succinic acid production will be developed for the first time in PERCAL. • The selected bacterial strains for lactic acid production will ensure simultaneous saccharification and fermentation at productivities higher than 1 g/L/h and final concentration higher than 80 g/L • The integrated ME-fermentation system will ensure the production of SA at productivities higher than 0.6 g/L/h and concentrations up to 60 g/L
The innovation should focus on production of intermediates at high yield and low impurity level.	• The acid form of SA is produced via Membrane Electrolysis (ME) in one step process that directly extracts from a broth into a low volume extract solution, offering a high extraction yield at low energy consumption around 2 kWh/kg. The integrated ME-fermentation system is expected to increase the SA production yield from sugars to values higher than 0.7 g/g. • Purity over 99 % will be achieved by a combination of ME based separation followed by evaporation (to remove acetic/formic acids), crystallisation & drying. Direct crystallisation of succinic acid from the concentrated extract will be

Challenges of BBI 2016 R08	How PERCAL addresses these challenges
	evaluated aiming to reduce production costs. • The ATB-based process for lactic acid production has been evaluated with numerous crude renewable resources using various bacterial strains resulting in yields higher than 0.9 g/g and production of the required isomer percentage and purity depending on the application.
Proposals should take into account seasonal and random variability of the MSW content and address several compositions of the organic fraction, focusing on representative case studies taking into account local economic, social and climate conditions.	There are three main factors affecting the organic MSW variability: waste collection system, geographic location and seasonality. In PERCAL project all these factors affecting MSW composition will be taken into account. Several representative bio-waste samples coming from different waste collection systems (separated and non-separated), different locations (Northern and Southern Europe; rural and urban areas) and seasons will be evaluated.
The projects should cover any TRL from 3 to 5. In the case of a pilot scale project (TRL 5), proposals should present a credible cost estimate for the proposed processes with a preliminary assessment of their competitiveness when scaled up.	PERCAL targets TRL3-5 as table 1.3.a shows A preliminary business model has been prepared by the different partners based on their current sales and consumptions, the estimated target cost, their current market share and the expected market evolution (table 2.1.d). However, due to the TRL at the end of the project, we expect a significant sales 5 years after project finish when TRL 8 will be achieved. More accurate calculations of profit are difficult to make due to the current low yield of some proposed routes.. Expected costs are determinate based on the expertise of the PERCAL's partners and the methodology proposed by Hermann et al[3] and further papers to calculate the techno-economic prospects of new routes for bio-based bulk chemicals.
Proposals should also include an environmental and socio-economic assessment.	WP8 will include all the environmental, economic and socioeconomic assessments, including LCA, LCC and S-LCA for the involved processes and the product lines. Outcomes will be comparable with current processes where applicable in order to demonstrate the environmental and social advantages of PERCAL project.

1.3. CONCEPT AND APPROACH.

1.3.1. Overall concept

Implementation of environmental policies, especially waste policies, is one of the European Commission's key priorities, as confirmed by its proposal for a 7th Environment Action Programme (EC, 2012) and the Roadmap to a resource efficient Europe (EC, 2011). While the EU's Waste Framework Directive (EU, 2008) and Landfill Directive (EU, 1999) set binding targets for recycling organic waste and diverting biodegradable municipal waste from landfill.

With the world population set to approach an estimated 9 billion by 2050, against a background of finite natural resources, Europe needs renewable biological resources - not just for securing healthy food and animal feedstuffs but also for raw materials to produce industrial products and bio-fuels. PERCAL is in accordance with these priorities by maximizing the value of components obtained from MSW.

On 13 February 2012, the European Commission adopted a strategy for «Innovating for Sustainable Growth: A Bioeconomy for Europe»[4]. A strong bioeconomy will help Europe to live within its limits. The sustainable production and exploitation of biological resources will allow the production of more from less, including from waste, while limiting negative impacts on the environment and reducing the heavy dependency on fossil resources, mitigating climate change and moving Europe towards a post-petroleum society

MSW biowaste contains large amounts of polysaccharides (30-40% cellulose, hemicellulose and pectin), lipids (10-15%) and protein (5-15%), depending on the origin and pre-processing steps. Efficient conversions of mixed, and variable feedstocks are challenging. To date, mixed wastes have been transformed by thermochemical processes, such as hydrothermal liquefaction (HTL) and gasification, and by anaerobic digestion, leading to methane and bio-hydrogen, fuels (ethanol and butanol), compost and fertilizers. Production of more complex chemicals is limited by large-scale economic feasibility due to pre-treatment and conditioning costs of the recalcitrant raw materials. *PERCAL's approach, proposes the fermentation of carbohydrates to ethanol, lactic and succinic acid, and the subsequent recovery of current fermentation flow, getting a valorization of the not fermented fractions in value-added bioproducts. PERCAL aims to increase the rate of carbohydrates hydrolysis in order to obtain high rate of sugars for the fermentation process and increase the bioproducts process yields.*

[3] Herman B.G. and Patel M. "Today's and Tomorrow's bio-based bulk chemicals from white biotechnology. A techno-economic analysis". *Appl. Biochem. Biotechnol.* 136: 361-388 (2007).
[4] http://ec.europa.eu/research/bioeconomy/index.cfm?pg=policy&lib=bbpp

To do it, PERCAL will use as a starting point the large experience of IMECAL and NOVO in the enzymatic hydrolysis of the biogenic fraction of MSW that are capable to convert more than 65% of the carbohydrates in fermentable sugars and similar percentage of protein (to be used as a nitrogen source and to reduce the addition of yeast extract) in the frame of PERSEO project as table 1.3.a shows.

Table 1.3.a Initial MSW composition, after PERSEO's enzymatic hydrolysis and after PERSO's ethanol fermentation process (Dry Fraction of Municipal Solid Waste-DFMSW).

Main component	MSW (g/kg DFMSW)	After PERSEO's Enzymatic hydrolysis (g/kg DFMSW)	*After PERCAL's Enzymatic hydrolysis (g/kg DOFMSW)*	To be used in:
Cellulose	302.0	106.0	*45.0*	Energy production
Hemicellulose	46.0	16.0	*7.0*	
Fermentable Sugars	----------	**225.0**[5]	***290.0***[5]	Fermentation to obtain ethanol, lactic acid & succinic acid.
Pectin	77.0	77.0	***77.0***	
Protein	105.0	105.0	***105.0***	
Lipids	141.0	141.0	***141.0***	Surfactants
Others (lignin, ashes…)	329.0	329.0	329.0	Energy production

PERCAL aims to establish new technologies for microbial conversions of MSW after enzymatic hydrolysis that permit to convert ***the 85% of carbohydrates in fermentable sugar****, to select microbial strains able to produce the targeted intermediate products with high productivities, and for their subsequent transformation into marketable products*. Additionally, PERCAL aims to use up to 600 g/kg of DFMSW in biochemical products, **more than double** than current PERSEO technology and other similar technologies currently available at pilot plant scale (see section 1.4.6).

PERCAL will focus in valorise in a cascade approach the greatest potential (carbohydrates, proteins and lipids) single compounds of biogenic fraction of MSW. The valorisation of the remaining fractions will be studied (based on LCA social and techno-economic analyses results) for post project further improvements of the overall process economic and environmental sustainability, non-useful components could be employed to produce biogas or other type of energy. Furthermore, a stakeholder group and networking with other complementary EU projects will be proposed to select the best technologies to valorise the separated components from MSW.

Figure 1.3.a shows the target products, processed involved and their selected feedstocks The MSW will be used as fermentation substrate in other to obtain higher-value products. Detailed information is presented in section 1.4.

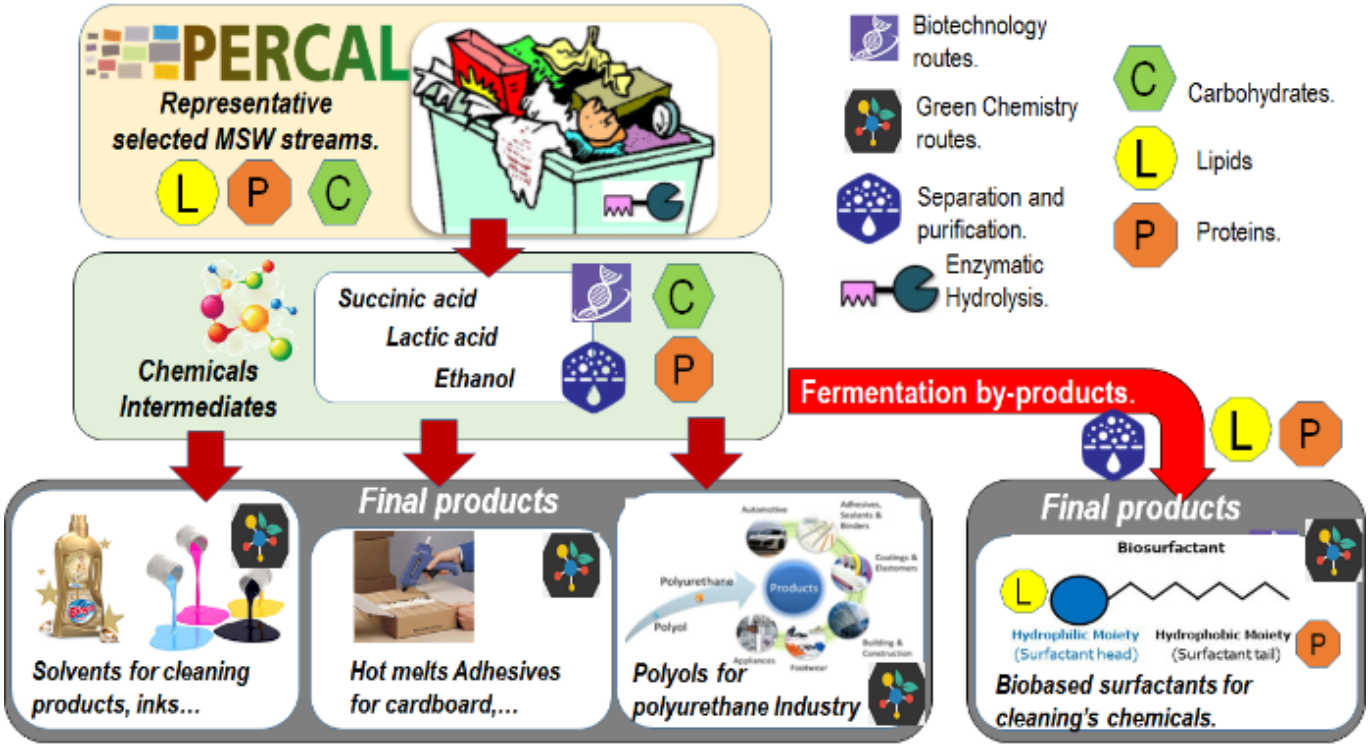


Figure 1.3.a PERCAL Flow Diagram including intermediates and sectors involved in each step.

[5] In the PERSEO SSF process, the fermentable sugars are continuously been fermented to ethanol, so this is an estimate of how many fermentable sugars will be produced in the hydrolysis step.

1.3.2. Positioning of the Project.

The activities developed in **PERCAL** project will be implemented at TRL3 with a TRL target of 5. Most of the proposed innovations are based on technologies studied at laboratory level, as it is shown in Table 1.3.b:

Table 1.3.b. TRL justification of PERCAL developments.

Technology proposed	Current→ target TRL	Justify current and final TRL
Improve the amount of fermentable sugars with new enzymatic cocktails.	4→5	IMECAL & NOVO have developed and validated physico-chemical and biochemical processes for the optimal conversion of different fractions of MSW and by-products into valuable organic compounds. New enzymatic cocktails will be tested supported by new fermentation routes to maximize carbohydrates use.
Simultaneous saccharification & fermentation of LA	3 → 5	ATB has validated lactic acid production up to pilot scale incl. full DSP using various agri-industrial resources via application of different fermentation regimes, e.g. batch, repeated-batch, continuous mode, SSF approach. MSW is a more complex system and required a in-depth knoleadge of main inhibitors and the interaction of the other MSW components.
Membrane Electrolysis (ME) to in-line impurities separation.	4 → 5	Membrane electrolysis has been demonstrated at laboratory level, at present a process for production of caproic acid from fermentation is operating at pilot scale in CMET. Lab scale demonstration will move to scale up pilot plant production for SA in collaboration with AUA.
AUA	3 → 5	AUA has demonstrated at lab-scale that *Basfia succiniciproducens* can produce more than 40 g/L of succinic acid from xylose-rich spent sulphite liquor derived from the pulp and paper industry (BRIGIT – FP7 project). Continuous cultures have also been developed using the same waste stream leading to around 25 g/L of succinic acid production. Preliminary fermentation on commercial xylose using the integrated fermentation-ME system carried out at AUA have led to more than 20% improvement of yield and productivity. Succinic acid production in an integrated system using ME will move from lab to pilot scale in collaboration with CMET.
Reactive distillation for production of ethyl lactate	4 → 5	Previous project ***Green Safer Solvent*** has demonstrated the technology at lab level. LA and ethanol are provided form MSW origin the esterification will be done at lab scale to validate the raw chemicals quality and catalysts. After the first phase a recently implemented prototype reactive distillation at pilot scale will be used to upscale and optimise process and product performance.
Lactic acid based HMA by reactive extrusion.	3 →5	AIMPLAS & SOFTER have great expertise in lactic acid reactive extrusion and petrol based HMA formulations. Several papers are found in the literature related to these issues. Scale up to TRL5 will be done in SOFTER facilities.
Polyols from Succinic acid.	3 → 5	COVESTRO has a vast experience with polyols (partly) made from non-fossil resources. Our Cardyon material is made with CO_2 based polyols. We are producing CO_2 based polyols using pilot plant that can also be used for SA based materials.
MSW biorefinery plant monitoring.	3 → 5	VISUM has developed tested a range of optical sensors in lab tests with successful results for specific applications. New tests tailored to monitoring the production of ethanol and acids will be designed and the instruments will be developed and under relevant conditions based on the expertise of IMECAL in PERSEO plant.

We were unable to identify public European projects with the same research objectives as those proposed in PERCAL. However, there are a number of national and European projects with strong synergy with PERCAL, and details are summarized in Table 1.3.c.

Table 1.3.c. Partners' projects related to **PERCAL**.

Description of experience/project	Scope	End Year	Website for information (PERCAL's Partner)
BRIGIT: New biopolymers produced from lignocellulosic residues for flame retardancy highly demanding applications.	FP7-KBBE	2016	www.brigit-project.eu (AIMPLAS, AUA).
BIOREFINE-2G: Production of Dicarboxylic Acids and Bio-based Polymers**.**	FP7-KBBE	2017	www.biorefine2g.eu/ (AIMPLAS,).

Description of experience/project	Scope	End Year	Website for information (PERCAL's Partner)
DAFIA: Biomacromolecules from municipal solid bio-waste fractions and fish waste for high added value applications.	H2020 Biotec	2021	In negotiation (AIMPLAS, VISUM).
OLI-PHA A novel and efficient method for the production of PHA polymer-based packaging from olive oil waste water	FP7-NMP	2015	www.olipha.eu (VISUM)
BREAD4PLA: Demonstration-plant project to produce poly-lactic acid (PLA) biopolymer from waste products of bakery industry	EU LIFE	2014	www.bread4pla-life.eu ATB & AIMPLAS
EUBis: Food waste valorisation for sustainable chemicals, materials & fuels	EU COST	2016	http://costeubis.org/ (ATB & AUA)
BranLact: Thermophilic lactic acid production utilizing rice bran in continuous cultivation	Bioeconomy	2019	Not available (ATB)
BIOFORSU: Valorisation of the organic fraction of municipal solid wastes by conversion to bioethanol and combustible biomass	Spain level project	2013	www.imecal.com/perseo/ (IMECAL)
PERSEO: Ethanol production from domestic solid waste organics.		2012	
Green Safer Solvent: Esterification of lactic acid and ethanol to ethyl lactate by means of reactive separation techniques.	FFG Austria	2016	TBWR
AgriMax: Agri and food waste valorisation crops based on flexible multi-feedstocks biorefinery processing technologies for new high added value applications.	BBI JU	2020	Not available yet (EXERGY, VISUM, AIMPLAS).
ELECTROTALK – impact of electricity on microbial metabolism, creation of interfaces.	ERC-StG	2017	www.electrotalk.org

EU and world-wide data bases have been consulted and there are no patents invalidating the proposed PERCAL project objectives. Besides product innovations, the innovation potential for the partners regarding the improvement of the municipal solid waste and fish waste fractionation and purification, microbiological processes, downstream, catalysis, green chemical and final production processes is substantial. In summary, a search of patents and registered intellectual property indicated that, if successful, we would be free to register the developed IP and to operate the developed system industrially.

1.3.3. Approach and methodology.

PERCAL is a **3 year project**, structured into 9 work packages. It will take into account all the development steps to obtain added value products from low value by-products to fit real industrial needs, including a work plan with a well- balanced structure that will combine research activities final products requirements definition and MSW selection and pre-treatment to maximize yield of selected intermediated (WP1), production of intermediates and production and validation of final products (WP2,3,4&5), plant study redesign and process monitoring to produce the different acids (WP6) and environmental and economic studies in WP7. PERCAL has a well-balanced consortium covering all the value chain from waste valorisation to end-users.

In the WP1, led by IMECAL, several MSW fractions of the different countries where partners or support companies are present will be analyzed or data will be collected from recognized sources and three of them will be selected as the most representatives. Different enzymatic cocktails will be tested to increase the fermentable sugars. The best of them will be selected for each target fermentation process (LA WP2 and SA WP3). In WP2, ATB and TBWR will optimize the LA from the MSW hydrolyses. AUA, CMET and ATB will develop an optimised bioprocess for the production of SA. In WP4, IMECAL & HAYAT will develop new biobased surfactants using the protein and lipid fraction. In WP5, AIMPLAS and SOFTER will develop new lactic acid Hot Melt Adhesives, TBWR and HAYAT will develop ethyl lactate solvents and COVESTRO will develop new polyester polyols for the polyurethane industry using the acids and alcohols develop in WP2 and WP3. In WP6, led by VISUM, the focus will be to improve the control of the PERSEO plant and study the optimal integration of the new LA & SA fermentation processes and biosurfactants production (from lipid and proteins) in the current pilot plant.

The environmental and social analyses, safety, energy balance, standardization and economic assessment to validate the main objectives of the **PERCAL** project and will be carried out in WP7 by AUA with the support of all partners. Innovation management, business plan and Communication, Dissemination & Exploitation activities boosting the project results will be done in WP8 by AUA. EXERGY will be in charge of market analysis and elaborate the preliminary business plan for all the PERCAL partners and Stakeholders companies. WP9 will be dedicated to

management and risk assessment.

1.3.4. Sex and Gender analysis.

The technologies, processes and materials developed in PERCAL are gender-independent. All PERCAL Partners have internal gender equality jobs creation and promotion policies. The consortium will foster the presence of women in the technical and management teams and decision board implemented to govern PERCAL project: the coordinator of PERCAL is Ms. Coll, woman researcher is in charge of the general assembly (Ms. Carracedo) and Executive Board (Dr. Marta Tortajada). The project aims to contribute to the objectives of enhancing Gender Equality by involving numerous female researchers (see section 4) in the project and by and by promoting women researchers to play a leading role in work packages. To this end, WP2 (Dr. Donna Dietz), WP4 (Dr. Irantzu Alegria), WP5 (Dr. Natalie Rahmen), WP8 (Ms. Rocio Roldán) and WP9 (Ms. Caterina Coll) are managed by female researchers, and as it is described in section 4, near to 50% of the staff researchers involved in the project are women.

The Treaty on European Union, the European policy establishes equal opportunities between women and men. In particular articles 2 and 3 establish equality between women and men as a specific task of the Community and also to promote equality. Monitoring the progress toward gender equality in science has become a well-established activity of the European Union research policy. In accordance with these principles, the approach of PERCAL is fully in line with Directive 2002/73/EC of the European Parliament and of the Council of 23 September 2002 on the implementation of the principle of equal treatment for men and women as regards access to employment, vocational training and promotion, and working conditions.

1.4. AMBITION

1.4.1 State of the art and progresses in the field of enzymatic hydrolysis of carbohydrates from MSW.

Due to its composition (cellulose, hemicelluloses, pectin, lignin,...), MSW is considered as a lignocellulosic substrate. The main sources of fermentable sugars are cellulose and hemicellulose, and therefore enzymatic approaches targeting these 2 complex polysaccharides have been developed. The classical scheme for cellulose degradation involves the synergistic action of three classes of enzymes[6]: 1) endo-1,4-β-glucanases randomly cleave internal bonds in the cellulose chain, 2) Exo-1,4-β-glucanases attack the reducing or non-reducing end of the cellulose polymer (celobiohydrolases). 3) β- glucosidases convert cellobiose, the major product of the endo- and exo-glucanase mixture, to glucose. The commercial cocktails available are made from fungal cellulases[7]. In addition the cocktails are enriched with xylanases, and oxidative enzymes that promote the efficiency of the classical cellulases. Commercial enzymatic cocktails available are: Cellic Ctec3® (Novozymes), AlternaFuel® CMAX™3 (Dyadic International Inc), Jtherm® (JBEI), Accelerase®TRIO™ (Genecor-DuPont), CodeXyme®Cellulase (Codexis), Accelerase® 1500 (DuPont), Accelerase® BG (DuPont), Accelerase® XC (DuPont), Accelerase® XY (DuPont), Biogazyme (ASA Enzymes). IMECAL has already tested some of them in the PERSEO plant and has identified that the yields can be improved with additional enzymes. These additional enzymes are mainly based on amylases which are able to break alpha-1,4 and alpha-1,6 glucosidic bonds of starch. Similar to cellulose, starch molecules are glucose polymers linked together by these two chemical bonds, as opposed to the beta-1,4 glucosidic bonds for cellulose. The amount of starch in OFMSW is significant, since it contains food such as bread, pasta, rice or potato with a relatively high percentage of starch. In the PERCAL project, combination of additional enzymes will be added to commercial cocktails in order to increase the yield from 65% to 85%.

Beyond of the state of the art, **PERCAL** will focus on the improvement of the hydrolysis process by the synergic combination of the most effective commercial cocktails with additional enzymes with the support of NOVO (one of the worldwide enzyme producer) and the experience of nine years of PERSEO technology.

1.4.2 State of the art and progresses in the field of valorisation of the proteins and lipids from MSW

To date most protein wastes are primarily used as animal feed or fertilizers. For example, Distiller's dried grains with solubles, a by-product from the bioethanol industry, is mostly sold as cattle feed, although it has poor nutrition source. Vinasse, a by-product from sugarcane-based bioethanol production, has been used as a fertilizer.

Valorisation in higher added value products of the protein fraction coming from biorefinery approaches is very crucial in order to increase the yield of the whole process[8]. Due to the origin of the substrate, its use in food/feed is disregarded, but the aminoacids and peptides are valuable for other fields such as agriculture (as biofertilizer) and

[6] Horn *et al* (2012). Novel enzymes for the degradation of cellulose. *Biotecnol. Biofuels* 5:45
[7] Brethauer et al.(2015). Biochemical conversion processes of lignocellulosic biomass to fuels and chemical-A review. *CHIMIA* 69, 572.
[8] Villegas-Torres et al. (2015). The protein fraction from wheat-based dried distiller's grain with soluble (DDGS): extraction and valorization. New Biotechnology, 32: 697-611

industry (as a source of chemicals). Hydrolysis is the main approach for the valorisation of proteins[9]. Depending on the target application, different methods are employed: acid/base chemical catalysis when the aminoacids are intended for industry; or enzymatic methods, less intensive and destructive, for their use in health or as biofertilizers. *To the best of our knowledge their use as synthons for biosurfactants has not been reported so far*.

The lipidic fraction is also an important byproduct in biorefinery. It is made of triacylglycerides, diacylglycerides, monoacylglycerides, phospholipids and free fatty acids. Additionally the cellular membrane of microorganisms from the fermentation processes is primarily composed of phospholipids. No references of the use of lipids from MSW have been found in scientific or technical literature.

Protein extraction can be achieved by employing aqueous-ethanol solutions between 50 and 90% (v/v), pH variation and hydrolytic enzymes. Proteases have proven to be an alternative method, but current enzyme costs limit the industrial application of this approach. For lipid isolation[10], solvent extraction, mechanical pressing and subcritical water technology are the most common processes. It should be also necessary to consider the environmental impact of the process and its sustainability. This will probably restrict the choice of chemicals used and the energy requirements of individual units operations.

The value of amino acids and vegetable oils as raw materials for the preparation of surfactants was recognized as soon as they were discovered early in the last century[11]. The combination of polar amino acids/peptides (hydrophilic moiety) and non-polar long-chain compounds (hydrophobic moiety) for building the amphiphilic structure has produced molecules with high surface activity. The amino acid or peptide moiety determines the main differences of adsorption, aggregation and biological activity between the amino acid/peptide-based surfactant. Hence, cationic, anionic, non-ionic and amphoteric surfactants can be obtained depending on the free functional groups. Further modification of these groups allows a fine-tuning of their properties to meet almost every particular application. Another way to revalorize the protein fraction within the project will be its application as nitrogen source in LA and SA fermentations, as a substitute of expensive yeast extract. After protein extraction, the protein will be hydrolysed using well-known proteinases[9] trying to achieve a yield of close to 90% depending on the enzyme[12] used and the MSW protein composition similar values obtained for other proteins wastes coming from the alcohol industry.

Beyond of the state of the art, **PERCAL** will focus on the valorisation of the protein and lipid content coming from the by-products of the fermentation processes of MSW (more than 300 g/kg of DFMSW) in order to produce new biobased surfactants as a novel application of these fractions. Hydrolysed proteins will be used in the fermentation of LA and SA to reduce the consumption of expensive yeast extract.

1.4.3 State of the art and progresses in the field of production lactic acid production from MSW.

Biotechnological processes and bio-based products are nowadays in focus of various studies as promising alternatives to petrochemical routes and products. Lactic acid (LA) assigns to bio-based organic intermediates of high interest. Up to date it is produced on industrial scale and is available on the market at a competitive price level. In chemical industry it serves as a building block for a variety of key chemicals or bulk polymers, e.g. acrylic acid, polylactic acid (PLA), or green solvents. Fermentative LA production is an interesting alternative with respect to the conventional petro-chemical route enabling the generation of a product of high purity[13]. Today, lactic acid is widely used in the food, cosmetic, pharmaceutical, and chemical as a monomer for the production of PLA. The global lactic acid[14] market was estimated to be 700 kt in 2013 and it is expected to reach near to 2 Mt by 2020, growing at a CAGR of 15% from 2014 to 2020 up to 18.6% reaching a projected market[15] of €3.4Billion by 2020.

The worldwide research is advancing focused on the use of renewable raw materials as carbon substrates as well as nutrient additive sources[16]. In this context, there is a strong interest to reduce costs for raw materials and to use renewable resources. With respect to cost aspect of bioprocess feedstocks the utilization of residues and waste materials[17], MSW[18] in particular, and agricultural by-products[19] became the focus of public attention.

[9] Kumar et al. (2015). Valorisation of protein waste: an enzymatic approach to make commodity chemicals. *Front. Chem. Sci. Eng.* 9,295-307.
[10] Abdelmoez et al. (2014), Oleochemical industry future through biotechnology. J. Oleo Science 63 545-554.
[11] Infante MR et al. (2004). Amino acid-based surfactants. *C.R. Chimie* 7: 583-592.
[12] Bals B. et al. (2011), Protease digestion from wheat stillage within a dry grind ethanol facility. Biotechnology Progress, 27(2): 428–434
[13] Neu, A.-K. et al. (2016) Fermentative utilization of coffee mucilage using Bacillus coagulans and investigation of down-stream processing of fermentation broth for optically pure L(+)-lactic acid production. Bioresource Technology. 211, 2016, 398–405
[14] www.specialchem4bio.com/news/2014/05/23/global-lactic-acid-market-to-grow-at-a-cagr-of-15-5-from-2014-20-grand-view-researc -2014
[15] MarketsandMarkets (2015) LA & PLA Market by Application & by Geography - Global Trends & Forecasts to 2020.
[16] Pleissner et al, Utilization of protein-rich residues in biotechnological processes. Applied Microbiol & Biotech, 100,2016, 2133-40.
[17] Zhang, L.et al. (2015) Simultaneous saccharification and fermentation of xylo-oligosaccharides manufacturing waste residue for L-lactic acidproduction by *Rhizopus oryzae*. Biochemical Engineering J., 94, 92-99
[18] Probst M et al. (2015). A closed loop for municipal organic solid waste by lactic acid fermentation. Bioresource Technology 175:142-151.
[19] Pessione A et al. 2014 Enantioselective lactic acid production by an Enterococcus faecium strain showing potential in agro-industrial waste

Lignocellulosic biomass represents the most abundant global source of biomass, and for this reason it has been largely utilized in many applications. Lignocellulosic materials can be used to obtain sugar solutions that may be usefully exploited for the production of lactic acid through the following steps: a) pre-treatment to break down the lignocellulosic structure, b) enzymatic hydrolysis to depolymerize lignocellulose to fermentative sugars, c) sugar fermentation to LA by lactic acid bacteria (LAB) and d) separation and purification of lactic acid. In recent years, one of the most used processes to obtain LA from lignocellulosic materials is the simultaneous saccharification and fermentation[20], which is able to prevent enzyme inhibition by the product[21].

At the ATB's lactic acid fermentation group there are long-term experiences with several feedstocks and bacterial strains, respectively. During the last months the application of different substrates formulations came into interest from other project activities. Therefore, in **PERCAL**, it will be possible to start further investigations with already existing strains (partly self- isolated bacteria) for the here desired utilization of the C6/C5 sugars after pre-treatment of the organic biowaste fraction. One of the main environmental-related advantages of the overall technology at ATB is the downstream processing step based on membranes[22] to avoid wastes like gypsum.

Besides the development of high-performance fermentation processes (batch, fed-batch, repeated batch, continuous mode) their upgrading into pilot scale will be part of the demonstration activities in relevant environment TRL5.

Beyond of the state of the art, **PERCAL** will focus on the performance improvement based on cheap, but probably complex and inhomogeneous feedstocks (MSW) compared to already existing approaches for other biogenic raw materials. We expect furans and other phenolic substances and acetic acid as a main inhibitors. Pre-treatment steps available in ATB facilities will be applied in combination with the use of more tolerant strains

1.4.4 State of the art and progresses in the field of production succinic acid production from MSW.

Simultaneous saccharification and fermentation of cotton stalk hydrolysates (65% glucose, 10% xylose) with *Actinobacillus succinogenes* reached 63 g/L succinic acid with 0.64 g/g yield and 1.17 g/L/h productivity [23]. AUA has demonstrated in a FP7 project (www.brigit-project.eu) the efficient production of succinic acid using the waste liquor produced by the wood sulphite pulping process[24]. Results at lab level in the BRIGIT project are showed in Table 1.4.a. *A. succinogenes* and *Basfia succiniciproducens* can consume all the sugars available in the MSW hydrolysate used in PERCAL. The BRIGIT feedstock is far more inhibitory than the MSW hydrolysate employed in PERCAL due to the high concentration of lignosulphonates, phenolics and acetic acid. Thus, it is expected that the organic acids produced during fermentation will be the main inhibitors to achieve high productivity and final concentration (33 g/L total produced acids for *A. succinogenes* and 35 g/L total produced acids for *B. succiniciproducens* [25]).

To tackle this problem, PERCAL will integrate succinic acid production via fermentation with membrane electrolysis to simultaneously remove the produced organic acids. The system will consist of a bioreactor coupled to a continuously recycling side-stream that passes through a high-rate electrochemical cell which performs the extractions.

Table 1.4.a Main results of SA production in Brigit Project.

Strain Type	*Fermentation Type*	*Yield (g SA /g sugar)*	*Productivity (g SA /L/h)*	*[SA] (g/L)*
Actinobacillus succinogenes	Fed-Batch[22]	0.52	0.39	27.4
	Continuous	0.42	0.7	22
Basfia succiniproducens	Fed-Batch[26]	0.54	0.31	39
	Continuous	0.49	0.78	21

Membrane electrolysis (figure 1.4.a) is an electrochemical extraction technique in which electrodes are present in the fermentation broth. Membrane electrolysis stands apart from electrodialysis in that the electrochemical reactions and products of the process are utilised in the process, either biologically or physico-chemically. This tends to allow lower input power and lower resistance relative to electrodialysis due to a smaller distance between the electrodes, and fewer membrane pairs in-between the electrodes. In so-called electro-fermentation water reduction (cathode)

bioconversion: Physiological and proteomic studies; J. Biotechnology 173, 31-40.

[20] Tang Y et al. (2013) L(+)-Lactic acid production from furfural residues and corn kernels with treated yeast as nutrients; European Food Research and Technology 236, 365-371.

[21] Hu J, et al (2016), High-titer lactic acid production by *Lactobacillus pentosus* FL0421 from corn stover using fed-batch simultaneous saccharification and fermentation. Bioresource Technology 214: 74-80

[22] Lin H et al. (2014), Lactic acid recovery from fermentation by bipolar membrane electrodialysis and ionic migration; CIESC J. 65, 4830

[23] Li et al. (2013) Efficient decolorization and deproteinization using uniform polymer microspheres in the succinic acid biorefinery from bio-waste cotton (*Gossypium hirsutum*) stalks. Bioresouce. Technology 135 : 604-609.

[24] Pateraki et al. (2016) Pretreatment of spent sulphite liquor via ultrafiltration and nano-filtration for bio-based succinic acid production, Journal of Biotechnology 233 : 95-105.

[25] Pateraki et al. (2016) Modelling succinic acid fermentation using a xylose based substrate, Biochemical Engineering Journal, 114: 26-41

[26] Alexandri et al. (2016) Evaluation of an integrated biorefinery based on fractionation of spent sulphite liquor for the production of an antioxidant-rich extract, lignosulphonates and succinic acid Bioresource Technology 214 : 504-513

typically occurs in the fermentation broth, producing hydrogen gas, driving fermentation, and hydroxide ions, which replaces alkaline pH control; whereas water oxidation (anode) occurs in an extract solution, producing oxygen gas and protons, which replaces the addition of acid[27].

The current drives the charged carboxylic acids across an anion exchange membrane (AEM) by electro-migration from the high-volume fermenter into a low-volume acid concentrate extract. AEM, also used in electroldialysis, is impermeable to cells and solids, resulting in a combined extraction, clarification, acidification and concentration step in a single unit. CMET have shown in synthetic solutions that low concentrations of acetic acid (10 g/L) can be extracted into a concentrated solution, in excess of 90 % electrical efficiency at 20 A/m^2, resulting in an acetic acid extraction of around 1 kg/m^2.d (membrane area). Succinic acid was shown to be driven into the extract solution at 1.4 kg/m^2.d at 40 A/m^2 for a final concentration of 69.4 g/L in the extract solution.

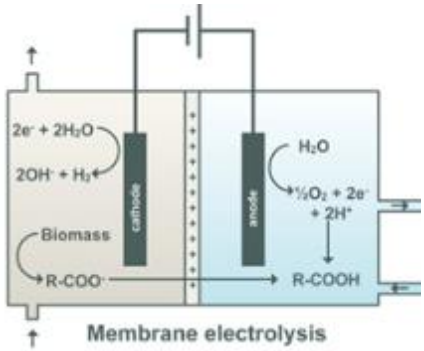


Figure 1.4.a. Membrane electrolysis technol.

Bacterial succinic acid production requires pH control close to neutral levels resulting in the formation of succinate salts that later have to be recovered in their acid form. In membrane electrolysis, the applied current drives the succinate anion from the fermentation broth into an extract solution, whereas the hydroxide ions generated in electrolysis replace pH control, and thus dosing the fermentation with costly sodium or potassium hydroxide is no longer necessary. CMET has extensive experience in the coupling of fermentation processes to in situ electrochemical extraction, and has achieved production of mid chain fatty acids at >80% purity from complex waste streams such as thin stillage or grass [28,29].

CMET has proved that the hydrogen gas that is generated in the fermentation broth can be metabolised by many fermenting organisms, theoretically replenishing NADH and thus potentially generating more reduced, higher value compounds, improving substrate yields and minimising side-products [26]. Hydrogen (H_2) supply can act as an electron donor in *Actinobacillus succinogenes* and *Basfia succiniciproducens* cell metabolism providing enough energy for the reductive TCA cycle pathway and possibly diminishing by-product generation. Furthermore, the cells and the colour do not pass through the membrane. This means that some downstream separation steps including the removal of cells, the removal of proteins/colour and acidification of the succinate salt are not needed. Another advantage of electrolysis is that the succinic acid concentration in the aqueous extracted solution can reach high concentrations reducing the energy requirements for evaporation or enabling the direct crystallisation from the concentrated extract (the level of the purity achieved by this method should be verified during PERCAL).

The removal of organic acids during fermentation and the supply of H_2 is expected to lead to higher final SA concentration, productivity and yield. AUA and CMET have carried out preliminary experiments with *Basfia succiniciproducens* in a recycled fed batch fermentation coupled with electrolysis membrane extraction leading to the overall production of 22 g/L of succinic acid with the extract solution containing 60 g/L of succinic acid. The control experiment carried out without in-line SA extraction led to 20 g/L of final succinic acid concentration at 80% lower productivity and 40% lower yield than the integrated system. These preliminary results demonstrate the high potential of the integrated system.

Beyond of the state of the art, in **PERCAL** bacterial succinic acid production will be optimised in an integrated fermentation-electrolysis system leading to higher bioprocess efficiency compared to conventional systems. This process will be compared with sequential succinic acid production via fermentation followed by downstream separation using an electrodialysis-based process available at ATB.

1.4.5 State of the art and progress in the field of lactic acid hot melt adhesives (HMA) by reactive extrusion.

The global hot melt adhesives market[30] is projected to reach USD 9.64 billion by 2020, registering a CAGR of 5.73% between 2015 and 2020.Hot melt adhesives (HMAs) are solids at room temperature, but when heated form a liquid adhesive layer that cools and bonds rapidly to a substrate. HMAs can be useful for high speed manufacturing and applications that require bonding versatility, large gap filling, and minimal shrinkage. HMAs do not require a carrier fluid such as an organic solvent or water, which eliminates the need for drying the liquid adhesive layer once it is

[27] Schievano et al. (2016) Electro-fermentation - Merging electrochemistry with fermentation in industry. Trends in Biotechnology *In Press*
[28] Stephen et al. (2015) Electrolytic extraction drives volatile fatty acid chain elongation through lactic acid and replaces chemical pH control in thin stillage fermentation, Biotechnology for Biofuels, 8:221
[29] Stephen et al. (2014) Electrolytic membrane extraction enables production of fine chemicals from biorefinery sidestreams, Environmental Science & Technology, 48: 7135–7142
[30] www.researchandmarkets.com/reports/3607761/hot-melt-adhesives-market-by-product-type-eva

applied to a substrate. Elimination of the drying step reduces solvent usage, increases production line speeds, and lowers transportation costs.

HMA compositions have historically been based on petroleum-derived polymers, and these compositions are further tackified, plasticized, and reinforced with a variety of resins, oils and waxes that can be derived from both petroleum and naturally occurring feedstocks such as wood, gum and tall oil rosin, and terpenes. These HMA compositions are subject to the cyclical price cycles common to all petroleum-derived materials, and also are generally very resistant to degradation once the articles employing them (such as cardboard boxes and the like) are disposed.

Hot melt adhesives based on polylactic acid[31] are claimed to have excellent hot tack strength along with long open time and moderate setting time. It can be used as a high performance sealant for heat sealing packaging (up to 80°C). To improve flexibility of PLA hot melts, other biopolyesters have been added to the formulation.

Examples of renewable HMA are detailed in section 1.4.6. However, their commercial acceptance is relativity slow due to the high cost which is expected to be corrected once these adhesives move into higher volume production or raw material cost reduction

Low cost and medium-high purity LA are a suitable raw material to produce HMA as alternative to current EVA or renewable HMA. Furthermore, reactive extrusion[32] will be suitable to produce low molecular weight polymers as in the case of polymer for hot melts adhesives[33], it is a continuous & low investment process to tailor-made polymerization. It describes a special extrusion process in which individual components are first bonded by a chemical reaction. Twin screw extruders are particularly suitable for the reactive extrusion process due to the excellent mixing properties and modularity. The liquid raw materials are fed into the twin screw extruder infeed in the stoichiometric ratio. The reaction of polymerization takes place in the twin screw extruder under constant mixing and kneading at short residence time and free of solvents. Surplus reaction heat is dissipated through the barrel wall. AIMPLAS & SOFTER has a broad experience in the production of polymers, including PLA, by reactive extrusion[34] using different types of catalyzers to obtain a several range of molecular weight and applications (figure 1.4.b).

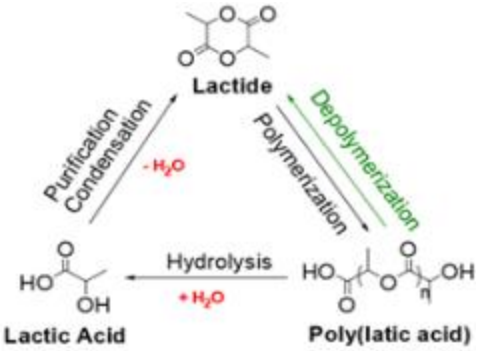


Figure 1.4.b Chemical routes to PLA & lactides from LA.

After the production of lactide (by traditional green chemical routes) and low molecular weight PLA (by continuous reactive extrusion) and in order to improve its adhesion capacities different reactive co-monomers such glycolic acid, maleic acid, methacrylate or glycidyl acrylate will be studied to be grafted with the PLA by reactive extrusion.

Beyond of the state of the art, **PERCAL** will focus on the production of hot melts by reactive extrusion of lactic acid and other co-monomers using suitable catalyzers and peroxides. The obtained copolymers will be formulated with suitable components as tackifier and wax. In addition, small amounts of various additives (e.g., antioxidants, UV inhibitors, plasticizers …) will be used to tailor-made final properties. An in-depth study, will be done to produce in one step (reactive extrusion (polymerization and grafting) plus components mixing) full renewable HMA.

1.4.6 State of the art and progresses in the field of production of ethyl lactate from renewable resources.

The worldwide solvent market is 15 Mt/y, almost all manufacturing and processing industries depend on the use of solvents. This large consumption means that eventually some products will leach into the ecosystem, most critically resulting in contamination of large amounts of water. Many petroleum-based solvents have the drawbacks of high toxicity and cause ozone depletion. Some examples include the dry cleaning industry's use of the toxic solvent perchlorethylene and degreasers such as trichlorethane and acetone. Ethyl Lactate is an environmentally friendly bio-solvent that can replace many petro-chemical derived solvents used in industrial and consumer applications, including inks, coatings, silicone oil, and grease removers. Its high solvency power, high boiling point, low vapour pressure and low surface tension enhances the performance of various resins, polymers and dyes. Ethyl lactate is not an ozone depleter, it is biodegradable, and is not a volatile organic compound (VOC).

In addition to the environmental benefits of these solvents, the economical savings from cheaper disposal make them an attractive alternative. Finally, ethyl lactate is not found on the Environmental Protection Agency's (EPA) Toxic Release Inventory (TRI) or listed in the Hazardous Air Pollutant (HAP) section.

[31] Viljanmaa, M., et. al., "LA Based Polymers as HMA for Packaging Applications", Int. J. Adhesion & Adhesives, 22, 2002, 219-226.
[32] Processing of acrylic hotmelts by reactive extrusion US 6753079 B2
[33] www.pstc.org/files/public/Maris08.pdf
[34] www.innorex.eu/; www.biorefine2g.eu/; www.bio4map.eu/; http://bread4pla.aimplas.es/;

The esterification reaction of lactic acid and ethanol to ethyl lactate kinetics can be highly accelerated using a catalyst and applying moderate temperatures in the range of 70-80 ºC. At equilibrium the 4 phase solution contains about 26% ethyl lactate only, therefore to accomplish much higher conversion rates a continuous separation of ethyl lactate and water is required. In addition to the main 4 molecules presented (ethanol, lactic acid, ethyl lactate and water) the esterification solution also contains significant amounts of azeotropes (water/ethanol, ethanol/ethyl lactate) and lactic acid polymerisates (such as di-lactides and tri-lactides). So, the esterification reaction on a detailed level is quite complex which is most likely the reason why there is up to now no detailed chemical model available which also includes azeotropes and LA- polymerisates and catalyst.

Reactive distillation process for the esterification of ethyl lactate is promising approach. PERCAL' proposal is to establish a catalytic zone by means of a built in solid catalyst within a distillation column. In theory such an apparatus can deliver a total transfer of introduced LA to ethyl lactate within the catalytic zone. The product ethyl lactate exits the system as bottom product accumulated in the column heater while water and ethanol exit at the head.

Gao *et al*[35] described an archetypal set-up of a reactive distillation of ethyl lactate without delivering any more detailed experimental results, several researchers[36-37] describing kinetics of esterification of ethanol and LA to ethyl lactate using ion-exchange resins as solid catalysts. Also basic results on the vapour-liquid equilibria in the reactive system have been published[38]. Other researchers described in detail: i) a numeric modelling for a reactive distillation, ii) the esterification model process and searched for optimal process parameters for high outputs of mono-ethyl lactate and iii) reactive distillation for esterification of lactic acid and methanol to recover methyl lactate[39].

Within the project, a new innovative technology for the production of ethyl lactate on the basis of reactive rectification will be developed. This allows performing the esterification of lactic acid with ethanol and the separation process of the target product ethyl lactate in one single apparatus. This technique enables significant advantages regarding efficiency. It is also foreseen that required investment and operative costs of such a technology are lower than for existing methods. Based on literature data, TBWR has been modelling the esterification process for ethyl lactate and designing a prototype reactive distillation for further experimental evaluation.

Beyond of the state of the art, **PERCAL** will use a prototype reactive distillation to upscale the production of ethyl lactate to a continuous process in the scale of 3-5Lh^{-1} ethyl lactate. The process will be specifically designed to the characteristics of PERCAL products ethanol and lactic acid from MSW. Also potential synergies to integrate esterification to an existing bioethanol demonstration plant will be assessed for further upscale potential. Through increasing the efficiency and developing a single process for ethyl lactate production it will make it easier for new, or existing, companies to enter the market. Continuous development and reduction of processing steps for ethyl lactate production will be developed from data acquired in this study. The production of highly pure lactic acid may also be examined by reversing the reaction after esterification.

[35] J. Gao et al., Investigation of Ethyl Lactate Reactive Distillation Process, Chemical Engineering Research and Design, 85, 525-9. 2007.

[36] C. S. M. Pereira et al.: Thermodynamic Equilibrium and Reaction Kinetics for the Esterification of lactic Acid with Ethanol Catalyzed by Acid Ion-Exchange Resin, Ind. Eng. Chem. Res. 47, 1453-63, 2006.

[37] P. Delago et al., Kinetic Study for esterification of lactic acid with ethanol and hydrolysis of ethyl lactate using an ion-exchange resin catalyst, Chem. Eng. J., 126, 111-8, 2007.

[38] D. T. Vu et al.,Vapor-Liquid Equilibria in the Systems Ethyl Lactate+Ethanol &Ethyl lactate Water, J. Chem. Eng. Data 51, 1220-25, 2006.

[39] S. Thotla, S. Mahajani: Reactive distillation with side draw, Chemical Engineering and Processing 48,927–937, 2009.

1.4.7 State of the art and progresses in the field of production of succinic acid (SA) derivate products.

Bio-succinic acid costs higher than its petroleum based counterpart. Although this is the case, microbial production of succinic acid has been attracting a lot interest mainly due to four main factors: (i) huge expected demand in the next years as section 2 shows (ii) The succinic acid produced through traditional chemical routes using petroleum-based starting materials cannot give high purity and yield without the usage of high temperature, high pressure, etc... On the other hand, microbial succinic acid production provides milder and environmental friendly conditions as well as good purity and yields. (iii) Potential use of land field organic waste as a feedstock for the production of SA, (iv) As the microbial production of succinic acid becomes a mature technology, the current bio-SA production costs will further decrease. It is highly expected that petroleum-based succinic acid will lose its cost advantage in the near future.

Microbial production of succinic acid also opens new opportunities to bio-based production of 1,4-BDO, THF, putrescine (Figure 1.4.c)[40] and polymers of these such as PBS, PBT, PTMEG, PA-4,6 and PA-4,10.

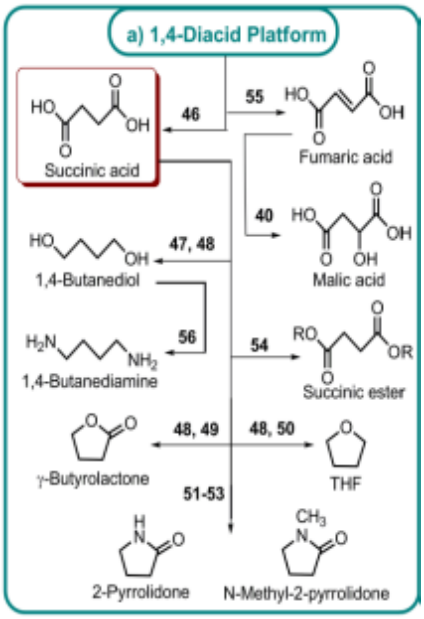


Figure 1.4.c Diacid platform.

Beyond of the state of the art, PERCAL will lay the foundation for a truly biobased industry to produce a new range of bio based building block for polyols, resins and polyurethane dispersions (PUD). PERCAL will avoid the competition with food for the production of polyols and truly focus on waste to resources.

1.4.8 Main competitors.

Table 1.4.a shows a list the main competitors for the different results that you have identified at worldwide, and identify those features (at the different levels: technological, obtained results, state of development, prices, market niche, etc.) that your results will have and compare them with the ones that your competitors have.

Table 1.4.b Main competitors and the main advantages and limitations of current solutions.

Result	*Current competitor*[41]	*Main advantages of the commercial solution*	*Main limitations of the commercial solution.*	*PERCAL Advantages*
Enzymatic hydrolysis of MSW.	Eberkem, Etanolix, Abengoa, Bionolix	No industrial plant for industrial production of chemical products from MSW. Mainly focused to biofuels production.		Use of PERCAL tech. for bioethanol, LA, SA process, increasing its competitiveness.
Lactic acid	PURAC B.V, Natureworks, Galactic, Henan Jindan, BBCA, Total	Good product quality, food grade product possible, highly developed fermentation & downstream.	All of the competitors are using first generation sugars (beet, cane, corn etc.) high dependency on sugar price.	Low cost sugars are feasible. High yield due in-line inhibitors removal.
No-petrol source Succinic acid	Reverdia, Succinity Myriant	Established at market. High purity	Genetic engineering of species to tolerate acidic fermentation conditions. Extensive dosing of chemicals to maintain neutral conditions is high cost, and requires recovery of salt. High downstream cost e.g. formic, acetic or lactic acids removal.	Extraction of succinic acid in-line with fermentation process.
Ethyl lactate.	Corbion, ADM, VERTEC, Arkema,	Established at market with suitable performance.	Mainly batch esterification, limitations in process efficiency to kinetics (equilibrium), high demand of product and by product separations. High cost versus petrol based.	Reactive distillation allows esterification and product separation in one single apparatus

[40] F. H. Isikgor et al. Lignocellulosic biomass: a suitable platform for bio-based chemicals and polymers, Polym. Chem., 2015,6, 4497-559.
[41] Full names of current worldwide competitors are detailed in section 2.1.2 Economic impact.

Result	*Current competitor*[41]	*Main advantages of the commercial solution*	*Main limitations of the commercial solution.*	*PERCAL Advantages*
No-petrol source HMA.	Collano Adhesive AG DaniMer Scientific	Hot melt pressure sensitive adhesives based on renewable sources Good temperature service 0°C to 55°C.	Lactic acid from sugar or food by-products (animal feed competition). High cost.	LA from MSW. Low cost polymerization process (reactive extrusion).
No-petrol based polyols.	NOVOMER	CO_2 based polyol – first products available (PU foams for seatings with FORD) – lower carbon footprint that oil based material	Mechanical properties are in question – only a fraction of fossil based raw material can be replaced by CO_2	Develop a 100% biobased near drop-in solution has the potential to further reduce the footprint of PUR products.
No-petrol based surfactant.	Jeneil Biotech, AGAE Techn., Rhamnolipid Inc., Sigma-Aldrich, BioFuture Ltd, Ecochem Ltd.,	Established at market with suitable performance.	Noble expensive materials as feedstocks	Cheap raw materials. Different range of properties, depending on the combination of lipid and amino-acid.

2. IMPACT

2.1 EXPECTED IMPACTS.

The PERCAL project is fully in line with the European BBI Call "H2020-BBI-R08.2016- Emerging technologies for conversion of the organic content of Municipal Solid Waste and improving waste-to-chemicals value chains" and with the wider objectives, guidelines and policies of several EU leading initiatives, as summarized in table 2.1.a.

Table 2.1.a Contribution to the expected impacts defined in the *BBI 2016 R08* topic text.

<table>
<tr><th>Expected Impact in the BBI.2016.R8 topic</th><th>Expected contribution of the BBI topic.</th></tr>
<tr><td>Improved yield of intermediate extraction/recovery from the organic content of MSW by 20% with respect to state-of-the-art or exceed 80% yield of intermediates.</td><td>With the application of improved enzyme cocktails for the carbohydrate fraction and the valorisation of the lipid and protein by-products, we expected to valorise up to 600 g/kg of Dry fraction of MSW, more than two times than current PERSEO technology or other similar technologies described in ANNEX I.
<table>
<tr><th>Main PERCAL yield impact.</th><th>Current intermediates yield from MSW as a feedstock.</th><th>PERCAL proposed target yield.</th></tr>
<tr><td>Lactic acid</td><td>Final titer of fed-batch fermentation up to 30 g/L[42].</td><td>High conversion yield (>85% based on fermentable sugars) for the production of up to 55 g/L LA with >97% enantiopurity will be achieved.</td></tr>
<tr><td>Succinic acid</td><td>No data available for MSW. Up to 40 g/L depending on sugar composition, up to 0.54 g/g yield and up to 0.78 g/L/h productivity (based on results from the BRIGIT project)</td><td>Up to 60 g/L final SA concentration with at least 20% higher yield from MSW sugars and up to 50% higher productivity (depending on MSW composition)</td></tr>
<tr><td>Biosurfactants</td><td>No data available for MSW. 40%-90% yield of enzymatic hydrolysis using as a by-products of well-known fermentation process (depending on the enzyme) (reference 9)</td><td>Target is to achieve the same hydrolysis yield than described in the alcohol industry (reference 9), taking into account the complexity to manage and control feedstock derived from MSW.</td></tr>
<tr><td>Ethanol</td><td>26g/L (current values in PERSEO plant)</td><td>30g/L (expected due to increase % of fermentable sugars) i.e., 26% higher.</td></tr>
</table>
With the pilot plant at ATB a new biotechnological procedure is realized, allowing an efficient refinement of ingredients from plant biomass and residues into industrial products according to the process steps of a standardized sugar-based fermentation technology. Besides the feedstock costs which should be much lower here due to the utilization of waste the expenses for the down streaming of a bio-product, for example, amount to up to 90% of the entire production costs, depending on the value of the product.</td></tr>
<tr><td>Validated removal of inhibitors to such an extent that it allows cost-effective downstream processing.</td><td>Membrane electrolysis extraction will be used to remove the production of organic acids during succinic acid production in an integrated bioprocess leading to enhanced final succinic acid concentration, yield and productivity. The succinic acid will be concentrated in the extract solution in the acid form leading to reduced unit operations for the purification of succinic acid. The integrated system will be compared with sequential succinic acid production via fermentation followed by succinic acid recovery using the electrodialysis-based downstream process available at ATB.
Due to the importance of separation and purification of the product the investigations concentrate on the down-stream processes of lactic acid in the form of lactate salts made by fermentation of the different substrates. Conventional processes are based on</td></tr>
</table>

[42] Probst, M., Walde, J., Pümpel, T. et al. Appl Microbiol Biotechnol (2015) 99: 3029. doi:10.1007/s00253-015-6414-7

Expected Impact in the BBI.2016.R8 topic	**Expected contribution of the BBI topic.**
	precipitation steps that generate large amounts of chemical effluents. Consequently the environmental impact and the operating costs of traditional precipitation processes can be reduced by using alternative technologies, such as two-stage electro dialysis.
Validated process by comparison of the resulting product yield with the one obtained from lignocellulosic biomass. The difference in yield should not exceed 10%.	AUA and CMET have carried out preliminary experiments with Basfia succiniciproducens demonstrating that a conventional fed-batch fermentation using xylose as carbon source led to 80% lower productivity and 40% lower yield than the integrated ME-fermentation system. The in-line extraction of organic acids is expected to improve succinic acid production when compared to lignocellulosic biomass. Similar approach will be proposed in the production of lactic acid by microbiological fermentation in order to optimize the yield avoiding the negative effect of inhibitors.
Contribution to the BBI JU Key Performance Indicators (KPI), specifically: - *Creation of at least 1 new cross-sectorial interconnection in bio-based economy clusters (KPI 1).* - *Set the basis for the creation of at least 1 new bio-based value chain (KPI 2).*	In PERCAL a new cross-sectorial interconnection will be proposed to produce novelty industrial valuable products from biogenic fraction of MSW. Chemical companies (HAYAT, SOFTER, ARKEMA & COVESTRO) will collaborate with biotechnological companies (ENERGO & NOVO) and engineering companies (IMECAL, EXERGY & VISUM). Several industrial sectors will be addressed such as solvents, adheives, polyols and surfactants, that are the raw material for a huge number of products (household and cleaning products, cardboard, packaging, paints, varnish, etc) Three bio-based value chains will be developed using MSW as a main feedstock: 1) Lactic acid from fermentative process to produce organic solvents and hot melt adhesives, 2) Succinic acid from fermentative process to produce polyols for the polyurethane industry and 3) Biosurfactants from the protein and lipid fraction as a by-product of both target fermentation processes. Although LA, SA and biosurfactants are already being produced from bio-based sources (usually from food competing sources such as sugars and oils), they have never been produced from MSW before. This represents a real challenge due its composition complexity and variability.

2.1.1 Enhancing innovation capacity and integration of new knowledge.

Valorising municipal solid waste (MSW) has an environmental rationale and creates an important competitiveness opportunity for Europe as described in the European Commission strategy for 'Innovating for Sustainable Growth: a bioeconomy for Europe'. Indeed, addressing our societal challenges is a major catalyst for increasing our innovation capacity and for the integration of new knowledge generated as a result of research into practical societal and industrial contexts.

The percentage of chemical production based on biotechnology is estimated to increase from less than 2% in 2005 to approximately a quarter of all chemical production[43] by 2025. In EU28 2013, the chemical and plastics represented a turnover of **42 billion €** and 200,000 employees[44]. Initiatives such as PERCAL are enablers for innovation in countless other sectors that can transform these wastes into sustainably sourced raw materials as inputs for countless discreet manufacturing and process industries in Europe and internationally.

To increase the innovation capacity of the industry, our approach is to capitalise on a valuable opportunity for the EU municipal waste management, engineering, chemicals, plastics and other end-user industries to work together in order to substitute currently used fossil-based counterparts with bio-based materials and to assume a leadership position on the world stage.

The innovation process is only complete when new innovative knowledge is put to industrial use. The innovative potential of PERCAL is enhanced by the **strong industrial composition of the PERCAL consortium (7 partners out of 12; five of them SMEs and three large companies**). Five of the industrial partners or support companies are BBI full members (**CENER, NOVO, ENERGO, ARKEMA and EXERGY**). Industrial partners are integrated into the project decision making structure to ensure that the involved industrial sectors drive project activities and benefit from the project results. To ensure success, industrial partners will lead all the technological WPs and enable the steering of project activities from the perspective of the whole supply and value chain. The success of PERCAL will be mainly based on the strong interaction and partnership of SMEs/industry and research organizations, all highly skilled in complementary fields as shown in figure 2.1. IMECAL and NOVO will be in charge to select European representative MSW streams based on placed and seasonal variables. The selected ones will be pre-treated with

[43] Smeets, E. et al. (2014) Evaluating the macroeconomic impacts of bio-based applications in the EU.
[44] http://biconsortium.eu/sites/biconsortium.eu/files/news-image/16-03-02-Bioeconomy-in-figures.pdf

enzymatic cocktails to maximize the production of fermentable sugars and minimize the formation of inhibitors.

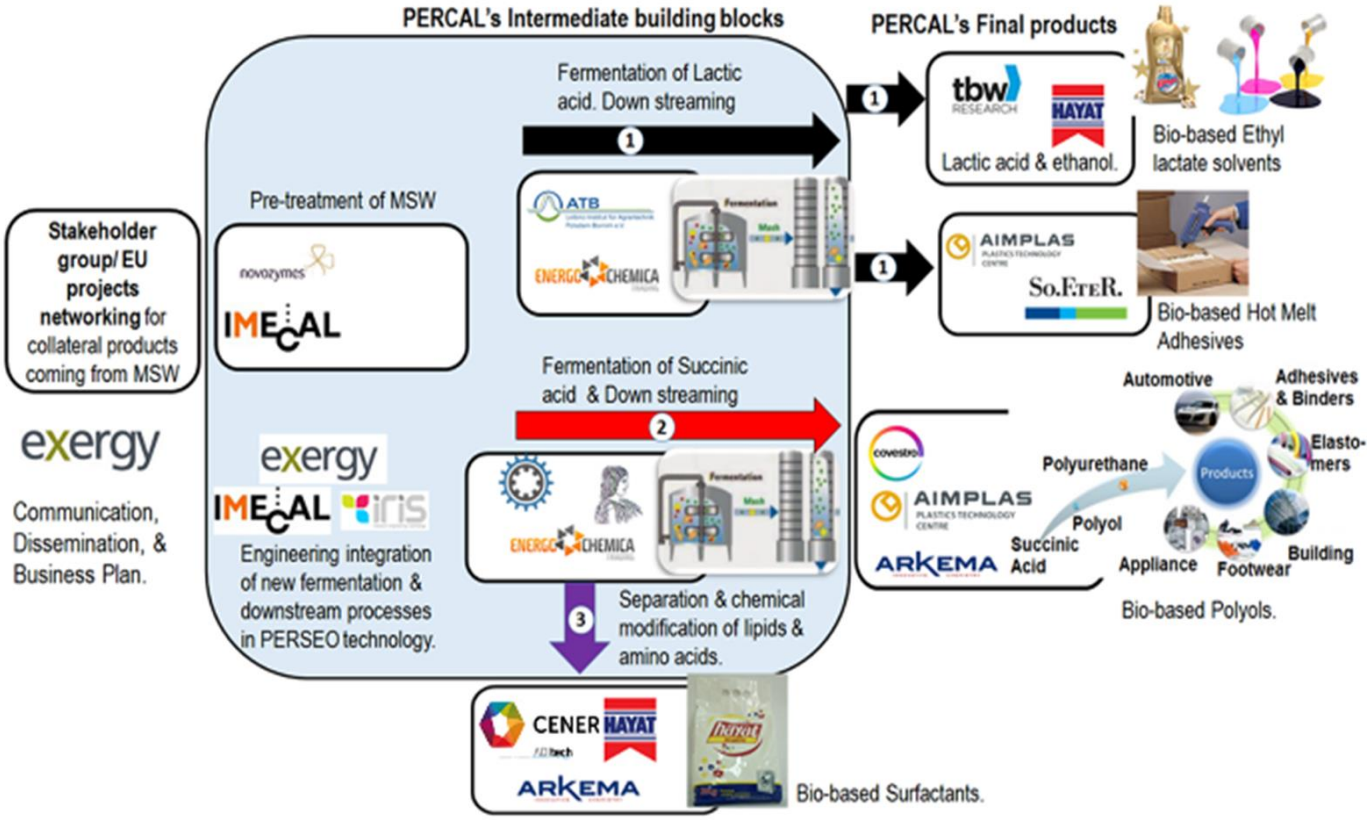


Figure 2.1.a. Structure and relation between partners in the consortium.

A fraction of the proteins will be employed as a nitrogen source of fermentation. After that three main routes are defined in PERCAL project: ① ATB and CENER will be in charge to select the best strains and fermentation process to maximize the lactic acid yield. The lactic acid will be chemically modified to produce: 1) Eco-friendly ethyl lactate solvents by reactive distillation from lactic acid and the ethanol produced in PERSEO pilot plant to be used in cleaning products and inks and 2) hot-melt adhesives by controlled reactive extrusion for cardboard and other non-food applications② AUA and CMET will be in charge to select the best strains and fermentation process (including its integration in a membrane electrolysis extraction system) to maximize the succinic acid yield. The purified succinic acid will be employed as an intermediate building blocks to produce polyester polyols, the main components of polyurethane. ③ Amino acids and lipids from the fermentation steps will be extracted and purified. After their chemical or enzymatic treatment, both components will be used to produce renewable and biodegradable surfactants as alternative to current petrol based ones.

The information of pre-treatment, fermentation and downstream of each acid will be used by the ***engineering companies IMECAL, VISUM and EXERGY*** for the scale-up process study in order to integrate in the PERSEO demo plant the key process parameters to maximize the yield of each one using complex feedstocks.

Cross-fertilization and synergies among partners with expertise in complementary and cascaded processes (VISUM, TBWR …), technologies and products will act as a booster for innovation placing better and more sustainable products on the market. The participation of large end-users guarantees high impact and the wide dissemination & exploitation of the project results at worldwide level.

2.1.2 Economic impact: strengthening the competitiveness and growth of companies

High economic impact will be achieved through strengthening the competitiveness of the Industrial Partners in the target market sectors and related ones (intermediate chemicals (SA, LA and ethanol), solvents, hot-melts adhesives surfactants and polyols) by: 1) promoting the use of PERCAL materials, components and systems in high value-added applications while keeping costs competitive, 2) strengthening and differentiating the EU Industry, 3) increasing workers' skills. ***Detailed information of each market is presented in the following pages:***

<u>Eco-friendly solvents.</u>

The solvent market production in Europe is estimated between 3.5 to 4.5 million tonnes per year, at prices ranging

from €2.00 to €3.75 per kilo[45]. This large consumption means that eventually some products will leach into the ecosystem, most critically resulting in contamination of large amounts of water. Many petroleum-based solvents have the drawbacks of high toxicity and cause ozone depletion. Some examples include the use in the dry cleaning industry of the toxic solvent perchlorethylene and degreasers such as trichlorethane and acetone.

Ethyl Lactate is an environmentally benign solvent. Lactate esters have found industrial applications in specialty coatings, inks, cleaners and standard cleaning products. It is produced from renewable resources and are neither toxic nor ozone depleting. They are biodegradable, and free of volatile organic compound (VOC). In addition to their environmental benefits, the economical savings from cheaper disposal make them an attractive alternative.

The current market share of bio-solvents in the EU is about 1.5% (around 60 kt/year-120 M€)[46]. However, bio-solvents produced from vegetable oils and from starch are progressively replacing petrochemical solvents. One of the main factors for their increased consumption is the EU legislation concerning the reduction of volatile organic compound emissions in the context of the air quality policy. Some estimations point at a potential share of approximately 12-40% of the solvents market (i.e., 1.6 Mt/year- 3,200 M€/year) in the next decade.

Packaging, healthcare sector disposables, detergents, hygiene products, cosmetics and paints are strongly influenced by health and environmental concerns and represent areas in which bio-based products could potentially reach a substantial market share. The sales of these products in the EU were roughly about € 250 billion in 2005 and the healthcare sector is expected to grow due to population ageing.

Currently all ethyl lactate is currently produced from ethanol and lactic acid made from renewable sources (e.g., starch or sugar). PERCAL aims to produce **Ethyl Lactate** from **ethanol and lactic acid coming from the organic fraction of MSW**. This is an unexplored route to produce bio-based solvents. The PERCAL's solvent will be used exclusively in **non-food applications** such as cleaning products and non-food contact inks.

Biosurfactants from modified amino-acids and lipids

The global surfactants market is projected to reach a volume of 24,000 kt and a value of €40,000 Million by 2020 with an estimated to grow at a CAGR of 5.5%, by value during the forecast period of 2015 to 2020[47]. The anionic surfactants segment dominated the global market, in terms of volume, with 8,000 kt in 2014, followed by the non-anionic surfactants, the volume of which was recorded at 6,500 kt in the same year. The amphoteric surfactants segment, with a share of 7.2% of the global market in 2014, is projected to grow at the highest CAGR during the forecast period of 2015 to 2020, as compared with the other surfactant segments.

The surfactants properties such as wettability, detergency, emulsion, dispersion, stabilization and foam/froth have vast practical applications in variety of industries. The market is largely driven by applications such as the soaps/detergents, personal care, and industrial and institutional cleaners industries.

Prolonged usage of surfactants obtained from synthetic sources generates toxic effects such as skin and eye irritation, neurotoxicity, mutation, and other health concerns in human beings. Several regulatory bodies such as Centers for Disease Control and Prevention (CDC), National Institute for Occupational Safety and Health (NIOSH), Organic Consumers Association, and National Institutes of Health (NIH) jointly work to observe the implication of detergents on humans. Investment in research and development to introduce new sources of raw materials for the manufacture of surfactants is projected to provide opportunities to the surfactants market.

The global Biosurfactants[48] market by revenue was €1,800 million (4% of the market) in 2015 is projected to reach €2,200 million by 2020. The high share of biosurfactants in Europe can be attributed to the growing awareness among consumers regarding the health hazards associated with chemical surfactants. In addition, the stringent regulatory framework of the region favors the use of bio based products in place of conventional petroleum based products.

Regardless of their greater biodegradability and reduced toxicity, cost competitiveness still remains the major concern for bio based surfactant production. However, recombinant or metabolically engineered, hyper producing strains, low cost feedstock could lead to a significant cost reduction.

Nowadays, 96% of surfactants are produced from petrol based components (e.g., alkyl-benzene sulfonate). PERCAL aims to produce **surfactants** from **the protein and lipid fraction of MSW by chemical and enzymatic routes**. The PERCAL's Biosurfactants will be used exclusively in **non-food applications** such as soaps/detergents and industrial cleaners industries.

Competitors and key stakeholders in the alternative solvents and surfactant market. An analysis of the structure of the market players and key stakeholders demonstrates that the market is controlled by large companies. Main

[45] http://cellulac.co.uk/en/ethyl-lactate/ and http://www.nrel.gov/docs/fy16osti/65509.pdf
[46] EU commission report, *Monthly Update on Industrial Policy Indicators and Analysis – June 2013-//ec.europa.eu/enterprise/policies/*
[47] www.marketsandmarkets.com/Market-Reports/biosurfactants-market
[48] www.mordorintelligence.com/industry-reports/biosurfactants-market

producers of ethyl lactate include Corbion, Galactic, ADM, Musashino Chemical Laboratory, Vertec BioSolvents, Somaiya Group, …The biggest players in the surfactant market are include BASF Cognis Henkel AG & Co. KGaA, Soliance, MG Intobio, Ecover, Kemin Industries. Identifying and effectively negotiating technology licensees will be a key activity for PERCAL partners. The untraditional source (MSW) and sustainability demands of final consumers will provide a strong market pull, supported by the new EU legislation.

Existing Competition: PERCAL offers a proprietary, eco-friendly and effective alternative over existing commercial products. Furthermore products will be sold at a lower price than existing solutions due to the raw cost of MSW as raw material produced in huge amounts each year. The market expected for VOCs free solvents is huge (1.6 Mt/y) and there are market possibilities for many different actors. The potential market for Biosurfactants is 22,000 kt/y with an average cost of 1,600-1,700 €/t.

Potential New Entrants: As the bio-based solvent and surfactants market is growing, with an increasing number of applications, it is inevitable that there will be additional companies interested in developing and selling these bio-based products. However, it is difficult to have a largely available and low cost feedstock for their production such as MSW and it is anticipated that the majority of uptake will be from existing manufacturers who will be interested in licensing the patented PERCAL technology. By allowing the technology to be licensed, TBWR and HAYAT will reduce the incentives for these companies to undertake the costly, slow and risky process of developing alternatives.

Materials Suppliers: the main ingredients of PERCAL bio-based solvents will be lactic acid and ethanol coming from MSW feedstock. MSW is available worldwide at very low price since it is very abundant. Additives required for the production of ethyl lactate are used in several processes and are available in the required quantities to meet the market demand, with established prices and several suppliers. Biosurfactants are produced by chemical treatment of the protein and lipid fraction of MSW, they also use standard reactants in the extraction of surfactants components and further chemical modification.

Main customers: The main customers of solvents and surfactants are large consumer goods manufacturers such as HAYAT and small or medium size companies that supply to local market. After PERCAL successful project, HAYAT will offer their own customers new non-toxic and VOC free "green" cleaning and related products at a competitive cost. Current consumption of solvents and surfactants in HAYAT is estimated in 750 t/year.

Political Groups and Consumers: Bio based solvents are beneficial in terms of health and safety aspects. Common solvents are both toxic and flammable, while ethyl lactate is not found on Environmental Protection Agency's (EPA) Toxic Release Inventory (TRI) or listed in the Hazardous Air Pollutant (HAP) section. EU legislation on reduction of VOCs is a driver for this adoption. PERCAL will offer the opportunity to provide bio-based solvents and surfactants that are both effective and safe. Effectively disseminating this information will be essential to estimulate non-governmental organizations (NGO) and consumer pull for PERCAL's solvents. Additionally, government funding for public-private partnership (PPP) schemes to develop non-toxic solvents & surfactants is likely to encourage new players to enter the market.

Hot-melt adhesives (HMA).

The global hot melt adhesives market[49] is projected to be worth €9 Billion by 2020, registering a CAGR of 5.73% between 2015 and 2020 with a consumption was estimated close to 1.5 million tons in 2014. HMA is a solvent-free and thermoplastic material. This market is driven mainly by the need substitution of solvent-borne adhesives. However, this market faces certain drawbacks, such as high cost associated with switching to hot melt adhesives and volatility in the raw material prices[50]. *Low cost raw material and processes, free of the cost volatility of petrol based chemicals, will act as a driving to the growth of the market*.

Ethylene-vinyl acetate (EVA) is the leading product segment and accounted for more than 50% of the total demand in 2014. Packaging was the largest application segment and accounted for more than 33% of the total demand in 2014 and is likely to continue this dominance up to 2020. They are used in packaging applications such as corrugated case and cartoon sealing. Can, bottle and glass labelling are among other major packaging application which this adhesive caters. Increasing hygiene awareness among consumers is likely to positively influence disposable product demand. Hot-melt adhesives find other significant applications in disposable application such as baby diapers, feminine hygiene and adult incontinence products.

It appears that all hot-melt is currently produced from petrol based polymers (e.g., EVA, polyurethane). PERCAL aims to produce **HAMs** from **low molecular lactic acid coming from the organic fraction of MSW by reactive extrusion**. The HMA from PERCAL will be used exclusively in **non-food applications**.

Competitors and key stakeholders in the alternative HMA market. An analysis of the structure of the market players

[49] www.marketsandmarkets.com/Market-Reports/adhesive-sealants-market
[50] www.mordorintelligence.com/industry-reports/europe-hot-melt-adhesives-market

and key stakeholders demonstrates that the market shared between large companies and small/medium local size companies. Main hot melt adhesives manufacturers include The 3M Company, Ashland Inc., Bostik S.A., Dow Corning, H.B. Fuller Company, Henkel AG & Company, Jowat AG, Sika AG and Toyobo Co., Ltd. (Japan) among others. Identifying and effectively negotiating technology licensees will be a key activity for PERCAL partners. The untraditional source (MSW) and sustainability demands of final consumers will provide a strong market pull, supported by significant environmental advantages and low expected cost volatility of the raw material.

Existing Competition: PERCAL offers a proprietary, eco-friendly and effective alternative over existing commercial products. Furthermore products would be sold at a similar or slightly higher price than EVA-based HMAs, given that MSW is a low cost raw material, but it is in competition with optimized petrol-based process. The potential market for VOCs free HMAs is 750 kt/y, i.e., the current market of EVA HMAs.

Potential New Entrants: As the HMAs market is growing, with an increasing number of applications, it is inevitable that there will be additional companies interested in developing and selling bio-based HMAS. However, it is difficult to have another feedstock for the production of bio-based HMA with the potential of MSW and reactive extrusion is a fast, low cost, low energy consumption and low investment process for HMA production. It is anticipated that the majority of uptake will be from existing manufacturers who will be interested in licensing the patented PERCAL technology. By allowing the technology to be licensed, SOFTER and AIMPLAS will reduce the incentives for these companies to undertake the costly, slow and risky process of developing alternatives.

Materials Suppliers: the main ingredients of PERCAL's HMAs will be lactic acid coming from MSW feedstock. It feedstock is available worldwide at very low price since it is very abundant. Other agents required for the chemical production of lactic based HMA (such as wax, oils, inorganic fillers, …) are used in several processes and are available in the required quantities to meet the market demand, with established prices and several global suppliers.

Main customers: The main customers of Lactic HMA are large consumer goods packers and small or medium size companies that supply to local markets. HAYAT, COVESTRO or ENERGO will use HMA in the packaging and labelling of their current products. This will drive to provide 100% biodegradable cardboard and labels.

Political Groups and Consumers: Solvent-free bio based HMAs are beneficial in terms of health and safety aspects in comparison with other types of adhesives (VOCs production). Common adhesives are both toxic and flammable, thus the major reasons for replacing them are, on the one hand, the reduction or suppression of volatile organic compounds in HMAs and on the other hand the long shelf life and easily disposability of HMAs. EU legislation on reduction of VOCs is a driver for this adoption and by effectively disseminating the benefits of HMA the required non-governmental organizations (NGO) and consumer pull will be achieved.

Production of SA and related polyols.

Succinic Acid market worth €400 Million and 650 kt by 2019. Succinic acid is an important intermediate chemical witnessing a high demand. It is a building-block chemical for producing many specialty and commodity chemicals that are used in a wide range of applications, such as polymers, food, metals, pharmaceuticals, coatings, fibers, solvents, lubricants, diesel fuel oxygenates & cosmetics. Europe bio-succinic market accounted for over 45% of the global volume share in 2013.

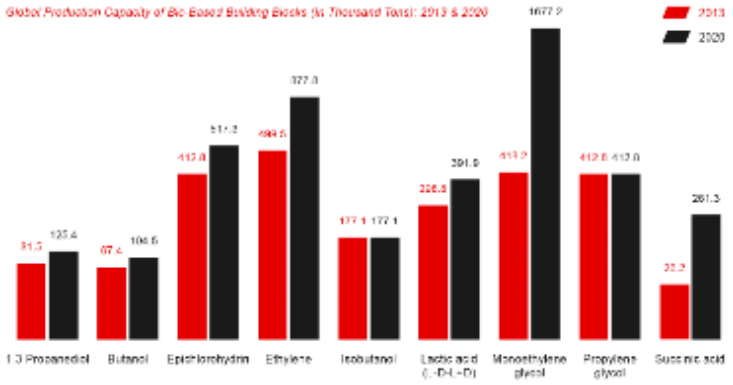


Figure 1.4-a Bio-Based Building blocks Market structure

Figure 1.4.b shows the main applications of bio-SA in Europe. Regulatory pressure exercised by numerous agencies for reduction of greenhouse gas by 20% by 2020 is expected result in augmenting market growth over the forecast period. Polyurethane is the largest application segment for succinic acid, as bio-based SA has the ability to reduce 94.0% greenhouse gas emissions in comparison to that of petro-based. Polyurethane foam is currently, the largest consumer of polyols. The polyurethane is used in textiles and fibers, auto motive industry, furniture, carpet cushion, bedding, insulation purpose in buildings and packaging. Its consumption is expected to reach 20

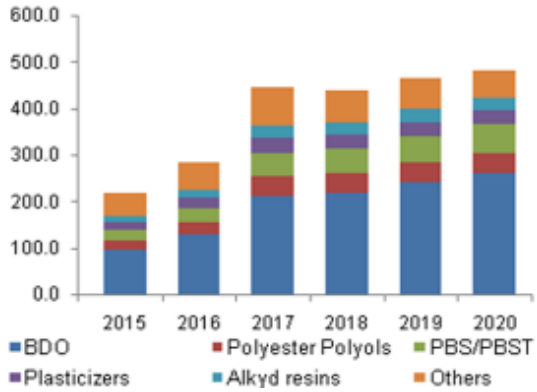


Figure 1.4.b. Europe bio-succinic acid market revenue by application, 2012-2020, ($Million)

billion € by 2019[51].

Manufacturers such as Cargill, Emery and Mitsui have distinguished themselves from other manufacturers by developing capacities for bio based polyols from plant oils[52]. Shifting consumer preference for bio-based materials over synthetic chemical is expected to be beneficial for the overall market growth. Major industry has invested heavily in R&D initiatives for superior and cheaper bio based products. Governments and regulatory associations such as Environmental Protection Agency (EPA), REACH etc. have implemented several guidelines favouring bio based polyols usage. Volatile organic compounds (VOCs) which are environmentally hazardous are used in conventional polyols manufacturing. Sustainable bio based polyols emit zero to minimal VOCs in their manufacturing process. All the above mentioned factors are expected to drive the overall market in the next years.

Competitors and key stakeholders in the SA market. Rising environmental concerns coupled with presence of domestic bio-based succinic acid manufacturers have shifted consumer focus towards sustainable and environment-friendly chemicals. The bio-based SA market is controlled by large players like Succinity GmbH, Reverdia and Myriant. All of them are using sugars as a fermentation feedstocks.

Existing Competition: The SA-based polyols from PERCAL will offer significant advantages over the existing competitive solutions. Some producers can offer a 100% bio-based eco-friendly polyester-polyols based on industrial oils (soy, castor, ..). However they are expensive and require land, fertilizers and water to obtain the raw materials. The SA based polyols from PERCAL will be competitive in cost and properties versus existing petrol based solutions, due to the use of the same chemical monomers as currently used but coming from low cost and renewable raw materials.

Potential New Entrants: As the bio-SA market is growing, it is inevitable that there will be additional companies interested in developing and selling bio-SA related polymers. By allowing the technology, COVESTRO will reduce the incentives of companies to undertake the costly, slow and risky process of developing alternatives. COVESTRO is worldwide leader in the PUR industry, it is difficult to find real competitors in this field.

Materials Suppliers: the main ingredient of PERCAL bio-based SA derivate are succinic acid, coming from MSW. Raw materials are abundantly available worldwide at very low price. Additional additives required for SA polymerization (catalysts, solvents, chain extenders, etc…) and processing (nucleants, lubricants, antioxidants …) are used in several processes and are readily available, with established prices and several suppliers.

Main customers: The main customers of polyols are large (such as COVESTRO) or medium size polyurethane manufacturer companies. COVESTRO will offer their customers new bio-based polyols grades at a competitive price and full from renewable sources. COVESTRO is collaborating with Reverdia (bio-SA producer) to develop bio-based thermoplastic polyurethane.

Political Groups and Consumers: Increasing use of bio-based SA on account of reduction in carbon footprints along with favourable regulatory support in light of rising awareness towards hazards associated with usage of petroleum-based chemicals is expected to augment market growth. Bio-succinic acid can be used as a replacement to phthalic anhydride in manufacturing alkyd resins and offers cost-effectiveness and environmental advantages, which in turn will propel its growth over the next six years, however, high cost for bio-succinic acid along with low availability of feedstock is expected to restrain market growth over the next years[53]. Extensive R&D in manufacturing process by various companies in terms of reducing production cost and increasing efficiency is further expected to offer new growth opportunities over the forecast period.

In summary, the potential impact of the products to develop in PERCAL project and the current and expected cost are summarized in the table 2.1.c. Expected costs are evaluated based on the expertise of the PERCAL's partners and the methodology proposed to systematically evaluate the techno-economic prospects of future production routes of bio-based bulk chemicals by Burk et al.[54]

Table 2.1.c Potential economic impact of PERCAL target products.

Target products.	Current// Expected price (€/kg)	Worldwide consumption (kt/y).
Solvents	2.0/1.8	4,500
Surfactants	3.7/3.5	24,000
Hot-melt adhesives	2.5/2.5	760
Polyester Polyols	4.0/4.0	4,000

2.1.3 Business model proposed by each partner for future commercialization of PERCAL developments.

Regarding the **segments in the value chain directly represented in the project**, and thus project partners, the expected impact is described below, which will result in an increase of the potential market and increased high added

[51] www.marketsandmarkets.com/PressReleases/succinic-acid.asp
[52] www.grandviewresearch.com/industry-analysis/bio-polyols-market
[53] www.grandviewresearch.com/industry-analysis/bio-succinic-acid-market
[54] M. J. Burk et al. Biotechnology for Chemical Production: Challenges and Opportunities, Trends in Biotechnology. 34, 187–190, 2016.

value product portfolio of PERCAL's partners:

- **Renewable chemicals by microbiological and green chemical routes (TBWR, AUA, ATB, AIMPLAS and CMET):** will select and optimize strains to maximize the yield of the bioreactor or develop new routes based on green chemistry. The combination of novel strains to obtain valuable products from complex by-products coming from Biogenic fraction MSW is the **main and innovative product to develop in the PERCAL project**. The typical business model for these partners would be to patent the modified strains, and use the validations in PERCAL to either licence out the strains and their applications or enter into joint ventures with companies which are already active in this area for the production of chemicals. Several examples exist in the industry that validate this business model: Genomatica, Metabolic Explorer, Global Bioenergies, BioAmber and Deinnove are all examples of companies that emerged as research-based start-ups after deriving novel engineered strains with differentiating traits enabling them to produce, for the first time, specific bio-based products from renewable resources. The PERCAL's partners are very active in the international industrial biotechnology market as service providers and are investing in the generation of their own IP in order to progress in the value chain and retain more value of their activities.
- **New biorefinery concept using the biogenic fraction of MSW (IMECAL, VISUM EXERGY).** IMECAL with the support of VISUM (specialist in sensors and monitoring) and EXERGY (energy and mass transfer modelling and simulations) plans to develop a new biorefinery plant concepts to produce multiple intermediates in parallel or cascade approach. Target products will be ethanol, SA, LA and purified proteins & lipids fractions. This turn-key plants will be sale by licencing agreement to large MSW plant managers (URBASER, URBAN, FCC, etc).
- **Production of intermediates from MSW such as ethanol, LA, SA (ENERGO).** ENERGO is a large chemical company interested to include in their portfolio new monomers and new sources of feedstock in their current second generation ethanol plant. ENERGO plans to study the adaptation of its ethanol plant to obtain new momomers (SA and LA) or to incorporate in a cascade approach the extraction and purification of rich MSW fractions such as proteins or lipids.
- **Additives to launch a new range of eco-cleaning products formulations such as ethyl lactate and surfactants**. . TBWR plans to either patent the formulations if novelty allows it or to keep them as trade secrets that could be licensed to HAYAT, to companies that produce coatings, ink and paints and other cleaning products companies out of the HAYAT portfolio products. HAYAT is a large company, which will use a combination of business models to introduce its new products in the market. The company expects to create a new range of eco-cleaning products with two main customers targets: i) selling directly to specialized retailers that look for products with low environmental impact and ii) Institutional customers thought Green Public Procurement[55] to be used in cleaning of schools, universities, conference centers, town halls, government buildings, etc. HAYAT does not exclude the possibility of joint ventures although this will be considered after establishing a strong market presence in other geographic regions by licencing the TBW patent. Production of new products will be considered within current assets or through toll manufacturing.
- **Lactic acid based hot-melt additives (SOFTER & AIMPLAS):** With a sales agreement with AIMPLAS, SOFTER aims to formulate hot melt adhesives using the lactic acid produced by ENERGO and the licencing companies of PERCAL biorefinery concept. The HMA will be sold directly by SOFTER to their customers in selected countries, using the current sales network of the company. Countries not covered by SOFTER will be reached by licencing the technology. Several compounders will be contacted in order to cover all the European market in a first step and USA market in a second step.
- **Polyester polyols for the polyurethane industry (COVESTRO):** based on the use of succinic acid to develop new polyols to be used in polyurethane formulations special designed for several industrial sectors. COVESTRO is a large company, which will use a combination of business models to introduce its new products in the market. The company expects moderate growth in these mature markets, strong growth in developing world therefore the exploitation focus will be on developing markets. Selling directly to customers and also utilisation of well-established existing distribution networks has proven a very effective business model in other lines of products and it is expected that these will be suitable for the polyurethane industry. COVESTRO does not exclude the possibility of joint ventures although this will be considered after establishing a strong market presence in other geographic regions. Production of new products will be considered within current assets or through toll manufacturing.

Business models and plans will be developed to support the commercialisation of each key project result and partner. The Business Model and Plan developed in WP9 will integrate IPR, innovation protection and exploitation strategies, and the accompanying techno-economic analysis and LCA developed in WP7. This will pave the way for the

[55] http://ec.europa.eu/environment/gpp/index_en.htm

successful exploitation of the project results. The expected impact, based on a conservative scenario of the value chains, is described in qualitative and quantitative terms in table 2.1.d. This results in an estimated benefit (an average 15% of turnover) of 86 M€ over the period 2025-2030 (5 years after the project end). It is important to highlight that the environmental benefits have not been included in these calculations.

Table 2.1.d. *Preliminary evaluation of PERCAL economic impact after 5 years project finish (when TRL8 is achieved).*

Potential applications	Expected price (€/kg)	World Market size (kt/y) /Market share[56]	Expected Turnover[57] (M€/y)	Direct Employ-ment[58]	IP management	Required investment
Lactic acid[59]	1.1	500 // 1%	5.5	16.0	Patented and develop a turnkey projects or licencing the technology to MSW managers or chemical companies.	MSW multiproduct (SA, LA, lipid/ protein based surfactant) biorefinery plant 10-15 M€/each.
Succinic acid	2.5	650// 1% (Needs of COVESTRO 80% conversion).	11.7	33.0		
Biosurfactants	3.5	24,000 // 0.02% (current HAYAT & other potential customer need)	16.8	48.0	IPR licencing or shared exploitation will be negotiated, depending on the target market and field.	
Ethyl lactate Solvents	1.8	4,500 // 0.15%(current HAYAT & other potential customer needs)	12.2	35.0	Development focus on a patent; licencing for commercial exploration to industrial up-takers	Reactive distillation ethyl lactate 1000t/a; 2,0 – 3,5 M€
Lactic acid HMA	2.5	760 // 1 % (current PERCAL's partners & other potential customer need)	19.0	54.0		Reactive extrusion HMA 350 kg/h, 0.5 M€
Polyester Polyols for PUR[60]	4.5	4,000 // 0.15% (current COVESTRO production capacity)	27.6	79.0	Own production and to sale in combination with di-isocianates and other chemicals.	Require moderate investment to adequate current chemical plant.

Additional market research will be carried out in WP8 to assess and quantify the needs of the market: existing competitors, target markets, customers' profiles etc., this analysis will confirm and validate market opportunities and exploitation routes. One of the issues of the envisaged fermentation processes is the fact that only a fraction of the carbon will be converted into the envisaged final products and a significant amount of microbial biomass (i.e. the microbes which are grown in the fermenters) will be produced which should also have some type of valorisation at the end of the process, e.g. used in anaerobic digestion for biogas production or others, energy cogeneration, etc.

2.1.4 Environmental and socially important impacts.

On 13 February 2012, the European Commission adopted a strategy for «Innovating for Sustainable Growth: A Bioeconomy for Europe»[61]. This strategy proposes a comprehensive approach to address the ecological, environmental, energy, food supply and natural resource challenges that Europe and the world is currently facing.

With the world population set to approach an estimated 9 billion by 2050, against a background of finite natural resources, Europe needs renewable biological resources -not just for securing healthy food and animal feedstuffs but also for materials to produce several bio-based products.

Environmental impact: will be reduced because new bio-based products will be marketed using environmentally friendly technologies from the cultivation to the final production, showing a positive CO_2 balance relative to existing fossil-fuel derived products or other bio-based sources. Table 2.1.e shows the estimated differences between current (fossil and Bio based) and PERCAL approaches in terms of Global Warming Potential (kg CO_2-Eq.), land use (m^2a) and Non-Renewable Energy Demand (MJeq.) per kilo of produced polymer or intermediates. Due to the use of waste by-products out of agricultural, preferential use of microbial synthesis, highly effective catalysts, low reaction temperatures, green chemistry routes, we expect a significant reduction in the three environmental parameters.

For example, the polyol commercialised today to produce polyurethanes is obtained through the catalytic

[56] Expected conservative market share (%) based on current worldwide consumption and production capacity of PERCAL partners. In brackets t/y of PERCAL products that will be produced for target market share.
[57] Only considers the turnover the amount of material that PERCAL industrial partners can produce and commercialize (HAYAT, COVESTRO, ENERGO, SOFTER & NOVO) or licensing (VISUM, TBW, IMECAL,).
[58] Job calculation 350 k€/job. Average turnover increase in the chemical industry to create a new employee.
[59] www.nnfcc.co.uk/publications/nnfcc-renewable-chemicals-factsheet-lactic-acid
[60] www.plasticsnews.com/article/20130715/NEWS/130719967/report-polyurethane-consumption-set-to-grow-4-5-percent-per-year
[61] //biobs.jrc.ec.europa.eu/sites/default/files/generated/files/policy/EU%20Bioeconomy%20Communication%20Strategy%20Brochure.pdf.

polymerization of petrol based intermediates. A few percentage of polyols can also be obtained from vegetable-based oils (triglycerides) such as castor oil or soybean oil. An LCA study reported that the production of bio-based polyols (soya-based) had successfully reduced 36% of the global warming emissions, 61% less non-renewable energy use and 23% less total energy demand compared to the petroleum-based polyols, and the most significant impact comes from the energy use for soya harvest.

Table 2.1e PERCAL expected impact on renewable source content and energy demand.

	Biobased or Fossil based materials[62].			Potential alternatives using PERCAL concept.					
	GWP (kg CO_2-Eq.)	Non-Renewable Energy (MJ/kg).	Land use (cm^2a/kg)	GWP (kg CO2-Eq/ % less).		Non-Renewable Energy (MJeq./kg/%less).		Land use (cm^2a/kg/% less)	
Fossil/Bio solvent	3.7/0.7	81.2/ No data	78/No data	0.5	>28%	28	>65%	20	>75%
EVA HMA	2.0	76.4	89	1.2	>40%	18	>75%	20	>78%
Fossil/Bio SA	1.9/0.8	No data/40	66/20,000	0.5	>30%	30	>25%	20	>70%
Fossil/Bio Polyol	3.7/1.3	88.2/4.97	95/4205	1.0	>25%	4.0	>20%	20	>80%
Fossil/Bio Surfactant	2.4/1.7	78.7/50.5	78/2500	1.2	>30%	35	>30%	20	>75%

The life cycle analysis of Myriant's bio-succinic acid process shows that the carbon dioxide emissions of the bio-based process are 94% lower than either the existing petroleum-based process to make succinic acid from maleic anhydride, or the process to make adipic acid from benzene[63]. The environmental impact of succinic was summarised in an EU study[64] based on LCA data for production through the fermentation of sugars using different raw materials and purification methods. However, bio-based products often show lower environmental performance for impact categories such as eutrophication or acidification. The higher impacts found for bio-based products in these categories are a consequence of the land use and agricultural activities (such as the use of fertilizers, pesticides and energy).

*In PERCAL, **higher GWP reduction will be expected** as the raw material is a true waste, currently without any valuable use, and does not require land for its production as opposed to biochemical products from wood or plants. Extraction, fermentation & purifications will represent the main environmental impacts in the selected products.*

Additional environmental and heath advantages are found by the use of bio-solvents instead solvents, lactic acid HMA versus EVA HMA and Biosurfactants versus surfactants as detailed in section 2.1.**2** in terms of low toxicity, full biodegradation and low or cero VOCs emissions.

Furthermore, the adoption of safe-by-design approaches will support EU competitiveness in this particular research and innovation field. Regulators can additionally use the well-defined predictive tools that underlie the concept of safe-by-design and thus improve the regulatory preparedness by timely interactions with innovators developing novel applications of polymers and intermediates coming from MSW and MRRM by-products. These interactions focus on sharing views and expertise on how best to deal with new safety aspects of innovative products for workers, consumers and the environment. A LCA of the key exploitable products will be undertaken in WP8 led by AUA, with the support of commercial software and the real data obtained at lab and pilot plant level.

Employment impact:

PERCAL aims at a contributing significantly in the large number of jobs that green chemistry will create in the coming years. We expect at least **245 qualified jobs directly** related to the sale of new bio-based products five years after the end of the project (when TRL8 will be achieved), based on a very conservative European market share as detailed in table 2.1.d. According the CEFIC, the sector generated an even greater number of indirect and induced jobs – up to three times higher than through direct employment, i.e., **PERCAL can create up to 750 indirect jobs.**

In order to remain competitive and maintain jobs in the light of major societal challenges and rising markets in the developing world, the European bioeconomy sectors need to innovate and further diversify. It is estimated that direct research funding associated to the Bioeconomy Strategy under Horizon 2020 could generate about 130,000 jobs and € 45 billion in value added in bioeconomy sectors by 2025.[65]

In 2014, according to Cefic[66], the European chemical industry produces 16.7% of the world's chemicals, employs 1.2 million workers and contributes €527 billion to the EU economy. While the value of chemical production grew about 45% from 1997 to 2017, direct employment in the EU chemical industry decreased by an average annual rate of 1.9% from 1996 to 2013. The number of employees in 2013 was 28% less than the 1997 level. i.e., a loss of more than 400,000 jobs. That works out at a 1% drop in jobs for every 1.6% increase in output of non-pharmaceutical

[62] a) Althaus H.-J. et al. (2007) Life Cycle Inventories of Chemicals. Final report ecoinvent data v2.0., 8. Swiss Centre for LCI, Empa - TSL.
b) Hischier R. (2007) Life Cycle Inventories of Packaging. Final report ecoinvent data v2.0., 11. Swiss Centre for LCI, Empa - TSL.
[63] www.myriant.com/pdf/myriant-polyol-technical-bulletin-2013.pdf
[64]//biobs.jrc.ec.europa.eu/documents/BISO%20EnvSust%20-%20Bioproducts%20-%20Succinic%20Acid_v3%20-%202016.pdf
[65] BECOTEPS (2011) The European Bioeconomy in 2030: Delivering Sustainable Growth by addressing the Grand Societal Challenges
[66]//fr.zone-secure.net/13451/106811/publication/contents/templates/Cefic_F_and_F_2014-Full_report_Blanc_150dpi.pdf

chemicals. Jobs have fallen, because sales have not kept up with increases in productivity that requires fewer workers.

Along with jobs created by gaining new customers and being on the leading edge of greener chemicals, additional jobs will be created by switching from chemical feedstocks to renewable raw materials. Renewable raw materials including wastes and residues will require local processing creating EU jobs and displacing imports of petrochemical raw materials. For example a US report states[67] that if around 20% of chemical production switched from making petrochemical-based products to plant-based products that would create/transfer 104,000 jobs.

2.1.5 Policy framework

PERCAL will contribute to reducing the EU's dependence on fossil fuels and imported biomass and contribute to meeting its energy and climate change policy targets for 2020. To achieve this, **PERCAL** contributes to the two essential elements identified by the **EU BioEconomy strategy**: a) by promoting the full valorisation of waste biomass for the production of resource efficient and sustainable bio-based products through biorefinery, enabling the substitution of fossil-based products and reducing biomass dependency in regions where the biomass availability is less plentiful[68] and b) by providing sound Dissemination and Business plans to support the commercialization of PERCAL bio-based products.

More generally, PERCAL fully aligns and supports the **EU Innovation Policy and the EU 2020 strategy**, as the outcome of PERCAL will have significant impact for the municipalities, fish processors, waste managers, biorefineries, downstream processors and bio-based product producers and all industrial application sectors with the potential of using bio-based products, contributing to the **EU Strategy for Key Enabling technologies**[69] by advancing the knowledge in biotechnologies in order to provide advanced materials. High economic impact will be achieved through increasing the competitive advantage of EU Industry by promoting the use of PERCAL bio-based products in high value-added applications. It will also consolidate the EU Industry and reduce the EU dependency on crude oil.

Regarding the **Lead Market Initiative (LMI) on bio-based products**, PERCAL will produce a number of high value-added bio-based products for several demanding sectors. All these products have been identified in the LMI Final Report[70] as being of high strategic impact and social interest. PERCAL technical approach, focused on sustainable and cost-competitive biotechnologies, and all the accompanying measures planned in the project to foster the market introduction of such bio-based products will have a major impact in the EU, protecting the environment and human health and strengthening industrial competitiveness. PERCAL will provide substantial elements that will contribute to the majority of the Priority Recommendations provided by the **LMI** Ad-hoc Advisory Group for Bio-based Products. In addition, the integration of the biotechnologies and methodologies developed in PERCAL into current industrial processes is fully in line with the Environmental Technology Action Plan.

Directive 2010/75/EU on industrial emissions[71] (IED): The IED sets out the main principles for the permitting and control of installations based on an integrated approach and the application of best available techniques (BAT). BAT is the most effective techniques to achieve a high level of environmental protection, taking into account the costs and benefits. On 7 January 2014, the IED repealed and replaced the following: **Directive 2008/1/EC** on integrated pollution prevention and control (IPPC), Directive 2000/76/EC on waste incineration and **Directive 1999/13/EC** on activities using organic solvents. New directives can support the production of bio based solvents, polyols and HMA.

The production of environmental friendly and cost competitive consumer products will be in line with the Green Public Procurement. As Europe's public authorities are major consumers, PERCAL's industry partners can offer the developed products for green procurement in a privileged position.

PERCAL will contribute to Circular Economy (Communication from the Commission to the European Parliament. Closing the loop - An EU action plan for the Circular Economy COM/2015/0614. Brussels, 2.12.2015.) creating new products related with secondary raw materials. Furthermore, PERCAL will promote the part of the 7thEAP (7th Community Environment Action Programme) which emphasizes the need for an EU strategy on management waste.

Finally, PERCAL aims to cover most of the key objectives of the **Strategic Innovation and Research Agenda (SIRA)** published in 2013 by the **Bio-Based Industries consortium**, related to the development of chemical products from biomass by-products, including complex biomass as MSW. PERCAL partners and support companies such as NOVO, EXERGY, ARKEMA and ENERGO are full members of BIC consortium and AIMPLAS and CMET are associated members. VISUM, COVESTRO and EXERGY are SPIRE members.

[67] www.greenbiz.com/news/2011/05/12/green-chemistry-holds-potential-reverse-job-losses
[68] http://www.crossborderbioenergy.eu/gis.html#6
[69] "A European strategy for Key Enabling Technologies – A bridge to growth and jobs ", COM(2012) 341 final.
[70] //ec.europa.eu/enterprise/policies/innovation/policy/lead-market-initiative/files/final-eval-lmi_en.pdf
[71] http://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32010L0075

2.1.6 Barriers.

The barriers which may affect the expected impacts of the project, and the proposed mitigation measures to remove these barriers, are presented in the table 2.1.f.

Table 2.1.f Main barriers identified and the proposed mitigation actions.

	BARRIER DESCRIPTION	MITIGATION ACTIONS
Technical	Technical risks inherent to an R&D project, 1) Developed bio-based products do not allow for the commercially competitive productivities, 2)High downstream process costs, 3) High variability of the raw materials, 4) Low quality polymers/additives and intermediate products in comparison with petrochemical based ones.	A risk and contingency plan has been outlined and risk monitoring strategies (see section 3.2) have been implemented in the work plan (see WP9, Task 9.2 Risk indicators definition and risk management).
Market	Initial Small Market Size for MSW based products: Low demand can lead to high unit costs and does not allow volumes to grow enough to lower the price (no economies of scale) and to be cost competitive. Current European policy and support for bio-based products and bioeconomy development is unclear.	Different market products will be proposed, with huge potential demand and medium cost (>1.8€/kg) such as adhesives, solvents, surfactants, polyurethanes and also chemical intermediates which are present in a huge industrial products. Through proactive communication PERCAL will provide information to policymakers on project results reinforcing the value of bioeconomy development and promote bio-based products consumption.
Social acceptance.	Imperfect flow of existing information about green chemistry safety: In ideal markets, buyers and sellers are fully informed on the features of the good or service under consideration; however, existing a lack of information about service life and durability of products coming from waste.	**PERCAL** will provide well documented results, dissemination materials and product datasheets including characterization and environmental data, etc., that will form a sound base to support high level development of regulations and green procurement policies. During the project extensive testing campaigns will be carried out to assess performance and long-term durability and realistic estimations will be provided by the main project players.
Industrial	There is a risk that the processes and technologies developed in PERCAL cannot be scaled up at acceptable costs and that no economies of scale are generated when industrializing the processes, due to, between others, limited demand, low yield or large investment requirements.	Transfer of PERCAL innovations to industrial partners during the project to test and validate the scalability of the processes. Economic projections of the full-scale production will be proposed in WP7, by the engineering companies.
Standards	Need to be fulfil the current standards. Since October 2011, CEN, CENELEC and ETSI are working towards the creation of a solid standardization framework for bio-based products, based on the outcomes of CEN/TR 16208[72]. The EC issued four standardisation mandates for bio-based products **ISO/TC 276** and **CEN/TC 411, 183 & 249/WG 17.** More information in section 4 (Annex 4.2).	**PERCAL** Consortium will follow closely the outcomes of the active WGs and whenever a project outcome is considered to be of relevance to the on-going work, **PERCAL** Partners will establish a liaison (HAYAT, COVESTRO & AIMPLAS is already participating in standardization) to present project results and if possible to contribute to the standards (e.g. CEN workshop agreements).

[72] www.cen.eu/cen/Sectors/Sectors/Biobased/Pages/documents.aspx

	BARRIER DESCRIPTION	MITIGATION ACTIONS
Regulation	Current legislation is concerned by the potential health, safety and environmental risks in relation to bio-based products obtained from biorefineries	**PERCAL** project will review the legislation applicable to this technology in its different areas: i) Regulations REACH and CPL (classification, labelling and packaging of substances and mixtures) and Legislative Decree EU 10/2011, ii) Directive 91/689 defining hazardous waste (update: 2008/98/EC), iii) Decree 52/97 defining hazardous substance: Assessment of the risk to the health and safety of workers from the presence of chemical agents, iii) Environmental impact assessment (EIA), strategic impact assessment (SIA), air and waste sectors, Legislative Decree 152/2006 and Decision 2000/532/EC, etc, to ensure compliance of the developed products, iv) VOCs regulations.

2.2 MEASURES TO MAXIMISE IMPACT.

Communication, dissemination and demonstration activities are critical for the successful exploitation of PERCAL research and innovation findings and accordingly a whole work package will be dedicated to these actions (WP8). Communication, dissemination and exploitation are linked activities and must be undertaken in a coordinated way, thus exploiting synergies and avoiding overlap between them; in PERCAL they will be coordinated by a **Dissemination and Exploitation Board** (**IPREB**, see section 3.2. for a detailed description). Activities and their impacts will be **continuously monitored** during the project and the resulting actions will be reported in all Contractual progress reports. Actions will be reported with subsequent plans for the next project period in the interim & final **Plan for Exploitation of the Results and Dissemination (PERD)**, due in months 18, 36 & 48.

PERCAL will implement a range of activities to ensure the optimal visibility of the project and its results, increasing the likelihood of market uptake of the knowledge it produces, and ensuring a smooth handling of the individual intellectual property rights of the involved partners to pave the way to knowledge transfer and exploitation. Table 2.2.a summarizes the focus of activities and how the dissemination and exploitation plan fit together to produce the overall PERD.

Table 2.2.a. Plan for exploitation of the results and dissemination (PERD).

Communication activities	Purpose – Raise awareness of project aims and outputs amongst broad stakeholder base. • Communication KPIs to measure effectiveness and efficiency. • Messaging and visual identity of the project and Stakeholder mapping. • Communication channels (website, social media, printed materials, scientific & trade journals)
Dissemination plan	Purpose – Support project exploitation through technical dissemination and industrial outreach. • Dissemination strategy: objectives, targets, activities and channels. • Dissemination KPIs to measure the dissemination effectiveness and efficiency. • Management of the three dissemination activities: – packaging knowledge for an effective take up, – reaching the selected early adopters to motivate for changes, – preparing the effective exploitation of the project results. • Dissemination administration (approval, reporting, deliverable). • Scientific and trade publications.
Exploitation plan	• IPR strategy and Exploitation plan for the project results • Business models and plans for key project results

2.2.1 Communication activities

Communication activities will use a range of communication channels to spread new knowledge and results generated by the project. They aim at reaching the dissemination objectives and reinforcing the plausibility of the exploitation plan. In order to carry out the project outreach in a comprehensive way, a detailed communication plan will be implemented at the beginning of the project (month 6). It will take the form of a full-fledged communication campaign to be designed and rolled out in parallel with the other project activities. It will aim to maximize the impact and visibility of the project among all relevant stakeholders in the fermentation, coatings and chemical manufacturing sectors at European level.

The communication leader will coordinate the upcoming activities and follow-up on the implementation of the campaign. Any successful campaign is based on a thorough, well-structured strategy. This entails the:

- Identification of the target audiences that should be reached (WHO).
- Tailoring of the messages towards these target audiences (WHAT).
- Decision on the best communication channels and tools to reach the target audiences (HOW).
- Development of a detailed implementation plan (WHEN).

The clear **categorisation of target groups** allows dedicated and targeted dissemination actions to be undertaken through specific communication channels and using targeted messaging.

Our communication plan encompasses a comprehensive series of actions (Table 2.2.b) to **broadcast the non-confidential results that are generated during the project to as wide and relevant an audience as possible** in order to heighten awareness about the results, to contribute to knowledge building in Europe, and to ensure that as many actors and citizens as possible have access to these results. This includes the broader European chemical, food and plastic industry, policy makers, researchers and the academic community, training institutes, early stage researchers, mass media (TV, radio and the written press) and multipliers for addressing the general public, as well as other stakeholders, such as other H2020, EC and national R&D initiatives, ESIF stakeholders, etc. In keeping with the general profile of each target audience, the most appropriate communication tools and channels will be selected for targeting each stakeholder group.

Table 2.2.b Outline of the PERCAL Communication Plan**.**

Target Audience	Tools & Channels	Expected Impacts
MSW plant management. Cleaning products manufacturer & their customers. Producers of HMA and their customers (cardboard, labels, ..). Producers of polyurethanes and their customers (car, packaging, coating, insulation, …) Environmental consultancies & engineering Machinery Retailers and consumers Industry Multipliers- Sectorial Associations (e.g. EuPC, EuropaBio, CEFIC, European Bioplastics, etc.) Standardisation bodies and policy stakeholders, national food safety authorities, EFSA, EC, national governmental bodies, ICC, AACCI, CEN, ISO, WHO, FAO, CODEX, AENOR, General public and Mass media Research and academic stakeholders Early stage researchers Government & policy makers Innovation programs Educational stakeholders and students	Project corporate identity tools (logo, tag line/slogan) Project Website, Project Leaflets, Project Posters, Powerpoint Presentations, Press releases Video clip (for YouTube, project website,...) ***Trade fairs/exhibitions***: PLAST (It), ACHEMA, Interpack, IFAT (Ge), Hispack, Smagua (Es), Eurocoat, Europrope (Fr), . ***Conferences, seminars and workshops*** (participation in organising, scientific committees, with poster, paper, presentation, or chairing): European Bioplastics, Biopolpack, Biopol, EFIB, etc. ***Mass media & Magazines***: Newspapers, local TV, Packaging today; Plastic additives, bioplastics magazine, etc. **Open Access Journal Articles:** J. Environ. Monitoring and Assessment; Green Chemistry J., J. Applied Polymer Sci., Bioresource Tech., etc. ***Involvement in networks/clusters***: Green chemistry, European Bioplastics, EuropaBio, Suschem, Municipal Waste Europe, European Federation of Waste Management and Environmental Services, Related Trade Associations, etc. ***Features in Newsletters and e-bulletins*** ***Visibility on third party web sites*** e.g. LinkedIn, Slideshare,, Pinterest etc ***Inclusion in EU dissemination resources*** such as the EU Horizon Magazine and the EIP Agri Website //ec.europe.eu/eip/agriculture	Generation of awareness about PERCAL and its mission to ensuring sustainable use of selected bio-waste in Europe and globally. Dissemination of new knowledge and technologies for transitioning polymer related industries to the green economy. Effective knowledge sharing among the research and scientific communities, which will contribute to the ERA, and to the EU recovery package and the EU 2020 strategy for smart, sustainable and inclusive growth. Sound communication towards policy-makers to increase the influence of the project and its results in policymaking (which will be reinforced by the presence of our expert Associations). Communication to retailers and consumers about the efforts being made by the MSW and Fish sector and the target products sectors to increase their sustainability for consumer and environmental protection. The general public should also feel they are getting value for money from public funded research that will ultimately benefit them as citizens. Attract more young people to scientific and research careers, enriching their curricula with knowledge from the project. Early stage researchers inspired by the results, will be motivated to remain

Target Audience	Tools & Channels	Expected Impacts
Other cross-cutting H2020 projects and international/ national projects	and EIP raw material //ec.europa.eu/eip-raw-materials/	in EU- research and possibly work in areas that are beneficial for society, sustainability and innovation.
PPPs and EIPs (e.g. EIP Raw Materials, BIOTECH, SPIRE, BBI, Suschem, etc.)	Social media: twitter, facebook, utube, LinkedIn, Google+ etc…	
Potential technology and bioproducts customers	Programme meetings	
	Policy roundtables	

2.2.2 Dissemination

The main communication activities are **focused on the dissemination of technical information to industry**. The goal is to disseminate **PERCAL** objectives and results among the interested industry sectors, demonstrating the added value, innovation and new technologies developed within the project. Due to the relevance of this target audience, the position of the project consortium within the industry is very important.

The goal of our communication activities will be to disseminate technical information both to the industry and the scientific community. Project results will be disseminated through participation in international conferences and trade fairs, and through publications in peer reviewed scientific journals and trade publications in the area of biochemistry, polymer processing, bioplastics, etc. Publication in **Scientific Conferences** related to the **PERCAL** R&D areas will be also issued. Table 2.2.c details the number of publications and the journals where the consortium plans to contribute. The project will disseminate the results to business stakeholders to make them aware of the expected impact of the project and promote the exploitation of its results. Presentations shall be made by the partners at business seminars and trade fairs. Additionally, each of the partners will disseminate **PERCAL** results in the fairs and exhibition they participate.

Table 2.2.c Activities related to participation in international events and publications in technical journals.

PERCAL's partner	Conferences & seminars.	Publications in high impact journals
IMECAL/ ENERGO	Five contributions as speaker or poster in conferences such as ISWA[73], Int. Petro chemistry & Chemical Engineering, Unconventional Resources Techn. and BioSpain.	Three publications in the field of waste treatment and scientific open access journal (such as Bioresource Tech., Environmental sci. & tech., Biomass and Bioenergy & Waste management & research)
AIMPLAS/ SOFTER/ COVESTRO	Four contributions as speaker or poster in conferences related to bioplastics, bioresins and adhesives. Yearly conference of CliB (Cluster Industrial Biotechnologies).	Two publications in adhesive and polyols related journals (Adhesion Sci. & Tech., Int. Adhesion & Adhesives, …)
AUA/CMET/ ATB	Six contributions as speaker or poster in European & international conferences related to monomers production from biomass[74].	Five in e.g., Applied Microbiology and Biotechnology, Biochemical Engineering Journal, Bioresource Technology, J. Biotechnology, J. Cleaner Production etc.
TBWR/ HAYAT	Two contributions in conferences related to Biorefining, green solvents or chemical processing as speaker	Two reviewed papers in Journal as Chemical Engineering Research and Design, J. Chemical Engineering Journal, J. Green Chemistry or alternatively open access Journals of similar focus.
VISUM	Participation in trade fairs and conferences such as EuroPACT 2017	One in a journal such as Chemometrics and Intelligent Laboratory Systems or "Chemical Engineering Science"
EXERGY	Participation in two conferences to promote the project findings and engage in outreach activities	None envisioned

[73] **ISAWA** International Solid Waste Association. **IFAT** World's Leading Trade Fair for Water, Sewage, Waste and Raw Materials Management
[74] World Congress on Industrial Biotechnology and Bioprocessing; International Conference on Environmental, Industrial and Applied Microbiology, BioMicroWorld, European Biomass Conference and Exhibition; European Forum for Industrial Biotechnology, EFIB; International Conference on Renewable Resources and Biorefineries, RRB or Symposium on Biotechnology for Fuels and Chemicals.

The project and its derived products will be presented in international conferences in order to gain attention from the industry and to contact target clients. In this last regard, the final products will be presented in conferences of specific professional sectors such as the petrochemical or consumer goods. When the results are subject to be published with no detriment to the economic side of the project, we will choose among the specialized technical peer-reviewed journals. However, dissemination of results will start with discussion of results between partners as well as within the research organization in order to determine whether confidential information could be disclosed or protection of intellectual property should be initiated. If no IPR conflict is found, we are aiming first at presentation of results at local and international conferences, which will be an important learning experience for young researchers. Being involved in writing the papers will prepare students and young researchers for their further career in academic or industrial research. We want to emphasize here that also if **part of the results will be protected by patents, we still intend to publish after filing of the patent if in agreement with the commercialization partner**.

Open Access will be provided to peer-reviewed papers published by PERCAL's Partners as table 2.2.c indicates, either in an open access journal repository ("gold" model) or for instance using the OpenAire orphan repository. Depending on the results published, the selection of the best suited and higher impact repositories will be made on a case by case basis in cooperation with **OpenAIR** provided that the IPREB does not identify issues conflicting with the industrial exploitation of project developments. Technical press platforms, generic or specific, will be continuously used, as well as the own partners communication channels.

Networking circles: In addition to the direct commercial contacts of PERCAL industrial partners, wider dissemination would be achieved through using the partner memberships of trade associations related to the aims of the project. Working with trade associations the Consortium would increase the potential for exploitation of project results and additionally create a mutual opportunity with the associations to promote green and sustainable chemistry within the relevant industrial sectors. Moreover, the know-how transfer of **PERCAL** to the European industry could be also provided by all PERCAL partners via existing contacts to other European research institutes and networks, such as European Bioplastic, BBI, CEFIC, SPIRE, ECP4 and EuPC.

Training and knowledge transfer: PERCAL will work to enhance the training of designers, plastics application technicians and education of young polymer engineers. In particular, the ***Guide to explore the conversion routes of municipal solid waste (MSW) to obtain high added value products*** (Task 8.6 and D8.8) delivered by the project will be a very useful tool to achieve this goal. This guideline will be published by the consortium and will include new important design rules for the various application areas, including the case studies of the demonstrator parts with all the necessary information for the adaptation of results to new product developments. As the guide will be published at the end of PERCAL, technical and economical evaluations for the developed prototypes will be included. This guideline will be a crucial tool for the potential end-users of PERCAL materials and technologies. PERCAL aims to Organise 2 workshops (anticipated at month 30 and month 36). The topic will be selected a year ahead of the workshop, depending on the results obtained in PERCAL across technical, economic and environmental areas. PERCAL will showcase the demonstrators developed during the project and to this end the Consortium will analyse the feasibility to organise a workshop in the context of a major fair/exhibition in order to maximise the impact.

2.2.3 Exploitation of results.

The exploitation routes and corresponding business models will be investigated in WP8. The route for commercialization of the results can be logically identified from the **PERCAL** WP structure and work plan, from the logical flow of the project activities and their clear role, as represented schematically in **figure 2.1.a**.

Exploitation expert EXERGY will act as coordinator of the exploitation activities and ensure that all IP-owning partners as well as potential and confirmed exploitation partners will work towards a successful technology and knowledge transfer to ensure efficient market uptake. The role of the exploitation manager will be to assess the commercial potential of the results and liaise with industrial companies to maximize their commercial use. The exploitation manager will follow the progress made in PERCAL and will develop full exploitation plans as the unique values of the project results are confirmed. The exploitation manager, with support of the members of the IPREB, composed of representatives of all partners, and will also be responsible for establishing targeted exploitation and dissemination strategies, when the project outcomes reach sufficient maturity. The entire project Consortium will discuss this strategy and the corresponding document will be delivered for the mid-term assessment (D8.3-M18 and 36). An updated and more detailed exploitation plan will be submitted thereafter (D8.6-M36).

In terms of further contribution to maximize the impact of this BBI action, significant **in-kind contribution** is planned as part of the industrial context of the action and will definitely contribute to increasing the impact and reaching the project objectives. *Co-funding by partners' in-kind contribution, the consortium covers close to 26% of the overall project budget.* This co-funding is justified by the real conviction from the partners (both RTDs and private companies in terms of new projects/services/turn-key plants) that they will on the long term obtain a return on their investment when the PERCAL's solutions reach the market or indeed to further foster its development. In

terms of **in-kind and own contribution**, most partners have planned a significant amount that is either calculated on a flat rate of their full budget. Furthermore, it is noteworthy that three LARGE companies are further contributing covering their full costs by themselves.

All technologies developed by the partners will be considered for exploitation, be it methods, processes or materials. A summary of anticipated ***Key exploitable results (KER)*** is reported in table 2.2.d. Exploitation issues will be brought to the attention of all partners at an early stage in the project ensuring that issues regarding market and commercialisation, identified during exploitation analysis, can be fed into the R&D actions of the project.

Table 2.2.d Summary of the key exploitable results.

WP	Key Exploitable Results (KER)	Partner Interested In Exploitation
1	New enzymatic cocktails for improved hydrolysis of carbohydrates and proteins presents in the biogenic fraction of MSW, in order to enhance the yield of SA and LA production by fermentative processes.	IMECAL/ NOVO/
2	Technologies for fractionation of MSW by products to revalorise protein and lipid fractions	IMECAL/ CENER
3	To develop the utilisation and upgrading of residues and process waste streams and their purification and upgrading of lactic acid (LA).	ATB/ IMECAL/ ENERGO
4	Membrane electrolysis for succinic extraction	CMET/ IMECAL/ ENERGO
5	Integrated continuous production and separation of succinic acid	AUA/ IMECAL/ ENERGO
6	Design novel and environmentally friendly processes for the production of biosurfactants from amino acids and lipids from MSW fermentation by-products.	IMECAL
7	Development a new lactic acid HMA modified with maleic anhydride as the same properties than current EVA HMA.	AIMPLAS/ SOFTER
8	Process for of ethyl lactate by reactive distillation and production of bio-solvents New range of printing inks and cleaning products free of VOCs and lower environmental impact.	TBWR/ HAYAT
9	Succinic acid as building block for polyols, resins and polyurethane dispersions (PUD).	COVESTRO
10	Plant simulation, sensors control and monitoring.	EXERGY/ VISUM/ IMECAL

The first phase of exploitation activities will aim to raise awareness among the project partners on exploitation issues In the first 6 months, EXERGY will take the lead in the development of an initial exploitation plan including a discussion on what each participant is hoping to gain from the project, and whether the expectations of all participants are compatible and coherent. This plan will be updated annually for the rest of the project duration. While the first exploitation plan is expected to be general and cover all activities in the project as a whole, the subsequent plans will focus more strictly of each of the identified exploitable results of the project.

The methodology to reach consensus on exploitation issues and to draw up a final exploitation plan is through **Exploitation Workshops**, described in detail in WP9. The exploitation strategy will answer crucial questions as:

- WHAT: clarification of exploitation results and development of exploitation routes.
- WHO: identification of lead partners to achieve each exploitable result and which project partners will exploit each exploitable result.
- HOW: which form(s) will the exploitation of these results take (direct industrial use, patenting, technology transfer, publications, etc.).
- BARRIERS: i.e. identification of (overlapping) IPR and commercialisation conditions - identification of risks actions to mitigate them; existing or emerging alternative solutions in competition with results; availability of knowledge and resources among participants and willingness to contribute to the result.
- IP issues and ownership of knowledge.

An exploitation plan will be prepared for each result and for the project as a whole and it will outline: i)

- Identification of all exploitable results & intellectual Property Rights (IPR) protection strategy.
- Analysis and evaluation of the exploitation potential of each result (market).
- Roadmap of exploitation activities during and after the project, exploitation routes.
- Technology implementation plan of the project results & the expected socio-economic impact.
- Factors influencing the exploitation (regulations, standards, etc.).

- Business plan for the key results. (cost-benefit feasibility, potential investors, transfer to other sectors)

Exploitation tracking: The use of Key Performance indicators will allow the consortium to monitor progress towards achieving the projects exploitation objectives for the target products. Table 2.2.e summarizes how the project will contribute to the KPI defined by the EC for Societal Challenges under H2020. More information about the contribution of each partner to the KPI (publications and conferences) see table 2.2.c.

Table 2.2.e PERCAL expected key Performance indicators (KPI) related to Societal Challenges.

PERCAL's partner	Patents and IPR protection	Products/technologies to develop.
IMECAL	Trade secret, new enzymatic cocktails to improve MSW hydroxylase.	Enzymatic cocktails for MSW pre-treatment.
ATB/ IMECAL	1 Patent to obtain LA from MSW hydrolyses.	Lactic acid production integrated in PERSEO platform.
CMET/AUA	1 patent to produce SA using the integrated system of fermentation coupled with membrane electrolysis extraction	Succinic acid production integrated in PERSEO platform
IMECAL	1 patent, Production of biosurfactants from the by-products of MSW fermentation	Biosurfactants for non-food related applications based on modified proteins and lipids.
AIMPLAS/ SOFTER	1 Patent, Production of hot-melts from lactic acid and other co-monomers by reactive extrusion	Hot-melts for cardboard applications.
COVESTRO	Use their own technology and patents to produce chemicals from bio-SA	Building block for polyols, resins and polyurethane dispersions (PUD) using bio-SA as a main raw material.
TBWR	1 Patent, Ethyl Lacate by reactive distillation	Production of biobased solvents for inks printing and cleaning household goods.
VISUM/ IMECAL	1 Patent or utility model, Global biorefinery configuration.	Process monitoring system to produce Bioethanol, LA and SA from MSW.

Intellectual Property Rights protection activities are managed under WP8. The overall IPR strategy of the project is to ensure that partners are able to fully exploit their market position, involving patent protection.

The Partners will sign **a Consortium Agreement** (CA) prior to the signature of the Grant Agreement with the BBI JU in which all the rules for ensuring confidentiality (procedure for granting publication of the results), ownership regime and IPR issues (background included and excluded, ownership, licensing terms, etc) are detailed. Some previous knowledge and background in detailed in the description of each partner in section 4. The CA will be based on the DESCA model. The CA will include the initial 'Key Exploitable Results' (Table 2.2.d) of the partners and patents interest list (Table 2.2.b), which will be the starting point for IPR protection and Exploitation Plans.

The following three areas of activity have been identified in order to ensure proper IP management and maximize the impact and market relevance of the knowledge generated in the project:

- Proper assessment of the pre-existing knowledge of the various project partners, their potential contribution to the project results IP, and potential overlap of IP. In shaping the IP strategy of the consortium these will be assessed based on the Consortium Agreement.
- The assessment report may be used to draft an IP and exploitation agreement, if needed, as an integral part of the final exploitation plan, in line with the consortium agreement, and used as an input for the exploitation plans. The document will be upgraded during the project to include new results unforeseen at the beginning of the developments as it is needed.
- The overall IPR strategy annually focuses on the knowledge generated by the project, the IP rights, the patentability and optimal IPR protection options. Any disputes over IPRs should be settled according to the consortium agreement. Legal requirements for commercialization are also explored, in particular through "freedom to operate" analysis as well as individual patent filing strategies and drafting of patent applications for the most promising project results, as it is needed.

As described in section 3.2, the SMEs partners will have a higher percentage of participation in the results exploitation, so the IPR management will respect at any time the objectives established in this type of Collaborative Project and thus watching over SMEs interests.

Exploitation through Standards development: Successful exploitation will only occur if the market is receptive to the products and processes developed in the project. Standards are a critical part of a functioning market place, standards for bio-based products are under development at CEN. After identification of the results with standardisation potential (initially focused on exploring the conversion routes of municipal solid waste (**MSW**), to

obtain **high added value products**), contacts will be made with relevant standardisation bodies in order to initiate, for instance a CEN workshop or contribute to existing standardisation activities in ISO. In the case of contributing to CEN, **PERCAL** will propose to CEN Workshop Agreement (CWA), a standardization document developed in a CEN Workshop (www.cen.eu/). All experts interested in developing the document will be welcome in the development of such CWA mainly for stakeholder members. The CWA document will likely be a technical specification or a technical report, but could also become a broader guidelines document that includes information like guidance for continuous-reinforced plastic materials, best practice and information on material performances and production processes, etc. Some **PERCAL** partners have a strong link with standardisation bodies such as AIMPLAS, COVESTRO and HAYAT.

Post project execution of exploitation plan: As discussed a central aim of the PERCAL dissemination plan is **to 'prepare' the market in order to reduce the incubation time for PERCAL new products and technologies to a maximum of 5 years** and to provide knowledge, methodologies and training materials that can contribute to increase competitiveness of the EU industry and workers skills in the field**,** facilitating the widespread adoption and market penetration of the developed materials and technologies.

Table 2.2.f summarises potential exploitation routes for the main key exploitable results. At the end of the project PERCAL technology will be at TRL5 and it is expected that a further 5 years will be required to achieve TRL8. Significant investment to move from TRL5 to TRL 8 is needed, i.e., additional budget outside of the project must be attracted using the patents and the trade secrets generated during the project and detailed in table 2.2.e.

Table 2.2.f Main target customers and knowledge licencing and sales strategy.

Product to be developed in PERCAL (and IPR owner)	***IPR licencing and sales strategy.***	***Main target Customers***
1. Cost-competitive technologies for the integral valorisation of MSW components		
Process monitoring system (VISUM & IMECAL).	Turkey projects, Licensing, joint venture, commercial agreements with companies active in treatment/separation/downstream processes.	Companies interested in the valorisation of components of MSW such Large MSW management companies (Urbaser, Urban, Engie, FCC, Solum, etc), local authorities, etc.
Global biorefinery configuration (IMECAL)		
Proteins and lipids from MSW (IMECAL).		
2. Production of intermediated products by microbial fermentation routes.		
Ethanol, Lactic acid and Succinic acid (IMECAL, NOVO, ENERGO)	ENERGO own production. Licensing agreements with large MSW managers for in-situ build of Turnkey plants to produce a specific product (SA, LA or Ethanol)	Current ethanol, SA and LA users for non-food applications such as detergent/soaps (HAYAT, UNILEVER …) or polyols (COVESTRO, DOW Chemical, …), . MSW managers and Bioethanol as building block (GI Dynamics, TOTAL, Clariant, etc...)
Modified proteins and lipids to produce high added value products (IMECAL).	Licencing or production agreements with companies specialized in biomass separation and chemical treatment of proteins and lipids.	Large chemical companies such as ARKEMA that are looking for applications of amino-acids and fatty acids in production of green chemical intermediates for polyamides and polyesters.
3. Production of additives and products by green chemistry routes.		
Lactic acid HMA (SOFTER)	SOFTER own production. Licencing or production agreements with HMA production companies.	Manufacturers of paper or plastic labels, cardboard packages, baby diapers, feminine hygiene and adult continence products. Companies such as Ashland, Bostik, Dow Corning, Henkel and Sika.
Ethyl lactate (eco-solvent) (TBW & HAYAT).	Licencing (or selling) of IPR to industrial users. In case no IPR is granted use knowledge to define a process design package in cooperation with an engineering company to implement & commercialize process.	Industrial partners within PERSEO process to integrate ethyl lactate production with bioethanol distillation Current producers of ethyl lactate and bulk chemical producers who are interested to implement the new production process such as Corbion, Galactic, ADM, Musashino Chemical Laboratory, Vertec BioSolvents, etc.., Producers of lactic acid or ethanol to further upgrade their product to generate a sustainable solvent.

Polyester polyols (COVESTRO)	COVESTRO own production. Licencing or production agreements with other chemical companies.	Producers of polyurethane parts or coatings for building, automotive, shoes, technical applications, etc. Manufacturer of polyurethane specialities or overseas polyurethane companies.
Biosurfactants (IMECAL)	Licencing or production agreements with surfactants companies.	Surfactant producers such BASF Cognis Henkel AG & Co. KGaA, Soliance, etc…

The partners have resources to cover the costs needed to implement the necessary co-funding for further developing and marketing the results of PERCAL. Besides internal resources in order to support the commercial exploitation of the project's results; EUREKA, national programs, investment funds (EU Investment Bank. EU Investment Fund), private investors and venture capital, SME instrument will be considered, as well as opportunities brought through Regional Funds (ESIF), which should be possible as PERCAL aligns perfectly to the priorities identified under the RIS3 strategy for most of the Partner's countries. To ensure that the maximum potential EU market is reached, the **IPREB** on behalf of the Consortium, will contact and involve Industrial Associations or SMEs in the new supply chain of PERCAL across EU thus achieving a multiplying effect and spreading the project results.

In summary, the protection of the interest of Industry, especially the SMEs, will be guaranteed as they are the majority represented in the various boards, and consequently an industry-driven vote will be guaranteed in all levels of the management structure. The existing **consortium represents the complete research, development, manufacturing and material supply chain**, thereby **ensuring that the IPR protection procedure to be outlined within the Consortium Agreement** will be effectively managed and that the project results will be protected at each stage. PERCAL will result in a number of innovative processes, materials, products and IPR with a high potential for commercialization. Therefore this consortium considers the issues of IPR management and protection as particularly important and the PEDR will be prepared in order **to reduce the time to market** to the minimum possible; although further developments will be needed after the project end to achieve TRL8. There are no known business agreements or commitments which may impose limitations on the exploitation.

The combined implementation of the proposed Communication and Dissemination Plan and all the IPR protection and Management procedures **will help to achieve the expected impact of the project** by facilitating the maximum market introduction of the project results. Additionally, it will ensure a significant impact on the general knowledge and skills of the related EU industry and scientific community through Open Access publications and contributions to standards, when possible. Consumers and general public will also be made aware of the productivity and environmental benefits brought by **PERCAL** technologies, fostering the acceptance of research on materials technologies, processing and final products, as a worthwhile activity with direct impact on the EU socio-economic & environment status.

3. IMPLEMENTATION

3.1 WORK PLAN - WORK PACKAGES, DELIVERABLES AND MILESTONES.

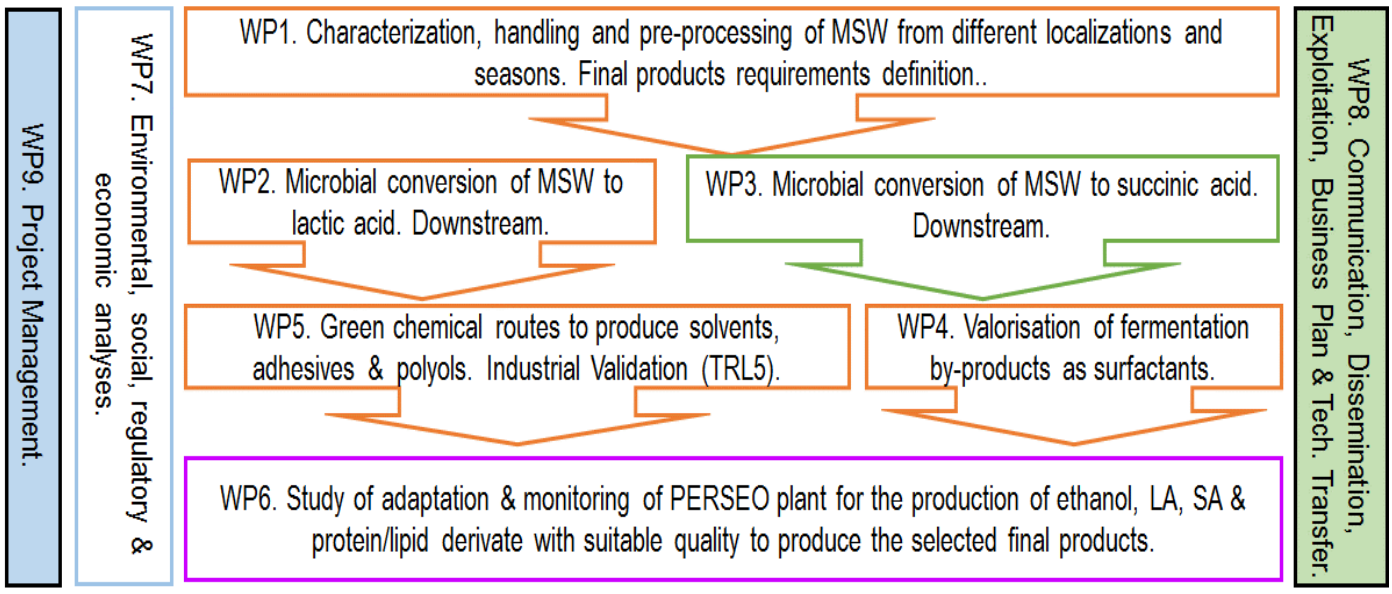
3.1.1. Overall structure of the work plan

This covers all the planned research activities from final products requirements and MSW characterization, selection and pre-treatment (WP1), optimization to obtain the most useful intermediates for the selected final products (lactic acid and succinic acid monomers in WP2 and 3 and surfactants from the protein/lipid fraction in WP4). WP5 aims to produce adhesives, solvents and polyols from WP2-3 monomers. WP6 will study the PERSEO plant adaptation and monitoring for the new proposed microbiological routes. The environmental and economic viability of the new products will be evaluated in WP8. Furthermore, WP9 activities are related to communication, dissemination, exploitation, business plan and technology transfer activities. WP10 will provide the full management of the project in order to make all the proposed developments in line with defined PERCAL objectives.

WP1: Final products requirements definition. Characterization, handling and pre-processing of MSW from different localizations and seasons. **WP Leader:** IMECAL
Definition of the main requirements of selected final products and intermediates. Selection of three representative MSW composition. Characterise and develop pre-processing methods ensuring availability and compatibility of the selected raw materials and their fractions for further conversions in industrial products (WP2 to WP4).
WP2: Microbial conversions to lactic acid. Downstream. **WP Leader:** ATB
Develop robust microbial strains for production of lactic acid, by fermentation of pre-treated MSW components. LA separation will be achieved by a membrane electro-dialysis based process. Fermentations will be optimized at laboratory scale. Downstream processing (DSP) for product extraction and purification will be developed.

WP3: Microbial conversions to succinic acid. Downstream. **WP Leader:** CMET
Select the most robust microbial strains for production of succinic acid, by fermentation of pre-treated MSW components integrated with membrane electrolysis for continuous removal of organic acids produced during fermentation. Fermentations will be optimized at laboratory scale. Downstream processing (DSP) for product extraction and purification will be developed.
WP4: Valorisation of fermentation by-products as surfactants. **WP Leader:** CENER
Develop novel biosurfactants by employing the lipid and protein fraction remaining after the fermentation processes to ethanol, LA or SA, of pre-treated and hydrolysed MSW.
WP5: Green chemical routes to produce solvents, hot melt adhesives and polyols and their further industrial validation (TRL5). **WP Leader:** COVESTRO.
Production and industrial validation (TRL5) of: 1) Ethyl lactate by reactive distillation from LA and ethanol, 2)HMA by reactive extrusion from LA, and 3) Polyester Polyols by conversion of SA with diamines. Polyester Polyols will be validated by PU formation using Di-isocyanates.
WP6: Study of adaptation and monitoring of PERSEO plant for the production of ethanol, LA, SA, protein/lipid derivate with suitable quality to produce the selected final products. **WP Leader:** VISUM
Study the adaptability of a versatile biotechnological plant to obtain different chemical intermediates and to determine the most relevant parameters to control and monitor in the PERSEO's process at industrial level.
WP7: Environmental, regulatory & economic analyses. **WP Leader:** AUA
Assess the environmental and economic impact of the full process proposed in PERCAL throughout each stage of development, to ensure compliance with legislation, to maximise market acceptance and facilitate increasing optimisation of these processes and products.
WP8: Communication, Dissemination, Exploitation, Business plan &Tech. Transfer. **WP Leader:** EXERGY
Complete IPR strategy, dissemination and technology transfer activities to ensure the full exploitation of the innovations and scientific results will be carried out. The active involvement of companies in the project, representing the different sectors active in the value chain will ensure wide exploitation with realistic objectives and plans. Communication, Dissemination and Exploitation Plans will be developed, including the PEDR.
WP9: **Project Management.** **WP Leader** IMECAL
Project activities of coordination, efficient grant management (resources optimization), communication with the EC, risk management

3.1.2. Graphical presentation of the components showing how they inter-relate (Pert chart).



3.1.3. Timing of the different work packages and their components (Gantt chart).

Num.	Work Packages and Related Tasks	Partners Num.	Delivera-bles	Project Months																	
				2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36
WP1	**Final Requirements. MSW Feedstocks.**	**All (1)**	**D1.1-D1.5**						H												
T1.1	Final products & intermediates requirements.	1,2,9,(10), 11-12	D1.1			♦															
T1.2	Study of variability of organic MSW.	All(1)	D1.2				♦														
T1.3	Characterization of MSW samples.	1,3,4,(6)																			
T1.4	Pre-treatment and enzymatic hydrolysis improvement from PERSEO process	(1),6	D1.3-1.4						♦	♦											
T1.5	Validation of fermentation processes.	1,3,4(6)	D1.5									♦									
WP2	**Lactic acid from MSW. Downstream.**	**1,(3),4,7**	**D2.1-D2.3**									H		H							
T2.1.	Screening & selection of bacterial strains.	(3)																			
T2.2	Fermentation processes into lactic acid.	(3),4	D2.1									♦									
T2.3	Downstream processing for recovery of LA	(3),7	D2.2													♦					
T2.4	Up-scaling of the entire process	1,(3)	D2.3																♦		
WP3	**Succinic acid from MSW. Downstream.**		**D3.1-D3.4**															H	H		
T3.1	Evaluation of MSW as feedstock.	(4),6	D3.1									♦									
T3.2	Optimisation of integrated bioprocess for SA production and in situ separation.	4,(5)	D3.2-3.3												♦						
T3.3	Succinic acid purification.	(4),5	D3.4														♦				
T3.4	Process scale-up.	4,(5)																	♦		
T3.5	Optimisation of bioprocess based on electro-dialysis downstream separation system.	4,(5)																			
WP4	**Valorisation of fermentation by-products**		**D4.1-D4.4**																H	H	
T4.1	Characterization of the fermentation residue of the ethanol/LA and SA processes.	(1),6	D4.1									♦									
T4.2	Hydrolysis of protein fraction	1,(6)	D4.2													♦					
T4.3	Extraction & hydrolysis of the lipid fraction	1,(6)	D4.3													♦					
T4.4	Synthesis of the amino-lipid biosurfactants.	1,(6)																			
T4.5	Biosurfactants in relevant ind. applications.	(10)	D4.4																	♦	
WP5	**Green chemical routes to produce HMA, solvent & polyols & Industrial Validation.**	**2,7,9-11**	**D5.1-D5.6**																	H_3	
T5.1	Ethyl lactate as a solvent.	(7),10	D5.1-5.2													♦				♦	
T5.2	LA based HMA by reactive extrusion.	(2),9	D5.1-5.2													♦				♦	
T5.3	Polyester polyol using green chemistry.	2,(11)	D5.6																	♦	
WP6	**Study of adaptation & Monitoring of PERSEO plant for the new intermediates**	**1,6,7**	**D6.1-D6.3**																H	H	
T6.1	Parameters of pre-treatment & fermentation.	(1),6,7																			
T6.2	Parameters of the separation/ purification.	(1),6,7	D6.1									♦									
T6.3	Monitoring of the processes parameters.	1,(7)	D6.2																♦		
T6.4	Integration and adaptation of processes to a biotechnological plant configuration	(1),6,7	D6.3																		♦
WP7	**LCA, S-LCA, LCC and Regulatory.**	**All(4)**	**D7.1-D7.5**																	H	H_2
T7.1	Evaluation of synergies with MSW projects	All(12)																			
T7.2	Modelling and simulation of the process	All (12)	D7.2																♦		
T7.3	Product life-cycle-assessment (LCA).	All(4)	D7.3																		♦
T7.4	Economic analysis (LCC).	All(12)	D7.1									♦									♦
T7.5	Social Analysis	All (12)	D7.4																	♦	
T7.6	Process inefficiencies Analysis	All(12)																			
T7.7	Regulatory and safety analysis.	All(2)	D7.5																		♦
WP8	**Communication, Dissemination, Exploitation, Technology Transfer.**	**All(12)**	**D8.1-8.9**		H							H									
T8.1	Communication	All(2)	D8.1-8.3		♦	♦						♦									♦
T8.2	Dissemination plan.	All(2)																			
T8.3	Exploitation plan	All(12)				♦															
T8.4	Business plans for key results	All(12)	D8.6																		♦

Num.	Work Packages and Related Tasks	Partners Num.	Delivera-bles	Project Months																	
				2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36
T8.5	IPR Management	All(12)	D8.4-8.5			♦															♦
T8.6	Technology transfer	All(12)	D8.7-8.9									♦								♦	♦
T8.7	Set up of PERCAL Stakeholders' Community	All(1)																			
WP9	**Project Management.**	**All(1)**			H																
T9.1	Management of the resources.	All(1)		╬		▲			▲			╬			▲			▲			╬
T9.2	Risk management. TMC's members	(1)																			
T9.3	Communication with EC.	(1)																			
WP10	Ethics Requirements	(1)		H																	

*EB: Executive Board, TMC: Technical management Committee

(1): (X) Leader Legend: ♦ Deliverable H Milestone ▲ Meeting ╬ Kick-Off, Mid Term, Final Meeting

3.1.4. Technical Risks.

The project team has carefully considered the technical risks associated with this project, summarising the potential risks, their likelihood and impact, and the actions which will be taken to mitigate them. Significant risks have therefore been addressed during the production of the work programme and partnership to ensure that the outcomes of the project are achievable (Table 3.1.b). The experience of all partners in their respective fields and a continuous co-ordination and risk assessment (T9.1) will make it possible to overcome or minimise potential problems if they arise. The following risks have been identified prior to the project starting.

Table 3.1.b. *Risks and mitigation proposals for different activities to be performed in the PERCAL project.*

Description of the Risk. (Risk Level*, and Likelihood (%)).	**Related WP(s)**	**Proposed risk-mitigation measures**
High variability of feedstock (M, 60%)	1	This variability has been taken into account in the strategies proposed in WP1 and mitigated already by collecting feedstock from diverse but related provenances. Additionally, a representative feedstock with a reasonable content of all the fractions required for further upgrading can be selected.
Interferences of fractions for the recovery of others (M, 30%)	1	To interact with WP4 to optimize the procedures.
Production of toxic compounds for further fermentation as a result of pre-treatment (M, 30%)	1	To implement detoxify mechanisms, depending on the target compounds
Undesired product formation (M, 40%)	1	To add industrial antibiotics if required (this is not possible if the objective is bacterial fermentation of the substrate, but for yeast transformation is recommended)
Inability of microbial strains to efficiently utilize all the sugars in MSW feedstock	2,3	To do all operations with the highest cleaning and sterilization measures.
Low titters and/or productivities of the fermentation products on MSW (M, 50 %)	2,3	Iteration with WP1 to improve pre-treatment. If not sufficient, optimize the medium further, e.g., by supplementing the MSW feedstock with fractions rich in minerals and vitamins. Consider other host strains.
Fermentation results are not always reproducible due to variations in the composition of MSW (medium, 50%)	2	Identify optimum nutrient requirements and supplement fermentation media with commercial sources of nutrients when there is lack of specific nutrients from crude fermentation media
Low performance (yield, productivity, titer etc.) of screened cultures in waste hydrolysates (low to medium, 25%)	2	Identify the biowaste fractions that lead to the required fermentation efficiency and optimise fermentations using only these fractions. Use alternative strains.
The presence of inhibitors may reduce fermentation efficiency (low, 15 %)	2	Adapt additional steps of purification during up-stream processing, select more tolerant strains or improve existing strains
The LA fermentation broth contains significant impurities originally coming	2	The option to encounter this shortcomings are: • Adapt the downstream processing DSP to handle the the

from the MSW feedstock (such as protein/ polypeptides, amino acids, high amount of ashes, other organic acids such as acetic acid (high 50%, expected technical challenge)		composition of the final fermentation broth by applying combinations of different DSP unit operations such as membranes (ultra and nano filtration), electro dialyses, ion exchange and recover a high quality LA product. • Evaluate and adopt the feedstock supply chain and first processing to insure impurities in feedstock are kept at an acceptable level for final end product. • Remove potential acetic acid in the fermentation broth by stripping • Use the esterification as an actual cleaning process for LA to deliver a high quality end product to the market
Sugar to succinic acid conversion yield is not improved with the integrated fermentation membrane electrolysis system (high 50%, expected technical challenge).	3	The expected increase in succinic acid concentration and productivity will suffice for the development of an efficient process
Other impurities are present in the succinic acid concentrated solution derived via membrane electrolysis (high, 30%)	3	A downstream protocol will be developed based on the impurities present in the concentrated solution
Excessive fouling of in situ membrane (Medium, 40%)	3	Different membranes and operational conditions will be tested, alongside operational regimes that can minimise fouling (e.g high cross-flow velocity, regular cleaning).
Negative impact of H_2 on the fermenting organisms due to a shift in partial pressure and available CO_2 (High, 50%)	3	H_2 is expected to have a positive impact on the fermentation, but this remains a hypothesis to be tested. If the H_2 strips too much CO_2, an increased flow of CO_2 may be required.
Low yield in the biosurfactants condensation reactions (Medium, 40%)	4	Different reaction approaches will be tested. Previous purification of some aminoacidos and lipids can be implemented.
Too complex mixture of biosurfactants (M, 50%)	4	Purification of the most interesting aminoacids and lipids in advance in order to perform the condensation reaction with pure species.
Impurities in the product LA can potentially interact or harm catalyst for reactive distillation process (low < 10%)	5	Different solid catalysts will be tested and evaluated;
Esterification rate in the reactive distillation process is insufficient and final ethyl lactate product contains significant LA amounts. (Medium < 25%)	5	Process parameters need adaption, especially increase average resident time within the reactive part of the distillation column; change feed parameters; increase reactive zone in column; in worst case redesign/change geometry by increasing total height of column.
Water amount within the reactor (coming from LA and ethanol raw material) plus water produced by esterification is getting to high to achieve high efficiencies (High, 50%, expected technical challenge)	5	The process needs to be specifically designed, and operated to handle the water in the distillation; options: (i) change the process parameters (temperature, feed rate…) decrease water content in raw materials (use only 96% ethanol and high concentrate LA product >80%); increase thermal input, change boiler; in worst case redesign column
Thermo-mechanical resistance of LA hotmelts is not enough for further processing Reduction in biopolymer properties, bad smell. (High, 30 %).	5	Different compatible antioxidants, lubricants and plasticisers will be tested to reduce the action of compounding extruder over the material. Use of low shear screw elements and minimize residence time thanks to the modular configuration of multi screw extruders.
Impurities in the product SA can potentially interact or harm catalyst for production of polyester polyols or other building blocks (H, 20%)	5	Covestro pilot plant have additional technologies to improve monomers purity, some of them will be studied/adapted to the new biobased SA. Alternative catalyzers will be proposed and evaluated.
Parameters of interest are difficult to study and monitor (M, 50%).	6	To this end a special effort in technology screening and preliminary tests will be done in order to select the most appropriate sensor technology.
Sensor development and testing taking a long time to achieve stable and satisfactory results (M, 40%).	6	Early stage sensor specification based on process requirements will be done and an iterative approach will be taken when starting test definition and sensor design in order to correct and foresee deviations.
Low versatility in the Process plant adaptation for Bioehanol, LA, SA production (L, 30%).	6	Study the best techno-economical case in each geographic situation having in account the biogenic fraction composition of each MSW.
PERCAL's products might be relatively expensive (M, 40%)	7	To revaluate and improve the production process to minimize energy consume or look for alternative and cheaper energy source. Look for

		alternative catalysts that improve yield or reduce energy consume.
LCA or Energy consume is higher than current process (H, 10%)	7	A comprehensive assessment methodology will be developed early in the project lifetime. If the target results cannot be achieved after the demonstration phase, strategies for further improvements will be developed and optimisation solutions will be applied. Modelling and simulation activities to be carried out after the demonstration phase will enable the consortium to choose the best and most efficient scenario. This will include the use of Aspen Energy Analyzer and exergy analysis as tools to optimise the energy consumption (heat integration) and to potentially reduce GHG and CO2 emissions.
Simulation software is insufficient to simulate, scale-up and model some parts of the process or even the entire process (M, 10%)	7	A deep research of the most accurate thermodynamic models for the simulation of the process will be carried out regarding each part of the process and the components involved in it. If not sufficient yet, a mathematic modelling will be developed based on the specification of the process and the experimental information by means of specialised software (e.g Matlab, Simulink…), especially where biologic processes are involved. The model will be then integrated into the AspenPlus or the required simulation software to have a unique final simulation flowsheet.
Lack of interest in the PERCAL outcomes (M, 50%)	8	Implementation of dissemination and exploitation activities focussing on the advantages of PERCAL benefits will be carried out.

*L=low risk, M=medium risk, H=high risk,

3.2. MANAGEMENT STRUCTURE AND PROCEDURES.

3.2.1. Organizational structure and the decision-making, including a list of milestones.

The Consortium of **PERCAL** will be coordinated by Partner IMECAL with the support of AIMPLAS. The project management structure has been conceived to ensure the competent industrial, scientific and business management of **PERCAL** project and to enable efficient and effective integration and communication throughout the management chain.

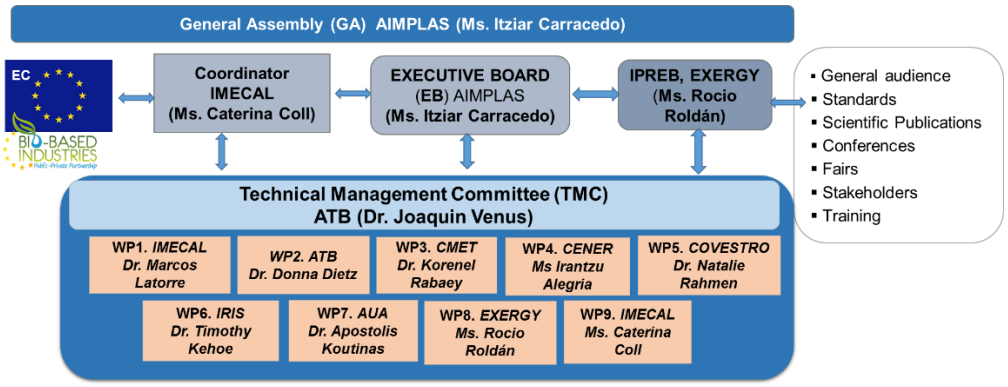


Figure 3.2.a Project Management Structure.

This management structure has been implemented successfully in similar EU Projects involving AIMPLAS (e.g. FP7 BRIGIT, BUGWORKERS or BISIGODOS, and H2020 JOSPEL or URBANREC and DAFIA), and as such, has been adopted by the **PERCAL** Consortium. A schematic representation of the management structure of **PERCAL** is shown in Figure 3.2.a the project management structure is clearly industry driven as the Coordinator is a SME (IMECAL), AIMPLAS chairs the Executive Board (EB) and the SME Partner EXERGY chairs the IPR, Dissemination & Exploitation Board (IPREB). All boards have a majority representation of Industrial Partners as described below. An overview of the governance structure and decision-making process is also detailed below. The composition of the boards and the decision making mechanisms have been designed in order to ensure that all decisions are taken in accordance with the Industrial interests, and SME's interests in particular.

The structure adopted is particularly business like (as opposed to bureaucratic) **to ensure that PERCAL is run as a "commercially" orientated project** with a very clear business focus. This has been achieved by limiting the number of partners represented on each management body, but ensuring that appropriate representation is pooled, in order to facilitate and speed up decision taking and ensure prompt reaction and implementation of countermeasures, if needed. Nevertheless, there is the opportunity for all partners to express their opinion and canvas support through the **General Assembly (GA)** which will meet at least twice a year to ratify progress and decisions made by the management bodies. The GA has the ultimate decision making power in the Consortium.

Decision –Making mechanisms: All Boards and Committees will typically meet at least twice per year. Any group will deliberate and decide validly provided a majority of its members are present or represented and each member will have one vote with the Chairperson having a casting vote if required. Decisions will be taken by a majority of two-thirds of the votes of the members present and the members that have delegated their vote. General Assembly meetings, meetings involving industrial demonstration or key technical meetings will be face-to-face, conference calls will be also encouraged.

All the partners have the responsibilities described below as a result of taking part in a specific decision making body**. In all of them, the SME and Industrial participants will have the highest level of decision-making competency (majority voting).**

The **General Assembly (GA)** consists of 12 representatives (one per partner), and is chaired by **IMECAL**. The GA is the ultimate decision-making and arbitration body of the Consortium (the final and highest authority in the project). However, it delegates short-term decisions-making actions to the **Executive Board** (EB). Such decisions do however require subsequent acceptance by the GA. The GA will decide on matters relating to: The political strategic and economic orientation of the Project, its objectives, definition and subsequent modifications, including the shift of budget of the Project. Actions towards defaulting parties, including their substitution. The nomination and election of EB members. The acceptance & exclusion of new parties or affiliates in the ECGA and in the Consortium Agreement. The Consortium Agreement and its Annexes. Description of Work, subsequent Project Implementation Plan and ECGA, including proposals and modifications. Ultimate approval of IPR Policies & Exploitation Plans proposed by the IPREB.

The **Executive Board (EB)** is responsible for the strategic and political guidance of the Project. It is **composed of one representative of each one of the 12 partners**. **AIMPLAS** will chair the EB. The **responsibilities** of the **EB** include: Overall guidance & supervision over strategic, political, economic & operational day-to-day issues regarding

the Project. Definition, approval and fulfilment of Project objectives. Scientific and Technological Quality Assessment. Definition and implementation of risk management methodology at all levels (financial, technical and administrative). Evaluation of results and definition of levels for dissemination relating to all kind of information and reports. Conflict Resolution. Constitution and administration of the funding. Presentation of the taken decisions to the GA periodically.

The EB will decide on matters relating to: The shift of tasks and responsibilities and budget of the Project. The regulations governing the decision-making and operational procedures of the EB and other Project bodies. The method of payment and management procedures of the Community's financial contribution. Requests for subcontracting, within the context of the Project. The actions to be taken in accordance with the evaluation of results obtained. The reporting, controlling and auditing procedures concerning the Project. The validation of reports and Project Deliverables to be submitted to the BBI JU within the reporting activity together with annual validation of the realised expenditure in accordance with the budget of the Project.

The **Technical Management Committee** (**TMC**) is the body responsible for ensuring the overall technical co-ordination of the Project work, in accordance with the Work Programme, subsequent Implementation Plan, and EB decisions, at Scientific, Technical, Financial and Administrative level. The TMC's main responsibilities also include: ensuring co-ordination between WPs, facilitating the take-up of research results, supervision of the implementation of the EB decisions, and the assessment of Project results. The **TMC, Chaired by IMECAL** will be composed by the WP Leaders: IMECAL (WP1), ATB (WP2), CMET (WP3), CENER (WP4), COVESTRO (WP5), VISUM (WP6), AUA (WP7), EXERGY (WP8) and IMECAL (WP9).

The **TMC** will be responsible for: Ensuring the implementation of the decisions taken by the EB at WP level; Making proposals to the EB for the re-allocation of material, financial and human resources, tasks and responsibilities. Preparing deliverables & other information requested by the EB, based on reports by the WP Leaders. Validating and seeking EB approval for subcontracting. Ensuring co-ordination between WPs facilitating the take-up of research results in the demonstration activities. Assisting WP leaders on day to day management. Monitoring the projects technical progress against the identified technical risks, as described in T9.3.

The TMC will **convene** meetings with the Work Package Leaders at least **quarterly** (typically 3 face-to-face meetings and 2 conference call meetings) and if applicable with the EB, for purposes of co-ordination and management. Where the TMC cannot settle decision making disputes, the decision shall be taken in the EB.

The ***Work Package Leaders (WPL)***, as identified above will report and be accountable to the TMC. The specific WPL will be responsible for their own WP organizational arrangements, work procedures, day-to-day management and time schedule, providing that such arrangements, work procedures and time schedule do not conflict with the ECGA, the Consortium Agreement, the Description of Work, subsequent Implementation Plans, or requests and other provisions made by the appropriate bodies within the Consortium or the BBI JU. Meetings of a specific WP will be convened at least quarterly, either face-to-face or by conference/video call. Extraordinary Meetings shall be: a) Called in and convened by the corresponding WP Leader; or b) by the TMC Chairperson.

The **Project Coordinator** (**IMECAL**) is the intermediate for communication between the Consortium and the BBI JU and shall report and be accountable to the EB and the GA. The Coordinator is in charge of the financial and administrative management (WP9) of the Project. The Coordinator will submit for EB validation, the reports which shall be submitted to BBI JU within the reporting activity. The reports covering project deliverables, supporting documents and audit certificates will be provided to the Coordinator by the TMC. For the avoidance of any doubt, the Co-ordinator has the capacity of representation of the consortium before the Commission. The Co-ordinator undertakes to keep the Joint Funds allocated by the EU Grant Agreement or by budget in accordance with the provisions which will be defined in the Consortium Agreement and in the Grant Agreement and decisions from the appropriate bodies. The main coordinators responsibilities will include:

a) Periodic Reporting: the project coordinator will be responsible for ensuring the collation and timely submission of the contractual reports required by the BBI JU, always according to the EC reporting requirements for the H2020 Programme. With regards to the project periodic reports, the project manager's activities shall include: the creation of report /information gathering templates, providing support to the project consortium and enabling each project partner to undertake their responsibilities for the provision of information to complete such reports and making sure that each partner is made aware of the deadlines for the timely delivery of any required information. Project reports will be submitted to the BBI JU on due time, including copies of the minutes, presentations given by partners and additional information. **b) Periodic Commission Questionnaires:** if the Commission requires a number of periodic questionnaires, the Project Co-ordinator will be in charge of coordinating the completion and reporting of these questionnaires. **c) To ensure the verification and submission of the project deliverable reports:** all deliverable reports have to be verified initially by the EB before submission to the project coordinator, who will ensure that they are according to the EC requirements and submitted timely. **d) Coordination of the Periodic and Review Project**

Meetings: the project coordinator will assume responsibility for the organization of those meetings, taking into account the attendance of the BBI JU representative(s) and the independent reviewer(s) appointed by the BBI. The project coordinator will coordinate: the meeting venue, agenda, minutes and other logistical arrangements where required, in cooperation with the hosting Partner. **e) Collection of Certificates on Financial Statements:** the project coordinator will ensure that all Partners who are required to provide a CFS, provide them in a timely manner at the end of the Project. **f) Reporting and Payment of Project Financing:** assuming the role of project banker and taking responsibility for holding and payment of the project funds in accordance with the agreement of the General Assembly and the Consortium Agreement. **g) Additional responsibilities of the project coordinator will include:** coordination for the updating and signature of the project Consortium Agreement and Contract amendments (when required); to provide a platform for communication between the BBI JU and the project consortium/ management structure; to provide a resource of support and information to the project consortium in fulfilment of the project contract and BBI requirements.

The SMEs and Industrial Partners will own the ultimate decision making in accordance with the organizational structure. As detailed above, all Consortium bodies have a majority of Industrial Partners, thus, in the end, the decisions are taken by majority voting, guaranteeing that the participating Industries needs and requirements are fulfilled and consequently, that the results of the project will be for the benefit of the sectors involved in the **PERCAL** value chain.

From the Technical point of view, **the Project Co-ordinator, delegating day to day work to the defined Management Bodies, will monitor the overall progress of the Project in all its aspects,** with specific responsibilities in this regard for:

Development of the Technical Report Procedures and the Technical Progress Monitoring and templates for technical activities. Assuring interaction between WP's and technical quality of (intermediate) project results. Assuring WP interaction and also for the project scientific and technical quality assurance. Co-ordination, integration and validation of the Technical Reports and Project deliverables to be subsequently submitted to the EC. Co-ordination of the other committees' activities (IPR and Exploitation Board, EB, TMC).

The **IPR, Dissemination & Exploitation Board** (**IPREB**) will be chaired by partner EXERGY, and the rest of Members will be key persons from project partners who are involved in the innovation and IPR generation and protection of key technologies for their companies. The **IPREB** will have the responsibility to support the Consortium on IPR issues. It will propose the IPR Policy and the Action Plan for the Exploitation of project results to the EB and GA. The **IPREB** will be responsible for the preparation, monitoring and updating of the list of exploitable results and innovations that will be used to monitor the progress and achievements of the project and to assess the Plan for the Exploitation and Dissemination of Results (PEDR, including both the Dissemination and Business Plan). The **IPREB** will look to increase the IP protection and ownership awareness, assimilation and implementation, supported by training and workshops given by the IPR helpdesk and other EC support services that might be considered necessary.

In summary, the protection of the interest of Industry will be guaranteed as they are the majority represented in the various boards, and consequently an industry-driven vote will be guaranteed in all levels of the management structure, allowing a *day-to-day management as in a processing chain at industrial level*, with protection of the SMEs interests and innovations.

3.2.2. Organizational structure and decision-making mechanisms.

The project management structure has been replicated from that used in FP7 and H2020 projects coordinated by AIMPLAS. This structure has been proven to be agile and has been successful in properly managing projects of a similar nature to PERCAL, solving conflicts and overcoming hurdles for their successful completion. In addition, strong cooperation already exists among the majority of PERCAL Partners within FP6, FP7 or H2020 projects, with an open management and cooperative approach which benefits from the relevant experience of all Partners. Moreover, the project partnership is highly complementary and industry driven; each Partner with sound experience in their respective fields of work, have clear objectives and responsibility of tasks, and each of them fully understands and is highly committed to exploit the project outcomes, ensuring that the high degree of cooperation achieved already during proposal preparation will continue during project execution.

Innovation Management: the **IPREB** will implement the **Innovation Management strategy** developed within **PERCAL**. All PERCAL's technology can be exploited by a huge number of additional applications by modification of microbial strains, catalysts or green chemistry routes. The Partners complementarities along the value chain and their highly innovative nature will foster cross-fertilisation and boost innovation within the project, simply by the collaborative execution of the work plan. In addition, the project will constantly look for internal and external opportunities in order to further promote innovation. The project will use the dissemination and business plan in order to detect needs of potential customers and future end-users of the results and will proactively look for feedback

from all stakeholders, entities and organisations (industrial platforms, public authorities, standard setters) that can be useful to improve **PERCAL** results, modifying accordingly the work plan if deemed useful. Similarly, the project partners will communicate internally the project development, fostering constructive criticism and feedback from staff, engineers, and designers and in particular managers. All this feedback will be gathered by the **IPREB** and discussed during the **GA** meetings to agree possible modifications of the workplan if improvements in the results can occur resulting from this innovation management strategy.

3.3. CONSORTIUM AS A WHOLE

3.3.1. Overall description: adequacy, complementarities, representatively of the value chain and synergies.

PERCAL consortium has a well-balanced composition. The PERCAL consortium brings together a wealth of expertise and resources within the areas, including partners that propose the best solutions to maximize the valorisation of the different components of MSW. The presence of main actors in the whole value chain demonstrates the critical mass of complimentary resources that will enable the PERCAL project to achieve its targeted societal, industrial, and scientific breakthroughs and commercial success. Each partner has a clearly defined role within the project and will contribute specific expertise that will enable the project success. **SME partners** (IMECAL, TBWR, VISUM and EXERGY) and the **IND partner** (SOFTER, HAYAT,COVESTRO) and the Industrial Support Industrial companies (ENERGO, ARKEMA and NOVO) will provide the fundamental knowledge, technologies and expertise in the abovementioned areas required to achieve the project objectives, as their activities within the project are fully in line with their own core business and research strategies. Figure 3.3.a shows the role/ expertise of each partner in the project and their distribution around Europe. **RTD Partners** (AIMPLAS, CMET, ATB, CENER and AUA) have the necessary research expertise and will provide support to the partners to achieve their objectives. Each consortium member has demonstrated a long term commitment to the development of their own research, policy and commercial activities within the industrial targeted by the project. All of them therefore offer excellent potential for a lasting integration of activities and the creation of a pan-EU culture for the technologies, materials and processes developed.

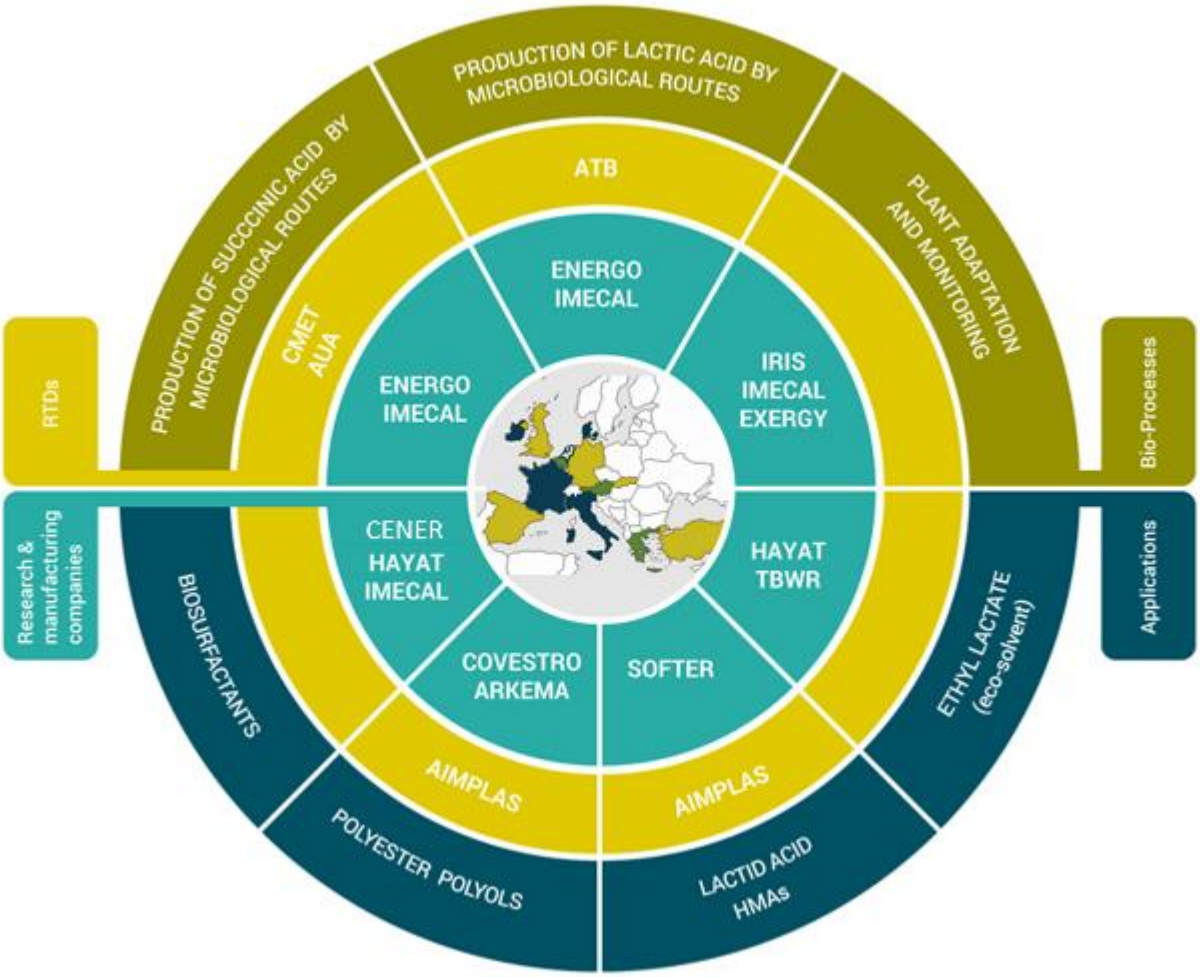


Figure 3.3.a. *Sectors and expertise brought by the PERCAL consortium.*

The mobility of resources and personnel will be encouraged throughout the project, not only helping to form strong bonds between the consortium members. Those actions are going to be implemented in order to lead towards an

integration of complimentary activities, services and future research projects to improve the composite multi stage processing at EU level. This is a complete storyboard of actors complemented by an advisory board to extend industrial and technical insight.

The table below (Table 3.3.a) gives a brief overview of each partner specific business and their expertise, and the main tasks in the project allocated to them based on that expertise.

Table 3.3.a DAFIA's Participant Expertise and Tasks Distribution.

Part num	Participant short name	Participant business / expertise-	Main role in the project
1	IMECAL	MSW biorefinery. Process engineering and expertise in biotechnological processes to obtain bioethanol from MSW.	Hydrolysis of MSW. Biosurfactants from amino acids and lipids. Re-engineering process for fermentation of acids. Project Management.
2	AIMPLAS	Research Technology Institute expert in polymer materials, nanotechnologies and processing.	Production of HMA from LA using reactive extrusion. Collaboration in Polyester polyols production.
3	ATB	Develop sustainable technologies for a resource-efficient utilization of biological systems to produce food, raw materials, & energy.	Bioconversion of the hydrolysed carbohydrates from organic municipal solid waste to lactic acid (LA).
4	AUA	University laboratory expert in bioprocess/biorefinery development, LCA and bioprocess design	Bioprocess development for coupled succinic acid production and separation using membrane electrolysis. LCA studies.
5	CMET	University laboratory expert on (bio-)electrochemical production and extraction of fermentation products	Bioprocess development for coupled succinic acid production and separation using membrane electrolysis
6	CENER	Technology center specialised expert in chemical processes, microbiology, fermentation process and analytical techniques	Characterization of MSW samples and fermentation by-products. Production of new biosurfactants from amino acids and lipids.
7	TBWR	Applied R&D in biorefining processing; separation of lactic acid and amino acids, prototype for reactive distillation of ethyl lactate.	Development of a simultaneous esterification and products separation process/reactive distillation for ethyl lactate.
8	VISUM	Advanced engineering company that specialises in process monitoring and control, and process optimisation.	MSW biorefinery processes monitoring and pilot plant control system study.
9	SOFTER	Production of plastic compounds and hot-melts.	Requirements definition & lactic HMA scale-up.
10	HAYAT	One of the largest FMCG (household cleaning & personal care, baby–care & feminine hygiene, and paper tissue & towels) manufacturers in Turkey.	Validation of surfactants, ethyl lactate & HMA as a components of their current Industrial products.
11	COVESTRO	A world-leading supplier of high-performance polymer materials for key industries.	Use succinic acid in polyester-based polyols, resins and polyurethane dispersions (PUD). To provide successful correlation between SA quality & final applicability.
12	EXERGY	Provide expert advice to business and the public sector on bioeconomy.	Simulations, plant engineering, energy balance and economic studies, business plan activities.

PERCAL partnership is composed by 12 organizations from 10 EU countries, having the required pan EU approach essential for achieving the objectives of the project since it is required to address the EU social and Industry needs in the field with a wide EU perspective. Furthermore, to ensure that the long-term and widespread take-up and commercial success of the PERCAL results, it is important that the project innovation and research activities are undertaken with a strong knowledge of the different industrial sectors addressed. The Consortium therefore ensures the participation of different EU regions, whose roles within the project are clearly defined and represent strategically important regional objectives, which will address the overall objectives of enhancing European industrial

competitiveness, moving towards a circular economy and hence **providing a clear EU added value**.

3.3.2. Industrial/Commercial involvement to ensure exploitation of the results.

PERCAL **project clearly has a number of benefits for the involved SME** as we explain on detail along the proposal. It enables the transfer and market replication of the new materials for complex waste (MSW) that contributes to the developed of four well-defined and high market products, supporting the take-up of the new advanced sorting, classification and production technologies and becoming a source of information, training and technical / regulatory support. Besides, PERCAL will create a new and more open market/supply chain for the materials, technologies and sales as previous figures shows.

The involvement of all the participants during and after the end of the project, in order to strengthen the industrial and commercial exploitation, is a key aspect of the consortium relationship, implemented through the innovation and IPR management strategies resulting in the Business Plan included in the Final PEDR.

During the project. Throughout the life of the project, all project partners will be responsible for updating the project 'Plan for the Exploitation and Dissemination of Results'. Dissemination will target 'take-up' actions within EU Local Authorities-Public Bodies, Stakeholders, SMEs and End-Users to promote the growth of the materials and technologies developed into new and existing market sectors; and to encourage SME participation within the PERCAL supply chain. Publication and dissemination of project knowledge and results will be conducted on agreement between the consortium partners, led by the **IPREB** chaired by EXERGY ensuring that the SMEs and industry interests are not only taken into account but are the real drivers for the exploitation strategy. Managed by the **IPREB**, each project partner will be required to develop their own 'Exploitation Plan' that will demonstrate how the project results will be integrated into their own forward business plan and research actions. Exploitation activities will include: protection and management of innovations, IPR and project results, gathering of reliable market intelligence and forecasts, feasibility studies for the creation of spin-off activities, evaluation of the project results for commercialization and application within new markets, steps to implement the proper industrial manufacturing policies, and the development of effective support structures to reinforce supply chain. Such activities are integral part of WP8.

After the end of the project. As explained on section 2.1, all the partners will obtain economic profits from the commercialization of the innovative PERCAL new materials and logistic implementation and will be able to introduce to the market cost competitive and sustainable products, increasing/opening markets and besides, to obtain profits by licensing the project results related to their knowledge field. However, as it will be detailed in the Business Plan and PEDR, the percentages of these profits will be different among the Partners depending on the exploitation agreement reached among co-owners which will reflect the percentage of contribution of each Partner during Project execution, the Background brought by them to achieve such result, and so on. Furthermore, these economic profits will encourage all the partners to contribute in the initial investment necessary to set up the exploitation process, once the project is finished. The SMEs participants are economically and managerially capable of performing the exploitation of the results, based on their turnover, their own investment in R&D and the level of economic contribution to the project. The other partners have also the right competences to guarantee the PERCAL exploitation of results in their areas. All partners are highly convinced of the potential impact of the project results in their businesses, and neither the SMEs, nor the other partners have conflict of interests regarding the areas of commercialization/exploitation, from a geographical, technological or market sector point of view. More details are provided in section 2.1., where apart from the exploitation interest, the steps to achieve the foreseen impact are described.

Industrial participation is very strong in this project. The consortium includes ten industry partners, with multiple and complementary technological, research and business expertise, covering the full spectrum of the activities proposed in PERCAL. They are dynamic enterprises with an active intention and outlook to expand their product/service portfolio into new markets. All project industrial partners planning to directly capitalize on the PERCAL results as we explain in detail in impact section, either by expanding their existing products/services, or by licencing technology. The geographical dispersion of the industrial partners ensures that they will cover an important proportion of European markets.

3.3.3. Other countries.

In the PERCAL consortium there is one participant from the Associated Countries, Turkey, that as the Article 7 of the H2020 Regulation says is fully entitled to participate in this call for proposal with exactly the same right as they are considered eligible partners.

3.4. RESOURCES TO BE COMMITTED

Each work package has been analysed to determine the personnel, equipment and financial resources required to undertake each task. The coordinator will lead the mobilization of these resources, ensuring or correcting any

deviations from the plan through constant monitoring of tasks. The budget distribution has been arranged in order to allocate the necessary resources to each partner taking into account their expected workload and involvement in the different WPs and their needs for materials and consumables, travel, equipment, other goods and services and subcontracting.

The description of those costs for partners whose 'Other direct costs' budget exceeds the 15% of the personnel costs is included this section (table.3.4a), altogether an overview of the main equipment's to be used during the project available at partners' facilities, which will be integrated through the WPs, has been included in section 4.

Table 3.4.a. Other direct costs (travel, equipment, other goods and services).

Partner		Cost (€)	Justification
Other goods and services	1. IMECAL	[illegible]	Laboratory consumables (12 k€), pilot plant consumables (8k€), Laboratory reactive (6k€), Shipment (5k€), WP8: IPR protection, dissemination material and events, conference fees, open access journal (4k€), Audit cost (1.5k€),
Equipment		[illegible]4	PERSEO plant equipment depreciation (plant cost 3 M€ during three years).
Travel		[illegible]	Travel to meetings (2 persons attending to 6 general assembly meetings, KOM and 8 trips for dissemination activities: fairs and conferences.
Total		[illegible]	
Other goods and services	2.AIMPLAS	[illegible]	Laboratory consumables (10 k€), pilot plant compounding consumables (15k€), Laboratory reactive (5k€), WP8:cost of IPR protection, dissemination material and events, and publication fees for open access journal (5k€),
Equipment			Co-rotating twin screw extruder (300 k€*24 month use at 25% time), equipment laboratory of synthesis (155K*24 moths use at 20% time).10 years depreciation.
Travel		[illegible]	Travel to meetings (1 persons attending to 6 general assembly and KOM meetings and 5 dissemination travels, attendance to conferences, fairs and technical demonstrations.
Total		[illegible]	
Other goods and services	3.ATB	[illegible]	WP4: Consumables, tools and small equipment (adaptation of existing devices, materials/chemicals/kits, sensors, analytical costs, membranes, lab ware etc.) (45 k€). WP8: IPR protection, dissemination material and events, conference fees, open access journal (5k€).
Travel			Travel to meetings (1 persons attending to 6 general assembly and KOM meetings and 5 dissemination travels, attendance to conferences, fairs and technical demonstrations.
Total		[illegible]	
Other goods and services	4.AUA	[illegible]	WP 3 and WP7 consumables and accessories, WP8: IPR protection, dissemination material and events, conference fees, open access journal (5k€).
Travel		[illegible]	Travel to meetings (1 persons attending to 6 general assembly and KOM meetings and 5 dissemination travels, attendance to conferences, fairs and technical demonstrations.
Total		[illegible]	
Other goods and services	5.CMET	[illegible]	Electrodes for reactors (~2k€/m^2) and membranes (regular replacement, ~150€/m^2) Components for peristaltic pumps (9.5 k€), sensors and heaters (2.5k€) to control the reactors. WP8: IPR protection, dissemination material and events, conference fees, open access (5k€).
Equipment		[illegible]	Belach fermenter, 5L; Electrocell Syncell, including 5 reactor frames (52 k€ depreciation time 5 years, 100% use during 2.6 years project.
Travel			Travel to meetings (1 persons attending to 6 general assembly and KOM meetings and 5 dissemination travels, attendance to conferences, fairs and technical demonstrations.
Total		[illegible]	

Partner		Cost (€)	Justification
Other goods and services	6. CENER		WP4: Laboratory consumables and reagents, microbial cultures and media, laboratory material, etc... WP8:IPR protection, dissemination material & events, conferences fees, open access (5 k€).
Travel			Travel to meetings (1 persons attending to 6 general assembly and KOM meetings and 5 dissemination travels, attendance to conferences, fairs and technical demonstrations.
Total			
Other goods and services	7. TBWR		WP5: Bulk chemicals for the reactive distillation (6 k€), membrans (3.5 k€), catalysts for lab/ pilot scale (9 k€), consumables for lab scale experiments (5 k€) and special analytics (7.5 k€), WP8: IPR protection, dissemination material & events, conferences fees, open access (4 k€).
Travel			WP5: Travel cost to cover mileage and allowance for travels from Vienna to Linz, where pilot plant is located: 30 travels @ 220€, 6600€; Other travel costs see bellow table.
Equipment			Depreciation for a reactive distillation plant to deploy to the project for 36 Months
Total			
Other goods and services	8. VISUM		Materials for the fabrication of 2NIR monitoring systems, and for their installation in laboratory facilities (optical systems 2x45k€, optomechanical parts: 8k€, mechanical parts: 8k€, vessels and tubing: 5k€, computer and software for programming control platform: 5k€, electronics & interface: 2k€)- Cost of lab consumables for calibration tests and analysis (4k€), transportation & insurance of prototypes (3 k€) IPR protection, dissemination material & events, and publication fees for open access journal (5 k€).
Travel			Travel to meetings (1 person attending the KOM and 6 general assembly meetings and 3 dissemination travels, attendance to conferences, fairs and technical demonstrations)
Total			
Travel	9.SOFTER		Travel costs attendance to General assembly meetings
Total			
Other goods and services	10. HAYA T		Inks, cleanning products, surfactants. laboratory consumables, reactives, … Conferences, seminars, fairs assistance fees.
Travel			Travel to meeting (1 person assist to 6 meetings in 3 years), visit potential customers, suppliers, fairs or conferences.
Total			
Other goods and services	11. COVESTRO		Consumables, auxiliary equipment, accessories in WP 5 activities
Travel			Travel to meeting (1 person assist to 6 meetings in 3 years), visit potential customers, suppliers, fairs or conferences.
Total			
Other goods and services	12. EXERGY		Software: AspenOne (License for 12 month. €16,676), SuperPro Designer (€15,234), IPR protection, Open access publication, Registration fees, attendances to conferences and fairs, dissemination materials and evaluation of patentability (€5k).
Travel			Travel to meetings (1 person attending the KOM and 6 general assembly meetings and 3 dissemination travels, attendance to conferences, fairs and technical demonstrations)
Total			

Table 3.4.b. Summary of subcontracting costs

Participant No.	Participant Short Name	Costs (€)	Justification (subcontracting)
1	IMECAL		Task 1.3: Laboratory analyses to determine the chemical composition of selected substrates. Task 1.4: Laboratory analyses for the evaluation of pre-treatment and saccharification processes. Task 1.5: Laboratory analyses to validate the fermentation processes for the production of target bioproducts (ethanol, LA, SA). Task 6.1: Study for the selection and optimization of the relevant fermentative parameters for the production of bioproducts at industrial scale. Analysis and control of parameters. Task 6.4: Study of the integration of fermentative parameters within the appropriate working ranges in the demo plant configuration. Analysis of parameters.
6	CENER	[illegible]	Task 3.1: Screening and selection of bacterial strain for succinic acid production and first stage of fermentation media optimization assays at lab scale according to selected strain. Task 4.4: Consultancy and support for the selection of analytical techniques to evaluate the activity of biosurfactant samples produced.

* According to article 13 of Grant Agreement the Best Value for Money will be applied, so that company BIOPOLIS will be subcontracted to carry out Table 3.4 b tasks, due to its know-how and consolidated experience in the biotechnological field and on these specific tasks in particular.

During the PERCAL three-year project, 7 technical meetings have been planned: kick-off, and one GA meeting each 6 months; all meetings will be held in EU countries (Spain, Germany, Austria, Turkey & UK). The Consortium has estimated around 700 €/trip/person in Europe (350€ flight, 250€ two nights hotel, 100€ meals, taxi, others). In consequence, the average cost per partner for PERCAL meetings assistance is 4,900 €, assuming that one person per partner attends. Additionally, 2 trips per partner per year are estimated for attendance to conferences, fairs, technical demonstrations, etc., this meaning 3,500€ more. **Full cost budget for industrial partners is 8,400 €**.

In the case of **Research centres** three additional dissemination travels will be proposed, **i.e., 10,500 €.** In the **coordinator´s case**, two people will attend all the meetings, resulting in 10 EU trips for project meetings and 8 EU trips for dissemination activities, this is around **12,600 €.**

In the WP8, for all industrial SME partners a quantity of 4,000€ in concept of IPR protection, Open access publication, Registration fees, attendances to conferences, dissemination materials and evaluation of patentability. This amount will be increase up to 5,000€ for the RTD, EXERGY and VISUM. These costs are included in other goods and services cost.

The budget proposed for PERCAL project is 3,394 k€, the BBI request is 2,519 k€ and the in-kind contribution 876 k€, IND will provide the 19% and SME the 39% of in-kind. The participation and leadership of SMEs and IND in PERCAL is clear as they represent the 58% of the Partnership and around 46% of the budget and 48% BBI JU contribution**.** In section 3.2 (management structure and procedures), it can be seen that the relevant Project Governance Bodies (General Assembly, Executive Board, Technical Management Committee, Innovation Management Board and IPR-Exploitation Board) **have a majority of SMEs**, having the decision-making power in all of them. Also 5 of 9 WPs are led by companies. All of them demonstrate a strong commitment to the project and understand the responsibilities that this brings with it and have assumed a fundamental role within the project work programme, as clearly summarized in Table 3.4.c.

Table 3.4.c *PERCAL Partnership Distribution.*

Partner type	Nº	Percentage of partnership (%)	Total effort (p/m)	Percentage of effort (%)	Total Budget	EU/ BBI request	EU/BBI (%)	In kind contribution	In kind contribution (%)
SME	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
IND	3	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	
RO	5	[illegible]	[illegible]	[illegible]	[illegible] €	[illegible]	[illegible]	[illegible]	
TOTAL	12		[illegible]			2[illegible]		[illegible]	

To obtain the complementary range of expertise and the critical mass needed (both in human and financial terms), it is necessary to bring together, companies and ROs representing the entire supply chain and R&D areas, working in close cooperation to solve a common problem, which cannot be found within a single EU Region, requiring the integration of resources from across EU.

4 MEMBERS OF THE CONSORTIUM

4.1 PARTICIPANTS (APPLICANTS)

PARTNER 1. Coordinator **Name**: Industrias Mecanicas Alcudia S.A. (IMECAL) **Web page**: www.imecal.com/perseo	**Type:** SME **Country:** Spain	IMECAL

Brief description of the organization: IMECAL is a SME which was founded in 1979 as an engineering company in the metal sector, working on demand and manufacturing all type of metal-mechanical equipment for civil works, petro-refineries and petrochemical, transportation, refrigeration industry, industrial auxiliary motor and ceramics industry, among others. R & D IMECAL Department has allowed the company to progress in the technological field, having made significant partnerships with companies, universities and public research organizations. The R&D line of production technology for second generation bioethanol, has developed a biotechnological patented technology for obtaining bioethanol from Organic Municipal Waste. In 2007 IMECAL built a second generation bioethanol plant with a capacity to process 25 ton/day of organic fraction of MSW (PERSEO Bioethanol®). Since this year IMECAL has participated in different projects to test and improve PERSEO technology to different MSW raw materials with different companies. IMECAL is member of the Biofuels TP Platform-European Union, and participates actively in several national and international conferences on biofuels.

Role in the Project: Project coordinator. In WP1 they work in the selection and analysis of the most representative MSW. Develop in collaboration with NOVO the new enzyme cocktails to increase the amount of hydrolysed sugars. Development a biosurfactants from the protein and lipid fraction of MSW. Evaluate with CENER, EXERGY and VISUM the best technologies to integrate the new products (LA, SA and lipid/protein) in the PERSEO technology plant. Contribution to communication, dissemination, exploitation and other related activities.

Scientific team:

Ms Caterina Coll Lozano, PMP® (F), Chemical engineer by the Higher Technical School of Industrial Engineers of the Polytechnic University of Valencia. Project Management Profesional (PMP®) certificate by Project Management Institute (PMI). Executive MBA by Bussines & Marketing School ESIC. Master degree in environment coursed in the Department of Chemical and Nuclear Engineering from the Polytechnic University of Valencia. Technical Industrial engineer in Chemist by the Technical School of Industrial Engineering of the Polytechnic University of Valencia. Stay in the "School of Science" of "Institute of Technology Sligo" in Sligo (Ireland). Stay on the Biomass Unit of the Division of Renewable Energy from the department of Energy of CIEMAT. Professional experience of over 15 years in process engineering and environment, mainly developed in CEAM (Centro de Estudios Ambientales del Mediterraneo) as Head of laboratory and environmental projects, and in IMECAL as Plant Manager of second generation bioethanol pilot plant PERSEO, and R&D Manager. Over the last years he has coordinated and participated in several projects at national and international level and performed scientific and technical publications in various journals, books and conference contributions.

Dr. Ing. Marcos Latorre Sánchez (M), Chemical Engineer by the Higher Technical School of Industrial Engineers of the Polytechnic University of Valencia. Masters in Design for Sustainability from Cranfield University (England). PhD in Green Chemistry at Institute of Chemical Technology (Polytechnic University of Valencia - Spanish National Research Council). Stay at Max Planck Institute of Colloids and Interfaces (Germany). He has more than 6 years of experience in new materials development at laboratory and pilot scale and advanced characterization for renewable energy systems. He has participated in a number of R&D nationally and internationally funded projects aimed at the valorisation of biomass into higher valuable products. He is author of 10 papers in peer-reviewed journals and has participated in different international conferences.

Ms María Frau Martínez (F), Technical Agricultural Engineering by the Polytechnic University of Valencia. International Welding Specialist (IWS) by CESOL. Senior technician in occupational risk prevention in the fields of occupational safety (CIEGS, Polytechnic University of Valencia), Ergonomics and Applied Psychology (IMF) and Health at Work (IMF). Extensive experience in handling graphic design programs like Auto-Cad, Inventor, Solid Works and Lantek among others. Professional experience in process engineering in Bonduelle Iberica SAU (Frudesa) and currently as a mechanical and process engineering in IMECAL, where performs mechanical and industrial design projects.

Mr Jaime Mesa Arguello (M), Industrial Engineer by the Polytechnic University of Valencia. Technical specialist in engineering design with CAD Software, manufacturing and inspection of pressurized tanks. Professional experience in R&D projects development and process engineering at IMECAL, where performs mechanical and industrial design projects.

Mr Juan Manuel Ferri Palop (M), Telecommunications technical engineer (Electronic Systems specialty) by the Polytechnic University of Valencia and Fachhochschule Ulm - Hochschule für Technik (Germany). Technical specialist in electronic systems, computer design and modelling programs. Professional experience in R&D projects development, and process engineering at IMECAL SA., where performs mechanical, electrical and industrial design projects.

Related publications, patents or products:

- Method for Recovering Energy from the Organic Fraction of Solid Urban Waste and Associated Facility. Signes Nunez, Vicente, ; Ballesteros Perdices, Mercedes, ; Ballesteros Perdices, Ignacio, ; Manzanares Secades, Paloma, ; Martinez Garcia, Jose Maria, ; Negro Alvarez, Maria Jose, ; Oliva Dominguez, Jose Miguel, ; Castaneda Sanchez, Rafael, ; Coll Lozano, Caterina. Spanish patent nº 2303792, European Patent EPO 2112226, USA Patent 8399228 B2 and international PCT WIPO 099038.
- Ethanol production from the organic fraction obtained after thermal pretreatment of Municipal Solid Waste. Mercedes Ballesteros, Felicia Sáez, Ignacio Ballesteros, Paloma Manzanares, Maria Jose Negro, Jose Maria Martínez, Rafael Castañeda, Jose Miguel Oliva Dominguez. *Appl Biochem Biotechnol* (2010) 161:423–431.
- PERSEO Bioethanol®: Obtención de bioetanol y combustible orgánico a partir de RSU. Caterina Coll y Vicente Signes. INFOENVIRO, nº 72, p. 49-42, 2012.
- Sistema de comunicaciones para el control de una planta de biocombustibles. Josep F. Signes Palop, Víctor M. Sempere Payá, Caterina Coll Lozano. *Automática e Instrumentación*, nº 444, p. 40-42, 2012.
- Proyecto PERSEO. Innovación tecnológica para el bioetanol de segunda generación. Rafael Castañeda, Caterina Coll, Vicente Signes, Mercedes Ballesteros. *RETEMA*, nº 121, p.50-56, Especial biomasa 2007.

Related Projects:

Project acronym/description.	**Scope**	**End Year**	**Website**
LO2X: Supercritical water co-oxidation of urban sewage sludges and wastes	LIFE 12 ENV	2016	www.lo2x.com
BIOFORSU: Valorisation of the organic fraction of municipal solid wastes by conversion to bioethanol and combustible biomass	Regional	2013	Not Available
BIOFUEL: Development for a new integral waste treatment process with production of liquid biofuel	Spanish Project	2009	Not Available
ATENEA: Production of bioethanol from citrus wastes	Regional	2008	www.imecal.com/perseo

PERSEO: Production of bioethanol from organic municipal solid wastes	Spanish Project	2006	www.imecal.com/perseo

Technical equipment:

- 25.000 m^2 of metal-mechanic engineering facilities to transform metallic plates and sheets to a wide variety of industrial equipment: pipes, chemical reactors, distillation columns… Machinery includes: Laser cutting, forming, machining, blasting, passivating, assembly and welding…
- 2000 m^2 semi-industrial biotechnology plant (PERSEO Bioethanol®) with a capacity to process 25 ton/day of organic fraction of MSW. The plant includes the following units: Pre-treatment, Fermentation, Solid-Liquid Separation, Distillation, Effluent Treatment, Auxiliary Systems, Final Product Storage, Control.
- Two physical-chemical laboratories included in the PERSEO semi-industrial plant, with different equipment: High pressure liquid chromatography, analytical balance, thermo-balance, centrifuge, microbiology fume extraction, pH meters, incubators with orbital shakers, solid grinders, oven, kiln, autoclave…

Pictures of bioethanol PERSEO semi-industrial plant and its laboratories

PARTNER 2. **Full name**: AIMPLAS-Asociación de Investigación de Materiales Plásticos y Conexas (AIMPLAS) **Web page**: www.aimplas.es	**Country:** Spain **Type:** RO	AIMPLAS PLASTICS TECHNOLOGY CENTRE

Brief description of the organization: AIMPLAS, Technological Institute of Plastics located in Valencia, is a private, non-profit Association with more than 500 associated companies created in 1990. AIMPLAS is formed by +120 highly skilled professional, more than 65% with a Masters, Engineering or equivalent degree in Chemistry, polymer engineering, materials engineering or equivalent, including 15 PhD.

AIMPLAS' fields of work are related to technological research and development on thermoplastic and thermosetting plastic materials & products, its transformation processes and their recyclability and sustainability. AIMPLAS generates new knowledge and technologies that can be transferred to companies in order to help them

to increase their effectiveness and competitiveness.

AIMPLAS has a broad expertise in the fields of recycling, compatibilization, reactive extrusion, biopolymers and renewable source materials, special assisted processing technologies and nanocomposites, gained during the last years in the frame of EU and national projects. Nowadays, AIMPLAS is involved in more than 25 European projects and has participated in 77 projects in FP5, FP6, FP7, LIFE+, CIP-EcoInnovation EU Programmes, among others, coordinating 27 of them. At National and Regional levels AIMPLAS participates in around 100 projects yearly.

AIMPLAS, as RO is focused to help companies in the plastic sector to develop new products and increase their competitiveness through innovation, has more than 20 pilot plants representing the most relevant polymer/plastics/composites production technologies present in the industry nowadays and has state-of-the art test facilities for chemical, optical, morphological, mechanical and physical characterisation. These pilot lines and laboratories are used by many customer's every year allowing them to test new materials, optimize production processes and launch new products to the market, supported by AIMPLAS technical staff, resulting in more than 5000 assays, 170 technical assessments and 120 skills training actions to more than 1500 clients per year. AIMPLAS has state-of-the-art 8500 m^2 facilities, including thermoplastics and thermoset pilot plants, analysis and testing laboratories (physical-mechanical, chemical, packaging, automotive and construction) and training areas.

Role in the Project: In WP5 Production of HMA from LA using reactive extrusion and Collaboration in Polyester polyols production. Leader of regulatory and safety analysis. Contribution to communication, dissemination, exploitation and other related activities.

Scientific team:

Dr. Amador García (M), PhD in Organic Chemistry by the University of Valencia, working in AIMPLAS as researcher in the chemical laboratory since 2009. He has been involved in several regional and FP7 European projects focused in nanotechnology areas, materials from renewable resources and coatings with enhanced properties, etc. (SUPIN, TRAINER, NMP INNOREX, RECAN, KBBE BRIGIT, KBBE BISIGODOS, BIOREFINE-2G). Before joining to AIMPLAS, he worked as researcher assistant in CIPF (Centro de Investigación Principe Felipe) and in the University of Valencia. During 2007 he was in the chemical group of Professor Dennis P. Curran (University of Pittsburgh) as visiting researcher. In 2010 he received the Plastic & Innovation award.

Dr. Alba María Ortiz Álvarez (F), European PhD in chemistry by the Universidad de Alcalá, Madrid (December 2012), working in AIMPLAS as PhD researcher in the chemical laboratory since 2013. She has been involved in several national and international research projects focused in different areas such as synthesis of new polymers, nanotechnology, materials from renewable resources and coatings with enhanced properties, and so on. Moreover, she has being involved in the technical work of on Food contact plastics as well as Pharmacopoeia. Before joining AIMPLAS, she worked at the Universidad de Alcalá, where she was involved in research projects. Her research focused on the design and synthesis of new enlarged-palladium complexes as recoverable catalysts by nanofiltration techniques, contributing in international peer-reviewed journals and in national and international conference papers in the field of catalysis, dendrimers and organometallic chemistry.

Ms. Raquel Llorens-Chiralt (F), Master in Material Engineering by the Polytechnic University of Valencia, working in AIMPLAS as Senior Researcher since 2005 in the Compounding Department, focused in new material formulations. She has been involved as main researcher in the technical execution of several FP7 Projects focused on nano-materials, biopolymers, flame retardants, blends, mineral and natural fibres composites and PVC. Before joining AIMPLAS she worked in different private industries (paints, ceramics and metallurgical).

Dr. Nuria Lopez (F), PhD in Materials by the Universidad Politécnica de Valencia, working as a researcher in the Extrusion Department at AIMPLAS since 2003. Extensive experience in polymer processing field, analysis of properties and development of new products. Technical advice, evaluation and optimization of materials using extrusion processing technology, new developments for private companies and close collaboration with other

Research Centres at Regional National and International level. Experience in sustainability developments, as the use of biodegradable and compostable materials and additives, materials from renewable sources, the use of natural fillers and studies of recyclability of materials.

Ms. María Llorens (F) graduated in Information Sciences from the Cardenal Herrera-CEU University. Then she holds a Master's Degree in Communication Management and a Master's Degree in Marketing Management from the Universitat Oberta de Catalunya. She has also a Master's Degree in Management Skills from the University of Valencia. She is currently working at AIMPLAS since 2004 and since 2011 she is the Head of the Marketing and Communication department, where she coordinates the relation with media, social networks, organization of events, digital marketing and graphic design of the company. As a result, AIMPLAS was awarded with the LIFE prize for the best textual communication for the project BREAD4PLA.

Mrs. Itziar Carracedo (F) has a University Degree in Chemical Engineering by Universidad Complutense de Madrid. She works in the International Projects Unit at AIMPLAS since 2007. She has been involved in international R&D projects funded under, the 6th (PEGASUS, FLOWFREE, INNOFOOT) and 7th (MAC-RTM, BUGWORKERS, PLA4FOOD, ESPRIT, NANOONSPECT) Framework Programmes, developing technical management and coordination tasks.

Publications:

- Signori, F., Boggioni, A., Righetti, M. C., Rondán, C. E., Bronco, S. and Ciardelli, F. (2015), Evidences of Transesterification, Chain Branching and Cross-Linking in a Biopolyester Commercial Blend upon Reaction with Dicumyl Peroxide in the Melt. Macromol. Mater. Eng., 300: 153–160.
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- R.M. González, M.P. Villanueva19 – Starch-based polymers for food packaging, Multifunctional and Nanoreinforced Polymers for Food Packaging,2011, Pages 527–570
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- Fang Jm, Fowler Pa, Escrig C, Gonzalez R, Costa Ja, Chamudis L: Development of Biodegradable Laminate Films Derived From Naturally Occurring Carbohydrate Polymers. *Carbohydrate Polymers* 2005, 60:39-42.

Projects:

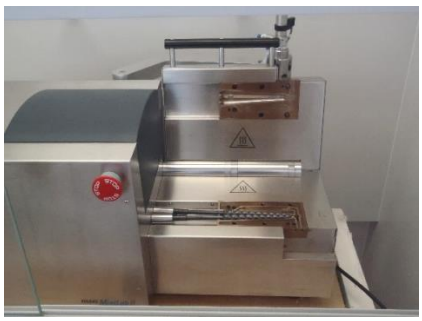
Project acronym/description.	**Scope**	**End Year**	**Website**
BUGWORKERS: New tailor-made PHB-based nanocomposites for high performance applications produced from environmentally friendly production routes.	FP7-NMP - 2009-SME	2013	www.bugworkersproject.eu/.
BRIGIT: New tailor-made biopolymers produced from sugars obtained from lignocellulosic residues for flame retardancy highly demanding applications.	FP7-KBBE-SME	2017	www.brigit-project.eu/
INNOREX: Realising the continuous, highly precise, metal-free polymerisation of PLA using alternative energies for an	FP7-NMP-2012	2016	www.innorex.eu/

innovative reactive extrusion.			
BIOREFINE 2G : Development of 2nd Generation Biorefineries Production of Dicarboxylic Acids and Bio-based Polymers Derived Thereof	FP7-KBBE 2013	2017	www.biorefine2g.eu/
BISIGODOS: High value-added chemicals and BIoreSIns from alGae biorefineries produced from CO_2 provided by industrial emissions.	FP7 KBBE 2013	2017	www.bisigodos.eu

Technical equipment:

- Equipment for mechanical, thermal, rheological and physical tests accredited to perform plastic characterization using the EN/ISO standards
- Rheometer, Optical and electronic Microscopy, DMA, DSC, Thermogravimetry. Raman Spectrometer.
- Twin-screw co-rotating extruders with gravimetric feeders (to feed solid and liquid additives) and other multiscrew planetary extruder.
- Well-established kilo laboratory for the synthesis of polymers in multigrams to kilograms level. During kilo-lab synthesis, organic chemists and chemical engineering work together to reach the appropriate cost effectiveness of the polymer synthesis process, optimizing parameters as reaction time, temperature profile, amount of catalyst and post-purification steps.
- The kilolab is equipped with 1, 2, 5 and 15 l glass jacketed reactors able to reach temperatures of 190ºC and another 10 l steel reactor able to reach temperatures up to 250ºC. Moreover, high volume Buchi rotavap and high capacity crystallization/filtration systems are available. Other auxiliary equipment as vertical stirrers for viscous polymers, vacuum pumps, water traps, nitrogen line and vacuum oven are currently available.

Reactor and Buchi 20 l rotavap



HAAKE™ MiniLab II Micro

AIMPLAS kilolab is also able to explore the polymer synthesis by means of controlled reactive extrusion process. In this context the Kilolab is full-equipped with HAAKE™ MiniLab II Micro Compounder which is ideal for reactive extrusion optimization process, compounding expensive materials or small scale materials such as nano-composites, bio-polymers or pharmaceuticals.

To perform all of these approaches, we take up several kind of strategies in polymer synthesis as:

- Ring opening polymerization (ROP)

Compounder	• Radical polymerization • Ionic polymeriation

PARTNER 3 **Name**: Leibniz Institute for Agricultural Engineering and Bioeconomy (ATB), Dept. Bioengineering **Web page**: www.atb-potsdam.de/en.html	**Type:** RTO **Country:** Germany	ATB Leibniz-Institut für Agrartechnik und Bioökonomie

Brief description of the organization: The Leibniz Institute for Agricultural Engineering and Bioeconomy (ATB) is a European research and development centre for agricultural engineering with an interdisciplinary field of activity at the nexus of biological and technical systems. ATB is funded by the Federal Government and the State of Brandenburg and receives additional third-party funds of around 35% of its budget (total 2015 budget >17.0 Mio).

ATB has multi-disciplinary research teams with currently about 250 staff members, 90 of them scientists and PhD students. In 2015 scientists published over 115 reviewed papers in international peer-reviewed journals.

The Institute's mission is to develop sustainable and highly innovative technologies for the resource efficient utilization of biological systems to produce food, raw materials and energy. Climate change and the incipient changes in the structure of agriculture represent particular challenges for agricultural engineering research.

ATB research activities are of interdisciplinary nature and follow a systems approach covering the entire spectrum from primary production and consumption of food, feed and non-food biomass including technology assessment.

The Institute's research work addresses topical scientific questions and contributes to the call for practical guidance and support in the domains of politics, industry, agriculture and horticulture. The institute owns outstanding infrastructure including pilot plants and experimental fields. Intensive collaboration with universities, other non-university research institutions and enterprises on national and EU/international level contributes largely to elaborate solutions for the agricultural and food industry.

Role in the Project: Bioconversion of the hydrolysed carbohydrates from organic municipal solid waste to lactic acid (LA), which is a main activity of WP2. Collaboration within the other WP's and support in view of scale-up (piloting) and downstream processing sub-tasks.

Scientific team:

Dr. Joachim Venus (M), Senior Scientist "Industrial Biotechnology", holds a diploma on Biotechnology (graduated from Anhalt University of Applied Sciences Köthen/Germany 1984) and a PhD on Bioengineering (TU Dresden). He is employed permanently at the Dept. Bioengineering of ATB Potsdam (http://www.atb-potsdam.de/en/institute/about-us/team/portrait/portrait/joachim-venus.html).

Joachim Venus is Program Coordinator for the research program "Material and energetic use of biomass" and head of the research group bioconversion/fermentation of biogenic feedstocks/residues at the ATB. His work emphasized on the development of continuous processes for the production of basic chemicals - in particular lactic acid - from biomass. He has published a great deal of papers in well-known journals of those research fields. Moreover, he has taken part in more than 100 presentations and proceedings, since 2003, such as in "World Congress on Industrial Biotechnology and Bioprocessing", "International Conference on Environmental, Industrial and Applied Microbiology/ BioMicroWorld", "European Biomass Conference and Exhibition", "European Forum for Industrial Biotechnology/EFIB2010", "International Conference on Renewable Resources and

Biorefineries/RRB" or "Symposium on Biotechnology for Fuels and Chemicals".

At ATB Joachim Venus is in charge of numerous research projects being carried out in the multi-functional pilot plant for the biotechnological production of lactic acid from plant biomass. His expertise and research areas are

- White Biotechnology, Biorefineries, Scaling-up of Bioprocesses
- pre-treatment of biomass for microbial conversion processes, bioconversion of renewable resources
- kinetics of cell growth/product formation and modelling of fermentation processes
- development of continuous mode processes for the production of basic chemicals (e.g. lactic acid) and biomass
- operation of a pilot plant facility for the optimization of biotechnological processes

He is heading the national advisory board "Biotechnology of Renewable Resources" at DECHEMA. On international level Joachim Venus is being also a Management Committee member for Germany in the COST Action TD 1203 (Food waste valorisation for sustainable chemicals, materials & fuels, EUBis).

Dr. Donna Dietz (F), PhD in Bioprocess Engineering by Hamburg University of Technology in Hamburg, Germany. During the time in Hamburg, she worked as scientist in FP7 Project Propanergy – Integrated bioconversion of glycerol into value added products at pilot plant scale. Since 2012 she is working as senior researcher in ATB. Dr.Dietz's expertise includes microbial fermentation of biobased raw material to value added products from lab to pilot scale.

Ms Kerstin Mehlmann (F), Diploma in biotechnology by the Beuth University of Applied Sciences (Germany). She has more than 15 years of experience in fermentations at laboratory and pilot scale and advanced characterization of several substrates. Over the last years she has participated in many projects at national and international level. Experience in microbiology works and analytical methods, HPLC in particular.

Related publications, patents or products:

- Turon, X., Venus, J., Arshadi, M., Koutinas, M., Lin, C., Koutinas, A. (2014) Food Waste and Byproduct Valorization through Bio-processing: Opportunities and Challenges. BioResources [Online] 9: 4, 5774-5777
- Pleissner, D.; Venus, J.: Agricultural residues as feedstocks for lactic acid fermentation. - ACS Symposium Series, Vol. 1186 "Green Technologies for the Environment" (2014) Chapter 13, pp 247–263, Chapter DOI: 10.1021/bk-2014-1186.ch013, ISBN13: 9780841230187/eISBN: 9780841230194
- Kwan, T.H.; Pleissner, D.; Lau, K.Y.; Venus, J.; Pommeret, A.; Lin, C.S.K.: Techno-economic analysis of a food waste valorization process via microalgae cultivation and co-production of plasticizer, lactic acid and animal feed from algal biomass and food waste. Bioresource Technology 198 (2015) 292–299. http://dx.doi.org/10.1016/j.biortech.2015.09.003
- Pleissner, D.; Venus, J.: Utilization of protein-rich residues in biotechnological processes. Applied Microbiology and Biotechnology 100(2016) Issue 5, 2133-2140, http://link.springer.com/article/10.1007/s00253-015-7278-6
- Pleissner, D.; Neu, A.-K.; Mehlmann, K.; Schneider, R.; Puerta-Quintero, G.I.; Venus, J.: Fermentative lactic acid production from coffee pulp hydrolysate using *Bacillus coagulans* at laboratory and pilot scales. Bioresource Technology 218(2016) 167–173, http://dx.doi.org/10.1016/j.biortech.2016.06.078

Related Projects:

- A.N.Bach Institute of Biochemistry of Russian Academy of Sciences, INBI, Moskau „Enzymes for lignocellulosic feedstocks degradation and production of sugars (C6, C5) and value added products", BMBF / DLR (Russia/Germany) – 2010-2013
- BREAD4PLA: Demonstration-plant project to produce poly-lactic acid (PLA) biopolymer from waste products of bakery industry, EU LIFE10/ENV/ES/479 – 2011-2014
- Food waste valorisation for sustainable chemicals, materials & fuels (EUBis,) – MC Member Germany, COST-Action TD1203 – 2012-2016
- Process optimization and investigations for the valorization of residues from the dairy industry (LACTOplus, Project-No: 848572), Program "Production of the future", Austrian Research Promotion Agency (FFG) – 2016-2017-

- Thermophilic lactic acid production utilizing rice bran in continuous cultivation (BranLact, Bilateral project with Beijing University of Chemical Technology/BUCT, German Call Bioeconomy International 2015) – 2016-2019

Technical equipment:

The ATB has a powerful research infrastructure. We can provide services for external research partners that include primary research data, bacterial strain collections, analytical instruments and methods, as well as application laboratories. ATB operates facilities for biotechnological and physico-chemical conversion processes to produce materials (lactic acid, a base chemical for bioplastics, and biochar) and energy (biogas) pursuing biorefinery and cascade concepts for the utilization of biomass.

R&D facilities description (including analytical suite at ATB)

- Upstream processing of raw materials & fermentation feedstocks (Equipment for pre-treatment, filtration, purification…; Micro-/Ultra-/Nanofiltration; Reverse Osmosis; Ceramic membranes; Hollow fiber membranes)
- Fermentation units from lab to pilot scale (1 litre up to 1000 litres) including continuous mode devices coupled with cell retention
- Downstream processing of raw lactates & lactic acid (Softening, Electrodialysis, Ion exchange cations/anions, Decolorization, Evaporation) which could be used/adapted for other bio-based products, i.e. organic acids
- Analytical methods for the main parameters of bioprocesses (e.g. cell numbers, dry weight biomass, strain characterization, HPLC for sugars and acids, IC, AAS, element analyzer, GC, ICP, Bio-LC, fibre analytics…)
- Pilot plant for lactic acid (own facility at ATB site for the demonstration of the entire process chain from the raw material(s) via bioconversion/fermentation up to final product) http://www.atb-potsdam.de/en/institute/about-us/research-infrastructure/pilot-plant-lactic-acid.html

From plant raw or waste materials lactic acid, a basic chemical for bioplastics, is produced in a continuous fermentation process. The practical scale of the mentioned pilot plant enables scale-up and optimization of the entire process: from testing various feedstocks to conversion and the production of product samples with specific characteristics. The aim is to make the process on basis of a wide range of raw materials stable and economically competitive.

PARTNER 4 **Name**: Agricultural University of Athens **Web page**: www.aua.gr/bioprocesses	**Type:** HE **Country:** Greece	

Brief description of the organization:

AUA serves agricultural sciences since 1920 and is involved in various research and educational projects including over 1500 research projects since 2002 mainly funded by the EC or the National Strategic Reference Framework. During the period 2010 - 2015, AUA has been involved in 40 European projects, four as the coordinator and in thirty six as partner in the corresponding consortia. The overall EU funding for AUA has been over 8 million euros. The staff members of AUA have established numerous collaborations with European/International researchers, with regular student exchange through the European Projects, bilateral actions with international universities or on-going long term collaborations.

The Department of Food Science and Human Nutrition (DFSHN) of AUA promotes the existing knowledge on Food Science and Human Nutrition through research and development programs conducted in collaboration with other Greek and foreign Universities and research institutes as well as with research and development departments of small and large Greek food industries. The Department consists of seven laboratories. The Laboratory of Food Engineering from the DFSHN is well equipped with adequate infrastructure to support a number of research projects funded by national and international research institutions. There are 3 post-doctoral researchers, 13 PhD students and 10 MSc students working in research topics related to valorization of renewable resources (including agri-industrial waste streams), industrial biotechnology, biorefinery development, platform chemicals production (including succinic acid), LCA and techno-economic studies and bioprocess design. AUA research team has received more than €1.500.000 since 2011 in the aforementioned research topics.

The group has more than 30 publications since 2013 on relevant topics. Since 2011, AUA has been involved in 3 national, 3 bilateral (collaborating with institutions in Spain, Brazil and Germany), 1 FP7, 1 Life+ research projects, and 1 project supported by the private sector (Petrobras, Brazil) focusing on biorefinery development from renewable resources and waste streams for the production of various products (including platform chemicals and succinic acid in particular). An MSc course entitled "*Food Bioprocesses and Biorefineries*" is organized by DFSHN-AUA covering topics in refining of renewable resources, industrial biotechnology and biorefinery engineering and design (18 month duration with 6 month dissertation). Dr Ioannis Kookos from the Chemical Engineering Department of the University of Patras is also member of the AUA team with expertise in process systems engineering, bioprocess design, and techno-economic evaluation.

Role in the Project: AUA will participate in PERCAL in the following topics: 1) Bioprocess development and optimisation for the production of succinic acid using MSW hydrolysates; 2) Collaboration with CMET in the development of an integrated bioprocess based on production of succinic acid via fermentation coupled with extraction of organic acids by electrolysis; 3) Development and optimization of downstream separation of succinic acid via an electrodialysis based process in collaboration with ATB; 4) Process design, techno-economic evaluation and LCA of the developed bioprocesses.

Scientific team:

Dr Apostolis Koutinas (male, Assistant Professor, DFSHN-AUA) is Chemical Engineer with PhD in biochemical engineering and 18 year experience in industrial biotechnology, biorefinery development and design of bioprocesses/biorefineries. He is currently supervising two post-doctoral researchers and three PhD students on succinic acid production, downstream separation and bioprocess optimization including life cycle analysis and techno-economic evaluation. He has 75 papers in peer-reviewed scientific journals (*h-index 21, non-self-citations*), 13 book chapters, 1 edited book and more than 90 international conferences and workshops. He has participated in 15 research projects since 2011 and serves as Associate Editor in *Biochemical Engineering Journal* (Elsevier). He is currently Guest Editor in a special issue on "*Advances on Biorefinery Engineering*" in the Biochemical Engineering Journal.

Dr Seraphim Papanikolaou (male, Associate Professor, DFSHN-AUA) is Agronomist with 2 MSc Diplomas and a PhD in fermentation technology. He has expertise in fermentation technology and bacterial metabolism. He has participated in 20 projects, published 85 papers in peer-reviewed journals (*h-index 35, non-self-citations*) and serves on the editorial board of 5 international journals.

Dr Ioannis Kookos (male, Associate Professor, Department of Chemical Engineering, University of Patras) is Chemical Engineer with MSc and PhD in Process Systems Engineering. He has expertise in the development of mathematical programming formulations for biorefineries including bioprocesses. He has 50 papers in peer-reviewed scientific journals with expertise in (bio)process design, techno-economic evaluation, modeling and control.

Mrs Chrysanthi Pateraki (female, PhD student) is Biologist with MSc in wine making and currently a PhD student at AUA to graduate at the beginning of 2017. She has carried out research on optimization of succinic acid production focusing on bioprocessing aspects and elucidation of microbial metabolism using spent sulphite liquor and other renewable resources as feedstock. She has closely collaborated with CMET in the development of the integrated succinic acid production system based on electrolysis membrane extraction coupled with fermentation.

Dr Vasiliki Kchrimanidou (female, post-doctoral researcher) is an Agronomist who acquired her Diploma from the DFSHN. She concluded her PhD at AUA in 2016 and has significant expertise in industrial biotechnology and biorefinery development

Related publications, patents or products:

1) Alexandri M, Papapostolou H, Vlysidis A, Gardeli C, Komaitis M, Papanikolaou S, Koutinas AA, 2016. Extraction of phenolic compounds and succinic acid production from spent sulphite liquor. *Journal of Chemical Technology and Biotechnology*, DOI: 10.1002/jctb.4880
2) Koutinas AA, Yepez B, Kopsahelis N, Freire DMG, Machado de Castro A, Papanikolaou S, Kookos IK. 2016. Techno-economic evaluation of 2,3-butanediol production from renewable resources. *Bioresource Technology*, 204: 55-64.
3) Alexandri M, Papapostolou H, Komaitis M, Stragier L, Verstraete W, Danezis GP, Georgiou CA, Papanikolaou S, Koutinas AA. 2016. Evaluation of an integrated biorefinery based on fractionation of spent sulphite liquor for the production of an antioxidant-rich extract, lignosulphonates and succinic acid. *Bioresource Technology*, 214: 504–513
4) Pateraki C, Ladakis D, Stragier L, Verstraete W, Kookos I, Papanikolaou S, Koutinas A, 2016. Pretreatment of spent sulphite liquor via ultrafiltration and nanofiltration for bio-based succinic acid production. *Journal of Biotechnology*, 233: 95–105.
5) Koutinas AA, Vlysidis A, Pleissner D, Kopsahelis N, Garcia IL, Kookos IK, Papanikolaou S, Kwan TH, Lin CSK. 2014. Valorization of industrial waste and by-product streams via fermentation for the production of chemicals and biopolymers. *Chemical Society Reviews* 43:2587-2627.

Related Projects:

Project acronym/description.	**Scope**	**End Year**	**Website**
FOODPRINT: Development of an integrated strategy for reducing the carbon footprint in the food industry sector	Life+ 2013	2017	www.foodprint.gr
Production of enzymes and building blocks (1,3-propanediol, 2,3-butanediol and fumaric acid) for petrochemical production from industrial residues	Industrially funded R&D	2017	-
BRIGIT: New tailor-made biopolymers produced from sugars obtained from lignocellulosic residues for flame retardancy highly demanding applications.	FP7-KBBE-SME	2016	www.brigit-project.eu/

BIO4OIL: New bioprocess for microbial oil from crude glycerol and cellulosic sugars	Greece-Germany bilateral project	2015	-
Nutri-Fuel: Development of novel bioprocesses for the production of biofuels from food industry waste streams	National funded project	2015	-

Technical equipment:

- The DFSHN-AUA has the adequate infrastructure to carry out submerged fermentations, qualitative and quantitative analysis of renewable resources, microbiological techniques and techno-economic evaluation.
- 7 bioreactors (1-7 L working volume), 8 shakers, 2 GC-FID, 1 GC-MS, 2 UV-Vis spectrophotometers, 1 microplate reader, 3 HPLC units with RI, UV and diode-array detectors, 1 FPLC, 5 incubators, 2 rotary evaporators, 8 balances with accuracy of 0.0001 g and 0.01 g, laboratory freezers (-40 °C), 7 water baths, freeze dryer, 3 centrifuges, 3 laminar flow cabinet, 3 microscopes and various generic equipment for microbiological techniques.
- one semi pilot scale spray dryer, UF/RO membrane system, double effect rising film evaporator and high pressure homogeniser.

PARTNER 5 **Name**: Center for Microbial Ecology and Technology **(**CMET). **Web page**: http://cmet.ugent.be	**Type:** HE **Country:** Belgium	

Brief description of the organization:

The **Center for Microbial Ecology and Technology** (CMET), formerly **LabMET (**www.cmet.ugent.be) is part of the Faculty of Bioscience Engineering at Ghent University (UGent). CMET houses a group of 80 fulltime personnel of which ~25 at a postdoctoral level. CMET is specialized in the fields of applied microbial ecology, microbial electrocatalysis, host microbe interactions, treatment and resource recovery of water and waste and biomaterials and nanotechnology. The principal research strategy at CMET aims at understanding phenomena of microbial ecology in order to manage the vast metabolic capabilities of complex microbial communities to solve practical problems. The group is also involved in a series of environmental biotechnology projects ranging from nitrogen recovery processes, anaerobic digestion, (trace) metal recovery and so on. CMET has always been a leading research group in the field of anaerobic digestion (AD). This is reflected in the number of publications on ADs (> 100) and several spin-off companies (e.g. Organic Waste Systems-OWS, EPAS, BIOTIM) and the coordinating role CMET takes in e.g. ManureEcoMine (FP7, 2013-2016). Since 15 years CMET also houses a (bio)electrochemistry group, with extensive experience in membrane electrolysis e.g. for sulphide and nitrogen recovery as well as production of caustic from waste streams. This experience puts CMET in an excellent position to perform the tasks within PERCAL. Furthermore, CMET has extensive experience in participation in EU collaborative projects as well as acting as coordinator e.g. for ManureEcomine.

Role in the Project: CMET is responsible for the development of a membrane electrolysis system coupled to the AUA succinic acid fermenter. The process will be optimized in terms of operational parameters, and in a second phase be implemented at pilot scale.

Scientific team:

Prof. Dr. Ir. Korneel Rabaey (M) is full professor and Head of the Department of Biochemical and Microbial Technology at Ghent University. Until mid 2011, he was senior lecturer and incoming director/founder of the Centre for Microbial Electrosynthesis at The University of Queensland where he recently received an Honorary Professorship. He is internationally recognized for his work in the field of bioelectrochemical systems, particularly with respect to novel process development. He has published over 150 papers in this field (>100 A1/international peer reviewed, > 12000 citations, H-index 46) for which he was recently designated as "ISI Highly Cited Researcher". Prof. Rabaey has driven technology development, by better integrating electrochemical concepts in the microbial or environmental engineering driven field.

Mr. Stephen Andersen is a graduating PhD student at CMET, and was the key developer of the membrane electrolysis approach for succinic acid extraction.

Related publications, patents or products:

- Desloover, J., De Vrieze, J., Van de Vijver, M., Mortelmans, J., Rozendal, R. and Rabaey, K. 2015 Electrochemical Nutrient Recovery Enables Ammonia Toxicity Control and Biogas Desulfurization in Anaerobic Digestion. *Environmental Science & Technology* 49(2), 948-955.
- Andersen, S.J., Candry, P., Basadre, T., Khor, W.C., Roume, H., Hernandez-Sanabria, E., Coma, M. and Rabaey, K. (2015) Electrolytic extraction drives volatile fatty acid chain elongation through lactic acid and replaces chemical pH control in thin stillage fermentation. Biotechnology for Biofuels 8.
- Logan, B.E. and Rabaey, K. 2012 Conversion of wastes into bioelectricity and chemicals by using microbial electrochemical technologies. *Science* 337(6095): 686-690.
- Rabaey, K. and Rozendal, R.A. 2010 Microbial electrosynthesis: revisiting the electrical route for bioproduction. *Nature Reviews Microbiology* 8: 706-716.
- Andersen, S. J., T. Hennebel, S. Gildemyn, M. Coma, J. Desloover, J. Berton, J. Tsukamoto, C. Stevens, and K. Rabaey. 2014. Electrolytic Membrane Extraction Enables Production of Fine Chemicals from Biorefinery Sidestreams. Environmental Science & Technology 48:7135-7142.

Related Projects:

Project acronym/description.	**Scope**	**End Year**	**Website**
ELECTROTALK – impact of electricity on microbial metabolism, creation of interfaces.	ERC-StG	2017	www.electrotalk.org
MRP Ghent Bio-Economy – fermentation of biorefinery sidestreams	BOF-UGent	2016	
ProEthanol – fermentation of vinasses for production of carboxylic acids	EU FP7	2014	www.proethanol.eu
Scale up of carboxylic acid production and extraction from thin stillage	IOF	2017	
Microelectronics for Microbiology – development of intelligent steering	FWO	2019	

Technical equipment:

CMET facilities are divided between analytical, microbial and molecular laboratories, temperature controlled labs, host interaction laboratories and a technology hall were several lab-scale reactors are available from millilitres to litres scale. The group is equipped with a broad range of analysis tools used in the determination of physicochemical and biochemical parameters (e.g. GC, HPLC, IC, AAS), microbial parameters (e.g. laminar

flows, anaerobic chambers, flow cytometers, multiwall spectrophotometers), as well as the molecular characterization, identification and quantification of microbial species or groups in diverse communities. Moreover, CMET has a broad knowledge and experience in reactor technology (e.g. SBR, CSTR, UASB, RBC, MFC, MEC) including also 5 fully automated, 5-L fermenter systems. Also due to its high activity in electrochemical research, CMET is provided with different tools for electrochemical characterization (e.g. ~60 potentiostats, rotating disc electrodes and a wide variety of test cells).

PARTNER 6. **Full name**: FUNDACION CENER-CIEMAT **Web page**: www.cener.com	**Country:** Spain **Type:** RTO	CENER ADItech

Brief description of the organization:

The National Renewable Energy Centre (CENER) is a technology centre specialised in applied research, and the development and promotion of renewable energies. It is highly rated and has acknowledged national and international prestige. CENER, as a foundation, is an organisation destined to support renewable energies sector competitiveness in order to help companies to reach their R&D needs: It offers high added value services and activities and transferring technology by means of a cutting-edge technological infrastructure, with the most modern laboratories and facilities in Europe.

CENER´s Biomass Energy Department is composed of a multidisciplinary staff, over 20 people, some of them with more than ten years' experience in the Bioenergy Sector and specific knowledge in biology, chemistry, agronomy and chemical engineering. There is also a full equipped laboratory with specific areas dedicated to: chemical processes, microbiology, fermentation processes and analytical techniques and the Second Generation Biofuel Centre (CB2G) which is a research infrastructure devoted to advanced biofuels and bioproducts process development. Since its foundation in 2002, it has been developing R&D activities in the bioenergy sector. These activities range from the evaluation of biomass potentials to conversion using different routes (biochemical and thermochemical) and also including the lifecycle analysis of all these processes.

Other relevant information:

- CENER's Biomass Department is an associated member of the BIC Initiative.
- CENER's Biomass Department is a member of the EERA Bioenergy Joint Programme Steering Committee

Role in the Project: CENER participates in WP1 in characterization of MSW samples and fermentation by-products; in collaboration with IMECAL, Validation of MSW as fermentation feedstock. In WP4, CENER will be leading the WP4 and production of new biosurfactants from amino acids and lipids. Contributes to WP7 (LCA and techno-economic assessment) and WP9 (Business plan, exploitation, dissemination and communication activities).

Scientific team:

Dr Javier Gil (M) took his PhD in Science (Chemical Engineering program) in 1999. He is the Director of the Biomass Department of CENER (National Renewable Energy Centre of Spain) since September 2015 coordinating the activities in several bioenergy fields: resources evaluation, sustainability studies, biofuels, bioproducts and thermochemical applications (torrefaction and gasification). Since 2002 to 2015 he also worked at CENER, prior he has occupied several R&D positions in Fraunhofer UMSICHT Institute and University of Zaragoza working in projects focused on biomass gasification and tar cleaning. In all these R&D organisations he has had the opportunity of coordinating and participating in several RTD - Biomass related projects, both

National and European funded

Ms Irantzu Alegria (F) holds a MSc in Biology by the University of Navarra. As Researcher of the Biomass Department since 2002, she has been working in second generation and advanced biofuels and biorefineries. Her main expertise is in biochemical conversion of biomass into biofuels and/or bio-based products, completed with a wide knowledge of analytics and biological conversion. She has participated in RTD projects focused on biochemical conversion of biomass into bioenergy and/or bio-based products. She also actively participated in the design of a pilot plant for biochemical conversion of biomass located in Aoiz (Navarra). She coordinates the technical work and monitoring the work from several EU funded (H2020, Interreg, etc.) as well as national and regional projects.

Ms Ibai Funcia (M) has a Degree in Chemistry and an Interuniversity Master's degree in Project Management. Since July 2003 has been working in CENER's Biomass Energy Department as a Biomass Researcher. Main research lines are thermo-chemical route: biomass torrefaction, biomass combustion (ash behaviour at high temperature) and biomass gasification (tar cleaning, sampling and analysis), biochemical route: biomass pretreatment, enzymatic hydrolysis and fermentation, biomass and Bio-fuel characterization. Chromatography methods development. Laboratory quality and assurance procedures.

Ms Mercedes Munárriz (F) holds a MSc in Biology by the University of Navarra. As Researcher of the Biomass Department since 2003, she works on the Development and Coordination of R&D Projects related to biodiesel production from microalgae, microalgae cultivation as well as in upscaling of biochemical processes. She has **Ms**

Ms Laura García (F) holds a MSc in Biology by the University of Navarra, and a master degree in Biomedical Investigation. From September 2009 to September 2010, she was research fellow at the Microbiology Dpt. in the University of Navarra Clinic. As Researcher of the Biomass Department since 2011, she works on the Development and Coordination of R&D Projects related to bioethanol production from residues and lignocellulosic biomass, syngas fermentation, as well as in microbiological processes for obtaining compounds of interest. She has developed his professional career both in laboratory and pilot plant, specializing in processes of enzymatic hydrolysis, fermentation as well as characterization of biomass.

Five relevant publications, patents or products:

- Alegría, I.; Garcia, L.; Ortigosa, I; Del Campo, I.; Sánchez, D.; Munárriz, M.; Iruretagoyena, J. Development of a biorefinery concept design for the Navarre region. 23rd European Biomass Conference. Vienna. June 2015.
- Toledano. A. Alegria, I, Labidi, J. Biorefining of olive tree (Olea europea) pruning. Biomass & Bioenergy. 2013.59 (503-511)
- I. del Campo; I. Ortigosa; I. Alegria; I. Echeverría. Experimental study of using butanol as an additive for reducing the vapor pressure of ethanol-gasoline blends. XIX ISAF- International Symposium of Alcohol Fuels. Verona. Italy.(October 2011).
- Funcia, I. Alegria, F.J. Iruretagoyena, M. Munárriz, M. Piera, F.M. Reyes-Sosa and I. Echeverría. Validation of a high throughput analysis of microalgae carbohydrates for bioethanol production "18th European Biomass Conference- Visual presentation. Lyon ,May 2010
- I. del Campo, I. Alegría, M. Zazpe, M. Echeverría and I. Echeverría. "Diluted acid hydrolysis pretreatment of agro-food wastes for bioethanol production". Industrial Crops and Products Vol 26 (2006), 214-221

Five Relevant Projects:

Project acronym/description.	Scope	End Year	Website
BUTANexT. highly efficient production process to convert sustainable feedstocks in the next generation of	H2020	2018	www.butanext.eu

biobutanol through and enzymatic hydrolysis and fermentation process			
BIOrescue. develop and demonstrate a new innovative biorefinery concept based on the cascading use of spent mushroom substrate (SMS) supplemented by wheat straw and other seasonal underutilised lignocellulosic feedstocks.	H2020	2019	www.iorescue.eu
AMBITION. develop a long-term joint European Community Research and Innovation Agenda on the integration of biofuels production and surplus grid electricity valorisation.	H2020	2019	Not available yet. http://cordis.europa.eu/project/rcn/206587_en.html
CYCLAG "A network of technology centers to develop an algae-based biorefinery". develop and validate the technological processes to obtain biodiesel from cultivating microalgae	Interreg	2019	hhttp://www.cyclalg.com/en/presentacion/
BIOPEST: Research study on fermentation processes for Biopesticides production at lab and pilot scale.	Regional project funded by the Navarra Government	2016	Not availabe

Technical equipment:

CENER owns a pilot plant for the production of second generation and advanced biofuels and bioproducts. This facility is called CB2G (Second Generation Biofuel Center). The general objective of this infraestructure is to develop and validate processes and components through R&D projects focused on pilot scale as an intermediate step towards industrial scale-up. The CB2G include several Process Development Units (PDUs) as pre-treatment, torrefaction, gasification and biochemical processes units. Some of CB2G capabilities include flexibility to handle a wide range of biomass feedstocks, ability to incorporate and test equipment from partners or third-parties into whole conversion processes and changing configurations and flexible scaling-up: from pilot to semi-industrial. Therefore, by means of this facility, CENER not only intends to produce biofuels but also other biobased products that could be interesting for the development of the biorefinery concept.

PARTNER 7 **Name**: Tbw research GesmbH (TBWR) **Web page**: www.tbwresearch.org/	**Type:** SME **Country:** Austria	tbw RESEARCH

Brief description of the organization: Tbw research (TBWR) is a R&D SME based in Vienna, founded in Nov. 2013 as a development spin off company of the IC-group, a highly experienced technical and engineering consulting company active in many fields mostly international with more than 400 employees. TBWR`s focus is on applied research only especially in cooperation with industry partners. Thematic fields of R&D activities are in general mobility, energy & renewables, construction and water management. Within the energy& renewables field TBWR is deeply involved in developing integrated processes for generating value added products and energy - biorefineries. There are several ongoing biorefinery R&D projects to develop valorisation pathways for various feedstocks (such as waste of food processing, grass, Lucerne or plants extracts for various product applications). Lactic acid has been a focus of several projects of TBWR's key staff such as developing a downstream processing (from lab upto pilot scale) for recovering LA from complex solutions such as grass silage juice. Furthermore TBWR has inhouse access to a reactive distillation apparatus at pilot's scale which was designed and integrated in cooperation with its sister company CES- Clean energy solutions. There is an ongoing research initiative to develop a simultaneous esterification and separation process for Ethyl lactate by means of a reactive distillation device. This pilot equipment and background expertise will also be available for the PERCAL project to test and develop a customized processing for the organic acids produced. TBWR also coordinates a national biorefinery network to enable and optimize interaction between academia and industry actors. Key staff of TBWR represent Austria within the international expert group of the IEA Bioenergy Task 42 - Biorefining.

Role in the Project: Involved in downstream processing of fermentation broth to lactic acid with partner ATB; Development and Assessment of a reactive distillation process for producing Ethyl Lactate from the two PERCAL intermediates LA and Ethanol.

Scientific team:

Michael G. MANDL, DI (M), Senior Researcher at TBWR and leader of the Biorefining group; Renewable Resources Engineering Graduate at University of Life Science and Renewable Resources (BOKU) Vienna. M. Mandl is key scientific staff and project manager of various national R&D projects. He has a comprehensive track record and expertise in applied research for more than 17 years. His specific field in biorefining is in green biorefinery e.g. separation of lactic acid and amino acids from grass silage juice. He joined the TBWR team in Aug. 2015, after he had worked for JOANNEUM RESEARCH for more than 17 years in the applied R&D. Currently he is also involved in modelling and upscaling a reactive distillation process for ethyl lactate and validating process with pilot plant performance data. M. Mandl is Austrian National Leam Leader of the IEA Bioenergy Task 42- Biorefinig and has been active within the IEA expert group form more than 6 years so far.

Dr Patrick J. Walsh BSc. MBA. D.Phil. (M) has worked in many countries, Scotland, Northern Ireland, USA, New Zealand, Australia and latterly the Republic of Ireland. His career covers the academic world (Canterbury University, NZ and the GMIT and NUIG in the RoI) as well as research institutes Foretry Commission Northern Research Station, Scion NZ). Recently he has entered private research and development by founding a company to commercialise some of his unpublished work. He is, at present, contracted by TBW to work on Biorefinery research and development. Dr Walsh has published widely on subjects dealing with a variety of subjects including Forest Health and Protectio, Viruses as species specific biocontrol, Bacterial control Methods, the use of plant extracts e. g; Terpenes and limonene. He represented Ireland at the IEA Task 42 Biorefining from 2007-2013. He has published over 100 scientific reports and papers and presented at conferences in many countries.

Ms. Stefanie Wong (F), Junior Researcher at TBWR; MSc degree in Biology, University of Vienna. Joined TBWR Energy and renewables in 2016.

NN – PhD (M/F) in Chemical Engineering, to be deployed to PERCAL Project

Related publications, patents or products:

- TBWR/ CES is currently defining a IPR for reactive distillation and separation process for ethyl lactate
- S.H. Sharma, M. Mandl, Green Biorefining in Sustainable Bioenergy Production, Edited by Lijun Wang, CRC Press 2014, Pages 535–564, Print ISBN: 978-1-4665-0552-0, eBook ISBN: 978-1-4665-0556-8, DOI: 10.1201/b16764-30
- De Jong, E, Higson, A, Walsh, P. and Wellisch (2012) Bio-based Chemicals: Value Added Products from

Biorefineries. International Energy Agency
- De Jong, E, Higson, A, Walsh, P. and Wellisch (2012) Product developments in the bio-based chemicals arena. Biofuels, Bioprod. Bioref. Accepted June 2012
- M.G. Mandl (2010) Status if green biorefining in Europe, *Biofpr*-Biofuels, Bioprod. Bioref., DOI:10.1002/bbb.219

Related Projects:

Project acronym/description.	**Scope**	**End Year**	**Website**
Green Safer Solvent – Esterification of lactic acid and ethanol to ethyl lactate by means of reactive separation techniques.	FFG Austria	2016	
Demonstration Green Biorefinery Utzenaich plant for the recovery of lactic acid and amino acids from grass silage by means of innovative separation techniques combined with an anaerobic digester.	FFG Austria	2012	//nachhaltigwirtschaften.at/de/fdz/projekten/demonstrationsanlage-gruene-bioraffinerie-utzenaich.php
GREEN BIOREFINERY - Technical, economic and ecological optimisation of value chains by the introduction and efficient use of sustainable raw materials.	SUSPRISE joint call	2010	//nachhaltigwirtschaften.at/en/fdz/projects/susprise/green-biorefinery-technical-economic-and-ecological-optimisation-of-value-chains-by-the-introduction-and-efficient-use-of-sustainable-raw-materials.php

Technical equipment:

- Pilot scale reactive distillation plant for the production and separation of ethyl lactate, established by national funds form the Austrian Research promotion Agency, Grant
- Strategic cooperations for analytic services with several university departments

PARTNER 8 **Full Name:** Visum Ltd.(Ex. IRIS Advanced Engineerging Limited) (VISUM) /Ireland **Web page**: www.seeingnewdata.com	**Country:** Ireland **Type:** SME	

VISUM is an optical engineering company from Dublin that specializes in the development and commercialization of Near Infrared Spectroscopy (NIRS)-based portable and inline analyzers and measurement systems for process monitoring and quality control applications in the food, pharmaceutical and chemical applications. The company was established in 2014 and currently employs 9 staff from its facilities in Nexus UCD (technology park at the University College Dublin campus). In 2016, the company launched the VISUMPalm (a handheld NIR measurement instrument) and has secured first sales. An integration platform (VisumNIR) for the inline version is currently under development in cooperation with early adopter clients from the food processing industry (dairy, meat and olive oil) and pharmaceutical industries. VISUM Ltd. is an SME Champion-awarded a Phase 1 and a subsequent Phase 2 Grant under H2020's SME Instrument.

The optical engineering and smart systems teams at VISUM Ltd. have specialized expertise in sensor integration, optical engineering, chemometrics, data analytics, cloud infrastructures and data management systems.

Role in the Project:

VISUM will lead WP6 where they will develop and test novel advanced sensors for monitoring selected process parameters and will design and evaluate the feasibility of a plant level integration of the sensors with a control system. They will also collaborate in the study of the relevant parameters to be analysed in order to select the most suitable sensor technologies.

Additionally, VISUM will contribute to WP8 (LCA and techno-economic assessment) and WP9 (Business plan, exploitation, dissemination and communication activities).

Scientific team:

Dr. Timothy Kehoe. *(male)* PhD in Engineering, University College Cork, Ireland- developed an IR photometric measurement system for thermal diffusivity in anisotropic materials. Masters in Optoelectronics, Queens University Belfast. Designed an Echelle spectrometer while based at Andor Technology Ltd. Worked at Tyndall National Institute, Cork, and Catalan Institute of Nanotechnology, Barcelona, developing diffractive optical and acousto-optical metrology techniques for nanotechnology applications, in particular for characterision of nanoscale structures fabricated by nanoimprint lithography and directed self-assembly. Other research activities: quantification of ordering in self-assembled structures by image analysis techniques and optical sensors using hydrogel materials. Tim is the VISUMPalm Product Manager, as well as the Technical Lead at VISUM Ltd. and he brings vast optical integration and device instrumentation expertise.

Dr. Severine Philippe *(female)* graduated with a 1st class Hons. B.Sc. in Applied Sciences (Physics and Physics Technology) from Dublin Institute of Technology, Ireland, in 2002 and received her PhD in Physics from Trinity College Dublin in 2007, for the study of polarisation effects in semiconductor optical amplifiers (SOA). She went on to work on the development and characterization of low noise SOAs, as well as THz generation by multimode lasers and their detection using four wave mixing in SOA, both in the Semiconductor Photonics Group, Trinity College Dublin, and the School of Electronic Engineering, Dublin City University. She previously worked as a R&D technician for the plastics industry and in quality control for electronic appliances. She joined VISUM in 2014 as an optical engineer and is currently involved in projects including various microscopy techniques, planar waveguide technologies and MIR spectroscopy.

Nikolina Mikulcic *(female)* User Interface Designer. Nikolina comes from web design and development

background with more than 15 years of experience in this field and she is an expert on UI/UX design strategy and front-end development (CSS/HTML). She studied graphic design and she started to learn web design due to her deeply interested for digital world. Since joining VISUM in September 2016, she has been creating a new User Interface for the VISUM product lines. Prior to VISUM, she has an extensive experience in the online world, having worked in companies such as EuroArt93 – digital agency, BlueBOX IT, Biroval. She also was a founder of design studio In Love With Blue and she worked as Project Manager, UI/UX designer and front-end developer. She was leader numerous digital projects for different clients as Erste Bank, Unicredit Bank, Telenor Bank, Komb Bank, ZAMP etc.

Chris Dermody *(Male)* Software Developer. Bachelor's degree in Product Design from Dublin Institute of Technology. Chris excels in creative problem-solving, using the latest technologies to transform ideas into real opportunities that provide exponential value to business operations. With a keen eye for user experience, Chris builds robust, responsive applications that stand the test of time. Since joining VISUM, Chris has been hard at work developing both front and backend features that push the VISUM product lines forward. Prior to VISUM, Chris enjoyed working across the full spectrum of business, founding startups and overseeing product development throughout the lifecycle of many successful businesses.

Ekaitz Hernandez Troyas *(Male)* Fullstack Developer at Visum. Bachelor Degree in Computing Science (5 years). Ekaitz comes from startup background focusing in web developing. He has 4 years of experience in the web industry. He started to work in Dublin for a small startup related with geolocation industry. He had the opportunity to create his own startup related with the cosmetic industry. This experience has given him the opportunity to learn cutting edge technologies as well to learn other facets of a company like business development or marketing. Also he worked as a freelancer implementing solutions with the Squarespace CMS. Ekaitz always wants to face new challenges and learn new things that are out his core competency.

Méabh Garrick *(female)* Optical Engineer. M.Sc Photonics, Friedrich Schiller Universität, Jena, Germany, B.Sc Physics with Medical Physics, National University of Ireland, Galway she worked as a research scientist at the Fraunhofer IOF in Jena, Germany after completing my Master Thesis at the institute. I specialised in the area of spectrally resolved laser-induced damage testing using a high-power tunable OPO laser system as well as angle resolved scattering (ARS) from surfaces and the unique combination of both techniques. My work was presented at the OSA Interference Coatings conference (June 2016).She speak Irish, German (and English). Motivated, team-orientated person with a flair for problem solving. She was awarded a scholarship to study at the Abbe School of Photonics, Friedrich Schiller Universität, Jena where she spent almost four years.

Five relevant publications, patents or products:

1. Development and commercialisation of the VISUM instruments- NIR spectroscopy devices- in-pipe, handheld
2. Publication in relation to Innovation Management: O. Mc Nerney, A. Stachowicz, P. Czuprynski, E. Philpott, D. Kolman, D. Martin, E. Herrmann (2010). Strategic Guide to Successful Use and Dissemination of the Results of Research and Development Projects. 1st edition, Steinbeis-Edition Stuttgart 2010. ISBN 978-3-941417-27-4
3. On 08/02/2016, VISUM was awarded the Seal of Excellence form the European Commission

Five Relevant Projects:

Project acronym/description.	Scope	End Year	Website
PREVIEW- PREdictiVe system to recommend Injection mold sEtup in Wireless sensor networks. (Grant Agreement number: 636892). VISUM Ltd. is responsible for the development of the Data Acquisition System and in in developing the Signal Conditioning, its integration with the	H2020	2017	www.preview-project.eu/

ARM main board, as well as implementing a prototype.			
Phase 1 SME Instrument: An IR-spectroscopy-based platform for the rapid integration and customisation of affordable, real-time monitoring solutions for the food and pharmaceutical industry (Grant Agreement number: 684767).	H2020	2017	-
Phase 2 SME Instrument: Cloud-enabled IR spectroscopy platform for the rapid integration and customisation of affordable, real-time chemical monitoring solutions for food and pharmaceutical industries (Grant Agreement number: 731043**)**	H2020	-	-

Technical equipment:

Spectroscopic Laboratory: equipped with several instruments, namely, an FTIR-ATR (Interspectrum), two UV-VIS (PerkinElmer and Ocean Optics), one Extended-SWIR (StellarNet) and one NIR (Mightex) spectrophotometers as well as a SWIR Hyperspectral imager (Specim-Xenics). The associated staff are trained and skilled in chemometrics modelling by using a commercial suite (SOLO+MIA from Eigenvector) as well as ad hoc non-commercial chemometric tools (Matlab and Python toolboxes).

Electronics, mechanical and optical workshops: equipped with up-to-date hardware and tools: signal generators, wide band oscilloscopes, PCB design [Altium] and building tools, development kits for programming microcontrollers, CNC-milling machines for electronic and mechanical prototyping, and CAD software [SolidWorks], among others. ZEEMAX software for the design of optical systems and design setup by means of advanced optical components and facilities (optical tables, windows, lenses, lasers and light sources and related "opticalware").

PARTNER 9 **Name**: SO.F.TER. SPA (SOFTER) **Web page**: www.softergroup.com	**Type:** IND (Large Enterprise) **Country:** Italy	So.F.TeR.

Brief description of the organization:

SO.F.TER. SpA (SOFTER) is part of SO.F.TER. GROUP (www.softergroup.com) a privately held multinational company active in the development and production of thermoplastic materials for over thirty-five years. Production facilities are located in Italy, Mexico, Brazil and USA.

SOFTER is an Italian large enterprise and one of the major independent compounders of thermoplastics in Europe. SOFTER has a great expertise in material design and production of polymers-based materials and is considered the ideal technical partner in co-design activities of final products thanks to its high operational flexibility and to its tailor-made approach to the design of materials.

The company currently supplies leading European manufacturing companies active in sectors such as automotive, building, household appliances, E&E, consumer goods, interior design goods, mechanical industry, sport&leisure items and footwear.

SOFTER employs over 270 persons and is certified according to ISO 9001 and ISO 14001. Its turnover in 2015 was about 220 million euros. In its more than thirty-five-years activity as developer and producer of thermoplastics

and hot-melts SOFTER has gained a strong knowledge in the areas of:

- material formulation and design
- continuous melt-mixing/compounding processes
- polymerization processes by polycondensation
- thermal, rheological, chemical and physical-mechanical characterization of thermoplastic materials

SOFTER has an established know-how about polymerization, production and characterization of co-polyesters, polyamides and co-polyester-based hot melts. This expertise will consent to support the partners in the project.

Role in the Project: In the framework of the project SOFTER will act as industrial discussion partner in the valorisation route aimed at the production of lactic-acid-based hot-melts. The company will have a guiding/advising role and will follow the project activities, evaluating results and giving opinions/suggestions/feedback in an industrial perspective.

Scientific team:

M.Sc. Raffaella BATTELLI (F) (**Contact person**), M.Sc. degree in Chemistry at University of Bologna and 2nd level European Master in Nanotechnologies of Polymeric Materials at University of Perugia, has been working at Softer's since 2010 as a researcher and project manager and has been involved in the development of thermoplastics for engineering applications. She is in charge of the management of international cooperation research projects related to the development & production of thermoplastic materials. In this project she will coordinate SOFTER's technical team and she will be active also as a researcher.

M.Sc. Michele PONTEVIVO (M), M.Sc. degree in Chemistry at University of Florence in 1981, has worked for Montedison and is co-inventor of three patents about high performance fluorinated polymers. He has been working at SOFTER's since 2004 in several different leading positions up to the present role of Technology Manager and Factory Manager. In the project he will be the person in charge for general and technical management.

Mr. Denis Grechi (M), Secondary school diploma as Industrial Expert in Phisics, has been working in the plastic sector since 2006 covering different positions (shift supervisor of the polyester polymerization process, analytical operator and coordinator of the CQ laboratory) up to the present role of researcher. He has a background in testing and analysing polymers and thermoplastic materials and in developing new co-polyester polymers and hot-melts.

Related publications, patents or products:

- In the last eight years SOFTER has developed the technology and the expertise for the **development and production of different bio-based thermoplastic materials for multiple applications**, with a focus on their technological processability and industrial implementation.
- SOFTER has developed the **technology for converting PHA into high performance biobased thermoplastic materials:** "Durable polyhydroxyalkanoate compositions" Ref: WO 2013124361 A1 "
- SO.F.TER. GROUP is the third largest co-polyester producer in Europe. SO.F.TER. GROUP is one of the top producers of thermoplastics in Europe and its products portfolio is one of the widest in Europe.
- SOFTER has been producing for more than thirty years a **wide array of elastomers** (i.e. thermoplastic elastomeric materials) and today is one of the top players in the European market with respect to both production volumes and variety of its products portfolio. Elastomers marketed worldwide under trend names FORPRENE®, PIBIFLEX®, LAPRENE®, SOFPRENE® are currently used for the production of parts supplied to several manufacturing sectors such as the automotive, household appliances, electronics, building, footwear, interior design goods and sport&leisure items industry
- The company has been developing, producing and marketing for more than fifteen years **engineering thermoplastic materials for high performance applications,** in sectors such as automotive, household appliances, E&E, building, interior design goods and mechanical industry (e.g. POLYFOR®, TECNOPRENE®, PIBITER® and NYLFOR® technical grades).

Related Projects:

Project acronym/description.	Scope	End Year	Website
TRASH2CASH, «Designed high-value products from zero-value waste textiles and fibres via design driven technologies»,	H2020 NMP18	2018	//trash2cashproject.eu/
BioGrapPA «Bioplastics from the grapes wastes to the package»,	EIT Climate	2016	
GREEN INDUSTRIAL SYMBIOSIS, value chain of valorization of the biomasses from agro-industrial waste and on its conversion into high-added value materials	Italian project	2015	
MINANO "New high-quality mined nanomaterials mass produced for plastic and wood-plastic nanocomposites,	FP7-NMP	2013	http://minano.vtt.fi/
LOTOTUBO "Development of a new generation of piping for construction sector based on nanomaterials, new formulations and new processes in order to replicate water-repellence and dirt-resistance of lotus leaf	Italian project	2014	

Technical equipment:

SOFTER's Technology Centre includes Laboratories and Material Prototyping Facilities dedicated to R&D activities. Laboratories are equipped with instruments to perform a wide range of mechanical, thermal and rheological analysis of thermoplastic materials. The Material Prototyping Facilities are equipped with continuous and batch melt-mixers and with pilot chemical batch reactors that are used for the preparation of experimental samples and for scaling-up activities. A CAE service for material flow simulation, mold virtual prototyping and predictive structural analysis is also part of the Centre.

PARTNER 10 **Name:** Hayat Kimya Sanayi Anonim Sirketi (HAYAT) **Web Page**: www.hayat.com.tr	**Type:** IND **Country:** Turkey	

Brief description of the organization:

Hayat Kimya was established in 1937 as a textile trading company. Company has entered the FMCG industry in 1987. Today, with 5200 employees Hayat Kimya is one of the largest FMCG (household cleaning & personal care, baby–care & feminine hygiene, and paper tissue & towels) manufacturers headquartered in Turkey.

Product portfolio of Hayat Kimya can be classified in three main groups:

i) Household laundry detergent/surface cleaning products,
ii) Hygiene products,
iii) Tissue products,

Company have several global manufacturing sites including the locations in Bulgaria, Ukraine, Russia, Iran, Algeria, Egypt, Nigeria and Turkey. Hayat Hygiene Group is the fifth largest player in global diaper industry and works with 8.6 billion baby diaper unit/year production capacity. Hayat also have an overseas distribution network, and thus they have sales operations in more than 100 countries in five continents.With its Papia and Familia brands,

as well as its non-home brand Focus and economic brand Teno, the company is the 7th largest tissue producer in Europe, and the largest in the Middle East and the Balkans. Hayat's cellulose and packaging paper plant in Bosnia-Herzegovina serves both the local market and international markets. Bingo, the favorite detergent of Turkish women, is produced in the Algerian market under the brand TEST.

Even though Hayat Kimya is originally an FMCG manufacturer, also produce several key raw materials and some mid products at production sites including Turkey (e.g. SLES and LABSA for detergent industry and flexible packaging films, nonwoven material for diapers and sanitary napkins) in commercial quantities that are indispensable to the detergent and hygiene industry.

All of the products of Hayat Kimya are certified by ISO EN 9001. Hayat Kimya has also the certificates of OHSAS 18001, ISO 14001, ISO 27001 and ISO50001.

Hayat Kimya has 5411 sqm facility in Turkey for R&D activities. More than 70 researchers are working in R&D team including 13 PhD and 18 MSc. The main objective of the R&D Center is to develop environment-friendly products in the FMCG segment.

R&D spending in 2015 is nearly 10 million dollar. In 2015 Hayat R&D center was selected the best R&D Center in chemical sector by ministry of science, industry and technology of the Republic of Turkey.

Role in the Project: They will be involved as an end-user in the project for the PERCAL project outputs in three different areas: as a solvent replacement in printing, as a surfactant replacement in surface cleaners, and as a white oil replacement in cardboard hotmelts. HAYAT also participates in the activities described in WP7 and WP8 related to environmental, economical, safety, regulatory, Business plan, IPR, dissemination and technological transfer.

Scientific team:

Dr Fatih Mehmet Erguney, PhD, R&D Manager (M). Fatih Mehmet Erguney is responsible for the product and raw material research and development of nonwoven and flexible packaging at Hayat R&D Center. He is also responsible for innovation management. After receiving his BSc degree in Chemical Engineering at the Bogazici University in 2000, Istanbul, he started working for a chemical company as a Process Development Engineer. After two years, he decided to pursue his PhD degree and joined the University of Akron, USA in 2003 to study Polymer Science. Afterwards, started working for a research oriented SME, called Sampas Nanotechnology. Dr. Erguney also has a lot of experience in project management. He is managing 20+ projects at any time at Hayat R&D Center.

Dr. Ipar Nimet Uzun, PhD, Specialist Researcher (F). Ipar Nimet Uzun studied Chemical Engineering at Yildiz Technical University (1998-2002). She carried out her MSc studies (2002-2005). After she got her MSc degree in 2005 she started her PhD studies (2005-2011) which are based on preparation of drug containing polymeric microspheres for controlled drug release using supercritical fluid technology (supercritical assisted and/or anti-solvent processing). She also worked as a research and teaching assistant (2002-2011) at the Chemical Engineering Department of Yildiz Technical University.

Dr. Sebnem Dogan, PhD, Specialist Researcher (F). She becomes her BSc degree with honours in Chemical Engineering at the Kocaeli University in 2007, Kocaeli, while also during her BSc degree she participate academical research studies and straight after the university she began her academic carrier as Research Assistant at the Kocaeli University. She started her PhD in 2009, its aim was to prepare polymer blends PLA / TPU with shape memory for a potential usage for biomedical applications. From 2007-2014, her academic carrier as Research Assistant in different universities has resulted with 6 projects funded by different national and international fundings, 10 publications in popular science journals and about 25 conference proceedings in all over the world. She joined Hayat authorized R&D Center as a research scientist in 2014 to create huge amount of benefits for the future with those new materials.

Publications: No relevant publications.

Projects: No previous relevant projects.

Technical equipment:

- Rheometer, Brookfield Viscometer, Tensiometer, High Resolution Optical Microscopy, Microtome
- FTIR, GC, GPC, HPLC, GC-MS, XRF, Raman-Microscopy, Cary Eclipse Fluorometer
- Filter Pressure Test, Lab Compounder, Drainage Tester, PFI Mill, Microfluidics M-110EH-30, CT SCanner

PARTNER 11 **Name**: Covestro Deutschland AG (COVESTRO) **Web page**: www.covestro.com	**Type:** IND **Country:** Germany	

Brief description of the organization:

Covestro is a world-leading supplier of high-performance polymer materials for key industries. Business activities are focused on the manufacture of high-tech polymer materials and the development of innovative solutions for products used in many areas of daily life. The main segments served are the automotive, electrical and electronics, construction and the sports and leisure industries.

Covestro, formerly Bayer MaterialScience, is headquartered in Leverkusen, Germany which is one of its biggest sites. It has around 30 sites throughout Europe, Asia and America, eight of which are world-scale production facilities. With sales of about €12.1bn in 2015 and 15,800 employees, Covestro is among the world's largest polymer companies.

As a global leader in polyurethanes, Covestro develops and produces the components for rigid and flexible foams as well as for solid materials. Diverse applications range from mattresses and car seats to insulating materials for refrigeration appliances and buildings.

Covestro is also the leader in the development and production of polycarbonates. Lightweight, tough and easy to shape, these high-performance plastics can be used in a wide variety of ways, and have large-scale use in industries such as automotive and construction.

As a world-leading supplier of raw materials for high-value coatings, adhesives and specialty chemicals, Covestro supplies these high-value niche products to protect and enhance the appearance of anything from cars, via furniture and flooring, applications in the sports and leisure sector through to cosmetic, textile and medical products.

Covestro seeks sustainable solutions for the major challenges of our times such as climate change, resource scarcity, population change and urbanization – providing the backdrop for favorable industry dynamics and opportunities for further growth.

Role in the Project: Succinic acid can be a potential bio-based substitute of conventionally applied fossil-based adipic acid or other diacids. Covestro could exemplary use succinic acid in polyester-based polyols, resins and polyurethane dispersions (PUD). Within this project, Covestro will give meaningful insights into the quality requirements for succinic acid and the correlation between product quality and final applicability. Furthermore, Covestro will analyze and test the provided succinic acid in various applications.

Scientific team:

Mr. Thomas Görgen (M) is heading the Alliance Management function, a part of Covestro Innovation Management. He coordinates the external R&D activities including the publicly funded projects. Thomas is a polymer chemist by training and experienced in H2020 due to his former appointment as German National Contact Point for FP6 (2002-2005)

Dr. Natalie Rahmen (F), graduated in Bioengineering at RWTH Aachen is responsible for scouting of bioprocesses relevant to Covestro

Dr. Raul Pires (M) is responsible for the New Technologies function within the Business Unit Coatings, Adhesives, Specialties. He will analyze and test the provided succinic acid in various applications.

Related publications, patents or products:

Succininc Acid:

http://press.covestro.com/news.nsf/id/covestro-and-reverdia-develop-bio-based-tpu

www.plasticstoday.com/covestro-and-reverdia-partner-development-bio-based-tpu/24790357423530

http://news.bio-based.eu/reverdia-covestro-in-succinic-acid-based-tpu/

www.chemanager-online.com/en/news-opinions/markets-companies/covestro-and-reverdia-develop-bio-based-tpu-foam

Other Biobased products:

http://press.covestro.com/news.nsf/id/Bayer-MaterialScience-develops-first-coating-hardener-from-biomass

www.wageningenur.nl/en/newsarticle/New-insights-in-furanic-intermediates-and-aromatic-derivatives-from-biomass.htm

Related Projects:

Project acronym/Title	Scope	End Year	Website
ThermoBIK-Thermoaktivierbare Bio-Klebstoffe	Biobased adhesives	2017	http://foerderportal.bund.de/foekat/jsp/SucheAction.do?actionMode=view&fkz=22018212

Technical equipment:

- Succinic acid will be used as a building block for polyols, resins and polyurethane dispersions (PUD). Equipment for polymerization is available. Furthermore, we have test facilities i.e. for urethane formation of polyols with isocyanates.
- All kinds of polymer analytic is available (GPC, NMR, IR, RAMAN, DMTA,…)

PARTNER 12 **Name**: Exergy Ltd (EXERGY) **Web page**: http://www.exergy.uk.com/	**Type:** SME **Country:** United Kingdom	exergy

Brief description of the organization: Exergy is one of the UK's leading renewable energy companies with the expertise to provide services, across the globe, to develop, engineer, construct and operate projects that contribute to our goal of a low carbon, sustainable future. We have been in the industry for over 10 years, contributing to the development of the global renewable energy market. We have helped to move the sustainable energy debate from the margin to the mainstream. Operating across a range of innovative low-carbon energy technologies, our capabilities cover everything from project inception to energy generation and operation. Sustainability lies at the core of our values; balancing commercial, environmental and social concerns to address the world's energy problems is what makes us different.

We are unique in our ability to offer vertically integrated skills in a range of sustainable energy technologies. Exergy can be involved at every stage: from project design and development, through planning and financing, engineering and construction to operation. Exergy ensures schemes are designed and built to generate energy in the most economical way and with minimal environmental impact.

Exergy is also member of SPIRE and BIC.

Role in the Project: Exergy will be in charge of the socio-environmental analysis (S-LCA) and LCC as well as

will lead the WP8 where Exergy will undertake the market analysis and the development of the business plans.

Scientific team:

Dr. Fernando Centeno, (M): degree in Physics, Master Degree and PhD in Renewable Energies and Energy Efficiency. Mr. Centeno has wide experience on the engineering and consultancy field related to the building construction & energy sector. Has been involved in two FP7 coordinations focused on the renewable energy sector. As Head of the R&D Group, he has participated in more than 38 international R&D project (i.e. H2020, FP7, IEE, Life+, Eco-innovation, Eureka, Eurostars, among others).

MSc.Eng Jose Luis Molto, (M): holds a master in Chemical Engineering by Rey Juan Carlos University. He has a strong background in simulation, optimisation and modelling of chemical and biochemical processes (Aspen HYSYS, Aspen PLUS and SuperPro) along with the feasibility and economical assessments of them. He is an active member of the European Federation of Chemical Engineers (EFCE) since 2013 and he is associate member of IChemE since 2016.

MSc.Eng Rocio Roldan, (F): is Chemical Engineer by Complutense University of Madrid. Her expertise area is related to the modelling and simulation of chemical and biochemical processes and unitary operations, especially focused on the AspenOne software handling and mathematic modelling. She has been involved in projects related to the valorisation of biomass into valuable products and related to circular economy. She is an associate member of IChemE.

MSc.Eng Sreenivaasa R. Pamidi, (M): is a bachelor in Mechanical and Energy Engineering from VIT, India and holds a master's degree in MSc and PhD Renewable Energy and Environmental Modelling from University of Dundee, Scotland. He has worked on several European research projects which have focussed on his main area of expertise in Energy Harvesting Infrastructure, solar Photovoltaics, solar thermal, Energy Efficiency Optimisation, and Modelling in the Area of Renewable Energies.

Related publications, patents or products:

i) Simulation of chemical pre-treatment process and Optimisation and scale up of the system based on simulation. Liquid and gas Fischer-Tropsch fuel production from olive industry waste: fuel from waste. Exergy Ltd, report for the European Commission. 2015.

ii) Sustainable techno-economic solutions for the agricultural value chain Scale-up and design of an optimised algae process for biodiesel and biogas production. Molto Marin, Jose L. Chemical Engineering, Thesis - Madrid, 2015.

iii) Scale-up and design of an optimised biomass to liquid process (BtL) for diesel production from agroforestal wastes. Roldan Aguayo, Rocio. Chemical Engineering, Thesis. Madrid, 2016

iv) Reducing the impact of forest plant production - design of a stand-alone PV-hybrid system for powering an innovative forestry incubator. Sreenivaasa R Pamidi, Marco Hernández Velasco. (2014)

Related Projects:

Project acronym/description.	**Scope**	**End Year**	**Website**
INNOBITE – "Transforming urban and agricultural residues into high performance biomaterials for green construction".	(FP7 ENV NO. 308465)	2015	www.innobite.eu
FFW - "Liquid and gas Fischer-Tropsch fuel production from olive industry waste: fuel from waste"	(FP7 ENV, No. 308733)	2016	www.fuelfromwaste.eu

AGROCYCLE - "Sustainable techno-economic solutions for the agricultural value chain"	H2020 No. 690142	2019	-
AgriMax - "Agri and food waste valorisation co-ops based on flexible multi-feedstocks biorefinery processing technologies for new high added value applications"	(BBI JU – VC3.D5 2015)	2020	Not available yet
CloseWEEE - "Integrated solutions for pre-processing electronic equipment, closing the loop of post-consumer high-grade plastics, and advanced recovery of critical raw materials antimony and graphite"	(H2020 WASTE No 641747)	2018	www.closeweee.eu

Technical equipment: The main office of Exergy is located in Coventry, United Kingdom, within the Technology Park where around 70 different business companies are placed. Exergy utilises industry-leading software for the simulation and optimisation of chemical processes (AspenOne software engineering), along with the specialised software for the measure and evaluation of the potential environmental impacts linked to processes and products SimaPro and GaBi.

4.2 THIRD PARTIES INVOLVED IN THE PROJECT (INCLUDING USE OF THIRD PARTY RESOURCES)

There are no third parties involved in the development of PERCAL project as shown in Table 4.2 from below.

There are two beneficiaries IMECAL and CENER subcontracting several activities in the project as described in the from below.

The subcontracted activities will also respect the article 13 of the Grant Agreement and the Best Value for Money, BIOPOLIS company will be therefore subcontracting by their specialisation and experience in the biotechnology sector and in particular in the tasks detailed below:

Table 4.2: Subcontracting and third parties breakdown

Partner		Cost (€)	WP	Task	Description
Subcontracting	1.IMECAL		WP1	Task 1.3. Characterization of MSW samples	Laboratory analyses to determine the chemical composition of selected substrates
			WP1	Task 1.4. Pre-treatment and enzymatic hydrolysis improvement from PERSEO bioethanol® process	Laboratory analyses for the evaluation of pre-treatment and Saccharification processes
			WP1	Task 1.5. Validation of fermentation processes	Laboratory analyses to validate the fermentation processes for the production of target bioproducts (ethanol, LA, SA).
			WP6	Task 6.1. Study of parameters of the pre-	Study for the selection and optimization of the relevant

Partner		Cost (€)	WP	Task	Description
				treatment and fermentation process for the production of bioethanol, lactic & succinic acid and the by-products stream.	fermentative parameters for the production of bioproducts at industrial scale. Analysis and control of parameters.
			WP6	Task 6.4. Integration and adaptation of processes to a biotechnological plant configuration	Study of the integration of fermentative parameters within the appropriate working ranges in the demo plant configuration. Analysis of parameters.
Third parties		0	N/A		
Total					
Subcontracting	2.AIMPLAS	0	N/A		
Third parties		0	N/A		
Total		**0**			
Subcontracting	3.ATB	0	N/A		
Third parties		0	N/A		
Total					
Subcontracting	4.AUA	0	N/A		
Third parties		0	N/A		
Total					
Subcontracting	5.CMET	0	N/A		
Third parties		0	N/A		
Total					
Subcontracting	6. CENER		WP3	Task 3.1.Evaluation of MSW as fermentation feedstock	Screening and selection of bacterial strain for succinic acid production and first stage of fermentation media optimization assays at lab scale according to selected strain.
			WP4	Task 4.4.Synthesis of the amino-lipid biosurfactant	Consultancy and support for the selection of analytical techniques to evaluate the activity of biosurfactant samples produced.
Third parties		0	N/A		
Total					
Subcontracting	7. TBWR	0	N/A		
Third parties		0	N/A		

Partner		Cost (€)	WP	Task	Description
Total		**0**			
Subcontracting	8. VISUM	0	N/A		
Third parties		0	N/A		
Total		**0**			
Subcontracting	9.SOFTER	0	N/A		
Third parties		0	N/A		
Total		**0**			
Subcontracting		0	N/A		
Third parties		0	N/A		
Total		**0**			
Subcontracting	11. COVESTRO	0	N/A		
Third parties		0	N/A		
Total		**0**			
Subcontracting	12. EXERGY	0	N/A		
Third parties		0	N/A		
Total		**0**			

5: ETHICS AND SECURITY

5.1 ETHICS

There are no foreseen negative ethical issues associated with the implementation of the project results.

Appropriate health and safety procedures conforming to relevant local/national guidelines/legislation are followed for staff involved in this project.

PERCAL concerns the use of Municipal Solid Waste (MSW) as feedstock to develop intermediate chemical products of industrial interest. These include lactic acid, ethyl lactate solvents, succinic acid and biosurfactants.

Ethical issues are raised as part of the participation of partner from Turkey: Hayat Kimya San A.S. This applicant has confirmed to the BBI that the ethical standards and guidelines of Horizon2020 will be rigorously applied, regardless of the country in which the research is carried out.

The specific authorisations for the import/export of materials will be provided in due time after being provided by the relevant authorities. Partner HAYAT will be involved in the validation of the final products therefore they will not receive any intermediate product nor MSW.

Companies have the responsibility of collecting information on the properties and the uses of substances that they manufacture or import materials

Regarding the protection of human health and the environment from the risks that can be posed by chemicals, all the substances used in the project will be evaluated to ensure that no risks are involved in their use. And in case anyone could have any hazard property, the specific substance will be sent to the destinatary accompanied by a document describing the risk management measures to be used to make safe use of it. For partners established outside the EU this requirement will apply as for the rest of partners, in this particular project.

There will be no experimentation on animals, with human cloning, genetic modification or human embryos or stems cells as a part of the project.

5.2 SECURITY.

Please indicate if your project will involve:

- Activities or results raising security issues: NO.
- 'EU-classified information' as background or results: NO.

Although the self-assessment identified no issues, we want to summarise here how certain aspects of the project that could lead to ethical concerns are addressed in the work plan:

- Regulatory and safety-related activities have been incorporated from the outset of the project with the intention of gathering data to fuel in a further safe and compliant development phase. In terms of product, safety aspects will deal with water and waste regulations similar than other chemical industries, where disposal of solid and liquid by-products must be safety for workers and environment and best practices and security measures must be a priority to avoid uncontrolled discharges to the environment and fires. In terms of possible harm to the environment, the use of a LCA approach, which identifies environmental as well as health impact, can help identify where the maximum impact could occur providing the opportunity for process or material modifications to minimise these impacts.
- Finally, the appropriate health and safety procedures conforming to relevant local/national guidelines/legislation will be followed by staff involved in this project. In this sense, all partners follow recognized hygiene and safety rules, including chemical regulations such as REACH, and, recommendations of good practices for workers safety will be given by ARCHA when checking the pilot processes in WP7. An Ethical & societal assessment covering those as well as further aspects is also included in WP7.

All in all, the developed bio based products from biogenic faction of MSW are in fact expected to be safer for both the consumer and environment (products with less toxicity) and ethically more acceptable than their standard (synthetic) counterparts or even than some of their bio-based relatives, often agriculture land competing (sugars and oils), which ethics are questionable, leading to fulfilling the will of consumer to make responsible choice not jeoparzing our future generation’s ability to access resources and to a clean environment.

ANNEX I- PERSEO® BIOETHANOL Technology

Bioethanol production. Following the pre-treatment, the PERSEO Bioethanol® Technology developed by IMECAL and CIEMAT for producing bioethanol will be demonstrated on a real-life environment with different organic fraction of municipal solid waste (OFMSW). **The demonstration plant is equipped with all the reactors, fermenters, and separation and distillation units needed for transforming the OFMSW to bioethanol. The typical ethanol yields obtained vary from 100 – 130 liters of bioethanol per tonne of organic waste in dry bases**. The bioethanol production involves several steps:

A. The raw material is subjected to a pre-treatment which aims to eliminate the inserts contained in the organic matter (stones, plastics, metals, glass, etc.).

B. A **thermochemical dilute acid hydrolysis** of the raw material will be done in order to produce a solubilizing of the cellulose and hemicellulose contained in the MSW.

C. After the feedstock is hydrolysed, a **simultaneous saccharification and fermentation process** is done. The cellulose saccharification and fermentation of glucose to ethanol is produced in the same fermenter, adding the enzymes, yeast and nutrients suitable for efficient conversion of sugars to ethanol.

D. The fermentation sludge obtained is subjected to a **distillation process** in order to concentrate the ethanol produced in the fermentation of the organic fraction of MSW.

E. **Dehydration process** is carried out of the mixture by passing through a molecular sieve to achieve dehydration required for ethanol (> 99.5%) as a final product.

Bioethanol production

State of the art: PERCAL will use an innovative patented process based on a thermochemical pre-treatment and a Simultaneous Sacharification and Fermentation (PERSEO Bioethanol® technology) to produce biofuels from municipal solid waste.

Second generation bioethanol can be produced by two main production routes, the thermochemical and biochemical.

Figure A1. The PERSEO Bioethanol® Technology plant

In thermal process, bioethanol can be produced via gasification of the lignocellulosic feedstock at a high pressure and in absence of inert gases and further conversion of the syngas into bioethanol. In the biochemical pathways, the lignocellulosic feedstock is firstly pre-treated in order to make the cellulosic component more accessible to the cellulases for the subsequent enzymatic saccharification step. During the pre-treatment a significant portion of hemicelluloses is converted into monosaccharides, mostly xylose (a C5 sugar). C6 sugars – mainly glucose is released after the enzymatic hydrolysis step and is then fermented with or without the C5 sugars depending on the design options. Several technology options are under development including separate hydrolysis and fermentation (SHF) or simultaneous saccharification and co-fermentation (SSCF) of C5 and C6 sugars with or without onsite production of the cellulases.

The main challenges of the biochemical route concern the integration of the system which requires a coordinated effort in improving each step to achieve an overall higher efficiency and a decreased production cost of bioethanol. In that sense, a particular attention should be paid to the feedstock cost. One of the strong assumptions in developing second generation bioethanol is the possibility to process low-cost feedstock. The concept of the process for enzymatic hydrolysis of cellulose and simultaneous fermentation, so-called Simultaneous Saccharification and Fermentation (SSF) is a promising technology. It was introduced by Gulf Oil Company, US and the University of Arkansas (Gauss et al. 1976; Huff et al. 1976) and various reports on bioethanol production have mentioned the

superiority of ethanol yield and productivity using the SSF process[75,76].The sequence of steps for the SSF is virtually the same as for the separate process, except that saccharification and fermentation steps are combined in one vessel, however, the presence of yeast or bacteria along with enzymes minimizes the sugar accumulation in the vessel, consequently higher rates, yields and concentrations of ethanol are possible using SSF at lower enzyme loading. Additionally, the presence of ethanol makes the mixture less vulnerable to contamination by unwanted microorganisms[77,78].

Linked National and International R&D Activities.

With regards to the production of bioethanol from MSW, there are some examples of industrial and pilot plants but none of these projects has validated a whole process able to transform the Stabilized Organic Fraction (SOF) available in MSW into not only biofuels (bioethanol and biomethane) but also other value-added co-products such as biofertilisers. There are a few industrial plants for conversion of MSW to bioethanol. The first international waste to bioethanol demonstration plant was built by IMECAL – CIEMAT in 2007, based in the biotechnological process PERSEO® Bioethanol. The **PERSEO** project in Valencia (Spain) started to obtain bioethanol from the organic fraction of the municipal solid waste. Another **demonstration plant was inaugurated by Abengoa in 2013**. Located in Spain, the organic matter is pre-treated to produce organic fibre, which is subsequently converted into bioethanol by using a fermentation and enzymatic hydrolysis of SOF from MSW. However, their strategy is not focused on the study of the most appropriate pre- treatment. Since pre-treatment has been viewed as one of the most expensive processing steps in cellulosic biomass-to-fermentable sugars conversion.

In Sweden, **St1 Biofuels Oy has completed a waste-to-ethanol project (Etanolix)** which recycles feedstock such as biowaste and process residue from local bakeries and bread from shops that is past its sell-by date into ethanol for transport fuel. In Finland, there is another plant **(Bionolix) that produces ethanol from biowaste from shops and households**.

Outside Europe, the **first industrial-scale** municipal solid waste to biofuel facility was opened in Canada in 2014 by **Enerkem**. Despite being the first industrial scale waste-to-biofuels facility of its kind to turn household garbage into biofuels and biochemical, **their technology is based on gasification** and subsequent catalytic synthesis, differing from the novel configuration of BIOWASTE where bio-ethanol will be produced more efficiently through alcoholic fermentation. Another example is Coskata a semi-Commercial Facility in Illinois (USA) which produces ethanol from both biomass feedstock and natural gas. In July 2014, ANDRITZ received an order to supply services **for Fiberight's cellulosic ethanol plant** in Iowa, USA. ANDRITZ Group´s biorefinery is focused on the transformation of municipal solid waste (MSW) and other organic materials into cellulosic ethanol as the primary product.

Linked plants

[75] Söderström J, Galbe M, Zacchi G. Separate versus simultaneous saccharification and fermentation of two-step pretreated softwood for ethanol production. J Wood Chem Technol. 2005;25:187–202.
[76] Neves, M. A. D., Kimura, T., Shimizu, N., & Shiiba, K. (2006). Production of alcohol by simultaneous saccharification and fermentation of low-grade wheat flour. Brazilian Archives of Biology and Technology, 49(3), 481-490.
[77] Roble ND, Ogbonna JC, Tanaka H. A novel circulating loop bioreactor with cells immobilized in loofa (Luffa cylindrica) sponge for the bioconversion of raw cassava starch to ethanol. Appl Microbiol Biotechnol. 2003 Feb;60(6):671-8. Epub 2002 Dec 19.
[78] T. Montesinos et al., Production of alcohol from raw wheat flour by Amyloglucosidase and Saccharomyces cerevisiae, Enzyme and Microbial Technology, 27, 2000, 362-370

Project (Start/End date)	Main Observations/ Advantages of BIOWASTE compared to these projects
Fiberight (2105 - In operation)	ANDRITZ's technology utilizes a steam heating concept to continuously preheat and cook the feedstock **at elevated temperatures**. BIOWASTE employs an innovative pre-treatment process at low temperature in order to avoid dioxins formation and reduce the production costs
Enerkem (2014 - In operation)	Their technology is based on gasification, a thermochemical process at high temperature. **Gasification presents several disadvantages in comparison with fermentation**: refuelling is often required for continuous running of the system and, in addition it handles with residues such as ash or tarry condensates.
Etanolix (2015 - In operation)	The ethanol plant is fully integrated into the functions and logistics of St1 oil refinery in Gothenburg and has an annual production capacity of 5 million litres of advanced bioethanol.
Abengoa (Babilafuente) (2013 - In operation)	Demonstration plant which has a capacity to treat 25,000 tons of municipal solid waste (MSW) from which it will obtain up to 1.5 million litres of bioethanol for use as fuel.
Bionolix (2013 - In operation)	The products of the Bionolix process are ethanol and solid biofuel which is used to produce energy part of which is used in the plant and waste water.
Coskata (2011 - In operation)	Semi-Commercial Facility Thermochemical / Biotechnical process: Pre-treatment, gasification to syngas, fermentation and separation. Gasification involves high temperature and pollutants in contrast with the BIOWASTE biological/chemical process.
IMECAL S.A. (PERSEO Bioethanol®) (2007 - In operation)	Demonstration plant based in biotechnological process to treat the organic fraction of MSW. Has a capacity to treat 25 t/d in fermentation to produce bioethanol and a solid fuel.

ANNEX II. Support letters.

novozymes

Letter of Interest and Support

Dear Ms. Caterina Coll, Chief Operating Officer of IMECAL Company.

I undersigned Johan Mogensen, acting as Vice President of Bioenergy and Feed Application Research at Novozymes, am pleased to declare the interest and support of our company to the project proposal entitled "***Chemical building blocks from versatile MSW biorefinery***" (Project acronym: PERCAL)" that you intend to submit to the H2020 call "BBI-2016-R08 Emerging technologies for conversion of the organic content of Municipal Solid Waste and improving waste-to-chemicals value chain" and coordinated by the company IMECAL.

Novozymes is a world leader in bio-innovation and specialised in industrial enzymes and microorganisms. As an enzyme supplier for various biorefining processes we would be interested in the results of the project in order to evaluate future collaborations with PERCAL's partners. Furthermore, Novozymes is interested to be involved in the PERCAL Advisory Group in order to give them more information about the materials and the processes we are developing, in regular meetings with the consortium or parts of the consortium.

Bagsværd, 30th of August 2016

Rethink Tomorrow

FORTISCHEM

Letter of Interest and Support

Dear Ms. Caterina Coll, Chief Operating Officer of IMECAL Company.

I undersigned Ing. **Ľubomír Kukučka,** acting as the **CEO and Member of the Board of Directors** of **FORTISCHEM a. s.,** am pleased to declare the interest and support of our Company to the project proposal entitled "***Chemical building blocks from versatile MSW biorefinery***" (Project acronym: PERCAL)" that you intend to submit to the H2020 call "BBI-2016-R08 Emerging technologies for conversion of the organic content of Municipal Solid Waste and improving waste-to-chemicals value chain".

FORTISCHEM a. s. as a chemical company is interested in the results of the project in order to evaluate future collaborations or commercial agreements with PERCAL's partners.

FORTISCHEM a. s. expects to attend communication and dissemination events, as well as contributing with advice and suggestions on industrial and market requirements during the project.

Nováky, 08.08.2016

Ing. Ľubomír Kukučka
CEO and Member of the Board of Directors

CONTACT INFORMATION

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FORTISCHEM a.s.
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JUDr. KATARÍNA JANOŠKOVÁ

BIN: 46 693 874
TAX ID No.: 20 23 52 82 54
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Bussines Reg.: Insert No. Sa, 10623/R
District Court Trenčín

Bank Details: Poštová banka, a.s.
Account Number: 5302801S/6500
Account Number: 20298844/6500
IBAN: SK 22 6500 0000 0000 5302 8015
BIC: POBNSKBA

ARKEMA
INNOVATIVE CHEMISTRY

Colombes, August 31st 2016

Jean-Luc DUBOIS
ARKEMA France
Corporate R&D,
420 Rue d'Estienne d'Orves
92705 Colombes, France

Ms. Caterina Coll
Chief Operating Officer of IMECAL Company

OBJET : PERCAL – letter of support

Dear Ms. Caterina Coll, Chief Operating Officer of IMECAL Company.

I undersigned **Jean-Luc DUBOIS,** acting as the **Scientific Director** of ARKEMA FRANCE, am pleased to declare the interest and support of our company to the project proposal entitled "***Chemical building blocks from versatile MSW biorefinery***" (Project acronym: PERCAL)" that you intend to submit to the H2020 call "BBI-2016-R08 Emerging technologies for conversion of the organic content of Municipal Solid Waste and improving waste-to-chemicals value chain" and coordinated by the company IMECAL

ARKEMA as a group operates already 5 biorefineries in the world, 4 of them using vegetable oils and fatty acids as starting materials. ARKEMA generates about 1 G€ of sales from products derived from renewable resources, and is looking for new and alternative sourcing of key starting materials for future productions. Arkema is a producer of several polymeric materials including polyamides, polyesters, adhesives, coating resins… and is interested in the results of the project in order to evaluate future collaborations with PERCAL's partners. Furthermore, **ARKEMA** is interested to be involved in the PERCAL Advisory Group in order to give them more information about the materials and the processes we are developing, in regular meetings with the consortium or parts of the consortium.

Colombes, August 31st 2016

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arkema.com

Associated with document Ref. Ares(2017)1[illegible]08069 - 28/04/2017

ESTIMATED BUDGET FOR THE ACTION (page 1 of 2)

Associated with document Ref. Ares(2017)2198069 - 28/04/2017

ESTIMATED BUDGET FOR THE ACTION (page 2 of 2)

(1) See Article 6 for the eligibility conditions
(2) The indirect costs covered by the operating grant (received under any EU or Euratom funding programme; see Article 6.5.(b)) are ineligible under the GA. Therefore, a beneficiary that receives an operating grant during the action's duration cannot declare indirect costs for the year(s)/reporting period(s) covered by the operating grant (see Article 6.2.E).
(3) This is the theoretical amount of EU contribution that the system calculates automatically (by multiplying all the budgeted costs by the reimbursement rate). This theoretical amount is capped by the 'maximum grant amount' (that the Commission/Agency decided to grant for the action) (see Article 5.1).
(4) The 'maximum grant amount' is the maximum grant amount decided by the Commission/Agency. It normally corresponds to the requested grant, but may be lower.
(5) Depending on its type, this specific cost category will or will not cover indirect costs. Specific unit costs that include indirect costs are: costs for energy efficiency measures in buildings, access costs for providing trans-national access to research infrastructure and costs for clinical studies.
(6) See Article 5 for the forms of costs
(7) Unit : hours worked on the action; costs per unit (hourly rate) : calculated according to beneficiary's usual accounting practice
(8) See Annex 2a 'Additional information on the estimated budget' for the details (costs per hour (hourly rate)).
(9) Flat rate : 25% of eligible direct costs, from which are excluded: direct costs of subcontracting, costs of in-kind contributions not used on premises, direct costs of financial support, and unit costs declared under budget category F if they include indirect costs
(10) See Annex 2a 'Additional information on the estimated budget' for the details (units, costs per unit).
(11) See Annex 2a 'Additional information on the estimated budget' for the details (units, costs per unit, estimated number of units, etc)
(12) Only specific unit costs that do not include indirect costs
(13) See Article 9 for beneficiaries not receiving EU funding
(14) Only for linked third parties that receive EU funding